TOPICALIZATION AND CLASH AVOIDANCE. ON THE INTERACTION OF PROSODY AND SYNTAX IN THE HISTORY OF ENGLISH WITH A FEW GLIMPSES AT GERMAN

Augustin Speyer

A DISSERTATION

in

Linguistics

Presented to the Faculties of the University of Pennsylvania

in Partial Fulfilment of the Requirements for the Degree of Doctor of Philosophy

2008

Anthony Kroch
Supervisor of Dissertation

Eugene Buckley
Graduate Group Chairman

ACKNOWLEDGEMENTS

The basic ideas of this thesis go back to my time as Visiting Scholar at the Linguistics Department of the University of Pennsylvania, where I subsequently entered as graduate student in order to pursue these thoughts further. That was in November 2002. Anthony Kroch taught a seminar on historical syntax at that time in which I discovered the basic impetus for this work – the decline of topicalization. The other participants were Silvia Cavalcante, Daniel Ezra Johnson, Beatrice Santorini and Laura Whitton. Their comments helped me immensely in the early steps of this work.

The biggest thanks go, of course, to the supervisor of this thesis, Anthony Kroch, for his commitment in supervising this thesis. Many of the ideas expressed in this study go originally back to him. But the contribution of the other committee members, Eugene Buckley, Rolf Noyer, Donald Ringe and Jiahong Yuan cannot be valued too highly either, and I wish to thank them for their commitment. I am also grateful that I could discuss parts of the work at various stages with Werner Abraham, Brian McHughes, Ellen Prince, Marga Reis, Arnim von Stechow, Hubert Truckenbrodt and William Barry. Their comments were extremely helpful. All remaining errors are my own, of course.

As the project went on, it became necessary to conduct experiments in countries with a strong supply of German and English native speakers. The preliminary German experiment was conducted in the Phonetics Lab of the Universität des Saarlandes at Saarbrücken in summer 2004. My thanks go to William Barry who made it possible for me to use the equipment and who also lent a willing ear to discussions, further Uta Panten and Dominik Bauer, who assisted me in doing the recordings. The preliminary English experiment was conducted in the Phonetics Lab of UPenn in spring 2005; many thanks to

Maciej Baranowsky for technical support. The main experiments, whose results are recorded in this thesis, have been made throughout the year of 2006 and in the beginning of 2007 at the Phonetics Labs of UPenn, of Tübingen University and ‘on the street’. Many thanks go to Somdev Kar, Marjorie Pak and Jonathan D. Wright for their technical support. In this context I also wish to thank my mother, Dietlinde Speyer, among other things because she supplied me with a large pool of linguistically naïve German native speakers among her colleagues.

Preliminary versions of parts of this thesis were read at various conferences, PLC 27 (February 2003, Philadelphia), NWAV 32 (October 2003, Philadelphia), International Conference on Linguistic Evidence (January 2004, Tübingen), TaCoS 2005 (June 2005, Stuttgart) and Interspeech 2006 (September 2006, Pittsburgh). I want to tank the audiences of these conferences for innumerable useful hints and comments in the discussion period.

Also I wish to thank my fellow-knights of the ‘Dr.-Cardona-Happy-Hour’, Jonathan Gress-Wright, Jean-Francois Mondon and Joel Wallenberg, for their friendship and for proofreading a draft of this opus and correcting my English, if it proved to be too baroque to be of any practical use to the reader. The substantial task of proof reading the completed study was undertaken by Anthony Kroch and Jean-Francois Mondon. Jean-Francois was also extremely helpful in preparing the printed versions.

Not unsuitably, this thesis was finished a century after John Ries’ *Die Wortstellung im Beowulf* appeared in print, the first study in which the Clash Avoidance Requirement was recognized.

ABSTRACT

TOPICALIZATION AND CLASH AVOIDANCE.

ON THE INTERACTION OF PROSODY AND SYNTAX IN THE HISTORY OF ENGLISH WITH A FEW SPOTLIGHTS ON GERMAN

Augustin Speyer

Anthony Kroch

The thesis has two aims: First, to produce compelling evidence for the so-called ‘Clash Avoidance Requirement’, a condition on the clausal grid level that no equally strong clausal stresses/ accents may stand adjacent to each other; second, to demonstrate how the Clash Avoidance Requirement influenced syntactic usage throughout the history of English. The first chapter gives a brief overview over the methods and theories used.

In the second chapter the decline of object topicalization in the history of English is presented. The reason is neither a general tendency of English word order to become more rigid (topicalization stayed grammatical), nor a loss of pragmatic environments in which topicalization was felicitous (the environments stayed the same). By closer look we see that only sentences with full noun subjects are affected. The interpretation pursued in the thesis is that the loss of the V2 word order option led to situations in which topicalization would easily lead to the juxtaposition of focused element (as in ‘*beans,*

John likes, but *peas*, *Mary* likes'). Since this situation conflicts with the Clash Avoidance Requirement, language users chose not to topicalize in such cases.

Chapter 3 shows experimental evidence for the Clash Avoidance Requirement: In both English and German the participants avoided use of constructions violating the Clash Avoidance Requirement. If forced, they inserted clearly measurable pauses between the clashing accents. As a consequence of these findings, the proper treatment of metrical prominence and focal emphasis – focal emphasis understood as emphasis on an element in narrow focus – in the framework of Metrical Stress Theory is discussed. The Clash Avoidance Requirement appears here as essential condition on the relevant grid construction rules.

The fourth chapter investigates topicalization and the Clash Avoidance Requirement in Old English. Among sentences with topicalization, the variation of V2 and V3 sentences is shown not to be strictly governed by the kind of subject – pronoun subject leading to V3, lexical NP subject to V2 – but to be sensitive to the information-structural function of both topicalized element and subject. We detect a clear correlation between V2 and focus on either the subject or the topicalized element which supports the theory of a prosodic motivation for the Middle English decline presented in chapter 2.

Table of Content

List of Tables

List of Figures

1. Introduction

1.1 Overview

The main concern of this thesis is to demonstrate how a general phonological, more specifically, prosodic requirement – the Clash Avoidance Requirement (= CAR) – can influence the syntactic usage of a given language, English. So it is, on a more abstract level, about the interaction of seemingly disparate aspects of the language, namely phonology (especially prosody) and syntax. The way they interact is highly dependent on principles of information structuring, and the effects of the interaction are observable over a given time span. It is consequently fair to say that this thesis touches on four linguistic disciplines, Syntax, Phonology, Pragmatics and Historical Linguistics.

Topicalization is an exemplary case for demonstrating this interaction and the power of the Clash Avoidance Requirement, and therefore much of this dissertation will be devoted to a discussion of topicalization in the history of English. In the second part of the dissertation we will see that the Clash Avoidance Requirement is responsible for a gradual decrease in the rate of topicalization in Middle and Early Modern English to a stable, yet low, frequency. This decrease in topicalization is observable only in cases in which the loss of the verb second word order option (= V2), which happened in the same time span, leads to potential violations of the Clash Avoidance Requirement. They can occur when two full noun phrases come to stand adjacent to each other, because then both noun phrases have a certain likelihood of bearing focal emphasis. The Clash Avoidance Requirement requires in this case, viz. when there are two phrases with focal emphasis in a sentence, that they must be separated by at least one element of minor prominence. In this thesis the decline of topicalization will be attributed to the danger of CAR-violations in the wake of the loss of the V2 word order option. Alternative explanations, such as the idea that the decline in topicalization has to do with the growing rigidity of word order in English, or that the decline in topicalization is due to the gradual loss of pragmatic contexts in which topicalization was used, will be argued against.

The thesis begins with some definitions and an overview over concepts mentioned throughout the study in chapter 1. After having shown in the second chapter how the Clash Avoidance Requirement influenced syntactic usage in earlier periods of English, the third part of the dissertation will be devoted to the Clash Avoidance Requirement in present day English and German and its technical description. I will present experimental data which shows that speakers of English and German prefer to avoid uttering two foci adjacent to each other, but if they are forced to do so, they rescue the Clash Avoidance Requirement by inserting a pause. I will discuss the reasons why speakers typically choose pause insertion and not other clash resolution mechanisms. The properties of rule-governed metrical prominence and semantic focal prominence are so different on a descriptive level that focus cannot simply be reduced to being a continuation of the metrical prominence system. Moreover, different rules are used to generate them which interact in a typical way, but remain quite distinct. A focus indicator is only assigned if there is a narrow focus on a word; otherwise, rule-governed metrical prominence takes care of the assignment of prominence up to the topmost level. The Clash Avoidance Requirement holds on this topmost level, the clause level, both in the presence and absence of focus, and can be easily formalized in the framework of Metrical Stress Theory, following Hayes (1995), as a ban on nonbranching feet.

In the fourth part, I will turn to Old English and show that here also the Clash Avoidance Requirement plays a central role in the interaction between syntactic usage and phonology. This is especially obvious in a hallmark of Old English syntax, the alternation of surface V2 and V3 word order. This alternation will be shown to be governed by the CAR: As we can observe, the alternation appears in such a way that the element with the least likelihood of bearing focus always immediately follows the topicalized phrase, either the subject if it is topical (most often realized as pronoun), or the verb if the clause has a full noun phrase, non-topical subject. The former case yields V3-sentences, the latter V2-sentences. This pattern suggests that Old English syntax was not a strict V2-syntax in the fashion of Modern German, but that Old English had two subject positions for subjects of a different (information-structural) shape, and thus resembled much more Modern English syntax than the classic West-Germanic (=German) type (cf. Haeberli 2002).

1.2 Some background

As it should be useful to give some preliminary definitions of concepts and ideas that are used in this study, let me briefly introduce some relevant concepts. I will devote two sections to this end; the first section (1.2) touches on the theoretical frameworks to be applied. In this section the pragmatic dimensions are introduced that will be discussed, the model of grammar and the metrical theory which I assume, and the German field-model, whose terms we will encounter frequently. The second section (1.3) discusses more specific concepts, viz. what we mean when we say "verb second", and how it is possible at all to determine prosodic properties in written texts, even written texts of a bygone stage of the language.

1.2.1 Pragmatic dimensions

One does not need to be a functionalist to recognize that in a number of languages one of the most important factors determining surface word order is discourse structure. Latin is certainly among those languages, but so is German, and, to some extent, even a language like English (Mathesius 1928).

But discourse structure is not a unitary notion that always influences word orders in the same way. The term *information structure*, or *discourse structure*, is rather a cover term for several ways in which information can be ordered. In the 1960s and 1970s, in the wake of the teachings of the so-called Prague school (e.g. Firbas 1974), it was assumed that there is only one informational structural dimension – a 'communicative dynamism', which subsumed theme-rheme, background-focus, given-new, frame-proposition, but at present most researchers assume that there are indeed several information structural ordering principles. Let us call these principles 'pragmatic dimensions'. It is important to note here that these dimensions are not reducible to one another (as the length of a physical object cannot be traced back to its depth, for instance), but exist independently and try to order the information in their own way, in consequence sometimes coming into conflict with other dimensions, of course.

Four dimensions are of relevance here. I do not wish to imply that there are not more dimensions, but these have been selected, partly because they proved to be of importance, partly because they influence the prosody of a clause directly. They are the following:

- <u>newness</u>: old versus new information,
- <u>topicality</u>: topic versus comment (roughly = theme versus rheme),
- <u>focus</u>: focus versus background,
- <u>scene-setting</u>: scene-setting versus proposition-internal.

Newness is a rather self-explanatory concept. Information that has been previously mentioned in the discourse counts as *old* (or, as Prince 1981 calls it, *evoked*), whereas information that is mentioned for the first time counts as *new*. There are several intermediate stages, either to be conceived of as different points on a scale, as in Gundel, Hedberg & Zacharski (1993),[1] or as different entities altogether, as in Prince (1981). One

[1] The authours use a very infelicitous term for the most salient (and necessarily given) piece of information, namely 'in focus'. When a conflation of theme-rheme, given-new and focus-background happens, it is

of the indermediate stages is the status that Prince (1981) calls ‘inferable’, which means that a given entity has not been mentioned in itself before, but other entities which are typically associated with this entity, are present in the discourse universe, so that the hearer can infer it via logical or plausible reasoning. Inferable information normally patterns with old information.

Old and new information are often encoded differently; old information tends to be realized by pronouns (if felicituous reference is guaranteed or at least likely), whereas new information is realized by phrases containing ‘real’ lexical material. The fact that old information patterns with pronouns in general will prove to be relevant in the further course of this study.

Let us turn to topicality. What counts as a topic has been a matter of debate, partly because there is a great deal of terminological insecurity connected with this concept. Some studies define ‘topic’ as the element which is at the leftmost position of the sentence (hence the term ‘topicalization’ for movement of elements to the left periphery).[2] This is not the sense in which the term ‘topic’ is used here. Other studies (e.g. Chafe 1976) equate topic with old information. As I have introduced old information as an independent notion, I obviously do not follow this usage either. In this study, topic is understood in a non-structural, pragmatic sense as the entity that the sentence is ‘about’ (following Reinhart’s (1981) definition, which is the standard definition of theme in the Prague school tradition and which in the end goes back to Paul (1875:125)); the rest of the sentence adds information to this particular entity. To determine what the ‘topic’ of

---

usually rheme and new and focus that are treated as identical. To call given and salient information ‘in focus’ is extremely misleading.

[2] E.g. Halliday (1967). This stems from the idea worked out by the Prague School that the topic (or theme) regularly precedes the comment (or rheme). Cf. e.g. Mathesius (1928); Daneš (1966); Sgall, Hajičová & Benešová (1973).

the sentence is, therefore, requires a certain amount of intuition, which most people however possess. An attempt to cast these intuitions into a more formal framework has been made by Centering Theory (Grosz, Joshi & Weinstein 1995; Walker, Joshi & Prince 1998), which makes crucial use of the fact that topics are usually old information, and that topics tend to be realized by predictable syntactic means. In English, for instance, topics tend to be realized as pronouns and frequently function as the subject of the sentence; this latter property probably is true for all Indo-European languages (cf. Lehmann 1976).

Focus is strictly speaking not a purely information structural term, but rather a semantic term, because we can identify a semantic operation that is associated with the presence of focus (Rooth 1985). We can distinguish several kinds of focus, e.g. presentational focus, contrastive focus (Rochemont 1986), verum-focus (Höhle 1992), and probably more. For English, focus is associated with prominence on the focalized element, and this prominence is the highest one in the sentence (see Jackendoff 1972). This means that focus is, in contrast to e.g. old/new information or topics, explicitly marked in the linguistic output. We assume, following Jackendoff (1972) and subsequent literature, that focus is realized by an abstract [+ focus]-feature that is associated at PF with an extra layer of prominence (more detailed see section 3.4.2). Other languages use other strategies to mark focus, e.g. focus particles (e.g. Japanese), pre-specified focus positions (e.g. Hungarian), or a combination of prominence and particle (e.g. German). A presentational focus falls on an element that is new to the discourse and whose newness should be emphasized at the same time. Contrastive focus falls on elements that stand in a partially ordered set (short: poset) relation to each other as members of a set that is either evoked previously in the discourse or is evoked by the first mentioning of one of its members. Verum-focus is a very specialized type of focus; it lies on the verb and emphasizes the claim that the proposition is true. All these different kinds of focus can, in the end, be reduced to contrastive focus, as Rooth (1985) showed; in all cases of focus a set, consisting of salient entities, is evoked of which the focused element is a member. The meaning of focus can be summarized as 'it is X, and not other members of the same set, although they would have been equally eligible'.

Scene-setting, finally, is an information-structural dimension, but with a semantic side to it. As opposed to, say, concepts such as topic-comment or newness, scene-setting elements have direct implications for the truth value of a sentence (whereas, e.g., it is irrelevant for truth conditional purposes whether a given expression is thematic or rhematic, for instance). We can define scene-setting elements as elements that specify the situation under which the truth value of the proposition has to be evaluated (definition after Jacobs 2001). They do not belong to the core proposition.

Although these four pragmatic dimensions are independent of each other, there are certain typical intersections (see also Speyer 2006a). Topics are, as a rule, also old information.[3] Not all old information functions as a topic, however. New information is in most cases focused, but it need not be. Foci can be new information or old information (this is often the case with contrastive foci). A phrase can be topic and focus at the same time under certain circumstances; we will encounter the intersection between 'topic' and 'focus' in chapter 2.2 of this study. Scene-setting elements tend to be old information; new scenes can be introduced, though, and in that case these usually receive focus.

---

[3] This is true if they are not in presentational sentences of the type 'Now let's talk about X!', but it is doubtful whether 'X' really counts as topic here. Such presentational sentences are better perceived as having no topic at all

The dimensions are often in conflict with each other. This is because each dimension poses certain requirements on the linguistic output: Old information prefers to stand before new information, topics prefer to stand before their comment, foci like to stand at one of the edges of the utterance, and scene-setting elements tend to stand before the proposition. All of these ordering requirements make sense independently from the point of view of sentence processing: it eases processing if old and new information are not jumbled together but are ordered somehow (Musan 2002). It also is more sensible to first evoke the 'filecard' (= topic) and only afterwards the stuff that has to be added to this filecard (= the comment), if we want to use Heim's (1982) famous metaphor.[4] It is better, if one wants to emphasize something, to put it in a position where it coincides with one of the clausal edges and therefore can be treated as separate processing unit. And if a situation's truth value is to be evaluated, it is more practical to know the situation before hearing the material that is to be evaluated. So each dimension has a certein claim on sentence structure and order. Which one of these claims determines the shape of the output varies by cases, although languages tend to have a ranking of the dimensions (see Speyer 2006a).

1.2.2 Modularity of Grammar

I assume a modular model of grammar in the tradition of Chomsky (1995; 2001). I assume the modified T-model (or rather: Mercedes-star-model; 0-1) in which there are

[4] This is true, although there are languages which put the topic behind the comment (Hockett 1963). The important point is that both parts of the information are separated from each other in an obvious way.

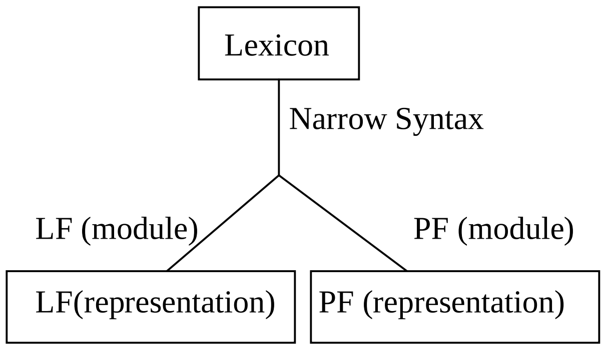
three components: Narrow Syntax, Logical Form (LF) and Phonetic Form (PF). In this study we are mostly interested in PF, in Narrow Syntax we are interested only insofar as it contributes to the PF-representation.

Narrow Syntax is the module in which material from the lexicon – at this stage represented as abstract concepts and feature bundles – is assembled and in which the first transformations take place, such as movement of the subject to SpecIP, for instance. At the place at which Surface Structure used to be in the Extended Standard Model (e.g. Chomsky 1981) is now a bifurcation that does not count as an independent level of representation. The output which narrow sysntax has produced feeds into two modules, LF and PF. LF is the module where movement operations take place that are not represented in the form of the sentence that is uttered (since the branch leading to the actual utterance is PF, and we have left this track at the bifurcation) and that concern mostly the correct semantic representation of the utterance, e.g. scopal properties. PF, on the other hand, is the module in which the syntactic structure is eventually flattened out, transformed into a linear string. Lexical Insertion takes place (see Halle & Marantz 1993) and purely phonological operations are performed, such as the assignment of prosody and the adjustment of the rhythmic structure. These are in principle not relevant for the semantic interpretation (with the apparent exception of focal emphasis, of course),[5] but they make possible the vocal production and give cues to the syntactic structure which,

[5] Focal emphasis is an apparent exception, because it reflects a semantically interpretable focus-feature (Jackendoff 1972) which is present throughout the derivation. As we will see further below (section 3.4), this property separates focal emphasis from other prosodic prominence assignments that become part of the derivation only at PF. As focal emphasis is pre-existent, compared to other stress marks, the whole prosodic parsing has to take it into account. The input into the prosodic parser, the grid assignment sub-module, is thus not simply a string but a string enriched by focus features on the relevant words. We will argue later (section 3.4.2) that the focus feature is translated into a 'credit' strong mark; this conception has the advantage that it explains also the fact why focal emphasis is always the strongest prominence in the sentence.

after reduction of the two-dimensional structure into a one-dimensional string, is no longer directly observable.

(1-1)



In contrast to Chomsky (1995), but in accordance with many other generative grammarians (see Fanselow (1991), Haider (1997), Rizzi (1997) and Haider & Rosengren (2003:206)), I assume that there are also movement operations that are not governed by strictly syntactic features, but that are discourse-structurally motivated. This implies that there are also functional projections that can host phrases with a cerrtain discourse structural status, such as the ones identified by Rizzi (1997). Movement to these projections is not warranted by Narrow Syntax, if information structure is not considered as part of the semantic representation (but cf. Asher & Lascarides (2003) for a ‘semantic’ view of information structure). For this reason it should be considered whether the place where such movement operations take place is perhaps PF rather than narrow syntax. We could view PF procedurally as consisting of several sub-modules, one in which additional, non-syntactically motivated and non-semantically interpretable movement operations take place, one in which the structure is reduced to a string, one in which Lexical Insertion takes place, one in which the rhythmical structure is assigned and one in which the well-known phonological rules of sandhi, assimilation etc. take place. But this question is beyond the scope of this study.

Also in contrast to Chomsky (1995), I assume that structures which are identified as ill-formed at PF – for instance, because they do not conform to PF-specific well-formedness conditions or because they are in conflict with the intended information structure – can be sent back to Narrow Syntax for repair. It is possible that Narrow Syntax has a standard catalogue of transformations that it performs in such a case. This cycling could perhaps be repeated until a version of the clause has been generated that satisfies PF. But this line of investigation will also not be pursued further in this study, although in the future it might be interesting to search for evidence bearing on the idea.

1.2.3 Prominence

Prominence in general is a concept which I need not define. I will, however, make a distinction between the phonological and the acoustic aspects of this concept. Acoustically, a syllable A is more prominent than a syllable B if A has higher values than B on certain measurements – pitch especially, but also volume and duration. Phonologically speaking, prominence can be represented by constructing a metrical tree and/or building a grid in which strong and weak marks are assigned; the more strong marks are assigned to a syllable, the more prominent this syllable is. The grid reflects the grouping of syllables and larger units into feet; the prominence that is assigned is dependent on the headedness of the feet. Later in this dissertation a distinction will be made between prominence that is assigned by rules and prominence that is the outcome of focus. I will distinguish these types of prominence terminologically in the following way.

On the phonological level, prominence assigned by the metrical calculus (the system that is described by rules of prosody and grid production) will be referred to as *metrical prominence* (or simply *prominence*). The rule-governed construction of metrical prominence can be disturbed by a *focus indicator*, which is prominence (or, as I will often call it in order to distinguish it from metrical prominence, *emphasis*) associated with a focus feature.

The highest prominence assigned by the metrical calculus of a given unit will be called its *prominence peak*. The highest clausal prominence will be called the *clausal prominence peak*.

On the level of phonetic representation, the term *stress* will be used for the acoustic correlate of metrical prominence, and the term *focal emphasis* or simply *focus* for the acoustic correlate of the focus indicator (for the usage of focus in this sense see e.g. Wells 2006). By use of these terms I do not wish to imply that one of these phonetic entities has fundamentally different properties from the other (e.g. that stress is louder than the rest, and focus is higher pitched than the rest, or the like); 'stress' in my usage can include pitch movement, longer duration etc. The phonetic correlate of the clausal prominence peak is called *sentence stress* or *nucleus*.

In making this distinction I follow Ladd (1996:160), who seems to be quite close to the consensus of the last few years. Ladd makes a distinction between 'normal stress' and 'focus-to-accent'. 'Normal stress' is rule-governed and thus prominence that can be calculated. Normal stress applies to all domains, including the clause. The highest stress of the clause is referred to by Ladd as *sentence stress*; Newman (1946:176) calls it *nucleus*, and this term has often been used to denote this comcept (e.g. Chomsky & Halle 1968 in their Nuclear Stress Rule). Ladd (1996:293 n.2) points out that often the term *default accent* is used. This usage is, however, due to misunderstanding of the term as he himself coined it in Ladd (1980), where it denotes a completely different concept.

The prominence associated with focus does not have an accepted designation; Ladd (1996:161) refers to it as *accent*, *focus*, or *emphasis*. This kind of prominence obviously has a semantic side to it, which metrical prominence does not have. Connected with this usage is the idea that every utterance has a focus somewhere, either a 'broad focus', meaning focus on the clause as a whole, the verb phrase or some other relatively large unit, or a 'narrow focus', meaning focus on just a word. A consequence of this perspective is that sentence stress always coincides with focal emphasis, as this is where the highest prominence of the sentence is, if the sentence or the biggest part of it is in broad focus. Ladd (1996:161) puts the matter in this way:

> Given the idea of broad focus, 'normal stress' rules can be seen as a description of where accent is placed when focus is broad.

If we have narrow focus, the rules for sentence stress cannot apply any more, as here the 'accent goes on the focused word' (Ladd 1996:161).

There are other definitions of 'stress' and 'accent'. Ladd's definition depends on Bolinger's (1961; 1972) distinction and is more or less identical to the distinction used by Sluijter (1995). For Bolinger, and the tradition of phonologists before him, *accent* is the term used for the highest prominence in a given unit, whereas *stresses* are the prominences on lower levels (the word, the phrase). He was perhaps the first to draw attention to the fact that it is exactly the highest prominence peak that often is not predictable by rules, but reflects semantic and pragmatic notions such as emphasis, newness, contrast, etc., what was termed focus soon thereafter (Jackendoff 1972). This development, of course, caused a certain terminological insecurity, as there were now two competing meanings of the term ‘accent’:

(1) highest prominence in the clause, or

(2) prominence associated with focus.

These meanings coincide exactly then when we assume that each sentence has a focus, and this is the line taken by e.g. Schmerling (1976), Ladd (1980), Selkirk (1984). Without the idea of broad focus, these meanings coincide only then when there is a narrow focus on some word. In other words: Only when a word is focused in a clause will this clause have focal emphasis. Otherwise it may have an accent in the sense of (1), but if we assume that both definitions must hold for focal emphasis, sentences without narrow focus do not have a focal emphasis at all, but simply sentence stress. It is line I will take in later sections.

One consequence of the terminological complexities sketched here is that there are many special uses of the terms *stress* and *accent*. Schane (1979:485), for instance, defines *stress* as the phonetic manifestation of prominence and accent as the underlying representation of it. In other studies, *accent* is the term used on the production side. Wells (2006) for instance uses *accent* only as the phonetic realisation of prominence associated with a pitch gesture, whereas the underlying prominence associated with focus is simply called *focus*. Sentence stress is called *nucleus*, which has the advantage that one does not have to commit oneself as to whether the nucleus is a kind of stress (rule-generated, no focus) or a kind of focal emphasis (broad focus).

1.2.4 Grid construction

The theory of grid construction used in this study is based on Metrical Stress Theory (Hayes 1995) with elements of Idsardi (1992); cf. also Halle & Idsardi (1995). The grid is constructed in the following way: each relevant element (in this study, the lowest relevant level is the word level, but the theory works the same way below the word level) is assigned a strong grid mark. In this study, asterisks are used for strong grid marks, and dots for weak grid marks. The next higher line adds alternating strong and weak marks following certain rules. This process is equivalent to the bracketing in Idsardi (1992). The lines are not simply a continuum, but (at least) three disctinct levels can be identified which serve as the domains for prominence assignment and for metrical rules. These levels are the word, the phrase, and the clause level (corresponding in conception, but not necessarily in detail, to the levels of word, phonological phrase and intonational phrase of the Prosodic Hierarchy, cf. Truckenbrodt 2007:436). In this introduction I use a simplified version with a continuous grid, for ease of explanation. The rules for the assignment of strong and weak marks are parametrized, that is, different in a limited way for different languages. In English and German, the two languages that are the focus of this study, the rules are different for the domains below the word and the domains higher than the words. The basic rule for grid construction on levels higher than the word in English is as follows, cast in terms of Metrical Stress Theory:

> *Iamb Construction Rule:*
> Assign iambs from right to left.

It is easy to see that this is an iterative version of the Nuclear Stress Rule, as we know it from e.g. Newman (1946:176) and Chomsky & Halle (1968:90). It means that the assignment process starts at the last word of the clause, assigning a strong mark to it, assigning a weak mark to the penultimate word, assigning a strong mark to the third-last word and so on, until the clause has been scanned completely. The next higher line uses the same assignment rule, and puts alternating strong and weak marks on the grid. It is not simply a copy of the line below, as the only positions that are available for assignment are the ones with strong marks on the lower line. The assignment process for this level goes on, until the clause has been parsed completely. In this fashion, line after line is added until further assignment would be vacuous, i.e. until a line is reached where only one iamb can be assigned. We will say that the parse is exhausted on this level. The relative prominence of the elements in the clause is a result of the relative number of strong grid marks each element has received. Schematically, the assignment process is shown in (1-2).

(1-2)
```
.               *
*       .       *
*   .   *   .   *
* . * . * . * . *
* * * * * * * * *
```

There are two factors that can interfere with this strict assignment. One is phrasing, the other eurhythmy. By ‘phrasing’ I mean the fact that not only the word and the clause are relevant domains for prominence assignment, but also the phrase. We thus need an intermediate level of representation. Each phrase must contain at least one strong mark (Truckenbrodt 2006), with the sole exception of functional elements such as pronouns. Besides the word, the (phonological) phrase and the clause (= intonational phrase), probably no other members of the Prosodic Hierarchy (cf. Nespor & Vogel 1986) are relevant for prominence assignment (cf. also Truckenbrodt 2006). And the ‘phrase’ I am talking about here is not necessarily the Phonological Phrase of Selkirk (1984) and Nespor & Vogel (1986), but rather a phrase that is roughly identical to a syntactic constituent: either an immediate constituent, that is, a syntactic phrase immediately dominated by VP (in its base-generated position, i.e. before movement of material to functional projections such as IP and CP), or the head of a VP, also in its base-generated position. Precedents for such a ‘direct correspondence approach’ are e.g. Cinque (1993) and Seidl (2001).

As pointed out above, I assume that there are three relevant levels for asignment of prominence: The word evel (ω), the phrase level (P) and the clause level (C). Each level consists of one or more lines. On each level, a different set of rules for grid construction applies. First, the grids for single words are constructed, by the general rules for grid construction as given in e.g. Hayes (1984:35), following Lieberman & Prince (1977: 315f., 322), and by the relevant rules for the word level. The peak mark of each word is projected on the next higher level, the starting point for phrase grid production. The relevant rules add lines to the grids of individual phrases, until the level is exhausted, i.e. until a line is reached on which only one foot can be assigned. The strong marks of the phrases are projected to the first line of the next higher level, the clause level, and serve as starting line for the production of the final grid, following the relevant rules on the clause level. Again, lines are added, until the level is exhausted. Every phrase that is dominated by VP and its extended projections IP and CP projects one strong mark onto the bottom line of the clause level (see Truckenbrodt 2006; with the exception of phrases that consist only of intrinsically weak elements, such as pronominal DPs). In this study, no higher unit than the clause is taken into account, although the sentence (= Utterance) constitutes a higher level.

The idea that for each level several lines can be constructed until the level is exhausted goes back to the notion that phrasal (and clausal) metrical prominence assignment happens cyclically (see e.g. Selkirk 1984). So the assignment process would proceed as in (1-3). Since what the metrical calculus basically does is assign feet, we may as well mark the feet in the grid.

```
(1-3)              (       .                                 *)
                   (.      * )       (  .                    *)
                    *      *            *                    *    clause level (C)
-------------------------------------------------------------------------------------
      ( .           * ) |       |           |    (    .      *)
       ( *)   (.     *) |  ( *)  | ( .    * ) | ( .    * ) ( .   *)
        *      *     *  |   *    |  *     *   |  *     *     *    *   phrase level (P)
-------------------------------------------------------------------------------------
        *   |  *  |  *  |   *    |  *  |  *   |  *  |  *  |  *  |  *   word level (ω)
      [word word word] [word] [word word] [word word word word]
```

This kind of representation takes account of two requirements on grid production that seemingly are in conflict with each other: On the one hand, there is the additive nature of metrical prominence, in the sense that every level builds on former levels, i.e. that a more prominent metrical prominence is the result of the addition of prominence marks on different levels. This implies that metrical prominence on the word level and on the phrase level must be represented in the same grid (cf. Truckenbrodt 2006), as the final resulting and audible gradation in prominence is the addition of metrical prominence marks on the word, phrase and clause level. On the other hand, there is the fact that the assignment rules are potentially different for each of the three levels. Take for instance metrical prominence in German: The rule for the assignment of metrical prominence on the word level is identical to the Latin rule, namely that the first moraic trochee, counting from the right, under extrametricality of the final syllable, receives the main prominence (Speyer 2006c). The rule for metrical prominence on the phrasal level, on the other hand, also counts from the right, but here it is iambs and not moraic trochees that are assigned – see the version of the Iamb Construction Rule above. The metrical prominence assignment rule for the clausal level, in the end, is similar to the rule for the phrasal level, but it treats verbal material at the edge as extrametrical.

We have to bear in mind furthermore that a metrical grid can be subject to another process, namely eurhythmy (cf. Hayes 1984). Eurhythmy is basically a well-formedness condition on grids; the basic rules are, freely after Hayes (1984), that the highest prominence marks should be kept as far apart as possible ('Phrase Rule'), and that in-between a strict alternation of strong and weak marks should be strived for. The grid in (1-2) would be perfectly eurhythmic. A grid like (1-3), on the other hand, would not be eurhythmic. Eurhythmy would first push the second highest mark to the first constituent (1-4a), thereby observing the Continuous Column Constraint (Hayes 1995:34ff.), then repair the equal heights of the intervening material by destressing the column which is closer to the next highest prominence peak – in that case the left of the two constituents (1-4b). Then the grid will be eurhythmic and an adequate metrical representation of an English sentence with the constituent structure given in (1-3). Note in this connection that certain function words such as the article or personal pronouns are not counted into the computation because they do not have word stress and therefore do not receive a strong mark even on the word level.

```
(1-4)  a.           .                                                   *       clause level (C)
                     *        .                  .                      *
          .          *   |    *    |             *    |          .      *       phrase level (P)
          *     .    *   |    *    |   .         *    |    .     *   .  *
          *  |  *  | *   |    *    |   *   |     *    |  * |  *  | *  | *       word level (ω)
       [word word word] [word] [word word] [word word word word]

       b.           .                                                   *       clause level (C)
                     *                           .                      *
          .          *   |    .    |             *    |          .      *       phrase level (P)
          *     .    *   |    *    |   .         *    |    .     *   .  *
          *  |  *  | *   |    *    |   *   |     *    |  * |  *  | *  | *       word level (ω)
       [word word word] [word] [word word] [word word word word]

       c.           .                                                      *
                     *                      .                              *
          .          *   |    .  |          *   |            .             *
          *     .    *   |    *  | .        *   |    .       *        .    *
          *  |  *  | *   |    *  | *  |     *   |    *   |   *    |   *  | *
       Some dark stranger gave blue flowers to the-mildly surprised girl
```

The theory as it is sketched out here is probably too simple and would need additional features if applied to other problems. But it seems to be accurate for the domain which we are mainly interested in: the domain of phrasal and clausal metrics, which I may call summarizingly *supraverbal metrics*.

It would have been possible to use Optimality Theory with the appropriate metrical formulations (for an overview over such systems in OT see e.g. Truckenbrodt 2007), but I decided to stay within the frameworks of classical autosegmental and suprasegmental theories, as a reformulation in OT terms would have no effect on the results to be described or on the explanations I will be proposing. The grid serves as input for the assignment of intonation contours, indicating the positions of the different low and high pitch accents and boundary tones (on intonational contours see e.g. Pierrehumbert 1980 for English, Féry 1993) for German. I assume that grid production and the assignment of intonational contours are two distinct processes (cf. also Truckenbrodt 2006). Therefore I will not treat questions of intonation proper (i.e. contour formation, pitch accent realization) in this study, but confine myself to the construction of the grid, as this is sufficient for the purposes of this study.

1.2.5 The syntactic field model

The *Feldermodell* (‘field model’) dates from the early years of Germanic linguistics; it was introduced in the 1820s by Simon Herling (Herling 1821; see Abraham & Molnarfí 2001), and gained momentum especially under the influence of Drach (1937). According to the most common version of the field model (cf. e.g. Höhle 1986; Grewendorf, Hamm & Sternefeld 1987; Reis 1987:147f.; Abraham & Molnárfi 2001), a sentence can be divided into the following parts which stand in the order given here:

Vorfeld	–	linke Satzklammer	–	Mittelfeld	–	rechte Satzklammer	–	Nachfeld
prefield		*left sentence bracket*		*middle field*		*right sentence bracket*		*back field*

Before the *vorfeld*, another – marked and very restricted – position (*vorvorfeld*, ‘pre-prefield’) can be introduced.

Each of these ‘fields’ has special properties:

- The verbal elements all stand in the s*atzklammern*. In main clauses the finite part of the verb is in the *left satzklammer*, the remainder of the verbal material in the *right* one. In subordinate clauses all verbal material is in the *right satzklammer*, the complementizer is in the *left* one.

- The *nachfeld* is usually filled with subordinate clauses or otherwise ‘heavy’ elements.
- Most of the non-verbal sentence material stands in the *mittelfeld*. There are no constraints whatsoever on what can stand in the *mittelfeld*, as long as it is not verbal. There are certain constraints on the order of elements, however (see e.g. Hoberg 1997, as summarizing representative of an abundant research literature).
- The *vorvorfeld* can only contain main clause connectives and material which can be shown to be left dislocated.

We are mostly interested in the *vorfeld*. The *vorfeld* in Modern German can contain exactly one constituent. There are some exceptions to that, and the further back in history one goes the more frequent these exceptions become, so that one is forced to assume that the one-constituent-only constraint of Modern German is a recent development, and that originally more than one constituent could stand before the left sentence bracket. This is going to be of immediate importance for Early German and English, and we will return to this question in section 4.3.

It can easily be seen that the *Feldermodell* translates directly into modern generative terms (cf. den Besten 1977, Vikner 1995): the *vorfeld* corresponds to SpecCP, the *left satzklammer* to the C-head, the *mittelfeld* to everything under C’ save for the – in German right-peripheral – V-head(s) and the I-head, which form the *right satzklammer*. The *nachfeld* contains IP and CP adjuncts to the right.

For Modern English, using the field model does not make much sense and does not offer great insights, although it could be done (the left sentence bracket contains all verbal material, the default filler of the *vorfeld* is the subject, although more than one phrase can stand in the *vorfeld*, and the distinction between *mittelfeld* and *nachfeld* is hard to draw as there is never overt material in the right sentence bracket). This is different for earlier stages of English in which the sentential structure shared some properties with Modern German. Therefore, terms of the field model will occasionally be used for Old and Middle English in the course of this study.

1.3 Further concepts

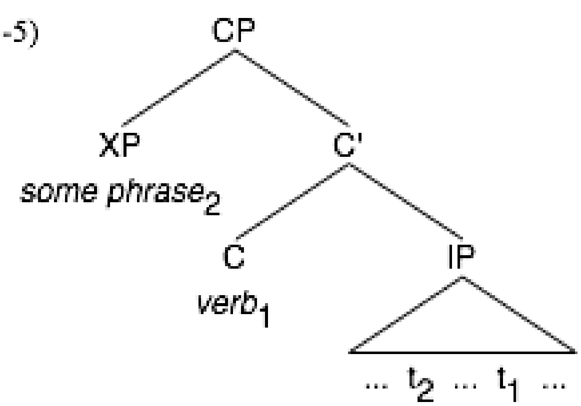
1.3.1 Verb second

In this study I will make use of the term *verb second* (V2) frequently. The usage of the label V2 has been rather imprecise in the past, and therefore it is perhaps useful to dwell a bit on this subject. V2 can be used in a more typological manner to express the property a language can have of putting the verb in the second position in the sentence, that is, the position after the first constituent. Note that whoever uses V2 in this sense does not have to commit him/herself to a specific analysis: S/he simply states that at the surface we have the verb in second position, no matter what the underlying analysis is that takes care of having the verb at exactly that spot.

A related notion is that of the *verb second constraint* which on a descriptive level says not much more than the following: Some languages (among which are the Germanic languages) show a tendency to build their sentences in such a way that the verb is in second position. The reasons for this tendency are unknown. Brandt et al. (1992) assume the presence of sentence type features that have to be saturated by movement of the verb to C and in some cases (with wh-questions and declarative sentences) also another phrase to SpecCP. Lately the hypothesis has been put forward that verb-seconding (and by that the creation of a 'vorfeld') serves to establish a topic-comment structure and the verb serves as marker which divides the sentence into these two parts (Hinterhölzl to appear). But this is of no concern for us here. The only thing to mention is that again, if one uses 'verb second constraint' on this descriptive level, nothing is said about the underlying structure.

There is however a less non-committal usage of the term. At least since Vikner (1995), ‘V2’ is often used to denote a special syntactic configuration, in which there is one functional projection above IP (which is usually referred to as CP). The V2-effect is derived by moving the verb into the head of that projection and some other constituent into the specifier projection of it (1-5). This corresponds closely to the analysis of the Modern German declarative sentence by den Besten (1977). When the term V2 is used, it is often implied that something like the structure in (1-5) is necessarily the underlying structure of any V2-sentence.

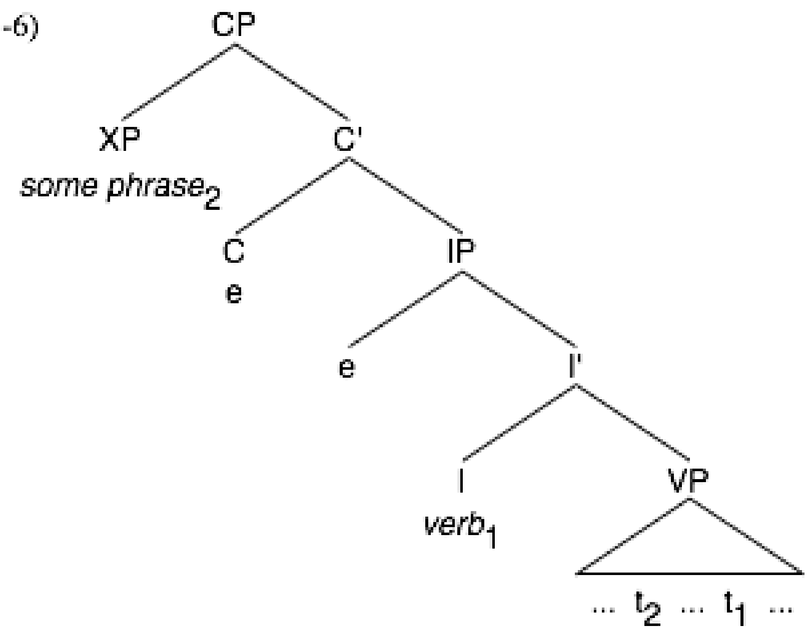
(1-5)



The problem is now, of course, that a surface V2 order can be the outcome of a variety of analyses, of which the one outlined under (1-5) is only one. For instance, a verb second order can also be the result of a structure as in (1-6).

It turns out that in Old English we have both kinds of V2: V2 by movement of the verb to C and of some phrase to SpecCP (I will hitherto refer to this kind of V2 as CP-V2) and V2 by movement of some phrase to SpecCP, but no movement of the verb from I to C and no element in the specifier position of the projection in whose head the verb has landed (e.g. Kroch & Taylor 1997; Haeberli 2002). I denote it here as IP for the ease of the exposition; we will get back to that question more precisely in part 4.

(1-6)



When I use V2 in this thesis I do not mean V2 by movement of the verb to C. For this special usage I use the term CP-V2. The structure of V2 I am mostly talking about is the version of V2 outlined in (1-6). It is important to note that this sentence structure is optional throughout the history of English (quite in contrast to CP-V2 in languages which have this structure, where it tends to be compulsory), and therefore it makes sense to speak of the ‘V2 word order option’ when talking about English. This implies, of course, that all cases of CP-V2, which was used in very limited contexts throughout the history of English (namely wh-questions, negative inversion and the like) are not covered by that term. The changes I describe do not affect CP-V2 (English has V2 in wh-questions today just the same way as it had 1200 years ago). They affect only V2 without movement of the verb to C. All instances of modern (and thereby also Old/Middle) English CP-V2 are not subject of this study.

1.3.2 The reconstruction of sentence prosody

Much of the discussion to follow hinges on the assignment of focus (and in the end focal emphasis) to several elements in the Middle English and Early Modern English texts that constitute the corpora which are used for this study (Kroch & Taylor 2000; Kroch, Santorini & Delfs 2004; Taylor, Warner, Pintzuk & Beths 2003). A problem with this investigation which will come immediately to mind is the fact that prominence is ordinarily not encoded in written texts. So how can we base any argumentation about written texts on focal emphasis?

While it is true that prominence – rhythmical and focal – is not encoded, however, it can be reconstructed. There are two facts that make this reconstruction possible: We can identify the pragmatically based focus structure of a written text, and in the case of older stages of English and German we may infer from the pragmatic analysis what the interaction of focus and prosody looked like.

It is true in general that we can analyse the information structure of any piece of text, written or spoken. This is obvious from the fact that we can read a book or a newspaper and follow the information structure without any problems, although there is no direct prosodic information available. To understand a text always means to be able follow its information structure. Now, the assignment of non-rhythmical (focal) prominence is always governed by information structure, mainly the parameters of newness and contrast. This means that we can make informed guesses as to which elements of a sentence would receive focal emphasis if spoken out loud (on this problem see e.g. Petrova & Solf 2007), just in case we can make informed guesses as to which elements are informationally focused. To identify informational foci in a written text, however, is not that difficult. If, for instance, the focus theory of Rooth (1985) is used, all one has to do is to hunt for elements that stand in contrast to other elements in the local discourse. So it is possible to identify, at least approximately, the focus structure of any extended written discourse.

How focus interacts with prosody, on the other hand, is a different matter. In living languages we can study the interaction directly. In 'dead' languages, we cannot, at least in principle, which is a possible objection to the method used in this study. And, one may object further, it is quite pointless anyway to map the focus structure of written texts to a hallmark property of spoken language, viz. prosody.

Both objections can be refuted. Let me begin with the second one. It is true that prosody is not written down, but we have to keep in mind that it is the same language faculty that generates spoken utterances and written texts. It is fair to assume that patterns of syntactic usage that manifest themselves in the spoken language can also be found in written texts of a low to middle stylisitic level (that is, excluding highly-stylized prose and poetry) since in such texts the same rules are responsible for generating sentences as in spoken language, and no other rules – rhetoric, stylistic, etc. – interfere. From this it follows that, if the normal usage of syntax in spoken language is prosody-sensitive – and this is especially true if there is optionality in the syntactic output – we may try assuming that texts of a low to middle stylistic level will show the same prosodic sensitivity. If this assumption leads to interpretable results in line with other aspects of our scientific understanding of language structure and history, we can take the assumption to be justified. Just this outcome is what we hope to present in the body of this work.

The objection that we do not know how focus interacts with prosody in a dead language is to some extent well-founded, but here we have to distinguish between languages that are really ‘dead’ – such as Sumerian, Egyptian or Hittite – and languages that may not exist in the form in which the records we are interested in are written, but for which close successor languages exist which we can study directly. Latin, for instance, is not as ‘dead’ as Sumerian, as there are several daughter and granddaughter languages of Latin in everyday’s use by almost a billion people.

I want to argue that the case is even stronger for Old English and Middle English, which are the languages on whose prosody some parts of the argumentation depend. The reason is the following: Modern German and Modern English are extremely similar with respect to the focus-prominence mapping: In both languages, focus is associated with a pitch accent on the focalized element itself. The realization of the pitch accent might differ in detail phonetically, but the basic system is the same (as one sees in comparing e.g. Pierrehumbert 1980 and Féry 1993). From this fact we can infer that focus was associated with a pitch accent also in the common ancestor of these languages, which is Proto-West Germanic. If this is so, however, we can also conclude that all stages between Proto-West Germanic and Modern English and German respectively had the same association. Pretty much the same goes for phrasal and clausal rhythmical prominence, by the way – the rules for nuclear stress or phrasal stress assignment are not identical in Modern German and English, but are so similar that they can be reduced to one another – hence it is fair to assume that Proto-West Germanic followed similar rules and constraints, too.

If these two conditions hold it should be possible to reconstruct with a fair degree of confidence both metrical prominence structure and focus indicator assignment in any text in English or German, from their respective earliest attestation on. I base my discussion in what follows (section 3.1; chapter 4) on this hypothesis.

2. Topicalization in Middle and Modern English – A prosodically induced change in syntactic usage

The second part of this dissertation is devoted to the exploration of an empirical fact, the historical decline of topicalization in English, which became evident only with the recent availability of parsed corpora of historical English (York-Toronto-Helsinki Parsed Corpus of Old English Prose (YCOE): Taylor, Pintzuk & Beths 2003; Penn-Helsinki Parsed Corpus of Middle English (PPCME): Kroch & Taylor 2000; Penn-Helsinki Parsed Corpus of Early Modern English (PPCEME): Kroch, Santorini & Delfs 2005). The rate of topicalization decreases in the course of the Middle English and Early Modern English periods. Section 2.1 is devoted simply to the demonstration of this decline. The following sections will discuss three alternative explanations for this decline, an obvious hypothesis (change in the pragmatic environments compatible with topicalization) which will be shown to be insufficient as explanation for the Middle and Early Modern English data (2.2), another simple hypothesis (decline of topicalization caused by rigidification of word order), which does not work at all (2.3), and a more complex one (complex in the sense that more parts of the grammar are involved), namely that a special kind of topicalization – double-focus-topicalization with focused full noun phrase subjects – was, for prosodic reasons, possible only as long as the V2 word order option was available. I will argue for this alternative (2.4). This explanation entails that prosody is more important than unambiguous pragmatic encoding; that this is true is shown by a comparison of German texts (with unconstrained topicalization) with their English translations (with prosodically constrained topicalization; 2.5).

2.1 The decline of topicalization

Topicalization in the non-technical sense of moving a constituent other than the subject to the left edge of the clause is one of the not-too-numerous examples of a construction that involves non-canonical word order in Modern English; that is, a word order different than S – V – O. The surface string scheme of topicalization in Modern English can be given as

X – S – V … ,

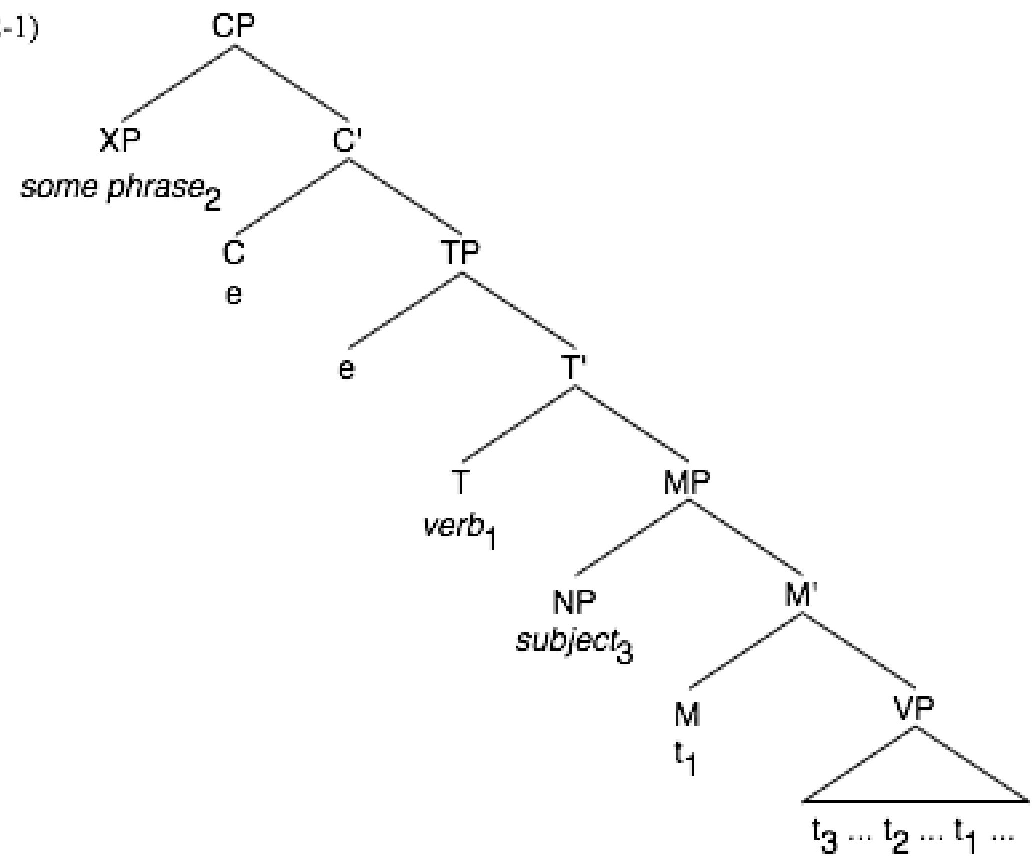
where S = subject, V = verb and X = any constituent. In Old and Middle English, topicalization could also be of the form

X – V – S … ,

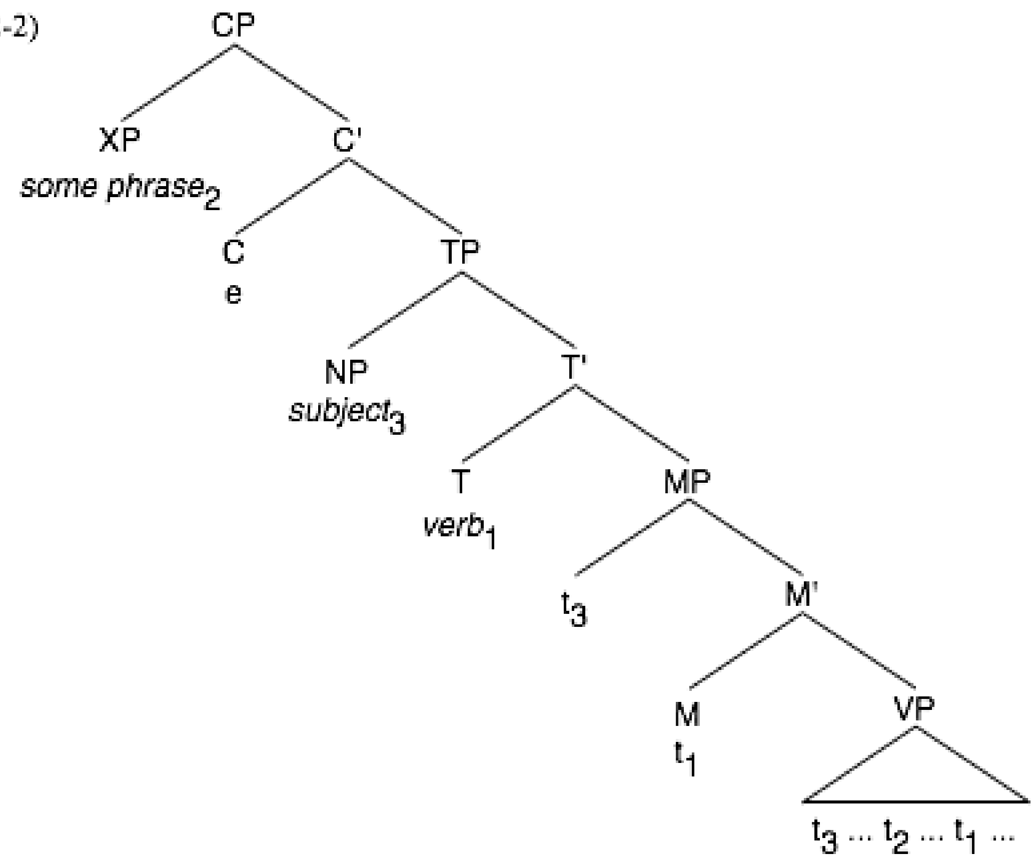
because in Old and Middle English the verb-second (= V2) option still played a role in that V2 was a common word order in declarative matrix clauses. Verb-second, which I mention here because it will become important in the further course of the dissertation, has been in its weakest form a common property of all Germanic languages, although some of the languages lost or at least modified it. In its most general form it says that the verb should occupy the place after the first constituent. It is easy to see that a sentence of the form X – V – S conforms neatly to this constraint. The structure of such a sentence would be as given in (1-6), repeated under (2-1) that is, with the verb moved not to C°, but to the highest possible projection of the I-architecture, let us say, T°. C° would be covertly filled (perhaps by a sentence type operator, as assumed by Brandt et al. 1992), so that movement of the verb to C° is impossible. Movement of the subject to SpecTP would be blocked because an empty expletive occupies that position (Haeberli 2002).

A topicalized sentence with V3 word order, the type familiar from Modern English, would have a similar structure, with the difference that movement of the subject into SpecTP would not be blocked (2-2). What I say here about the structure of topicalized sentences most certainly goes for Middle English; we will see (ch. 3) that Old English made use of identical structures.

(2-1)



(2-2)



Let us go back to topicalization. Typical Modern English topicalization cases are given in (2-3).

(2-3) a. In the afternoon, I usually go for a walk

b. Beans he likes, but peas he hates.

c. This proposal, we discussed at length

d. Pterodactylus, it is called.

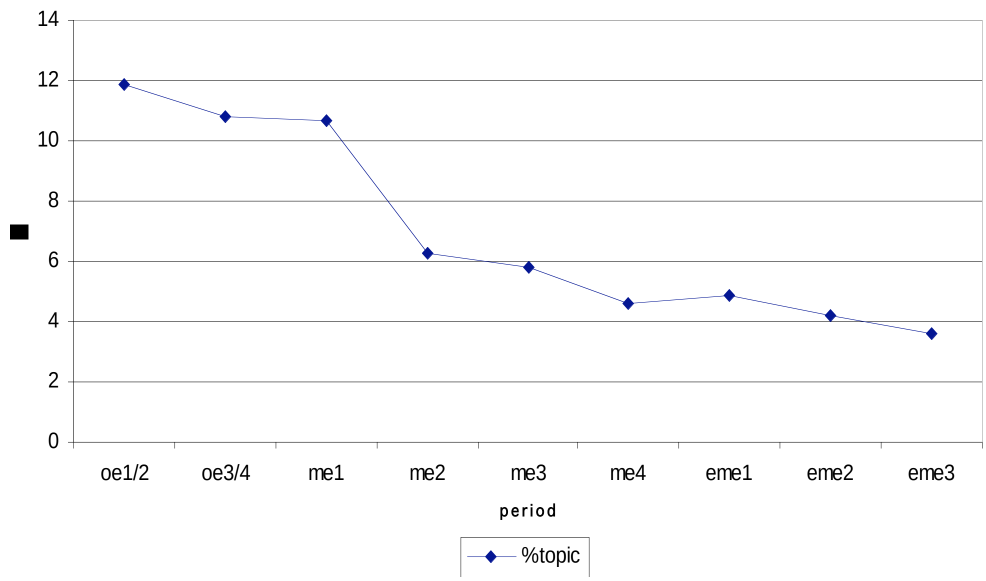
There are several types of topicalization that can be distinguished by the pragmatic-information structural and intonational properties of these respective types. Examples (2-3a-d) exemplify some cases, namely preposing of a scene-setting element (2-3a), topicalization in the stricter sense in double focus constructions (2-3b), preposing of non-contrastive, discourse anaphoric element (2-3c) and focus movement (2-3d). As the respective properties of these types will become relevant only in section 2.2, I will postpone a more detailed discussion. The only distinction which I wish to make here is the one between object topicalization (i.e. preposing of an argument of the verb, illustrated by (2-3b, c, d) and topicalization of a scene-setting element (in the form of a prepositional phrase or an adverbial phrase, see 1-3a). I will concentrate for the most part of the following on object topicalization.

If we look in the Penn-Helsinki parsed corpora of Middle and Early Modern English (Kroch & Taylor 2000; Kroch, Santorini & Delfs 2005) and the York-Toronto-Helsinki Parsed Corpus of Old English Prose (Taylor, Warner, Pintzuk & Beths 2003), we notice that there is a continuous secular decline in object topicalization from earliest Middle English into early Modern English. The rate of topicalization of direct objects in earliest Middle English is just over 11% – which means that in 11% of all main clauses that have a direct object this direct object is preposed – and declines to a rate of about 3.5% by the late 17th century, a rate comparable to though perhaps slightly higher than the rate in Modern English (Table, Figure 2.1).

Table 2.1: Rate of direct object topicalization.[6]

	oe1-2	oe3-4	me1	me2	me3	me4	eme1	eme2	eme3
# of sent. with DO	6184	10002	5329	3642	9608	5583	7719	10103	7057
thereof topicalized	736	1080	570	228	558	257	376	428	247
%	**11.9**	**10.8**	**10.7**	**6.3**	**5.8**	**4.6**	**4.9**	**4.2**	**3.6**

Figure 2.1: Rate of direct object topicalization.



The decline in topicalization is a feature of virtually all dialects. The texts of the North are deviant for reasons that will become evident,[7] but otherwise we see a clear tendency in all dialects towards a decline in topicalization (Table and Figure 2.2).

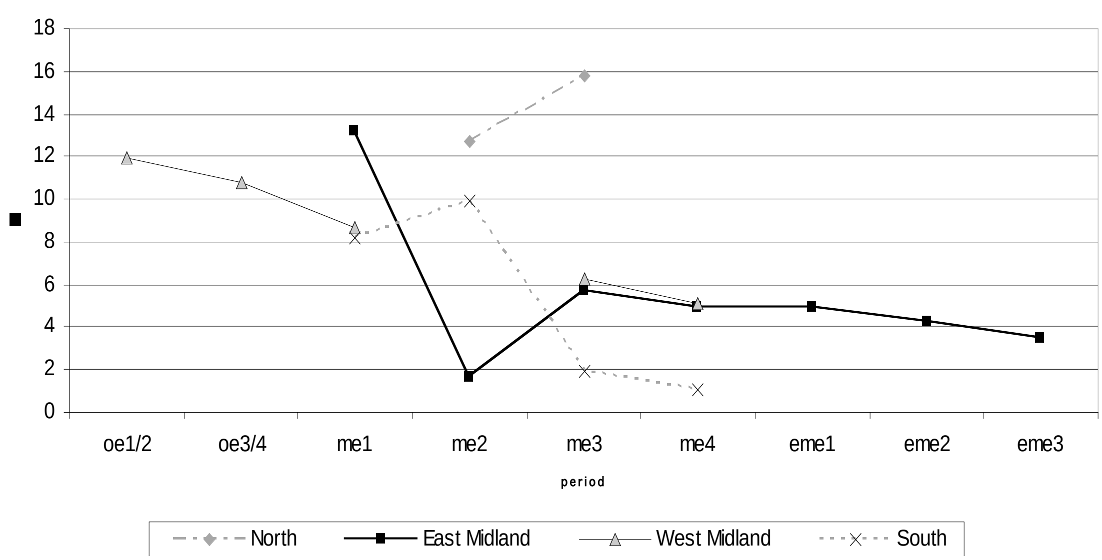
[6] The periods in this and the following tables and figures are those of the corpora Kroch & Taylor 2000, Kroch, Santorini & Delfs 2005, and Taylor, Warner, Pintzuk & Beths 2003. They cover the following timespans: oe1: x-850; oe2: 850-950; oe3: 950-1050; oe4: 1050-1150; me1: 1150-1250; me2: 1250-1350; me3: 1350-1420; me4: 1420-1500; eme1: 1500-1570; eme2: 1570-1640; eme3: 1640-1710.

[7] We will see later that there is a causal connection between the loss of V2 and the decline of the rate of topicalization. For the moment I want to summarize it in the following catch phrase: As long as V2 is an option in the language, topicalization is unproblematic. The North used V2-syntax much longer than the other dialect areas (Kroch & Taylor 1997), therefore it is not surprising that the use of topicalization stays stable there.

Table 2.2: Decline of topicalization in the different dialect areas.

	oe1/2	oe3/4	me1	me2	me3	me4	eme1	eme2	eme3
North									
all sent. with DO				652	759				
thereof topicalized				83	120				
%				**12.7**	**15.8**				
East Midland									
all sent. with DO			2392	1821	4718	3219	7719	10103	7057
thereof topicalized			315	29	267	158	376	428	247
%			**13.2**	**1.6**	**5.7**	**4.9**	**4.9**	**4.2**	**3.5**
West Midland									
all sent. with DO	6184	10002	2827		2109	1815			
thereof topicalized	738	1080	246		133	93			
%	**11.9**	**10.8**	**8.7**		**6.3**	**5.1**			
South									
all sent. with DO			110	1170	2022	549			
thereof topicalized			9	116	38	6			
%			**8.2**	**9.9**	**1.9**	**1.1**			

Figure 2.2: Decline of topicalization in the different dialect areas.



So we can say: Topicalization of direct objects (or rather accusative NPs, but most of these are direct objects) has declined over the course of English language history. In quantitative terms, it dropped from around 12% to around 3% between 1100 and 1700 AD.

2.2 The pragmatic properties of topicalization

The obvious question is: What can be the cause for this decline? In the remainder of this chapter I will attempt to answer this question. There are three candidate answers, two obvious ones (which prove to be wrong) and one less obvious one (which I will argue for). The second of the obvious answers will be the subject of section 2.3, so I will not discuss it here. The first of the obvious ones I do discuss in this section, however. It would go like this: We know that topicalization is sensitive to information structuring processes: In Modern English, topicalization is only possible if certain discourse structural requirements are met that will be summarized in this section. A hypothesis which comes to mind immediately is that perhaps in earlier periods of English there were more discourse structural contexts under which topicalization was possible. The decline in topicalization would thus be a gradual loss of contexts in which topicalization was felicitous.

In order to test this it is necessary first to examine the discourse structural contexts in which topicalization is felicitous in Modern English (2.2.1). In a second step the discourse structural properties of topicalization in Middle and Old English have to be examined and compared to those in Modern English. It will turn out that in Old English there was indeed one discourse structural configuration that was compatible with topicalization that was lost by earliest Middle English, viz. topic – comment in the strict sense. However, as the decline did not stop in Middle English but continued, this cannot be the explanation for the decline in Middle and Early Modern English (2.2.2).

2.2.1 The discourse-pragmatic functions of topicalization in Modern English

First we have to stop to think under what circumstances Modern English allows topicalization, or actually any kind of preposing. It turns out that topicalization can occur in a variety of rather disparate contexts in Modern English. The most important ones have been mentioned under (2-3) in section 2.1, repeated under (2-4).[8] First I will elaborate shortly on the distinguishing properties of these types:

(2-4) a. In the afternoon I usually go for a walk

b. Beans he likes, but peas he hates.

c. This proposal we discussed at length

d. Pterodactylus it is called.

Scene-setting preposing is the most common type. Scene-setting elements have been defined in section 1.2.1 as elements that describe or limit the situation, in which the proposition with which they are connected is evaluated. As they in some sense have scope over the proposition as a whole, at least in the semantic sense, it is conceivable that they are frequently moved to a position where they overtly take the remainder of the utterance into (structural) scope. Scene-setting elements are usually not in focus. The prosody of the clause is such that the scene-setting element bears metrical prominence of similar height as the element at the right edge of the clause that is realized as the nuclear stress.

[8] I follow Dryer (2005) in distinguishing two different types of 'topicalization', apart from scene-setting preposing and focus movement. Dryer's *Double Focus Preposing* corresponds to my 'topicalization in the stricter sense' or 'double-focus-topicalization'; Dryer's *Nonfocus Preposing* is narrowed here to 'anaphoric preposing'. I am aware that this construction is not as restrictive as it appears to be from my treatment here; however, it is fair to say that Nonfocus Preposing occurs most often in the guise of anaphoric preposing. A more differentiated treatment of Nonfocus Preposing is beyond the scope of this study.

Scene-setting preposing does nothing to establish a theme-rheme (= topic-comment) or a focus-background structure, it is on a completely different plane.

Topicalization in the stricter sense (= double-focus-topicalization) is exemplified in (2-4b). Topicalization in the stricter sense, as it is used here, is confined to preposing in sentences with two foci, one focus on the topicalized element, one on some other element in the remainder of the sentence. It is important to point out that the topicalized constituent is always in focus in this type of topicalization (Barry 1975:1; Dryer 2005; contra Birner & Ward 1998: e.g. 83). Focus is understood in the sense of Rooth (1985): By using focus on an entity, one evokes a set of alternatives to this entity. In the semantic representation of the utterance we have thus a variable in the place of the focus and an additional statement that identifies the variable. Informally in (2-5):[9]

(2-5) a. [[**he likes beans**]]^{s,g} = LIKE (he, beans)

b. [[**BEANS he LIKES**]]^{s,g} = $\exists$x [LIKE (he, x) | (x $\in$ M = {…, beans,…}) $\wedge$ x = beans]

The type of focus that most obviously conforms to this definition is contrastive focus, but other types of focus, like presentational focus (see Rochemont 1986) or verum-focus (see Höhle 1981) can be derived from that (see e.g. Rooth 1985:10ff., who explicitly relates new information together with the contrastive quality of his set-based proposal).

[9] I didn't represent 'he' with a variable but treated it as a proper name, in orderto avoid having too many variables.

This notion of focus is very general so far; in fact it is not enough for topicalization to have two focal emphases in general; the two foci have quite distinct semantico-pragmatic properties. For double-focus topicalization the generalization holds that the entity referred to by a topicalized constituent stands in a partially ordered set (short: poset) relation to a set evoked earlier in the discourse, but recently enough that it is still salient (Hirschberg 1986:122; Prince 1986:208ff.; 1999:7ff.).

This is, however, not yet a sufficient condition for topicalization. Often the poset-definition goes for both focused phrases in a clause with two foci. How does the speaker decide which one to topicalize?

Kuno (1982), working on multiple wh-questions, developed the idea that the wh-phrase selected to be fronted is the one that provides a 'sorting-key' "for sorting relevant pieces of information in the answer" (Kuno 1982:141; cf. also to the following van Hoof 2003). Let me illustrate this with an example, adapted from Kuno (1982:140f.). Let us assume a multiple question such as (2-6).

(2-6) Which students did they give A's to in which subjects?

An appropriate answer to that would be (2-7a), whereas (2-7b) sounds deviant. Sentence (2-7b), however, would be a perfectly felicitous answer to a question formulated as in (2-8), to which in turn (2-7a) would be an infelicitous answer.

(2-7) a. They gave an A to John Doe in geometry, biology and English, to Richard Roe in history and music,….

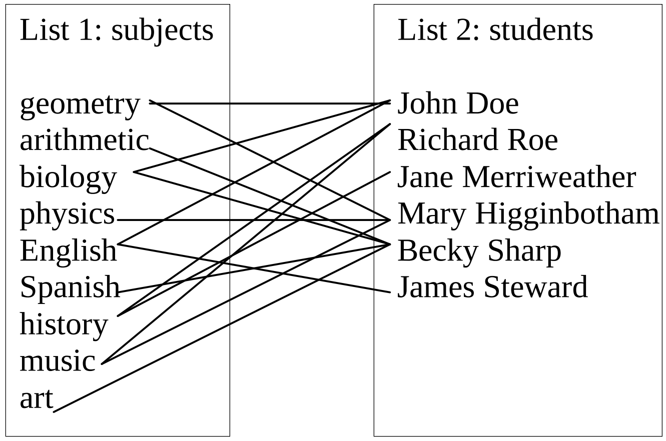
b. In geometry, they gave an A to John Doe and Mary Higginbotham, in history to Richard Roe and Jane Merriweather,…

(2-8) In which subjects did they give A's to which students?

It is easy to see that the surface order of the wh-phrases (and thereby their scopal relationship, see Kuno 1982:144) corresponds to the surface order of their respective answering phrases. The answering expression in the higher position is topical in the sense that the following expression is about the actual value of the fronted wh-expression and not about the value of any other wh-expression. This does not mean that it is an archetypical topic; it has, however, the characteristic of 'aboutness' in common with archetypical topics. I wish to point out that focus-background and topic-comment are two entirely different pragmatic dimensions; it is not the case that topics are automatically background or foci are automatically part of the comment-part. But with sorting-key elements, focus and topic intersect, such that a phrase can be focus and topic at the same time.

We can visualize this in the following way: Questions like (2-6) and (2-8) evoke two lists (therefore Kuno 1982:137 refers to such cases as 'multiple-choice questions'). The items on these lists are in a relationship to each other (2-9); the relationship, expressed by the question, is 'have grade A in (x,y)', where *x* is taken from list 2, and *y* from list 1.

(2-9)



The relationship is, however, not a trivial 1:1 relationship. In order to organize the information about the real relations holding, one first has to decide, which list organizes the information in the ongoing discourse. This is the sorting-key. Let’s say, I am more interested in the subjects list, so I order the information according to subjects (2-10):

(2-10)
$$\begin{bmatrix} \text{geometry} & \begin{bmatrix} \text{John Doe} \\ \text{Mary Higginbotham} \end{bmatrix} \\ \text{arithmetic} & \begin{bmatrix} \text{Mary Higginbotham} \end{bmatrix} \\ \text{biology} & \begin{bmatrix} \text{John Doe} \\ \text{Becky Sharp} \end{bmatrix} \\ \ldots & \ldots \end{bmatrix}$$

In some ways, this is like schönfinkelization of a function. In double focus constructions a non-permutable function between two sets is established, and the sorting keys are the elements in the set to which the function assigns members of the second set. The equivalent to this in natural language would be a sentence in which the sorting-key list has scope over the elements of the other list, that is, in which the information is ordered after the sorting-key.

What Kuno (1982) does not say explicitly, but clearly assumes, is that the answer sentences to such a multiple wh-question tends to have a form in which the scopal relationship is overtly expressed. In other words: The sorting key elements tend to stand before the elements of the other list. This surface ordering has the effect of enforcing the distributivity of the sorting key list over the second list.

At this point topicalization comes into play. The topicalized element in a double-focus topicalization sentence corresponds to the sorting key, and it is only the sorting key that can be topicalized. Let me illustrate this with an example: A question such as (2-11a) clearly has as its sorting-key ‘kinds of vegetables’. The answer (2-11b) is at least as felicitous as the answer (2-11c). If the question gave the persons as sorting-key (as question (2-11d) does) the sentence (2-11b) would no longer be a felicitous answer, only (2-11c). Note that both foci fulfil the condition of poset as stated by Prince (1999); poset is thus a necessary, but not a sufficient condition on topicalization.

(2-11) a. Which kind of vegetable does which person like?

b. Beans are the right stuff for John, and peas the right stuff for Mary.

c. John likes beans, and Mary likes peas.

d. Which person does which kind of vegetable like?

It is this type of topicalization which much of the discussion in this and the next part of my study is about, since this is the type that potentially produces prosodic violations as will be discussed in section 2.4.

All kinds of constituents can be topicalized (see Birner & Ward 1998: 45f.), as long as they conform to Prince (1999)'s and Kuno (1982)'s conditions on topicalization. The second focal emphasis can also be on any element. Birner & Ward (1998) distinguish between 'normal' topicalization involving two focused referential phrases (2-12a), and proposition assessment cases, such as proposition affirmation (2-12b), proposition suspension (2-12c) and proposition denial (2-12d), all of which have in common that the second focus is some kind of a verum focus. In addition to that I want to add split topicalization. This construction is special because both foci happen to be in the same (quantified) phrase; the quantifier is stranded whereas the content part of the phrase is topicalized (2-12e). Otherwise, conditions similar to normal topicalization hold. In Modern English the use of split topicalization is strongly restricted, and example (2-12e) is just barely acceptable. I want to introduce it here nevertheless as it will be relevant for the treatment of German in later sections.

(2-12) a. Baseball I like a lot better.

b. (It was necessary to pass,) and pass I did

c. (Mark submitted his report) if submit it he did

d. John Madden he's not.

e. He knows tons of endocrinologists, surgeons and laryngologists. But ophthalmologists he knows none.

(examples a-d from Birner & Ward 1998)

Anaphoric preposing is the preposing of a noun phrase whose referent is typically not an entity but a set of propositions, namely a relevant portion of the previous discourse. This noun phrase is most often a demonstrative pronoun (2-13a), but full NPs are possible, too, with (2-13b) or without (2-13c) a demonstrative pronoun.

(2-13) a. This we all know.

b. This discussion I find important.

c. The argument so far everybody could follow.

This construction is highly specific in function. The anaphoric noun phrase object serves to refer to one or more propositions of the previous discourse in a summarizing fashion. Preposing the anaphoric object promotes it to a sentence topic. It fulfils both conditions on topichood: Its referent is discourse-old, and it is what the sentence is about. Let us refer to this kind of topic as a *ϕ-topic* (for propositional topic). Its difference from an archetypical topic (let us call it an *e-topic*, for entity topic) is that the referent of the anaphoric noun phrase is not an entity but a proposition. In a way an e-topic is then a topic in first-order logic – an entity that can be an argument to a predicate; remember that topichood is nothing more than a property of an argument in a given proposition – whereas a ϕ-topic is *a priori* a topic in second-order logic; the proposition containing it must take a second-order-object as one of its arguments. The anaphoric process here is not much more than an act of reference to this second-order-object.

Typically, sentences with a (preposed) anaphoric noun phrase are comments on previous discourse segments. These sentences as a whole, as has been said above, are the subject matter of the comment made by the speaker/writer, and therefore can be regarded as topic of this comment. By virtue of being topic, they do not bear focal emphasis.

Normally such comments have wide focus and therefore no other focal emphasis appears in them. As the highest clausal metrical prominence peak falls on an element in the verb phrase, that is, close to the right periphery, we get the same prosodic contour as the ones which we observed with preposed scene-setting elements, namely the highest clausal metrical prominence peak on the right edge of the clause, another metrical prominence peak on the preposed anaphoric noun phrase and lower metrical prominence on the subject (2-14; 2-15).

```
(2-14)  scene-setting:     (*)        (.       *)
                            *          *       *
                           Yesterday John arrived

(2-15)  anaphoric:             (*)      (.         *)
                                *        *         *
                           This discussion I find important.
```

<u>Focus Movement</u>, finally, is the preposing of a single focalized element in an utterance. For Focus Movement the condition holds that the preposed element stands in a relation to an entity or set already evoked in the discourse (cf. Birner & Ward 1998:84ff.). This is only a necessary condition. More specifically, the focalized element in the preposed phrase often represents a value of an attribute (not in the syntactic sense) that applies to some entity already evoked in the discourse (Prince 1981b:259). The value itself is new information, the attribute (expressed by the predicate of the sentence in which focus-movement takes place) is explicitly stated in the previous discourse or is at least inferable (Prince 1981b:259). A typical example would be (2-16a; from Prince 1981b:259), where the value is *three* and the attribute is *cooking n meals a day*.

Furthermore, Focus Movement very often serves to specify the relevant member(s) of a more general set evoked earlier, such as in Birner & Ward's example (95b) in 1998:84 (here repeated as 1-16b below). It is important to note that the preposed element – the specification – is a referential expression that specifies the reference more precisely than anything else that was said earlier in the discourse about the relevant member of the set: Often it is exact dates (such as (2-16c) = Birner & Ward's (95c)), unique names (2-16b) or the like that are subject to Focus Movement. Thus it is not surprising that examples like (2-16d) in which a new term is introduced for a salient entity are very common, perhaps the most common type of examples. Another field of use for Focus Movement is in corrections (2-16e). We can view corrections merely as a special case of specification: The set and even the relevant member of the set have been evoked earlier, but the member has been referred to with the wrong term in the previous mention. By the correction the correct relationship between entity and referring expression is established; in that way it refers more precisely than the previous (wrong) referring expression (see Birner & Ward 1998:86 for more examples). Focus Movement is also used in confirmation questions such as (2-16f; adapted from Birner & Ward 1998:88) in which the person asking wants to make sure the precise identity of the relevant member of the set.[10]

---

[10] Intonationally similar, but pragmatically quite distinct is Yiddish Movement (see Prince 1981b; Birner & Ward 1998:90ff.) which I do not go into here as it is clearly a Yiddish substratum phenomenon and thus dialectally quite confined. I want to point out that this is the only kind of topicalization in the broader sense that is not common to all dialects of English. All other types of topicalization that have been presented here are grammatical in both American and British English. I point that out as a reviewer of a paper on the Clash Avoidance Requirement (then still under the catchy, but misleading name 'Trochaic Requirement') I wrote earlier (Speyer 2005) tried to convince me that topicalization in general is only a matter of Northeastern American English. The browsing of two randomly selected British English works of fiction (by Dorothy L. Sayers and P.G. Wodehouse) provided examples for each of these types of topicalization (see section 3.1.1) so that I regard the objection by the reviewer as unwarranted.

(2-16) a. You […] had to cook for ten childrens on Sunday. […] Three meals a day I cooked on Sunday.

b. A: Are there many black kids in that school now?

B: Not many. I had two really good friends. Damon and Jimmy their names were.

c. I promised my father – on Christmas Eve it was – to kill a Frenchman at the first opportunity I had.

d. A: Oh yeah, and here we see a fossil of this weird flying reptile… Pterodactylus it's called …

e. A: Now we've got this flying lizard… the peri… plero… plesiotantalus…

B: Pterodactylus it's called

f. A (= customer at fast food joint): Gimme a cheeseburger, large fries and a large Coke.

[five minutes elapse]

B (= employee at fast food joint): Large Coke you ordered?

Later, when the Clash Avoidance Requirement will be introduced, it will be easy to see that Focus Movement never produces an utterance at odds with the Clash Avoidance Requirement. This is because such sentences contain only one focus; consequently there are no other elements that can clash with it. We will see that a clash occurs if two elements with the same level of prominence are adjacent to each other. In a clause with focus movement this can never happen.

At least these four types of topicalization can be distinguished in Modern English. The list is not necessarily comprehensive, but it is fair to say that these types are the most common ones. Because in the later argumentation the prosodic properties of these respective types of topicalization play a role, I want to point out here that double-focus-topicalization is the only one of these types that can produce prosodic clashes. In the other cases there is either only one element with maximal prominence (focus movement) or no focal emphasis is involved (scene-setting preposing; anaphoric preposing). In these cases the rules will automatically generate a grid in which the metrical prominence peak on the preposed element and the nucleus will be separated by at least one phrase (viz. the subject) that bears lower metrical prominence.

2.2.2 The discourse-pragmatic functions of topicalization in Old and Middle English

From the preceding section we know when topicalization can occur in Modern English and we were able to distinguish four types. If we now turn to Old English, at first glances it looks as if the hypothesis that the loss of topicalization is a loss of environments in which topicalization is pragmatically felicitous hits home: Apart from the four pragmatically determined enviroments in which topicalization may occur in Modern English we find a fifth, namely topicalization of an aboutness e-topic. Two examples are given in (2-17). The topic of the discourse portion from which sentence (2-17a) is taken is the Holy Spirit, which is referred to by means of an anaphoric pronoun, as is quite usual in Old English. One could argue that the usage of a demonstrative pronoun could always have some deictic force and therefore this is perhaps not a simple e-topic. But we can easily find examples in which the topicalized element is a personal pronoun (2-17b), and here we can be sure that they are not deictic at all. Example (2-17b) has an accusative experiencer in the preverbal position. Some of the examples with topicalized personal pronoun can be related to accusative experiencer verbs and their tendency to put the experiencer first (like in German, where ‘Mir gefällt das Haus’ is less marked then ‘das Haus gefällt mir’, although vorfeld-movement of dative objects otherwise marked in German), but by far not all. (2-17c) shows an example of a sentence where it is a non-deictic personal pronoun and an e-topic beyond doubt that is topicalized.[11]

(2-17) a. Þone asende se Sunu,

this sent the Son

‘The son sent this one’

(coaelhom,+AHom_9:114.1350)

b. ne hine ne drehð nan ðing,

and-not him not troubled no thing

‘and nothing troubled him’

(coaelhom,+AHom_11:558.1780)

c. & hit Englisce men swyðe amyrdon.

and it English men fiercely prevented

‘and the Englishmen prevented it fiercely’

(cochronE,ChronE_[Plummer]:1073.2.2681)

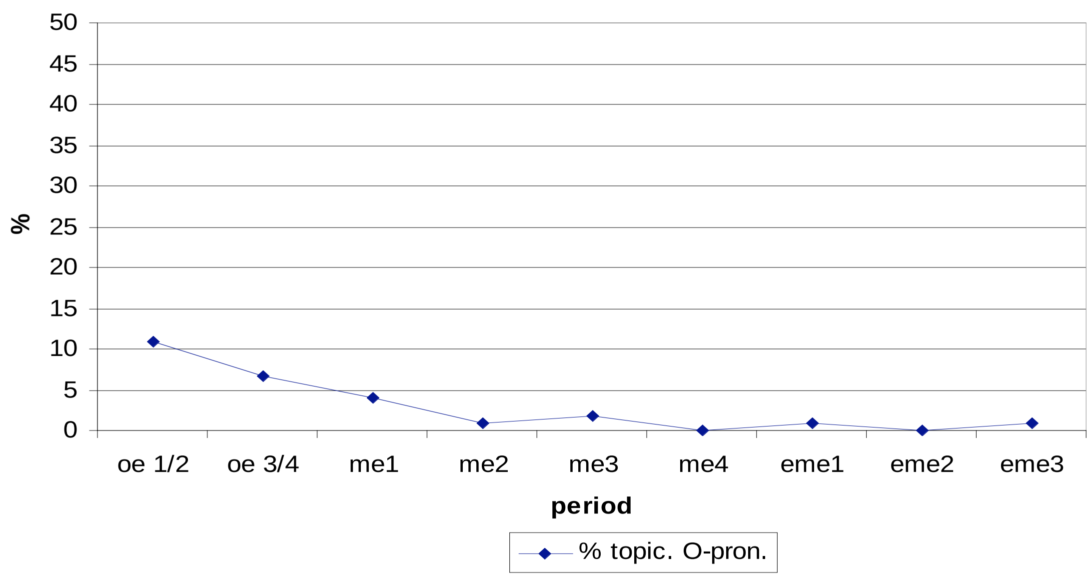
[11] I do not think that the loss of verbs with non-nominative experiencer is responsible for the decline in pronoun object topicalization. The rate of pronoun object topicalization declines during the Old English period. The relative portion of non-nominative experiencer does not decline, but, on the contrary, rises slightly. I took a random sample of 100 sentences with topicalized pronoun and looked what type of verb each sentence had. The proportion of non-nominative experiencer verbs was 27% in oe1/2, 32% in oe3/4.

The presence of topicalized personal pronoun objects can be used as indicator for the topicalization of e-topics. We know that e-topics tend to be realized pronominally, as e-topics are maximally salient and therefore reference by a pronoun is quite unambiguous and therefore unproblematic (see e.g. Walker, Joshi & Prince 1998). Thus, we may say that pronouns are the archetypical mode of realization of an e-topic. A consequence of this fact is that in periods in which we can find topicalized pronouns in a fairly large number, we can assume that e-topics could be topicalized in these periods. As pronominal objects can be found easily in syntactically parsed corpora we can measure the frequency of these likely cases. In Table 2.3 and Figure 2.3 we see what we get.

Table 2.3: Number and rate of topicalized personal pronoun objects

	oe 1/2	oe 3/4	me1	me2	me3	me4	eme1	eme2	eme3
all pr. obj.	200	603	285	213	454	316	107	155	96
thereof topic.	22	40	11	2	8	0	1	0	1
%	11.0	6.6	3.9	0.9	1.8	0	0.9	0	1.0

Figure 2.3: Number and rate of topicalized personal pronoun objects



The percentage is the rate of topicalized pronominal objects among sentences with pronominal objects. We see that it begins to trail down already in the Old English period, while it is practically gone after the first period of Middle English. The solitary examples after that are either consciously archaic (and this archaic use can be found even much later, see (2-18) or contrastive, hence not e-topics.

(2-18) 'Tis said, here lives a woman, close familiar to the enemy of mankind. <u>Her</u> I'll consult, and know the worst.

(Charles Jennens: Libretto to Handel's oratorio 'Saul', 3rd act, 1739)

So this data gives us evidence that topicalization of e-topics ceases after the end of Old English.

Topicalized e-topics were found frequently in Old English. By Middle and continuing through Early Modern English, however, the picture resembled rather closely the picture in Modern English: Here we find basically only examples of the four types of topicalization that are observable today, and no e-topic-topicalization (2-19). Focus movement is not as constrained as it is today (see 1-19g, where the only focalized element is *al þis contre*) but it stays unconstrained throughout the observed time span and therefore a putative narrowing-down of focus movement cannot be made responsible for the decline. Likewise, anaphoric preposing occurs more freely in Middle English (see 1-19e, where the referent is not a set of propositions, but of entities; they have been described in the previous discourse, so that this example in some ways meets the conditions on anaphoric preposing that were identified further above), but it stays unconstrained throughout the observed period and thus, again, a putative restriction on the conditions of anaphoric preposing cannot be the cause for the decline of topicalization either.

(2-19) a. <u>scene-setting preposing</u>:

& in þis batail ham mette Cattegren and Horn, Engistes broþer, so þat

and in this battle him met Cattegren and Horn, Engist's brother, so that

eueryche of ham slouġ oþer;

each of them slew other

'and in this battle Cattegren and Horn, Engist's brother, met him, so that each of them slew the other'

(cmbrut3,53.1561; m3)

b. <u>double-focus topicalization</u>:

and þat land Brut ġaf to Albanac his sone

and that land Brut gave to Albanac his son

'and that land Brut gave to Albanac, his son'

(Scotland, in contrast to England that Brut kept; cmbrut3,12.315; m3)

c. <u>double focus topicalization; proposition denial</u>:

Delue may I noġt,

delve may I not

'I may not delve'

(cmwycser,256.567; m3)

d. <u>anaphoric preposing, referent set of propositions</u>:

This heard þe king,

This heard the king

(capchr,152.3553-3559; m4)

e. <u>anaphoric preposing, referent not set of propositions</u>:

And alle þees foure miġtes & þeire werkes mynde conteneþ

And all these four powers and their works mind contains

& comprehendeþ in it-self.

and comprehends in itself

‘And all these four mights and their works the mind contains and comprehends in itself.’

(cmcloud,115.578, m3)

f. <u>focus movement</u>:

A "fortunat ascendent" clepen they whan that no wicked planete […]

A fortunate ascendent call they when that no bad planet

is in the hous of the ascendent.

is in the house of the ascendent

‘A fortunate ascendent they call it, when no bad planet is in the house of the ascendent.’

(cmastro,671.C1.268, m3)

g. <u>focus movement:</u>

for al þis contre þe Danois hauen gete, & take þe cite of Ġork;

for all this country the Danes have got and taken the city of York

'For the Danes have conquered all this land and have taken the city of York.'

(cmbrut3,105.3185, m3)

What about the question from which we started? We can partially explain the decline in topicalization with the hypothesis that the number of pragmatic configurations under which topicalization was possible was diminished. This however can only explain the first bits of the decline, viz. the decline from the beginning of Old English to the first period of Middle English. If this were the only solution, the rate of topicalization should stabilize after me2 at around 6.3%, as this is the rate of topicalization in me2, after the topicalization of e-topics has finally gone out of use. But this is not what we get. The rate of topicalization continues to drop, to a value almost half as high at the end of eme3 (3.6%). So it is fair to say that the pragmatic-loss hypothesis fails to account for the decline at least in Middle and Early Modern English.

Let us summarize this section: A possible and obvious explanation for the decline in topicalization would be that fewer discourse structural configurations came to be compatible with the usage of topicalization over time. The four prime uses of topicalization in Modern English are: preposing of scene-setting elements, preposing of the sorting key in a double focus construction, preposing of an anaphoric element (or ϕ-topic) and focus movement, that is: preposing of a contrastive focus. If we turn to Middle

English, we see that topicalization was used to express the same discourse structural configurations as today, no more and no less. In Old English, on the other hand, e-topics as well could be topicalized. This option ceased to be possible in Middle English, so we can say that a pragmatic explanation can account for the decline in topicalization from Old to Middle English. It does nothing, however, to explain the subsequent decline over Middle and Early Modern English.

2.3 Another possible explanation: rigidity of word order

A second possible explanation for the decline of topicalization which immediately comes to mind is that the decline of object topicalization might have something to do with the general tendency for English word order in the Middle English period to become more rigid.

In section 2.3.1 I will explain what is meant by this: As a consequence of the loss of case marking English word order gradually became stricter. This was because grammatical functions that were formerly marked by the case morphemes could not be marked that way any more, and therefore another way of marking grammatical function was needed. Starting from Middle English, English chose to express grammatical functions by word order: the subject comes first, then the indirect object (if present), then the direct object. Such a word order is rigid in that deviations from it results in ungrammatical sentences. I will demonstrate the power of rigidification by presenting several word order options that are non-canonical from a modern point of view, but were possible in Old English – viz. scrambling of full noun phrases and of pronouns. All became ungrammatical in the course of Middle and Early Modern English, as they interfered with the subject-before-object constraint.

A simple guess would be that topicalization is just another one of those constructions that declined because they interfered with the marking of grammatical functions by word order.

In section 2.3.2 I will argue that this cannot be the right explanation. Bear in mind that topicalization did not become ungrammatical. It only became less frequent over time until it reached a stable, yet low, level of usage on which it stayed to the present day. The scrambling operations presented in 2.3.1 however did become ungrammatical. A possible objection could be that topicalization is perhaps simply taking longer to become ungrammatical than the other constructions. This can be easily refuted by the fact that the usage of topicalization should have reached a rate of close to zero by now, if we assume that the speed of the decline remains stable.

2.3.1 The rigidification of English word order

Old English was a language with reasonably rich inflectional morphology. The part which is of interest here is the nominal morphology. Old English distinguished four cases by purely morphological means, nominative, accusative, genitive, dative (a fifth case, the instrumental case, was on its way out even in Old English), and two numbers, singular and plural. To the respective combinations of these parameters suffixal morphemes were assigned. Old English had several declension classes, so there were several morphemes for each given case-number-combination, depending on which declension class the word belonged to (see e.g. Brunner 1965).

Rich nominal morphology is usually connected with free word order (cf. e.g. Kiparsky 1997:461). By ‘free word order’ we do not mean that anything goes, but that the constituents are free to be organized in a way that is suitable to other modules of the language faculty, e.g. information structural needs, because the grammatical relations are expressed sufficiently by the respective case morphemes and thus need not remain in their base-generated place. The model case for this situation is Latin. In Latin there are few fixed points with respect to surface sentence structure (pretty much the only fixed point is that the verb is normally at the end, which we might translate such that Latin had an Infl-final structure). The arguments are ordered for purely pragmatic reasons, whereby the main factors seem to be the establishment of a theme-rheme-structure and focusing by preposing (see Speyer 2007c).

In Old English the assignment of cases to grammatical functions was relatively straightforward. For most verbs it holds that their subject bore nominative case, their direct object accusative case, and their indirect object dative case. The genitive case served foremost to express argument relations within the noun phrase. Prepositions governed cases as verbs did, as a rule the dative or accusative.

After the silence of English sources following the Norman Conquest, the picture is radically altered. Among pronouns, dative and accusative are no longer distinguished. Among lexical nouns, the only case that is marked is the genitive (although it is replaced often by a prepositional phrase headed by *of*) and only in the singular, at that. The only distinction that is marked in a morphologically stable way is the singular/plural distinction.

What caused this collapse in morphology is not the subject of this thesis. Several theories have been put forward; probably it is a conspiracy between the phonological weakening of endings that is observable already in Old English and creolization processes due to the heavy influx of Norse settlers in the 9th and 10th century (cf. Kroch & Taylor 1997: 317ff. for the loss of verbal morphology).

The point that is important for our purposes is what consequences this morphological collapse had. After the collapse, it was no longer possible to identify the grammatical function of a noun by its case morphology. Therefore other ways had to be found to unambiguously express grammatical functions of nouns. In English two strategies came to be used:

- The replacement of case forms with prepositional phrases. This is a very common way to make up for the loss of case forms and has been employed e.g. by the Romance languages. In English the dative case could be replaced by a *to*-phrase, the genitive case by an *of*-phrase. In contrast to the western Romance languages, where all noun phrases in the functions that in Latin had been expressed by the dative and genitive case must appear as prepositional phrases with the respective outcome of the Latin prepositions *ad* and *de*, respectively, in English the use of prepositional phrases as ersatz cases has been variable.
- The rigidification of word order. In generative terms, rigidification means that the language renounces movement operations that alter the structure in such a way that the arguments are serialized in a different way from the way they would be serialized if they were still in the structural configuration in which they were base-generated. Concretely this means: An English clause is base-generated in such a way that the order of the arguments comes out as subject – indirect object – direct object. Whereas in earlier stages of English it was possible to perform operations that moved e.g. one of the objects or both over the subject, this option ceased to be possible in Middle English. This is because after the loss of case endings the only way to identify a noun phrase as direct object was by its position after the subject (and the indirect object, if present).

If we look in the available corpora of historic stages of English, we can directly see how the word order did become more rigid. It was not a change that happened suddenly, but took a bit of time. Let us demonstrate it with full noun phrase scrambling.

Scrambling is a process in which elements are moved from their base-generated position to a position below the C-architecture (cf. e.g. Haider & Rosengren 1998; 2003). There are certain constraints on scrambling in Germanic languages, e.g. that pronouns have to precede full noun phrases, but otherwise the operation is relatively unconstrained syntactically.

Whereas in earlier Old English scrambling was possible, it becomes less frequent in later Old English. We then see a rapid decline at the beginning of the Middle English period to a marginal rate, until it goes out of use. Table 2.4 shows the data for fronting – most of which is due to scrambling – of full noun phrase objects over full noun phrase subjects.[12]

Table 2.4: Rate of fronting of full noun phrase objects over full noun phrase subjects, only subordinate clauses.

	oe 1/2	oe3/4	me1	me2	me3	me4	eme1	eme2	eme3
clauses with full NP obj. and subj.	935	923	434	204	1549	510	1461	1591	1401
thereof scrambled	68	18	1	0	4	1	2	2	0
%	**7.3**	**2.0**	**0.2**	**0.0**	**0.3**	**0.2**	**0.1**	**0.1**	**0.0**

[12] The tables and figures give the numbers for all subordinate clauses, because there the fronting cannot be the result of topicalization (which in the case of Infl-final main clauses would be possible in principle).They are not confined to unambiguous Infl-final-cases and do not separate argument and adjunct noun phrases. As not all instances of object-before-subject in subordinate clauses are instances of scrambling, these numbers are to be taken with a grain of salt and read as scrambling (in Infl-final subordinate clauses) plus some process (possibly more like topicalization to a C-prjection below the position hosting the complementizer) in Infl-medial subordinate clauses.

The reason why scrambling and other fronting mechanisms went out of use in the history of English can easily be explained by rigidification: in a scrambled sentence, the object stands before the subject. As case marking has been lost in Middle English, the hearer of the scrambled sentence cannot identify the object as such; s/he might even think that the scrambled object is the subject as the normal situation would be that the first of two noun phrases is the subject. So scrambling would inevitably lead to misunderstandings.

The rigidification process gained such momentum that it was even applied in situations in which scrambling would not have led to misunderstandings. Such situations occur when pronouns are involved, since pronouns still distinguish between nominative and oblique case. The situations under consideration are when object pronouns scramble over subjects, both pronominal and full noun phrase subjects. Table 2.5 shows the rate of fronting of object pronouns in front of full noun phrase subjects, whereas Table 2.6 presents the rate of fronting of object pronouns over subject pronouns. Figure 2.4 graphically displays the data in Tables 2.4, 2.5, 2.6.[13]

[13] Here a little word about the rigidification hypothesis in general might be in order. The rigidification process seems to have set in already in the middle of the Old English period, when case morphology was still available. This is problematic for the hypothesis as a whole, as the consequence seems to precede the cause. Also, the fact that it appeared with pronouns too, although they kept their case marking, suggests that the connection between rigidification and the loss of case marking is not as immediate as one might want to believe. Finally, scrambling could have disappeared simply because the head-final character of IP and VP went out of use in the course of Middle English. If we believe that scrambling is only a property of head-final structures (cf. e.g. Rosengren 1994), the disappearance of scrambling would come for free with the exodus of head-final IPs and VPs.

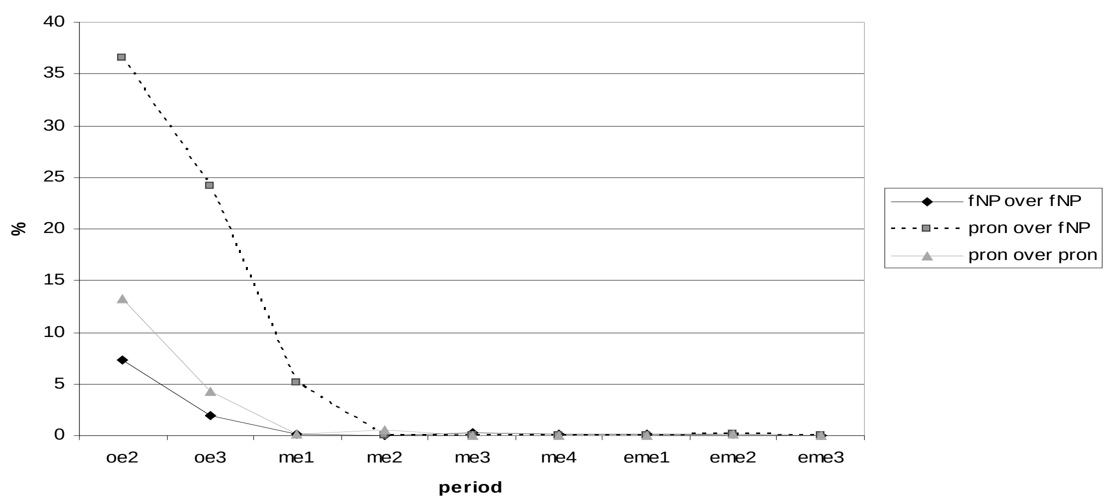
Table 2.5: Rate of fronting of full noun phrase objects over pronoun subjects, only subordinate clauses.

	oe1/2	oe3/4	me1	me2	me3	me4	eme1	eme2	eme3
clauses with full NP obj. and pron. subj.	669	895	470	118	538	257	338	449	297
thereof scrambled	244	216	24	0	0	0	0	1	0
%	**36.5**	**24.1**	**5.1**	**0.0**	**0.0**	**0.0**	**0.0**	**0.2**	**0.0**

Table 2.6: Rate of fronting of pronoun objects over pronoun subjects, only subordinate clauses.

	oe1/2	oe3/4	me1	me2	me3	me4	eme1	eme2	eme3
clauses with pron. obj. and subj.	1641	1840	898	355	814	517	732	850	816
thereof scrambled	217	80	2	2	0	0	0	1	0
%	**13.2**	**4.3**	**0.2**	**0.6**	**0.0**	**0.0**	**0.0**	**0.1**	**0.0**

Figure 2.4 Rate of fronting of different types, only subordinate clauses.



A nice side-effect of this discussion is that the loss of scrambling in the beginning of the Middle English period provides us with a theory why the topicalization of e-topics went out of use. Let me briefly point out the argument here, before we go on to show why rigidification of the word order could not have been the factor responsible for the decline in topicalization.

In recent theories about vorfeld-movement (and I should point out that this process is comparable to English topicalization as it is the same target position in both cases, viz. SpecCP) the importance of so-called 'formal movement' has been recognized (Frey 2004a; 2006). Frey assumes that SpecCP can be filled by two distinct movement processes, either informal (operator-driven) movement of e.g. focalized phrases, or formal movement. In the latter case the highest element in the mittelfeld is moved automatically to the vorfeld. If the language allows for scrambling, the highest element in the mittelfeld is the highest scrambled constituent. For German at least it can be shown (e.g. Frey 2004a) that the element marked as aboutness topic is the element that scrambles highest.

If the same constraint holds for Old English (and there are no reasons to assume the contrary, as far as I can see), topicalization of e-topics would be nothing else but formal movement of the highest scrambled constituent – the topic – to SpecCP. If scrambling becomes impossible – as happened at the transition from Old to Middle English, as we have seen – there is no way for formal movement to operate either, at least not in the way that it can select the e-topic, being the highest scrambled constituent, and move it to SpecCP, since nothing scrambles any more, including the e-topic. Formal movement then could only move the subject from SpecIP to SpecCP, but this is pointless, therefore we may assume that as soon as scrambling went out of use, formal movement died out too, especially as it seems to have been much less categorical than in German anyway (SpecCP must be occupied in Modern German, tends to be occupied in older stages of German, but is very frequently unoccupied in Old English). The fact that the loss of scrambling and the loss of e-topic-topicalization happened simultaneously suggests that this scenario is correct.

2.3.2 Rigidification as an explanation for the decline of topicalization?

Let us go back to the main question, namely, whether the decline in topicalization can be explaned by the general tendency toward rigid word order. At first glance it looks promising, because for object topicalization, the same description holds that holds for scrambling: it is a construction in which the object stands earlier in the sentence than the subject. Consequently, it is conceivable that the same argument that goes against scrambling could also apply to topicalization: Because the normal order subject >> object is reversed in a topicalization sentence, the listener is likely not to be able to detect the grammatical functions correctly. Consequently, topicalization brings up the same difficulties as scrambling and is therefore abolished.

An immediate objection to this explanation is the fact that topicalization is still grammatical today, whereas scrambling is not. This difference is crucial, as it shows that topicalization and scrambling are not subject to the same restrictions and therefore an explanation put forward for the loss of scrambling need not hold for the 'loss' of topicalization.

Someone who defends the hypothesis that topicalization became infrequent because of rigidification may raise the objection that perhaps topicalization simply takes longer to disappear than scrambling, for whatever reason.

The main argument against this view is an empirical one. In contrast to the rate of scrambling, the rate of topicalization seems to settle to a stable, if low, frequency after the decline. If topicalization were on its way out, the rate should not stabilize, but continue to decline, and we should detect this continuous decline in our data. Unfortunately the available parsed corpora cover only the time until 1710. We can however extrapolate what frequency of topicalization we would expect today if the decline continued at the same rate at which it went throughout Middle and Early modern English. Later in this argumentation we will see that the rate of topicaization declines parallel with the loss of the V2 word order option (Figure 2.9). This means that they can be modelled by the same equation with the same values (see Kroch 1989 for the mathematical details of the logistic function that can be used to model language change). If the decline of topicalization were only the first portion of a 'loss', and the intercept and the slope of the respective curves are more or less the same, the third important value, the end point (that is the time at which the change has reached completion) should be the same, too. This is obviously not the case: V2 has gone out of use completely– in fact, there is hardly any V2 left even in the period eme3 – but topicalization has not.

Another argument against rigidification as an explanation for the decline in topicalization is that the decline is not a wholesale process but only affects a fraction of the cases. The rate of topicalization in sentences with subject pronouns, for example, remains stable, which strengthens the point. Compare the case of scrambling where pronouns and full noun phrases are affected equally.

We see that the explanation that topicalization went out of use because it interfered with the configurational marking grammatical functions cannot be correct: Speakers obviously tolerate the non-canonical argument order object >> subject in the case of topicalization, otherwise it should be judged ungrammatical and not be used. The reason why they tolerate this order is probably because it is obvious that the topicalized constituent belongs to another domain. Either (with V2) the verb intervenes (as it would not do in the case of scrambling), and so it is obvious that the topicalized constituent is not in the ‘normal sentence’ domain, that is, under IP, or the order X – S – V arises, and in this configuration the subject can be clearly recognized as the NP left adjacent to the verb. So there is no danger of mistaking the subject and the object because the listener knows that in a configuration of the type X – S – V, where X and S are NPs, the first NP must be a non-subject and the second NP must be the subject. But if English speakers tolerate topicalization in general, the impetus for rigidification has been removed.

Let us summarize this section. A potential explanation for the decline in topicalization is that English word order became more rigid, as a consequence of the loss of case morphology. The reality of this rigidification can be demonstrated by the loss of the fronting of objects over subjects in subordinate clauses (‘scrambling’) for which virtually no examples can be found after a certain time (~1250 AD). Rigidification of word order cannot be the right explanation for the decline in topicalization, however, as topicalization should have become ungrammatical then too, which is not the case.

2.4 A third explanation: the Clash Avoidance Requirement

If one looks closely at the data summarily presented in 2.1 one notices that the decline of topicalization affected only a limited part of the data. This will be demonstrated in section 2.4.1 and 2.4.2. In section 2.4.1 it will be shown first that the type of subject matters: the decline is clearly observable if the subject is a full noun phrase, but hardly to be seen if it is a pronominal element. The question is: What is the special quality of pronominals as opposed to full noun phrases? The answer, suggestive of the explanation put forward later, is that they are likely not to carry focus. In fact, they normally do not bear any prominence, be it metrical or indicating focus.

In section 2.4.2 we will leave the area of focus. I will show that the rate of preposed scene-setting elements remains stable, too. In the same line of the demonstration as the preceding section I present an observation that will turn out to be helpfully suggestive: scene-setting elements rarely carry focus. In contrast to the pronominal subjects of section 2.4.1, they do have a phrasal stress, but we can show that the next phrasal stress of comparable (that is: high) prominence is normally not on the subject but at the right edge of the sentence. This means that, under normal (wide focus) intonation, scene setting elements are usually separated from the next element having an equally high level of prominence by an element of lower prominence, namely, the subject.

From this, I will deduce in section 2.4.3 that topicalization remains unproblematic, if at least one non-prominent element is in the left area of the clause. Topicalization declines only in the case where two elements are on the left edge of the clause that have a certain likelihood of receiving focus, that is, a full noun phrase object and a full noun phrase subject. A hypothesis that is introduced here tentatively but will be defended in detail in later chapters is that topicalization of full noun phrase objects in clauses with full noun phrase subjects is problematic because full noun phrase objects can carry focus and therefore topicalization of such elements can lead to situations in which two elements of high prominence are adjacent to each other, namely, when the subject is also a full noun phrase and focused. Let us assume that a structure with two equally highly prominent elements adjacent to each other is ill-formed. This is in line with the observation on scene-setting elements in 2.4.2. From all this it follows that a structure, containing two focused or otherwise equally highly prominent elements, is only well-formed if the two highly prominent elements are separated by at least one less prominent element. This requirement will be referred to as Clash Avoidance Requirement (= CAR).

In section 2.4.4 we will see the CAR in all its power at work. A syntactic change that took place in the same period as the decline of topicalization was the loss of the V2 word order option. This change will first be decribed quantitatively. It is not far fetched to assume that the loss of the V2 option and the decline of topicalization are somehow related. By means of the CAR, we can give an explanation for this relationship: As verbs are less likely to be in focus than noun phrases, V2 is a good way to avoid CAR-violations. I will show that throughout Middle English V2 word order was employed especially frequently in cases where the topicalized element and the subject both bore focus. As the V2 option disappeared, however, potential problem cases for the CAR would rise, as now all topicalization cases in which the topicalized object and the subject both are highly prominent will end up with two adjacent elements of equally high prominence, an ill-formed structure. Therefore speakers chose not to topicalize any more in such problem cases; hence, the rate of topicalization decreases in such cases. This means that the CAR is powerful enough to influence the choice between syntactic structures and to override the pragmatic requirements that led to topicalization in the first place. The decline of topicalization is thus an epiphenomenon of the loss of the V2 word order option.

2.4.1 Type of subject

To justify the above explanation for the decline in topicalization the obvious next step is to look at the data in more detail.

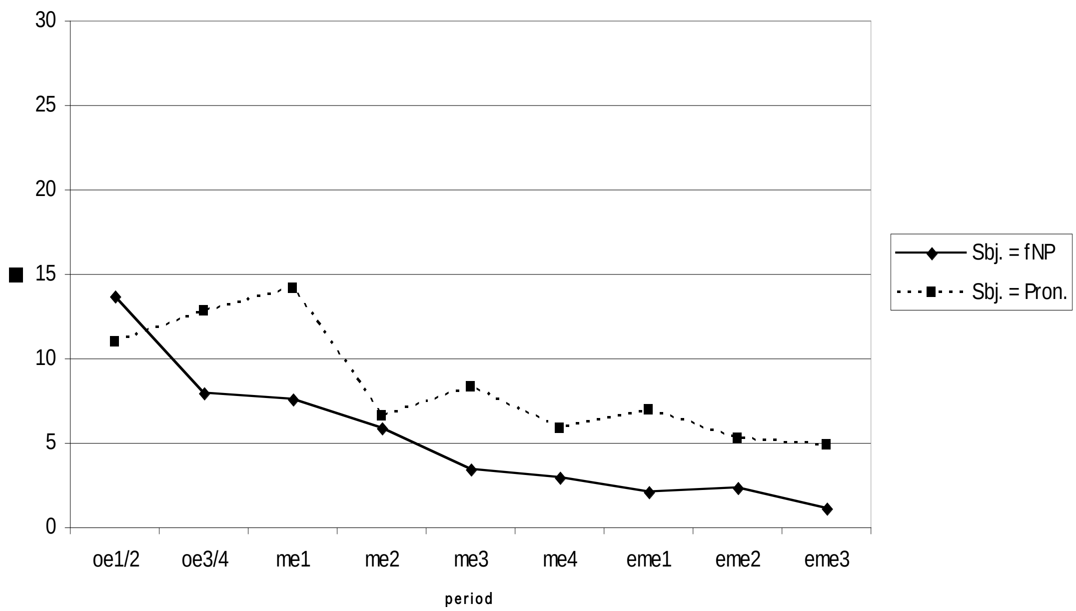
When we examine the Middle English examples more closely it is noticeable that very often the subject in a topicalized sentence is a pronoun. If we separate sentences with pronominal subject from sentences with full noun phrase subjects, we see that the decline in topicalization discussed above is really for the most part a matter of sentences with full noun subjects. In sentences with pronominal subject the decline is much less dramatic: After a sharp fall between me1 and me2,[14] the rate of topicalization remains more or less stable. In contrast, the decline in sentences with full noun phrase subjects is much more dramatic than it would appear from the overall topicalization data in Table 2.1. Table 2.7 and Figure 2.5 illustrate this divergence clearly.

[14] The sudden fall exists because of two facts: First, in the period me1, there was still a certain wreckage of knowledge of Old English syntax present among the scribes, even after the Norman Conquest. Second (this was pointed out to me by Joel Wallenberg, p.c.), the dialect base in the period me2 is much broader than the one of me1, including also more ‘progressive’ dialects to a greater extent than me1. Period me1 is very much biased towards a Northern syntax because it contains the Orrmulum, which forms a large part of the corpus for that period. The dip in Old English full noun phrase subjects is there because most of the material of oe3 is by one author, Aelfric, whose language is particularly progressive.

Table 2.7: Rate of topicalization, full NP and pronoun subjects separated.

		oe1/2	oe3/4	me1	me2	me3	me4	eme1	eme2	eme3
full NP sbj.	# of sent. with DO	2017	4165	2855	1582	4925	2271	3229	3584	2544
	thereof topic.	277	330	219	92	167	66	67	82	28
	% top.	**13.7**	**7.9**	**7.6**	**5.8**	**3.4**	**2.9**	**2.1**	**2.3**	**1.1**
pers . pr. sbj.	# of sent. with DO	4167	5837	2474	2061	4683	3312	4490	6519	4513
	thereof topical.	459	750	351	136	391	191	309	346	219
	% top.	**11.0**	**12.8**	**14.2**	**6.6**	**8.3**	**5.8**	**6.9**	**5.3**	**4.9**

Figure 2.5: Rate of topicalization, full NP and pronoun subjects separated.



The different behaviour of pronouns and full NPs with respect to topicalization leads us to the supposition that there is something in the nature of pronouns and full noun phrases respectively that is responsible for this difference. So we must look for intrinsic differences between these two types of noun phrases.

Instead of giving a list of possible candidate for intrinsic differences between full noun phrases and pronominal noun phrases, I just start with the difference that turns out to be the crucial one. This property is the difference in prosodic status between full noun phrases and pronouns. Pronouns are naturally unstressed elements (see e.g. Ries 1907:95; Truckenbrodt 2006), whereas full noun phrases have at least word stress or phrasal stress. Connected with that is a different cognitive status: pronouns always represent old information, whereas full noun phrases can represent both old and new information (cf. Gundel et al 1993). The topic of the sentence tends to be expressed pronominally (Grosz et al 1995). Thus pronouns normally do not bear any focal emphasis, since topics normally are the very referential expressions that do not attract focus.[15] Full noun phrases, on the other hand, are more likely to receive focal emphasis, as they can easily represent new information or stand in contrast to some other referential expression.

It can be seen quite easily (by determining the focus structure of a sample set of topicalization sentences with pronominal subjects) that pronominal subjects almost never bear focal emphasis. If in a case of topicalization in the stricter sense – that is, in a double focus construction – the second focus lies on the subject, this subject is essentially never realized pronominally. This is because, if the second focus lies on the subject, the subject must be in a contrastive relationship to something else, and a merely pronominal reference is not sufficient to allow the correct reference in such a case, if there are no deictic means available to make the reference unambiguous (which in written texts is usually not the case). Let me illustrate this with an example: Suppose a couple, let us call them Rich and Alyssa, has invited two old friends, John and Bill, and is busy deciding which dishes to serve. They are especially uncertain with regard to the vegetables. Alyssa knows that one of the friends likes beans, and the other likes peas. As Rich knows them

[15] As a matter of fact, ordinary topics never receive focal emphasis. There are however elements, namely the sorting-key expressions in double focus constructions, that share certain properties with topics but are obligatorily focused. Therefore I qualified my statement somewhat in the main text.

better, Alyssa asks him: “Well, I know one of them likes beans and one likes peas. Do you know who likes beans and who likes peas?” Rich’s natural answer might be something like “Well, JOHN likes BEANS, and BILL likes PEAS,” or even topicalized: “BEANS, JOHN likes, and PEAS, BILL likes.” The point here is not whether it is natural to use topicalization or not in this case, the point is rather that we have two foci in each sentence, one on the subject, one on the object. If Rich were to use a pronoun in the answer (“Well, BEANS, HE likes, and PEAS, HE likes”) the reference would crash as both potential referents of ‘he’, John and Bill, are equally salient in the discourse. By the mere fact that they are in contrast to each other they are promoted to equal salience, no matter whether they have been equally salient in the preceding discourse or not. The reference with a pronoun could only work if the referents are present and can be pointed at in such contrastive cases. As we are dealing with written texts, however, direct deixis can be excluded. So we can say: if we have a double focus construction, and one focus is on the (topicalized) object, and the second focus is on the subject, the subject must be realized as a full noun phrase. This entails that in topicalization sentences with a pronominal subject, the second focus cannot lie on the subject but must lie somewhere else.[16]

So, it is clear that for topicalization it matters most whether the subject is prosodically weak (= pronominal) or not. For the case of double focus topicalization, we can formulate certain hypotheses: That there is a decline if the subject is a full NP may indicate that a potentially focalized subject is incompatible with the obligatorily focalized object that stands together with the subject in the left periphery. That there is no decline

[16] This is of course rather strong and does not cover cases in which pronominal reference is unambiguous, as in contrasts like ‘I ↔ you’, ‘we ↔ you’ etc. These cases are rare in our texts, however, and therefore can be neglected for the argument.

when the subject is a pronoun might likewise reflect the fact that pronoun subjects almost never are focused, and therefore are almost always compatible with the focused topicalized object.

We can perhaps tentatively conclude that the rate of object topicalization remains stable if at least one of the elements in the left periphery does not bear focus. From this it follows that there must be something ill-formed about two focalized elements in the left periphery. Foci have as their main prosodic property that they constitute prominence peaks. In section 2.4.2 I will present a bit of evidence that indicates that the problem is not the foci per se but the associated prominence peaks that come to stand adjacent to each other when both the topicalized object and the subject are focused. Following the eurhythmic theory of Hayes (1984) we may state tentatively that the factor that makes two high prominence peaks in the left periphery ill-formed is the eurhythmic desire for alternation. Two adjacent prominence peaks violate this principle, and therefore we perceive such a sentence as unnatural, simply wrong, if there is nothing that intervenes.

The idea that the desire for alternation plays such a central role and, following from that, that sentences with pronominal subjects are systematically different from sentences with full noun phrase subjects because of the different prosodic properties of these subject types essentially goes back to John Ries (1907). He observed that the variation between V2 and V3 in the Old English of Beowulf was dependent on whether the subject was a pronoun or not, and deduced from this that it was the prosodic weakness of the subject pronoun that accounted for it: The second element in a clause had to be prosodically weak, and since pronouns are elements that are prosodically maximally weak (in Ries' view, 1907:95), the pronoun naturally occupied this position. He formulated a rule, which sounds a lot like the CAR, which I will propose in section 2.4.3 below. His *Rhythmisches Gesetz* (‘rhythmic law’) runs as follows:

> *Rhythmisches Gesetz* (Ries 1907:91):
> „Auf die erste Satzhebung folgt, wenn möglich, eine Satzsenkung.“
> (‘The first peak in the clause is followed by a trough, if possible’)

This law he recognized as being so powerful that it could directly influence the way the sentence was built (Ries 1907:92). It followed directly from a eurhythmic principle which foreshadows Hayes (1984) and, again, the CAR, to be proposed below, which he words as follows:

> “rhythmischer Wohllaut eines Satzes kann weder durch unvermittelte Nebeneinanderstellung mehrerer starkbetonter Worte erzeugt werden, noch verträgt er sich mit der Zusammenhäufung vieler unbetonter, er beruht vielmehr in erster Linie auf der Abwechslung.“ (Ries 1907:91f.)
> (‚rhythmic euphony of a sentence cannot be generated by juxtaposition of several strongly emphasised words, nor is it compatible with the accumulation of many unemphasised ones. It depends first and foremost on *alternation* [my emphasis, A.S.]’

Thus, the crucial element of this investigation, the CAR, directly goes back to John Ries, whom we can regard as the true discoverer of it.

2.4.2 Scene-setting elements

Up to this point, our discussion has been mostly about focal emphasis and we have concluded tentatively that it is somehow problematic to have two focalized elements in the left periphery of a sentence. There is another relevant type of element that was fronted frequently in Middle English and that is still fronted commonly in Modern English, viz. scene-setting elements. Scene-setting elements can be defined semantically as expressions that limit the situation in which a given proposition is judged with respect to its truth value (definition following Jacobs 2001). They are usually locative or temporal prepositional phrases or adverb phrases.

A typical property of scene-setting elements is that they do not receive focal emphasis (unless in a contrastive context). Consequently, if the explanation put forward above for the decline of topicalization is correct, we should expect the rate of scene-setting elements not to decline. The reason for this is as follows: In a corpus, most examples of sentences with a preposed scene-setting element will have wide focus, simply because such sentences are much more common than sentences which contain a narrow focus. Note that the scene-setting element does not attract narrow focus per se, quite in contrast to topicalized objects in double-focus constructions. This means that a configuration such as in a double focus construction – namely that there are two adjacent high peaks in the left periphery, both a reflex of a narrow focus indicator – normally does not occur in such sentences: In the absence of a focus indicator, they will receive normal metrical prominence by the nuclear Stress Rule and the rules of eurhythmy and thus end up having the main metrical prominence peak at the end of the sentence and the second highest peak on the scene-setting element. Thus, such sentences will be more or less automatically metrically well-formed (because the rules generate only well-formed sentences, unless contrastive focus disturbs the picture), and this means that it does not matter for the metrical structure whether a scene-setting element is preposed or not.

Tables 2.8 and 2.9 and Figure 2.6 show that, as expected, the rate of scene-setting preposing does not decline between Old and Modern English. Table 2.8 shows the rate of topicalized temporal adverbs, whereas Table 2.9 show the rate of topicalized locative adverbs; Figure 2.6 illustrates both tables. In both cases, the rate does not show a consequent behaviour; it declines slightly during the Old English period, then rises significantly during the Middle English period, then drops off somewhat. This is not the place to go into the reasons for this rise; the main point is that we see no continuous decline.[17]

Table 2.8: Rate of topicalization of temporal adverbs.

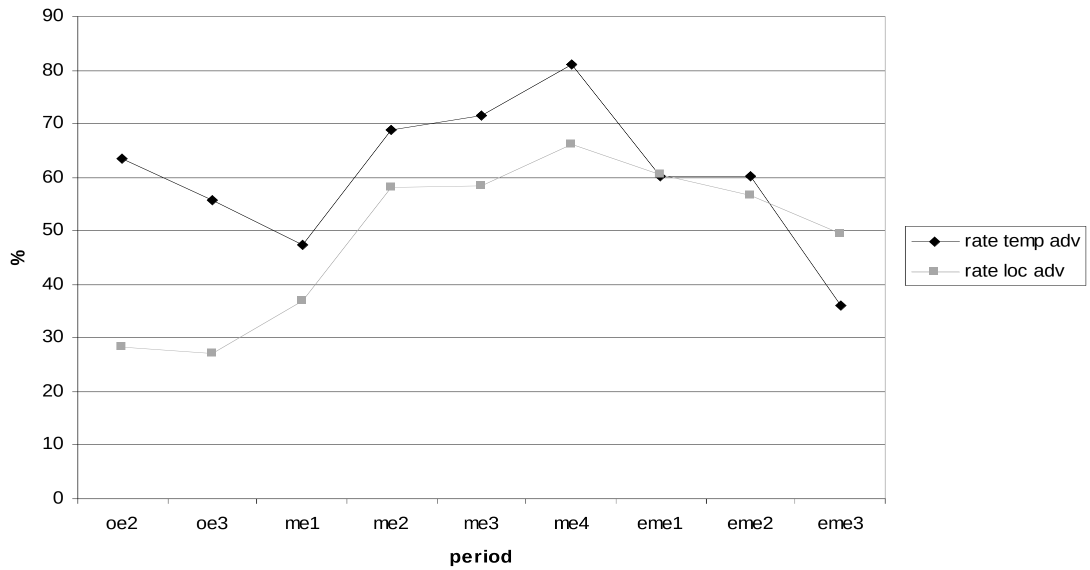
	oe1/2	oe3/4	me1	me2	me3	me4	eme1	eme2	eme3
# of sent. with temp. adv.	2404	3776	1678	578	3545	2510	2153	3101	1620
thereof topicalized	1524	2107	794	398	2536	2038	1294	1864	584
% of topic.	**63.4**	**55.8**	**47.3**	**68.9**	**71.5**	**81.2**	**60.1**	**60.1**	**36.0**

Table 2.9: Rate of topicalization of local adverbs.

	oe1/2	oe3/4	me1	me2	me3	me4	eme1	eme2	eme3
# of sent. with loc. adv.	289	361	491	124	922	753	866	712	588
thereof topicalized	82	98	181	72	539	498	523	400	291
% of topic.	**28.4**	**27.1**	**36.9**	**58.1**	**58.5**	**66.1**	**60.4**	**56.2**	**49.5**

[17] It is not clear why the rate of the topicalization of local and especially temporal adverbs drops so sharply in eme3. If a syntactically parsed corpus for the time after 1710 were available, we might have a chance to find out whether this is an accident or reflects the start of a real development. At any rate, it has nothing to do with the changes described here. During the decline of object topicalization, the rate of adverb preposing remains stable.

Figure 2.6: Rate of topicalization of temporal and local adverbs.



We could also have used all PPs, as they for the most part are scene-setting elements or other (unstressed) adverbials. Prepositional objects are comparatively rare. In Table 2.10 the functions of a random sample of prepositional phrase from the periods me1 and me4 are listed. It becomes immediately clear that prepositional objects (which are the only prepositional phrases that could become prosodically 'dangerous' if topicalized, as they could be focalized) account for less than one tenth of all prepositional phrases.

Table 2.10: Grammatical functions of prepositional phrases.

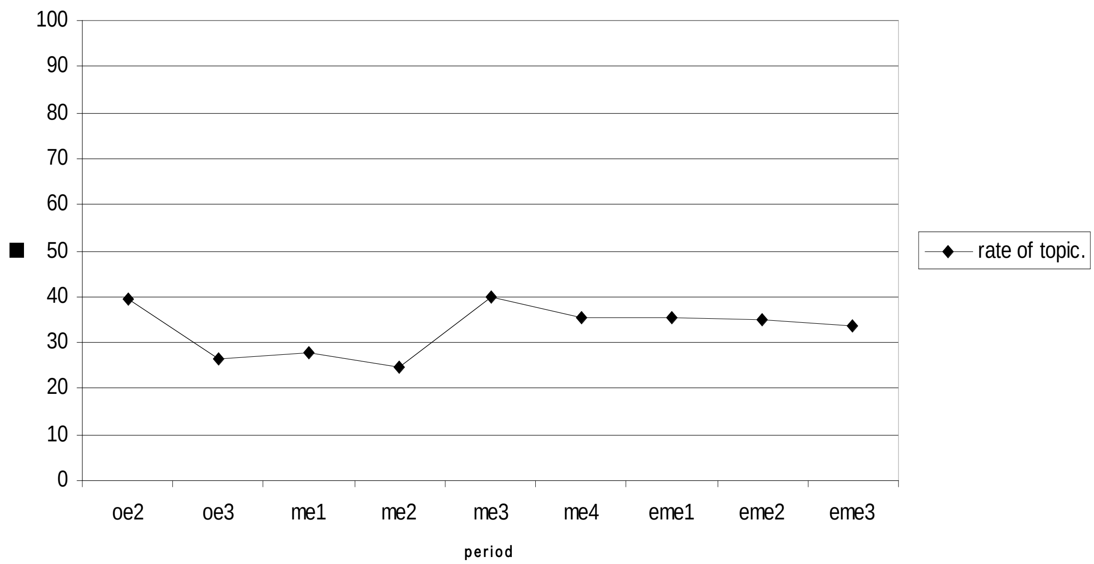
	me1 (n = 45)	me4 (n = 82)	total (n = 127)	**%**
scene-setting	25	44	69	**54.3**
other adverbials	16	30	46	**36.2**
prep. objects; directional phr.	4	3	7	**5.5**

If we, therefore, produce the rate of topicalization of prepositional phrases, we are not surprised to find that it patterns more with temporal and local adverbs, in that the rate stays stable and does not show any decline (Table 2.11 and Figure 2.7). Note that this rate is based on much more data than the rates in Tables 2.8. and 2.9 and is much more stable.

Table 2.11: Rate of topicalization of prepositional phrases.

	oe1/2	oe3/4	me1	me2	me3	me4	eme1	eme2	eme3
# of sent. with PPs	1448	1271	7709	4628	16518	9596	13309	15503	10494
thereof topicalized	569	335	2142	1134	6566	3399	4729	5437	3516
% of topic.	**39.3**	**26.4**	**27.8**	**24.5**	**39.8**	**35.4**	**35.5**	**35.1**	**33.5**

Figure 2.7: Rate of topicalization of prepositional phrases.



To summarize, we saw earlier that the decline in topicalization is observable only if the element in the left periphery of the clause can be focused. From that we concluded that there was some problem with having two foci in the left periphery. Scene-setting elements typically do not bear focal emphasis; and, in accordance with our hypothesis, their rate of topicalization does not decrease. In this respect, they are comparable to pronominal elements.

There is however a crucial difference between scene-setting elements and pronouns: Pronouns are naturally weak elements. This means that they do not bear metrical prominence and thus do not receive either word stress or phrasal stress. Scene-setting elements, on the other hand, do bear metrical prominence, by virtue of being phrases with lexical words in them. A question which might arise from this is: If it is ill-formed to have two focalized elements in the left periphery, should it not be ill-formed, as well, to have two elements with high metrical prominence on the phrase level assigned to them in the left periphery?

This question foreshadows the discussion of part 3. For the moment it is sufficient to point out that usually, when we have topicalization of a scene-setting element, the phrase after the scene-setting element (that is: the subject) is less prominent than the scene-setting element.[18] This follows automatically from the rules for metrical prominence assignment. Note that these rules would exclusively generate alternating structures, if there are no semantically governed foci which hinder the 'normal' prominence assignment. The next phrasal metrical prominence of comparable (in fact, higher) prominence is the prominence peak that corresponds to the 'nuclear stress' at the end of the clause.[19] As a sentence with a scene-setting element contains at least two other constituents, viz. the subject and the verb, and as the highest metrical prominence is

[18] It is important to point out that this is not due to the presence of any focus features or the like, but reflects just the normal stress as assigned by the prosodic rules; see chapter 2. Of course, if a focal emphasis were present, as in e.g. 'Yesterday, only BETTY slept', the focalized emphasis is more prominent than the scene-setting element. Such sentences are unproblematic, because there is no focal emphasis on the scene-setting element; it just receives whatever prominence the grid assignment machinery assigns to it.

[19] This fact is what leads many researchers of prosodic constituency to assume that topicalized elements and the rest of the sentence form two separate intonational phrases (e.g. Nespor & Vogel 1986, Taglicht 1998).

assigned to some element in the verb phrase positioned at its right periphery, which is the verb, some object or adverbial, but never the subject,[20] we can conclude that under wide (maximal) focus the two phrasal metrical prominence peaks on the scene-setting element and the subject will never have the same prominence (2-20).

```
(2-20)  …
         ( * )          ( .        * ) (Clause level)
           *          |   *    | *   (Phrase level)
        …
        Yesterday, Betty slept.
```

What does this tell us about focal emphasis? From the discussion of scene-setting elements we can conclude that it is not the presence of two elements of some prominence in the left periphery that is problematic, but the presence of two equally prominent elements. If we never have two peaks of equal prominence on the phrase level in the left periphery, as is normally the case with scene-setting preposing, no problem arises and therefore the rate of preposing / topicalization does not decline.

Note now that foci do not come in different levels of prominence: There is a strictly bipartite distinction: Either an element is focused or it is not (cf. Ladd 1980:46). This means, however, that whenever two focused elements wind up adjacent to each other, these two adjacent elements have equal prominence (namely the prominence given by focus). So the problem with topicalization of a full noun phrase object in clauses with full noun phrase subjects is perhaps not so much that we have two neighbouring foci per se, but rather that we have two equally strong prominence peaks adjacent to each other.

[20] Strictly speaking, the subject does not belong to the verb phrase at least at the surface, as it has been moved out to SpecIP. So, if the default highest phrasal stress assignment takes place within the verbal phrase, the subject is ruled out on the outset as candidate for receiving it.

2.4.3 Formulation of the Clash Avoidance Requirement

Now we are in a position to summarize the findings of the preceding sections: A structure is metrically ill-formed if it has two elements of equal prominence adjacent to each other. The ill-formedness can be interpreted as the violation of a principle or at least requirement. The requirement in question – the Clash Avoidance Requirement (= CAR), which has been introduced in an informal way already – can be formulated in the following way:

> *The Clash Avoidance Requirement (descriptive form for foci):*
> If there is more than one focus in a clause, at least one non-focused element must intervene.

or, more generally:

> *The Clash Avoidance Requirement (descriptive general form)*
> Two elements of an equal, given prominence must be separated by at least one element of lesser prominence.

As it stands, the CAR looks rather similar to the Rhythm Rule of Liberman & Prince (1977) and we will see in chapter 3 that this is no coincidence. It is also highly reminiscent of Ries' (1907:91) 'rhythmisches Gesetz' (rythmic law), which is also no coincidence, as the whole train of thought is heavily influenced by Ries (1907). There are parts of the requirement as it stands that do not necessarily follow from what has been said in the previous sections. It will be the task of chapter 3 to show that this is indeed the right form of the requirement. So let us accept it tentatively for the moment. The alternating quality that is the essence of the requirement is suggested by the scene-setting cases in 2.4.3 that can be said to be well-formed because there the phrasal peak on the scene-setting element is always followed by a lesser prominence. Likewise, if we have two focalized elements adjacent to each other, the ill-formedness is a direct consequence of that adjacency. So, the requirement as it stands does cover the facts thus far considered. In chapter 3 we will look for further, independent evidence for the reality of this requirement.

In the light of the Clash Avoidance Requirement, the obvious unease speakers began to have with topicalization with full noun phrase subjects can be explained: It is because in topicalization often focal emphasis is on the topicalized element, and a full noun phrase subject is more likely to have focal emphasis than propnominal subjects, and if they are adjacent, it results in a violation of the CAR.

It is, however, still not clear at the moment why there was a decline in topicalization, in other words: why in the beginning of the observed period topicalization with full noun phrase subjects seems to have been unproblematic. If the only factor we have to keep in mind is the focus on a topicalized object and on a subject in a double-focus-sentence, we need an explanation for why such structures were not avoided at all times. Shouldn't we expect a constant, low frequency throughout the history of English? Obviously something changed in the course of Middle English, to cause topicalization in sentences with full noun phrase subjects, which obviously was perfectly in harmony with the CAR before this putative change, now to be at odds with it. It will become clear that indeed we can identify a change that did exactly that. This is the subject matter of the next section.

2.4.4 The loss of the V2 word order option

In hunting for the change that made double-focus topicalization awkward, we should start by searching for syntactic changes of which we know that they happened in the crucial time span, that is, the Middle English period, and examine whether we can see a possible way how they could interact with the CAR.

What we find is the following: Middle English is characterized by two major syntactic changes that took place during this period, namely the change from OV to VO word order and, towards the end of the period, the loss of the verb-second word order option (V2; cf. van Kemenade 1987). It is not conceivable that the change from OV to VO word order should have made any difference for object topicalization, as there the object ends up at the left edge of the sentence anyway, no matter where it was base-generated. Therefore we can ignore this change and turn to the second change, the loss of V2.

Let me repeat that in purely descriptive terms, V2 is a word order option familiar to Old English and Middle English. The V2 constraint is a strong tendency common to all Germanic languages to form sentences such that the verb is at second position in the sentence. Of course there are several ways that this constraint can be implemented syntactically, and different languages choose different structural configurations to produce V2 sentences. Therefore it should be clear that the term V2, at least in the way I use it here, does not necessarily entail movement of the verb to C. In English this version of V2, CP-V2, is the structure that underlies wh-questions and certain other constructions, but not normal declarative sentences. In declarative sentences, the verb is not moved to C, and 'V2' word order is a by-product of movement of some constituent to SpecCP, movement of the verb to T and movement of the subject to a position below T, with C and SpecTP empty (which is the well-known analysis by Kroch & Taylor 1997; Haeberli 2002). It is this sense of V2 I use in this section and all other sections of this study, unless indicated otherwise. V2 was optional in Middle English (as it was in Old English, as we will see in chapter 4), which means that movement of the subject to SpecTP was possible, too, thus producing V3 sentences.

In a V2-sentence, some constituent stands at the left edge of the sentence, and is followed by the verb. In (2-21) there are some Middle English examples from Aelred of Rievaulx's *De Institutione Inclusarum* (ca. 1450, Southern West Midlands). Modern German has grammaticalized the V2 constraint as the unmarked case for declarative main clauses and wh-question main clauses by making movement of the verb to C obligatory, so that all sentences of these sentence types are automatically V2 sentences. In (2-22) the Modern German glosses of the Middle English sentences in (2-21) are perfectly grammatical. Note that the Modern English glosses, on the other hand, are ungrammatical. This illustrates that the V2 word order option is basically gone in Modern English declarative clauses.

(2-21) a. Al this say I not oonly for the but for other

* *All this say I not only for you-sg. but for others*

'All this I say not only for you but for others'

(Aelred 4.104)

b. In that tyme shuld euery cristen man adde somwhat moor to his

* *In that time should every Christian man add somewhat more to his*

fastyng

fasting

'In that time, every Christian should do more with respect to fasting'

(Aelred 8.194)

c. but sikernesse might he noon gete

* *but certainty might he none get*

'but he could get no certainty'

(Aelred 12.313)

(2-22) a. Alles das sage ich nicht nur für dich, sondern für andere.

b. In dieser Zeit sollte jeder Christenmensch hinzufügen ein bisschen mehr zu seinem Fasten.

c. Aber Gewissheit kann er keine bekommen.

The standard subject-verb sentence of Modern English is of course a sentence that superficially conforms to the V2-constraint. But this is accidental; it is well-formed because of another constraint that is at work in Modern English, namely that the subject has to precede the verb. Since it is this constraint that Modern English follows rather than the V2-constraint, the Modern English translations of (2-21a,b) are not V2- but V3-sentences.

The verb second option remained possible throughout the Middle English and Early Modern English periods, but it showed a significant decline in use, at least in the dialects of the South and of the West Midlands. These dialects had also a certain impact on the dialect of London that was to become the written standard dialect towards the end of the Middle English period. In the North, V2 stayed alive for a longer time. At a certain point, however, even Northerners started to write in the London dialect, so that we have no evidence as to when V2 was finally lost in the spoken discourse of the Northern dialects as well. Leaving aside wh-questions, which still exhibit V2, in today's English V2 occurs only marginally in certain well-defined special contexts (e.g. Locative Inversion, 1-23a or certain presentational sentences, 1-23b). So the outcome of the change is undisputable. The change did not affect CP-V2, but only V2 by movement of the verb to T and the subject to a position lower than.

(2-23) a. Down the hill rolled the ball.

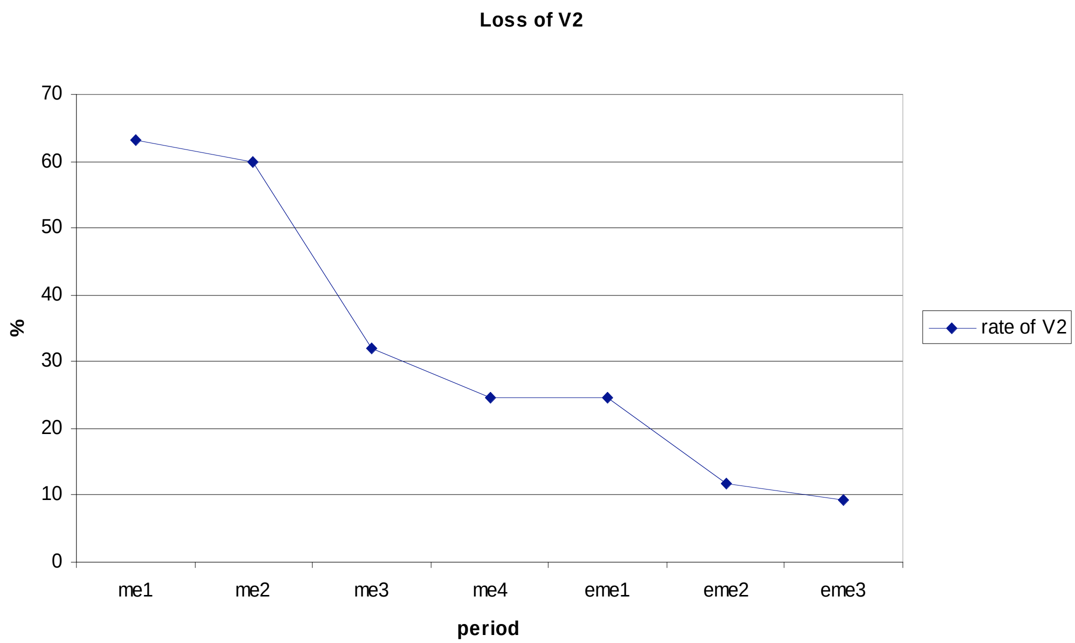
b. Here comes the judge.

The reason for the change and its exact process are more of an issue (see e.g. Haeberli 2001). For the moment however it suffices to say that the loss of verb-second (V2) can be demonstrated e.g. by showing that in topicalized sentences the rate of sentences in which the verb follows the topicalized constituent (creating an X-V-S-order) decreases as opposed to sentences in which the topicalized constituent is followed by the subject, which in turn is followed by the verb (creating an X-S-V-order). Table 2.12 and Figure 2.8 show this development for sentences in which the preposed constituent is a prepositional phrase and the subject is a full noun phrase. Due to heavy dialectal differences (especially in the North, which sticks to V2 much longer than the rest of the English language community), texts from the North have been excluded; furthermore two East Midland texts which show a deviant pattern (both by Capgrave) have been excluded.

Table 2.12: Loss of Verb-Second.

	me1	me2	me3	me4	eme1	eme2	eme3
# topical. PPs	659	250	2116	639	1168	1113	684
thereof PP-V-S	416	150	674	158	289	131	63
% PP-V-S	**63.1**	**60**	**31.9**	**24.7**	**24.7**	**11.8**	**9.2**

Figure 2.8: Loss of Verb-Second.

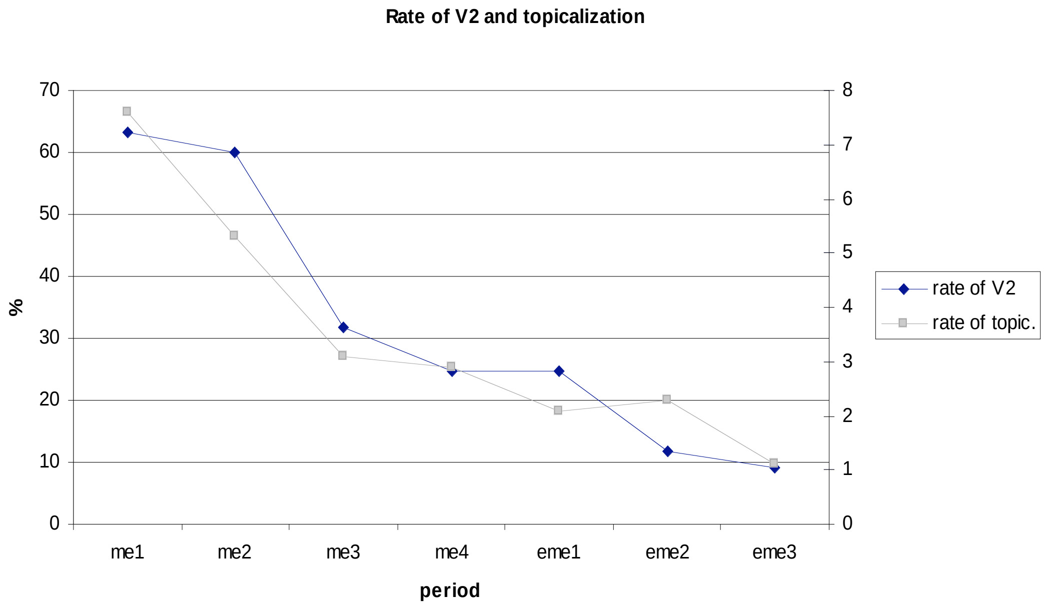


If we superimpose the curve with the decline in object topicalization (Table 2.13; here confined to the same texts that formed the basis for Figure 2.8; only full noun phrase subjects), we see that the two developments follow the same time course (Figure 2.9).

Table 2.13: Decline in topicalization in Southern and West Midland texts.

	me1	me2	me3	me4	eme1	eme2	eme3
# sent. with DOs	2855	1300	4615	2271	3229	3584	2544
thereof topical.	219	69	145	66	67	82	28
% topic.	**7.6**	**5.3**	**3.1**	**2.9**	**2.1**	**2.3**	**1.1**

Figure 2.9: Decline in topicalization in Southern and West Midland texts.



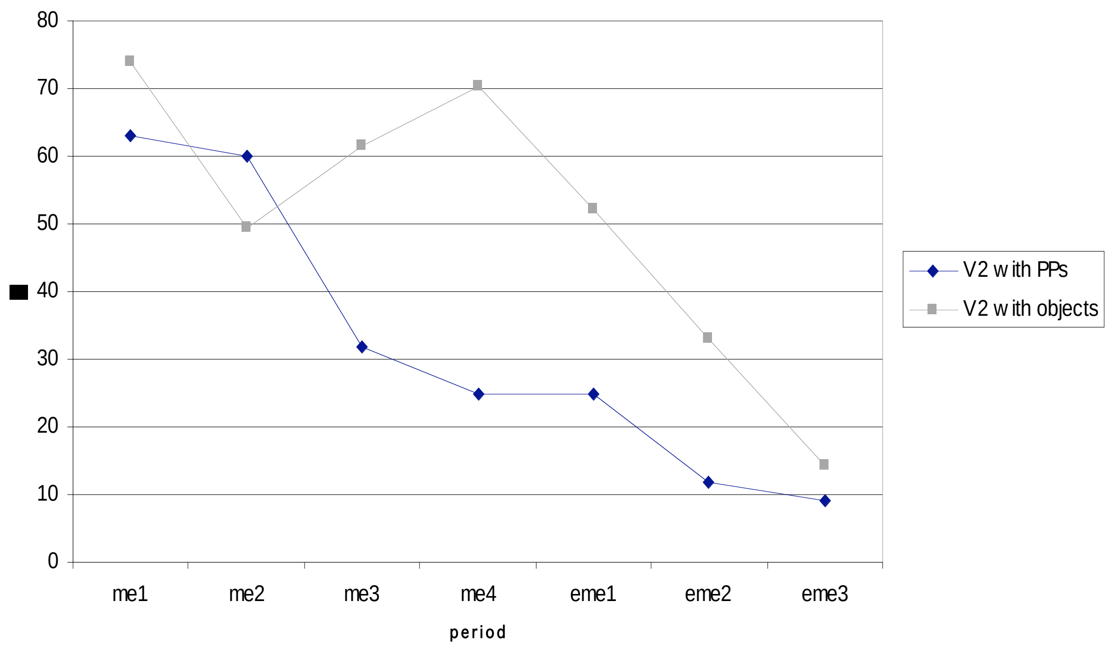
So it is tempting to assume that these two processes are related. In the following I attempt to give an account of what this relationship might look like.

First it is interesting to note that the rate of V2 in sentences with object topicalization remained much higher throughout the Middle English and Early Modern English period, until it reaches a low rate which is comparable to the rate of V2 in sentences with preposed PPs in eme3 (Table 2.14; Figure 2.10; texts from the North and Capgrave excluded). This observation will play a role further along in the argumentation, so it should be borne in mind.

Table 2.14: Rate of V2 in main clauses with topicalized PP, compared to main clauses with topicalized direct object.

	me1	me2	me3	me4	eme1	eme2	eme3
# topical. PPs	659	250	2116	639	1168	1113	684
thereof PP-V-S	416	150	674	158	289	131	63
% PP-V-S	**63.1**	**60**	**31.9**	**24.7**	**24.7**	**11.8**	**9.2**
	me1	me2	me3	me4	eme1	eme2	eme3
# topical. DOs	219	69	145	47	67	82	28
thereof O-V-S	162	34	89	33	35	27	4
% O-V-S	**74.0**	**49.3**	**61.4**	**70.2**	**52.2**	**32.9**	**14.3**

Figure 2.10: Rate of V2 in main clauses with topicalized PP, compared to main clauses with topicalized direct object.



An interesting property of this data is that it seems to constitute a violation of the Constant Rate Effect (Kroch 1989). The loss of V2 is really the product of a competition between a grammar (basically the Old English grammar) that allows for V2 (more precisely: movement of the subject to a position below T) in afirmative declarative sentences, and one that does not (basically the Modern English grammar in which the subject obligatorily moves higher than T). The ‘loss of V2’ happens when the new grammar, which does not allow for V2, wins out. The curve of the decline of V2 with prepositional phrases shows the classic picture of such a grammar competition situation: The decline of the ‘old’ grammar (and the rise of the ‘new’ grammar) follows an S-shaped curve. The Constant Rate Effect would now predict that V2 should be lost at the same rate – that is, with the same starting point, the same slope and the same ending point – regardless of what kind of an element is in the preverbal position. The data of Table 2.14 and Figure 2.10 seems to refute this, since here it looks as if the grammar with the V2 option comes into competition with the grammar without the V2 option later, and the slope is steeper.

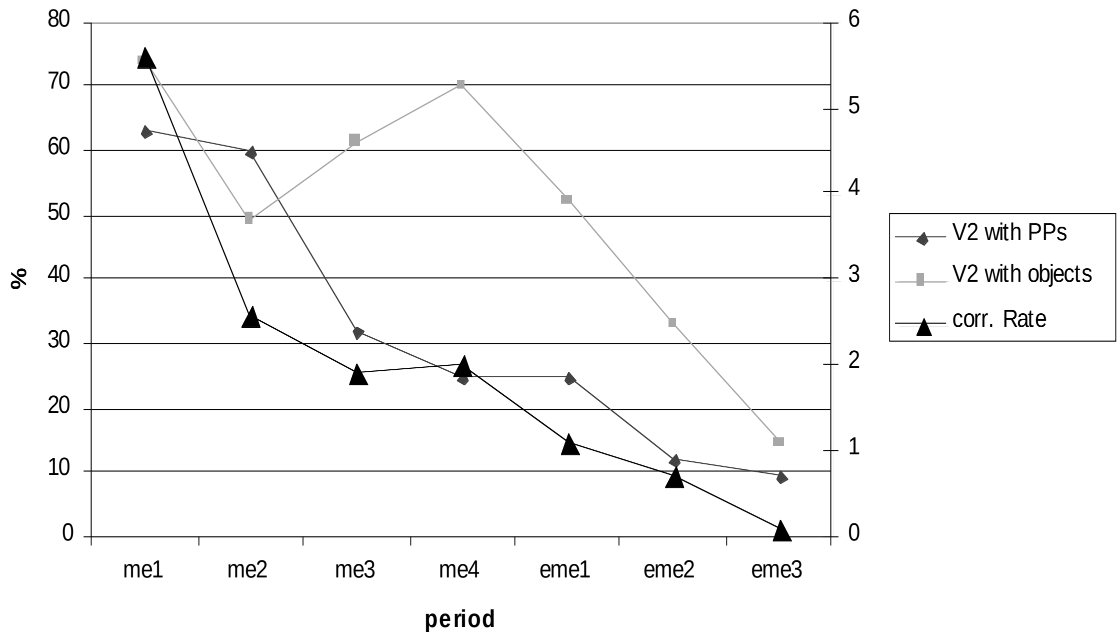
This impression is, however, erroneous. We have to bear in mind that a certain number of cases – the cases with potential V3 word order that would violate the CAR – are not included in the data because, in these cases, the object will not have been topicalized. Note that we do not have the same problem with preposed prepositional phrases as here the CAR has no effect. This means that the two frequencies of V2 in Table 2.14 and Figure 2.10 are not directly comparable.

We can, however, make them comparable. We can factor out the effect of the CAR with direct objects simply by multiplying the rate of V2 by the rate of topicalization. The result is in Table 2.15 and Figure 2.11. We see that now the curve for direct objects is parallel to the curve for prepositional phrases. This means that V2 disappears at the same rate in both environments, just as the Constant Rate Effect would predict.

Table 2.15: Corrected rate of V2, including an estimate of the CAR violating cases

	me1	me2	me3	me4	eme1	eme2	eme3
rate of V2	74.0	49.3	61.4	70.2	52.2	32.9	14.3
rate of obj. top.	7.6	5.3	3.1	2.9	2.1	2.3	1.1
corr. rate (%)	**5.6**	**2.6**	**1.9**	**2.0**	**1.1**	**0.8**	**0.1**

Figure 2.11: Rate of V2 in main clauses with topicalized PP, compared to main clauses with topicalized direct object and to the corrected rate.



Now let me turn to the second observation mentioned above, which is as follows: If we consider the German sentences in (2-22), we see that there any focal emphases are reliably separated by at least the verb. In sentence (2-22c) with split topicalization this is perhaps clearest as split topicalization usually occurs with double focus. The special property of split topicalization is that there the two focal emphases are base-generated in the same phrase (with indication of focus repeated in 1-24d). If we look at other examples of German double focus sentences (2-24), we see that this intuition is indeed true: The first element in focus usually stands before the verb, the other element somewhere else (cf. also Jacobs 2001, Steube 2003).

(2-24) a. /ALLE hab ich natürlich \NICHT gelesen.

b. Die neuen /SCHUHE hat sich Susi in \MÜNCHEN gekauft, die neuen /HAARSPANGEN in \GARMISCH.

c. Den /HUNDERASIERER hat \ULLER gekauft, nicht ich!

d. Aber /GEWISSHEIT kann er \KEINE bekommen.

It is perhaps fair to say that V2 sentences are seldom in danger of violating the CAR, as the second element, the verb, is only very rarely in focus. A possible conclusion is that speakers of Middle English, in which V2 syntax was still possible, could easily use V2 if they wanted to topicalize and the second focus was on the subject. Now, it is frequently the case that the second element with focal emphasis is the subject; in a sentence like (2-24c) it could hardly be elsewhere. But generally expressions in which both the subject and the object are put in contrast with some other subject and object engaging in the same activity (such as 1-24b,c) are quite natural.

That there is a clear relationship between the position of the second focus and V2 word order is shown in Table 2.16. A subset of Middle English sentences with topicalized object and full noun phrase subject has been randomly selected and the focus placement in these sentences has been reconstructed according to pragmatic properties of the phrases involved. This gives three types, dubbed here patterns A,B,C respectively. Remember that object topicalization with full noun phrase objects indicates most often a double-focus-topicalization, in which the topicalized object bears focal emphasis.

- Pattern A: The second focus is on the constituent directly following the topicalized constituent (**<u>O</u>** – **<u>1</u>** – 2 – X)
- Pattern B: The second focus is on the second constituent after the topicalized element (**<u>O</u>** – 1 – **<u>2</u>** – X)
- Pattern C: The second focus is on some constituent after that (**<u>O</u>** – 1 – 2 – **<u>X</u>**)

These patterns can be cross-categorized with whether they are V2 or not. If we now measure the rate of V2 out of all sentences that have the second focus on the subject (that would be Pattern A without V2 plus pattern B with V2) and compare it with the rate of V2 out of all sentences that have the focus on the verb (that would be Pattern A with V2 plus Pattern B without V2) we see that V2 is preferred when the second focus is on the subject, but dispreferred when it is on the verb. The comparison data shows that the rate of V2 is higher if the second focus is on the subject than we would expect from the non-critical group (pattern C) and the rate of V2 when the second focus is on the verb is much lower.

Table 2.16: Rate of V2 depending on focus, without bare demonstratives.

	Second focus on subject $^{\text{B-V2}}/_{\text{(B-V2+A-nonV2)}}$	Second focus on verb $^{\text{A-V2}}/_{\text{(A-V2+B-nonV2)}}$	Comparison data $^{\text{C-V2}}/_{\text{(C-V2+Cnon-V2)}}$
numbers	101/113	4/29	46/65
inversion (%)	**89.38**	**13.79**	**70.77**

So it is quite obvious that Middle English speakers used the V2 word order option, as long as it was still available in their grammar, exactly in situations in which using the V3 option would create a focus clash and avoided it where using V2 would create such a clash. In other words: The variability between V2 and V3 surface syntax was utilized in order to avoid violations of the CAR, which are likely when the subject is a full noun phrase, and the topicalized element the object.

Further support for the hypothesis that the V2 option was used preferably in contexts in which CAR-violations loomed, is the fact that the rate of V2 remains comparatively high in sentences in which the topicalized element is the object, as opposed to sentences in which the preposed element is a PP. The reason is that PPs, being mostly scene-setting elements or other adjunct elements, only seldom attract focus, whereas with object topicalization the topicalized object in most cases does attract focus, as most instances of object topicalization are double-focus-topicalizations. It is also true that the preposing of a PP is usually not part of a double focus construction, but rather scene-setting preposing (see section 2.2), where normally no focal emphases are involved at all. If this is so, neither the V2 nor the V3 word order option would lead to significant advantages with respect to the CAR, if a PP is preposed. Consequently, there is no bias in favour of the V2 option, and the decline in V2 is much ‘faster’ than with direct objects. With direct objects, on the other hand, there is a strong bias in favour of V2 as here to choose V2 leads to a real advantage, namely the guarantee of a metrically well-formed sentence, and therefore people prefer V2 in these contexts as long as they can. And at the time, in which the V2 option was practically gone – in the period eme3 – both rates meet at a comparably low level (Table 2.14, Figure 2.10).

The loss of V2 brings speakers in conflict with the CAR. Topicalization of an focused full noun phrase object will in many cases lead to a CAR-violation, namely in all cases in which the subject bears the second focus. Pronominal subjects usually do not bear focus, as we observed earlier, therefore topicalization here will as good as never lead to a situation in which the CAR is in jeopardy of being violated. Full noun phrase subjects, on the other hand, can bear focus and do it more frequently than pronominal subjects, thus creating possible CAR-violations.

The speakers of Middle and Early Modern English who had to produce such sentences now face a dilemma, so to speak. Object topicalization is not done for its own sake; it rather helps to encode certain types of pragmatic information, as we have seen. But the only reasonable way to interpret the data is that the speakers of late Middle and Early Modern English gave up on topicalizing in cases in which the CAR is violated. That means that the CAR is more important to these speakers than the unambiguous pragmatic encoding of objects by topicalizing them. Renouncing topicalization was apparently not a big deal and topicalization seems not to have been the crucial strategy for marking direct objects for their pragmatic function. They had their function, regardless, whether they were topicalized or not, and the addressees unterstood the function more or less clearly, independently of topicalization. By topicalizing, however, sentence processing is presumably made easier: If an element is topicalized, it can only have a particular pragmatic function out of a small set of functions, basically the ones presented in section 2.2. That means, as soon as a speaker topicalizes an element, the choice of potential functions for this element is limited and by that the chance increases that the addressee gets the intended meaning correctly. But nothing hinges on topicalization. In some ways, it is luxurious to topicalize: It is not necessary, but makes life easier.

So we can see clearly the force behind the decline of topicalization: The loss of V2, whatever its cause was, led to situations in which topicalization would produce structures that are ill-formed under the CAR because in them two focal emphases end up adjacent to each other. To avoid violation of the CAR, speakers compromised their desire to encode pragmatic information and ceased to topicalize when it was problematic for the CAR.

Let us summarize this section: We have shown that the type of subject matters: The decline of topicalization is only visible when the subject is a full noun phrase. Otherwise, the rate of topicalization stays stable. A comparison with scene-setting elements shows that the rate of fronting of scene setting elements does not change either. The crucial property that the case of fronted full noun phrase object and full noun phrase subject as opposed to the other cases has is that only there can two focal emphases end up adjacent to each other. From this demonstration we proceeded to the formulation of the Clash Avoidance Requirement, which says that two focal emphases must be separated by at least one non-focused element, or more general: Elements of equally high prominence have to be separated by at least one element of lesser prominence. Topicalization of full noun phrases can lead to conflict with the Clash Avoidance Requirement, if there is a focus on the topicalized element and if there is another focus on the subject. This poses a problem only in a syntactic system of the style of Modern English where the subject always has to precede the verb and topicalization therefore automatically leads to a V3-sentence. In Middle English, V2-syntax was still an option. With V2-syntax, any focal emphases on the topicalized element and the subject can be separated by the verb, and most often are separated in fact. V2 seems to have been utilized to avoid violations of the Clash Avoidance Requirement; once it had ceased to be a grammatical option, topicalization of full noun phrase objects in sentences with full noun phrase subject will ordinarily lead to violations of the Clash Avoidance Requirement. Language producers therefore refrain from using it.

2.5 Prosody beats Pragmatics

We concluded in section 2.2 that an alleged reduction in the range of information structural configurations that allowed for topicalization cannot be the reason for its decline, at least not in Middle English. It is true, however, that topicalization (which is, after all, optional in all cases) was employed less often in one particular case, namely in the case of double-focus-topicalization.

Double-focus-topicalization is the only type of topicalization with which violations of the CAR can arise, simply because it involves two equally strong foci, one of which is obligatorily at the left edge of the sentence, viz. on the topicalized constituent. After the loss of the V2-word order option, a CAR-violation will arise automatically if the second focus is on the subject (2-25a). This, as we mentioned earlier, does not happen in cases in which the subject is a pronoun, but it does happen frequently in cases in which the subject is a full noun phrase.

The easiest way to avoid a CAR-violation in such cases would be to leave the object in its base-generated position. As example (2-25b) illustrates, a sentence with foci on both the subject and the object is perfectly well-formed under the CAR if they are in their base-generated positions.

(2-25) a. $\boxed{*\quad\ *}$. C
BEANS JOHN likes

b. * . * C
JOHN likes BEANS

Obviously, this is the strategy that speakers of Middle and Early Modern English chose. The CAR-conformity of a sentence like (2-25b) has its price, of course. By not using topicalization, it is not immediately obvious which of the focalized phrases serves as the sorting-key, thus presumably the sentence is less easy to compute. But the speakers of English since the Middle Ages seem to have put up with that: They point out the sorting-key by topicalization only if topicalization does not lead to conflicts with the CAR; otherwise they are content not to mark the sorting-key explicitly and trust that the addressee will get it right from context.[21]

Modern German, on the other hand, is a language in which double-focus topicalization is possible and used frequently in order to indicate the sorting key explicitly. As German observes the verb-second-constraint rigorously, no clash of focal emphases will arise if the second focus does not happen to be on the finite part of the verb (a rare case). Moreover, contrastive foci alone have a high likelihood of occupying the *vorfeld* (cf. Steube 2001; Speyer 2004; 2006a).[22] This seems to be the German equivalent of English focus movement, but it is employed much more often. If we now take a number of German texts and collect all sorting-key and focus topicalizations from it, and compare it with the English translation of these particular sentences, we should get a sense of how sorting-keys are represented in Modern English.

---

[21] This is typical for pragmatic indicators in general: They tend to follow what Liedtke (1997) calls a ‘resultative’ usage mood, which means that the speaker trusts the pragmatic property they are meant to encode can be deduced from the context; pragmatic markers are therefore usually optional. His example are illocutionary particles in German. Topicalization can be regarded as just another structural pragmatic marker: It helps to ‘encode’ a certain function (in this case: the sorting-key), but the interpretation (here: the sorting-key interpretation) is available, regardless of whether the pragmatic marker (here: topicalization) is present or not.

[22] This has nothing to do with prosody. In general elements that are members of a set evoked in the local discourse have a strong tendency to move to the left edge position, regardless of whether they are in focus or not. They compete with scene-setting elements and aboutness topics for the vorfeld-position, and in conflict cases beat aboutness topics, but are beaten by scene-setting elements (see Speyer 2006a for a more detailed account). See e.g. Frey 2006, who argues for a prespecified ‘ContrastP’ on the left periphery of the clause.

Here a brief word is appropriate about the comparability of German and English focal emphasis, which also bears upon the question of how far focal emphasis in historic stages of English is reconstructable. Parts of this paragraph repeat what was said in 1.3.2, but I repeat it here, as it is relevant for this section. It is remarkable that in both Modern English and Modern German focal emphasis (as the acoustic realization of focus) has basically the same properties: It is realized in the same way, that is, as a mixture of pitch, loudness and duration, and it is triggered in the same fashion by narrow focus. This makes the two languages directly comparable for our purposes. Moreover, the rules for assignment of the sentence nucleus are rather similar. We can thus conclude that these three points – realization of focal prominence, mapping of focus to prominence, rules for metrical prominence assignment – must have been as they are in Modern German or English already in the common ancestor language of German and English. Otherwise it would be difficult to explain why the two systems should be so similar. But if that is so, we can easily infer that the main points of today's prominence system were already applicable to Old or Middle English or Old or Middle High German.

Let us go back to the direct comparison of German and English topicalization. I chose three well-known German novels written in the 1950s and early 1960s whose authors come from different dialect areas. The texts were *Katz und Maus* (Cat and Mouse) by Günther Grass (Pommern, now northern Poland), *Der Richter und sein Henker* (The Judge and his Hangman) by Friedrich Dürrenmatt (Swiss) and *Ansichten eines Clowns* (The Clown) by Heinrich Böll (Northern Rhine Valley). I took the examples of object (accusative, dative ad prepositional) in the *vorfeld* from these texts and categorized them into three categories:

- No second focus,

- second focus on some element other than the subject,
- second focus on the subject.

Second focus on the verb did not occur in the sample. These examples, furthermore, were cross-categorized according to whether the subject was a pronoun or a full noun phrase.

Then I took English translations of these texts and looked at the equivalents of the topicalized sentences. The focal structure should most often be the same in both languages, so that the cases of double-focus in German should correspond to double focus in English. Whereas the German examples are all of such a kind that the double-focus corresponds to *vorfeld*-movement of the focalized phrase that serves as sorting-key – in that point it is the functional and formal near-equivalent of English double-focus-topicalization – we expect the corresponding English sentences to make use of double-focus-topicalization only if no focus clash arises. That is, we expect double-focus-topicalization only if the second focus is not on the subject.

I counted the examples from the novels in the different categories. First, I simply made a distinction between pronominal and full noun phrase subject to give evidence that the underlying assumption is correct that in sentences with double focus the subject is rarely focalized if it is a pronoun but frequently focalized if it is a full NP subject. Table 2.17 shows the comparison.

Table 2.17: Rate of pronominal subjects among sentences with different focus structure types.

	only one focus (= 1)	2nd focus on non-subject (=2X)	2nd focus on subject (=2S)	*total*	**% of 2S**
pronoun subject	38	66	2	*106*	**1.9**
full NP subject	10	17	23	*50*	**46.0**
total	*48*	*83*	*25*		
% of pron.sub.	**79.2**	**79.5**	**8.0**		

Table 2.17 shows clearly, in my opinion, that the assumed correspondence between pronominal subjects and lack of focus is correct. In double focus constructions, if we get a full noun phrase subject, in about half of the cases it is focalized.

Having given evidence for one of the assumptions underlying this study, let us return to the main thread and demonstrate that sorting-key topicalization in Modern English is only done if it does not result in a focus clash. As I said, I looked at the three categories of examples separately and simply checked how the linear structure of the English sentences corresponded to the German focus examples. The main comparison is between categories 2X and 2S; categorie 1 serves as control data of topicalization (in the form of focus movement) if no CAR violation is possible. Tables 2.18, 2.19, 2.20 show the results.

Four strategies have been found to translate a German sentence with object in the *vorfeld* into English:

- T = keeping the German topicalization and translating as English topicalization (2-26),

- S = transforming the sentence so that the topicalized object in German corresponds to the English subject (2-27),
- N = translating the sentence with canonical word order and forgoing the sorting key effect (2-28),
- F = dissolving the sentence into sentence fragments (2-29).

(2-26) Mahlkes Haupt bedeckte dieser Hut besonders peinlich (Grass, p.127)

Mahlke's head$_{OBJ}$ *covered this hat*$_{SUB}$ *particularly embarassing*

'This hat covered Mahlke's head in a particularly embarassing manner.'

Translation: On Mahlke's head this hat made a particularly painful impression.

(2-27) Studienrat Treuge […] sollte eins ausgewischt werden (Grass, p. 28)

Teacher Treuge$_{OBJ}$ *should one*$_{SUB}$ *swept become*

'A practical joke was to be pulled on teacher Treuge.'

Translation: The intended victim was Dr. Treuge.

(2-28) Den findet keine Sau. (Grass, p. 71)

this$_{OBJ}$ *finds no sow*$_{SUB}$

‚Nobody will find it'

Translation: Nobody'll ever find it.

(2-29) Zu den sechs […] kamen noch drei weitere (Grass, p.61)

To the six came yet three others

'Three others joined these six in the afternoon.'

Translation: Six that morning and three more in the afternoon.

Table 2.18: Translations of sentences with object in the *vorfeld*, case 2S.

type	**T**	S	N	F	total
number	**0**	7	15	3	25
% of total	**0**	28.0	60.0	12.0	

Table 2.19: Translations of sentences with object in the *vorfeld*, case 2X.

type	**T**	S	N	F	total
number	**26**	15	42	0	83
% of total	**31.3**	18.1	50.6	0	

Table 2.20: Translations of sentences with object in the *vorfeld*, case 1.

type	**T**	S	N	F	total
number	**5**	8	33	2	48
% of total	**10.4**	16.7	68.8	4.2	

Whereas topicalization is sometimes translated as such in cases 2X and 1, that is, the cases, where no focus clash is pending, there is no such example in the critical case 2S. This suggests strongly that we have been correct in our assumption that topicalization is not becoming marginalized in general, but that topicalization still can serve to express the sorting key in a double focus construction; speakers however are prone to forgo topicalization and the pragmatic clarification connected with it[23] if it stands in conflict with a more powerful constraint; that is, the CAR.

[23] I wish to remind the reader once again that topicalization does not have an semantic or pragmatic effect in the sense that to topicalize something means to mark it as sorting-key; it is marked as sorting-key semantically no matter where in the sentence it stands, but topicalization only serves to make the sentence easier to compute in that the sorting key element is promoted to an exposed processing unit of its own

Let us summarize this section: The Clash Avoidance Requirement puts restrictions on double focus topicalization, but not on the other types of topicalization. It can be shown that in English translations of German texts (where these restrictions do not hold, because German bypasses conflicts with the Clash Avoidance Requirement by its V2-syntax) German topicalization is translated as English topicalization only if it does not cause a violation of the Clash Avoidance Requirement.

2.6 Summary

In section 2.1 it was shown that topicalization has declined over the course of English language history. This was demonstrated by calculating the rate of topicalization of accusative noun phrases (most of which are direct objects) in the available parsed historical corpora of English, which dropped from around 12% to around 3% between 1100 and 1700 AD.

Section 2.2 presented a possible explanation for the decline in topicalization, namely that the range of discourse configurations compatible with topicalization narrowed. This was shown to be wrong: Topicalization is used to express the same discourse structural configurations today as in Middle English. The four prime uses of topicalization are: preposing of scene-setting elements, preposing of the sorting key in a double focus construction, preposing of an anaphoric element (or ϕ-topic) and focus movement. The Clash Avoidance Requirement puts restrictions on double focus topicalization, but not on the other types of topicalization.

Section 2.3 dealt with another potential explanation for the decline in topicalization, which is that English word order became more rigid, as a consequence of the loss of case morphology. This can be demonstrated by the fronting of objects over subjects in subordinate clauses ('scrambling') for which no examples can be found after a certain time (~1250 AD); this indicates that such fronting became ungrammatical at that time. This cannot be the right explanation for the decline in topicalization, however, as topicalization should have become ungrammatical at the same time, which was not the case.

In section 2.4 it was shown that the type of subject matters: The decline of topicalization occurs only if the subject is a full noun phrase. Otherwise, the rate of topicalization remains stable. A comparison with scene-setting elements shows that the rate of fronting of scene setting elements does not change. The crucial property that the case of fronted full noun phrase object and full noun phrase subject as opposed to the other cases has is that only there can two focal emphases end up adjacent to each other (as pronouns and scene-setting elements tend not to be focalized). From this follows the formulation of the Clash Avoidance Requirement, which says that two focal emphases must be separated by at least one non-focused element, or more generally: Elements of equally high prominence have to be separated by at least one element of lesser prominence. Topicalization of full noun phrases can conflict with the Clash Avoidance Requirement, if there is a focus on the topicalized element and if there is another focus on the subject. This is only a problem in a syntactic system like that of Modern English where the subject always precedes the verb and topicalization therefore produces a V3-sentence. In Middle English, V2-syntax was still an option. With V2-syntax, focal emphases on the topicalized element and the subject are separated by the verb. V2 seems, at that historical stage, to have been utilized to avoid violations of the Clash Avoidance Requirement; once it had ceased to be a grammatical option, topicalization of full noun objects in sentences with full noun phrase subjects can easily lead to violations of the Clash Avoidance Requirement and speakers of Early Modern English therfore refrained from using it. The argument hinges on the hypothesis that prominence patterns in historical texts are detectable, and I reasoned that this is true.

In section 2.5, finally, evidence was presented that prosodic requirements trump the pragmatic convenience of topicalization. It could be shown that in English translations of German texts (which can avoid CAR-conflicts easily, having a rigorous V2-syntax) German topicalization is translated as English topicalization only if it does not cause a violation of the Clash Avoidance Requirement.

3. Phonological Aspects of the Clash Avoidance Requirement

The third part of this thesis is devoted to the development of the Clash Avoidance Requirement from a theoretical and an experimental perspective. Section 3.1 deals with double foci from an empirical perspective. After having established the fact that topicalization in general, even double focus topicalization, is acceptable in Modern English I present data from several experiments that offer direct evidence for the Clash Avoidance Requirement in utterances with double foci. Section 3.2 puts the Clash Avoidance Requirement into a broader theoretical context by comparing it to similar proposals made earlier in the literature. In section 3.3 a short overview of the Clash Avoidance Requirement in German is given, followed by a treatment of more theoretical issues in sections 3.4. and 3.5; namely, the relationship between metrical prominence and focal emphasis (3.4) and the motivation for a pause to be the preferred repair mechanism in cases of focus clash (3.5).

3.1 Double foci

In section 3.1.1 I show that in Modern English we see the same patterns that evolved in Middle English and Early Modern English, namely that topicalization is natural only in sentences in which the subject is unfocused (this follows of course only if it is topicalization in the stricter sense, that is, in double focus constructions). As pronominal subjects are nearly always unfocused, we see that topicalization in Modern English occurs mainly in sentences with pronominal subjects, but only seldom in clauses with a full noun phrase subject.

The unease in having two equally strong elements adjacent to each other is not confined to focal emphasis: We get similar effects with locative inversion, a construction in which V2 is still possible in Modern English, but only, if the V2-structure conforms to the CAR.

Section 3.1.2 presents data from several experiments on English and German in which the participants were forced to put two foci adjacent to each other. There are two groups of experiments: production experiments and reading experiments. In the production experiments, the participants could hardly be made to utter a clause with two adjacent foci. In the reading experiments, they inserted a pause between the two adjacent foci that they were forced to read. Adjacent foci were always treated in the same way, regardless of whether they were connected with topicalization or not.

3.1.1 The acceptability of topicalization in Modern English

Topicalization in general is acceptable in all dialects of English. It is avoided in written discourse for purely stylistic reasons (cf. e.g. Copperud 1960:64f.), but it can be heard in spoken discourse of both British and American English. Sometimes one gets judgements of British speakers that claim it is a purely Northeast American phenomenon (and due to Yiddish influence). This claim, which in reality is due to a bias in favour of the written style in which topicalization is avoided, can easily be refuted; under (3-1) to (3-4) there are example of all four types of topicalization that have been identified in section 2.2. They come from two books by the British theological scholar and mystery author Dorothy L. Sayers, one novel ('Whose Body?') and one collection of short stories ('In the Teeth of the Evidence'), both written between 1920 and 1940, and one novel by P.G. Wodehouse ('Bill the Conqueror'), published in 1924. Note that most examples, notably those of double focus topicalization and focus movement, are taken from direct speech passages of one of the characters (mostly London policemen and villains).

(3-1) <u>Scene-setting preposing</u>

In the afternoon he found himself in Harley street (Body 211)

(3-2) <u>Double-focus topicalization</u>

a. on her it had looked rather well; on him, it would be completely abominable. (Teeth 323)

b. My trousers I dried by the gas stove in my bedroom […], My pauper's beard I burned in the library (Body 249)

c. but drunk I was not (Teeth, 308)

(3-3) Anaphoric preposing

(He must be equipped with india-rubber legs, a chest like an ice-box and the shoulders of a prize-fighter.) These qualifications Bill possessed (Bill, 107).

(3-4) Focus movement

tall, narrow houses they are (Teeth 310)

So we can say that the decline in topicalization, which is really for the most part a decline in the usage of double-focus-topicalization and which we have commented on in the preceding chapter, has not led to a total loss of topicalization in Modern English, but to a situation in which this construction is used only marginally. But double-focus-topicalization is still perfectly grammatical. This can be seen by the fact that sentences like the ones in (3-2) or (3-5) are perfectly acceptable with the appropriate intonational contour and in the appropriate context.

(3-5) /ABernathy he \LIKES. But /HIGginbotham he \HATES.

The appropriate contour is the Hat Contour (also sometimes called Bridge Contour; on the contour see e.g. Büring 1997; Féry to appear). The Hat Contour is not directly associated with topicalization, but rather with double foci in general. This becomes clear if we look at sentences like the one in (3-6a), which is the non-topicalized counterpart to

(3-5), or (3-6b). In (3-6b) the foci are on the subject and the verb and thus double-focus-topicalization would not even be possible.

(3-6) a. He /LIKES \ABernathy. But he /HATES \HIGginbotham.

b. /BOB \LIKES beans. But /RICK \HATES them.

This is not an English idiosyncrasy, but can also be found in German at least. As I will resort frequently to German data in the forthcoming sections it is perhaps worth pointing out that what has been said for English topicalization also applies to German. In German, double focus constructions are as (in)frequent as in English. German, being a strict V2-language, has no topicalization in the sense that English has, but it has a comparable property in that in double focus constructions there is a strong tendency to put one of the foci into the '*vorfeld*', that is the position before the landing site of the finite part of the verb form (3-7).

(3-7) a. /BOHnen \MAG er. Aber /ERBsen \HASST er.

b. /RObert \MAG Bohnen. Aber /RIchard \HASST sie.

It is worth noting that in German at least movement of one focus in a double focus construction to the *vorfeld* is completely unconstrained and does not hinge at all on the choice of subject. If the properties of German *vorfeldbesetzung* and English topicalization are comparable to a certain extent, then this suggests that it is not the property of being a full noun phrase subject or pronominal subject per se that has a positive or adverse effect on topicalization. Rather, it is the epiphenomenon of being prosodically strong and weak, respectively, that is characteristic of these two types of noun phrases.

Let us return to English double-focus topicalization. The intuitions about the acceptability of topicalization sentences change as soon as the pronoun is replaced by a full noun phrase. Sentence (3-8a) has been judged as marked by native speakers of English that had to read sentences similar to it for an experiment (on which see below), sentence (3-8b) showed real processing difficulties and provoked the sentiment that "you can't say it like that", i.e. unacceptability. It is necessary to point out that all sentences were put in contexts in which topicalization is natural.

(3-8) a. ?ABERNATHY Decker LIKES (both persons having been introduced earlier in the discourse)

b. *ABERNATHY DECKER likes. (in the sense: Decker (subj.) likes Abernathy (obj.))

Note that the syntactic structures underlying (3-5) and (3-8a, b) are identical. Therefore the acceptability cannot have anything to do with ungrammaticality in the stricter sense, viz. that there is some violation of a syntactic well-formedness principle at work.

Even if the reader has forgotten all that s/he has heard in part 2 on topicalization in Middle English, s/he cannot help suspecting from this data that it might have something to do with the form of the subject NP, i.e. whether it is a pronoun or a full noun phrase. There are several observations which suggest that this is true.

One observation is that, if we look at a corpus of topicalization sentences, like the one put together by Ellen Prince and Gregory Ward (Prince & Ward, unpubl.),[24] we see that topicalization in sentences with pronominal subject is much more common than with full noun phrase subjects. Table 3.1 shows the subject types of the first 200 sentences from this corpus (leaving out examples of preposed scene-setting element).

Table 3.1: Subject types in a corpus of naturally occurring topicalization in Modern English.

full noun phrase subject	demonstrative pronoun subj.	personal pronoun subject
17	2	181
8.5 %	**1%**	**90.5%**

To evaluate these numbers, one has to compare it to the rate of personal pronoun subjects in sentences with canonical word order. In order to do this I composed a small corpus that had the same proportions of genres as the portion used in Ward/Prince's corpus. The part of the Prince/Ward corpus that was used was composed of the genres in the percentage shown in Table 3.2:

Table 3.2: Genres and their proportion in Ward/Prince's corpus.

	oral (incl. radio, film, TV)	newspaper	literature (belletristic and scholarly)
N= 200	101	69	30
%	50.5	34.5	15

The comparison corpus was put together with randomly selected texts from the same genres (Table 3.3):

[24] I want to express my warmest thank to Gregory Ward for letting me use this corpus.

Table 3.3: Genres, texts and their proportion in the comparison corpus.

	oral (Switchboard Corpus 020001A, -5A, -6A; from LDC online)	newspaper (Metro Philadelphia, 11/13/06)	literature (P.G. Wodehouse: Uneasy Money)
N= 302	152	105	45
%	50.3	34.8	14.9

The rate of personal pronominal subjects, demonstrative pronoun subjects, and full noun phrase subjects in the comparison corpus is shown in Table 3.4:

Table 3.4: Subject types in a corpus of naturally occurring canonical word order sentences in Modern English.

	personal pronoun	demonstrative pronoun	full noun phrase
oral (N = 152)	98 (64%)	17 (11%)	37 (24%)
newspaper (N = 105)	21 (20%)	1 (1%)	83 (79%)
literature (N = 45)	21 (47%)	2 (4%)	22 (49%)
total	140	20	142
percentage total	**46.4**	**6.6**	**47.0**

If we compare Table 3.4 to Table 3.1, we see a remarkable difference. The overall proportion of full noun phrase subjects among random canonical word order sentences is roughly 1:2. The proportion of full noun phrase subjects in topicalization sentences, on the other hand, is merely 1:10. This suggests that topicalization and full noun phrase subjects do not go well together. This is what we would expect, given the discussion in part 2.

Another observation is the following: In (3-9a) we have an example of a topicalized full noun phrase object and full noun phrase subject from the Ward/Prince corpus. It should be noted, that here the second focus lies on the subject, but that there is a phrase intervening between the two focalized phrases. The other examples of topicalization with full noun phrase subject were such that the topicalized phrase was very heavy (e.g. a wh-clause), or that the focus in the remainder of the sentence was on a constituent different from the subject. Only in two cases (which had, by the way, a pronominal subject) was the second focus in the clause adjacent with the one on the topicalized phrase (3-9b).

(3-9) a. I made two minor mistakes. {**One** apparently **everyone** in the class made.} (K. Gottschlich, talking about his just-returned math test; noted by M. Pollack)

b. Don't pout! If you want to fight, we'll fight. Just don't pout. {**Pouting YOU** win}, {**fighting I** win.} (The Odd Couple; Oscar to Felix)

So the pattern is rather clear: Topicalization is acceptable in sentences with a pronominal subject, but rare and sometimes even unacceptable in sentences with a full noun phrase subject. The question we have to ask now once again is: What is the crucial difference between pronouns and full noun phrase subjects?

Knowing the discussion about the decline in topicalization in Middle English, we easily can pinpoint the crucial difference: It is the prosodic weakness of personal pronouns, which shows up in their typical property of counting as intrinsically weak for the purposes of metrical prominence; consequently, they bear no independent stress phonetically. They can be focalized (as every element can, see below), but situations in which this plays a role are seldom.

Thus it is fair to say that the same property of full noun phrases (namely being able to bear metrical prominence, as opposed to pronouns) that was found to be responsible for the decline in topicalization in Middle and Early Modern English, accounts for the fact that topicalization with full noun phrase subjects is only marginally acceptable in Modern English. A full noun phrase subject with its phrasal metrical prominence – in double focus topicalization sentences often even with focal emphasis – cannot be adjacent to a topicalized constituent which bears focal emphasis. The reason for that has been identified as the Clash Avoidance Requirement (= CAR): If a focused full noun phrase subject immediately follows a focused topicalized object, the Clash Avoidance Requirement is violated, since it requires a weak element to intervene between the focused phrases.

Leaving the area of focal emphasis, we get similar effects with a construction in which two phrasal prominence peaks are in danger of clashing, namely locative inversion. Example (3-10a) shows a typical example of locative inversion. Note that under a more canonical word order (i.e. subject before verb) the two phrasal prominence peaks are in clash (3-10b). English native speakers highly prefer version (3-10a) over (3-10b). If the subject is however a pronoun, the judgments are reversed: The version with locative inversion (3-10c) is judged as unacceptable, whereas the version with subject-before-verb is the version everyone prefers.

(3-10) a. down the HILL rolls the BALL.

b. ?down the HILL the BALL rolls.

c. *down the HILL rolls it.

d. down the HILL it rolls.

The only difference between (3-10a, b) and (3-10c, d) is the realization of the subject noun phrase. Structurally, (3-10a) and (3-10c) are identical, likewise (3-10b) and (3-10d). Nevertheless we get the opposite acceptability judgments. This means that, as is the case with double focus topicalization, the judgment cannot be due to syntactic differences, but to other, presumably more surface-oriented, that is, phonological constraints. That the relevant constraint is the CAR is easy to see: We have clash of phrasal prominence peaks in exactly the version (3-10b) that is judged unacceptable, although it shows canonical word order.

So it is clear that the CAR is an important factor, if not the most important one, in the acceptability of clauses containing more than one prominence peak on the highest level.

3.1.2 Experimental data on double foci

So far we have seen only indirect evidence for the CAR in Modern English: The marginality of topicalization sentences with full noun phrase subjects has been attributed to the CAR. But we have not seen direct evidence for the CAR so far. Thus the applicability of the CAR in Modern English has been only an assumption up to now; there has been no data that can be explained exclusively by the application of the CAR.

The following section is devoted to presenting several pieces of direct evidence of the CAR in Modern English and Modern German.

The CAR in the form in which it has been introduced in section 2.4.4 (repeated below) makes certain predictions on the natural production of clauses.

> *The Clash Avoidance Requirement (descriptive general form)*
> Two elements of an equal, given prominence must be separated by at least one element of lesser prominence.

One prediction is that speakers, if nothing hinders them, will strive to produce only sentences in which the CAR is observed, that is: in which, if there are multiple foci, these foci are separated by at least one unfocused element respectively. It is not the case that CAR-violation can happen only when foci are involved: in sentences without special prominence, CAR-violations can occur on lines lower than the topmost line (as in the scene-setting cases mentioned above, (3-10)). Normally, however, phrasal prominence peaks are graded in such a way that the CAR is fulfilled (for more on this see section 3.4).

Another prediction is as follows: When speakers are forced to utter two adjacent foci, they will do so reluctantly and try to ‘rescue’ the CAR somehow by manipulating the intonational structure. There are basically three ways to manipulate the intonational structure of a clause with adjacent foci in such a way that the CAR is observed:

- One of the foci is not realized at all (3-11a)
- One of the foci is shifted to another constituent (3-11b)
- A weak element is inserted between the adjacent foci (3-11c).

(3-11) a.	*	*	.		*	.	.	
	*	*	*	➔	*	*	*	
	A	A	a		A	A	a	
	[+foc]	[+foc]	[-foc]		[+foc]	[+foc]	[-foc]	
b.	*	*	.		*	.	*	
	*	*	*	➔	*	*	*	
	A	A	a		A	A	a	
	[+foc]	[+foc]	[-foc]		[+foc]	[+foc]	[-foc]	
c.	*	*	.		*	.	*	.
	*	*	*	➔	*	()	*	*
	A	A	a		A	a	A	a
	[+foc]	[+foc]	[-foc]		[+foc]	[-foc]	[+foc]	[-foc]

In the schemata under (3-11), the top tier is the tier that is relevant for the focus indicator. The lower tier is the tier relevant for metrical prominence on the phrasal level; its exact nature does not concern us here. The [focus] features are semantic features; optimally they should be translated to an obligatory strong mark in the grid for elements with the feature [+foc], the focus indicator.

We will see, that the possibility schematized in (3-11c) is the one that is usually employed. Tampering somehow with the realization of focus, what (3-11a) and (3-11b) boils down to, is dispreferred for reasons that we will discuss in section 3.5.

Thus speakers are basically stuck with the strategy to insert extra material, weak material at that, between the two foci in clash in order to reach a CAR-conforming structure. It is perhaps important to point out that this is a performance phenomenon and has nothing to do with syntactic well-formedness, which is given regardless of whether the CAR is fulfilled or not. But phonological requirements such as the CAR may reject even syntactically well-formed sentences or may play a role in the selection of several syntactically well-formed alternatives. Examples of this are the kind of sentences that were produced in the experiment which will be described below. Here, we have an opposition between (3-12a) and (3-12b). Both are syntactically well-formed, as left-dislocation in (3-12b) is an operation that is covered by syntactic rules.

(3-12) a. Bert$_{[+foc]}$ bastelt$_{[+foc]}$ gerne.

Bert tinkers gladly

b. Bert$_{[+foc]i}$ der$_i$ bastelt$_{[+foc]}$ gerne.

Bert this tinkers gladly

It is perhaps not the most obvious alternative, if we follow basically everything that has been written on movement (e.g. Chomsky 1981; 1995); for instance that movement is only done under pressure of the satisfaction of certain features. In recent years, a discussion concerning proper and improper movement has begun (e.g. Rizzi 1997; Frey 2006), which we can summarize as follows: optional movement is allowed, as long as we can detect an operator that triggers the optional movement. Let us assume that left dislocation is such a case of improper movement and that it is therefore syntactically well-formed, but that it is not the 'first choice' for syntax. Further, let us assume that what happens is that Narrow Syntax generates several alternative outcomes, all of which are gramatical, of course. One of these is the 'normal' alternative (3-12a). It is rejected by the PF-component as it does not conform to PF-specific requirements. Other alternatives that have been generated are evaluated, until an alternative has been found that satisfies also the PF-requirements.[25]

It can be noted that speakers usually insert semantically empty elements in such cases. We will see that mostly a more verbose reformulation (left dislocation, changing a simple verb into an object-verb complex, using periphrastic verb forms) is employed to rescue the CAR. The constructions that are used conveniently are of such a form that weak material is placed into the critical spot automatically (see examples (3-16) through (3-18) below).

To test these predictions, I conducted three experiments with native speakers of English and three experiments with native speakers of German; the experimental designs for both languages were comparable. In the following I will present the experiments and the results, and show how they offer direct evidence for the activity of the CAR in English and German.[26]

The first experiment was a production experiment. I constructed several scenarios and asked the participants to pretend that they were part of the scenarios. The scenarios were drawn up as situations in which two persons interact in a dialogue; the participant had to act the role of one of the participants. All scenarios were built in such a way that at

[25] As another example in which phonological requirements regularly influence syntactic usage observe object clitics in Old Irish. Here lexical material is inserted for purely phonological reasons. Object pronouns in Old Irish are obligatorily cliticized. As opposed to e.g. Romance clitics the spot where they are cliticized is not to the right of the verb but to the right of the verbal prefix, e.g. *do-beir* 'he brings' but *do-n-beir* 'he brings us'. The form *do-beir-unn*, which would be the form if the clitic were attached at the end of the verb form is possible only in marginal contexts. The question arises what to do with verbs that do not have verbal prefixes, e.g. *caraimm* 'I love'? In this case a dummy-prefix *no-* is inserted whose only purpose is to provide something for the pronoun to lean on, producing *no-t-charaimm* 'I love you'. The alternative *charaim-ut* is possible only in marginal contexts (Thurneysen 1946:255ff.). So here we have at least one example in which the lexical array has to be accomodated to a phonological requirement, namely the enclitic property of object pronouns.

[26] The sheets that were used for the experiments can be viewed under www.ling.upenn.edu/~speyer. All analyses (especially duration measurements) were done using Praat.

several points the virtual dialogue partner asks a question opening up multiple foci, and the participant had to answer this question as naturally as possible.

In the German experiment the foci were chosen in such a way, that under a word order that is adequate for the communicative situation, the two foci would wind up adjacent to each other, if the speaker does not re-build his/her sentence drastically. The three tested cases were: Focus on subject and verb (coded SV), focus on subject and object (coded SO; the participants were biased in producing a subordinate clause for this case, as only then would these two constituents end up adjacent to each other), focus on object and verb (coded OV). In each case schemata were also given in which the ‘easiest’ realization would not lead to an focus clash. The purpose was to see whether a difference was visible between pending clashing cases and non-pending clashing cases; in other words, to look whether the reformulations, destressings etc. that are observable in pending clash cases are really due to the clash or are just due to a habit of individual speakers to speak ponderously.

In the English experiment there were two cases: The foci were on both the subject and the verb (coded SV), or the foci were on the subject and the direct object (coded SO), but in a situation in which the object gives the sorting key and thus could undergo topicalization.

The content of the desired answers was given in little schemata, such as in (3-13); B is the participant and all s/he saw was the schema with the arrows or the circles, respectively. These schemata were drawn such as to avoid biasing the participants; if anything, there was a bias in the direction of the ‘easiest realization’ that is not CAR-conforming.

The purpose of the experiment was to observe how the participants would build the answer sentences containing the two foci. The CAR predicts that the ‘obvious answers’, such as under (3-13), should be avoided; it predicts that either the sentence is changed in such a way that the foci are separated by some material, or a clearly perceivable pause is inserted.

(3-13) a. A: “Wie ist es mit Pitt und Bert? Was machen die so hauptsächlich?“

how is it with P. and B. what make they so mainly

‘How about Pitt and Bert? What is it they mostly do?’

B: Pitt → gerne basteln

Pitt gladly tinker

Bert → gerne brüllen

Bert gladly shout

easiest realization of the schema in B would be:

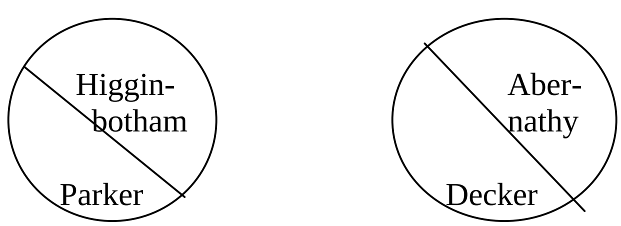
“/PITT \BAStelt gerne, aber /BERT \BRÜLLT gerne”

Pitt tinkers gladly but Bert shouts gladly

‘Pitt likes to do things, but Bert likes to shout.’

b. A: “Is it now Higginbotham or Abernathy? Whom did the bosses choose?”

B:



This experiment is much easier to conduct in German (as there the word order is much less fixed than in English); therefore, there is much more to say about the German experiment.

A preliminary experiment with 6 participants was conducted in July 2004 in Saarbrücken, Germany. The main experiment, from which the data is presented here, was conducted in December 2006 and January 2007 in Tübingen and Stuttgart, Germany. There were 15 participants. The participants were 5 male persons aged 20-70 and 10 female persons aged 20-65. The recording was done with a portable digital voice recorder (Olympus VN-480 PC) and, to improve the recording quality, an external microphone (Philips SBC ME 570). *In toto*, each participant had to produce six sentences with pending focus clash and six without pending focus clash; with 15 participants this adds up to a database of 90 sentences with and 90 sentences without pending focus clash. 'Pending' focus clash means cases in which an focus clash would occur if the participant would produce a sentence with the easiest possible realization, that is, without insertion of material or movement or the like.

Tables 3.5 and 3.6 and Figure 3.1 show the results, separated by subcase. Principally I found several ways in which the participants built and manipulated their sentences. They are listed below (where the codes are also explained). The source, i.e. which of the participants uttered the example sentences, is indicated after each example; the participants are numbered #1 through #15.

U (= unmarked): Easiest realization. Clause directly reflects the schema (3-14)

(3-14) Schema: Most → Mick

Wein → Mark

Clauses: MICK mag MOST am liebsten, und MARK mag WEIN am liebsten. (#5)

Mick likes cider at dearest and Mark likes wine at dearest

'Mick likes cider best and Mark likes wine best.'

Very often the participants built their sentence such that extra lexical material followed the first target word. This corresponds to the theoretical case (3-11c) from above. As can been seen from Tables 3.5 and 3.6 and Figure 3.1, this strategy was predominantly chosen if an focus clash was pending.

There are several subtypes which are listed below:

I (= interjection): Insertion of particles etc. (3-15).

(3-15) Schema: ‚Ja doch, ich weiß genau, dass…

Pitt → Blau

Bert → Gelb

Easiest realization: dass […] BERT GELB mag

that Bert yellow likes

'that Bert likes yellow'

Produced clause: während BERT eher so auf GELB steht (#13)

whereas Bert rather so on yellow stands

'whereas Bert rather goes for yellow.'

L (= left dislocation): The first focused element (or the topic in general) is left dislocated, leaving an (unfocused) pronoun at the original site (3-16).[27]

(3-16) Schema: Pitt → gerne basteln

Easiest realization: PITT BASTELT gerne

Pitt tinkers gladly

'Pitt likes to do things'

Produced clause: Also der PITT, der BASTELT gern (#1)

well the Pitt this tinkers gladly

'Pitt, he likes to do things.'

N (= noun): The sentence is slightly reformulated and built in such a way that non-verbal material intervenes. Notably the predicate is expressed with a noun-verb combination instead of a simple verb (3-17: 'die Lieblingsfarbe haben' instead of 'mögen').

---

[27] The distinction between left dislocation and hanging topic construction is not relevant here (on the distinction see Frey 2004).

(3-17) Schema: ‚Ja doch, ich weiß genau, dass…

Pitt → Blau

Bert → Gelb

Easiest realization: dass PITT BLAU mag.

that Pitt blue likes

'that Pitt likes blue'

Produced clause: dass PITT die Lieblingsfarbe BLAU hat (#4)

that Pitt the favourite colour blue has

'that blue is the favourite colour of Pitt'

V (= verb): If a clash between the verb and one of its arguments was pending, some participants ameliorated the problem by changing the simple verb form into a complex verb form, such that the unfocused auxiliary stood in the position immediately after the first focused element (3-18).

(3-18) Schema: Bert → gerne brüllen

Easiest realization: BERT BRÜLLT gern

Bert shouts gladly

‚Bert likes to shout'

Produced clause: BERT möchte gerne BRÜLLEN (#2)

Bert would gladly shout

'Bert would like to shout'

The other possibility is to manipulate the focus structure of the sentence. This corresponds to the theoretical cases that were demonstrated in (3-11a/b). Focus shift (coded S, 2-19) manipulates the focus structure locally, but leaves it intact on the whole, by moving the focal emphasis to another part of the predicate (like an adverb, such as in (3-19). With defocusing (coded D, 2-20) it is different; one of the foci fails to correspond to emphasis. Whereas focus shift was employed relatively rarely, defocusing was found comparably often, but only in the case when a clash was between verb and object (see Table 3.5, 3rd row). It is not clear how to interpret this result; it might be that verb-object clusters often count as complex predicates and therefore it is normal to realize only one focus, although, strictly speaking, both parts are in focus separately. Interestingly, often a second focal emphasis was realized nevertheless, but on an element that was not in focus (namely the topic). In (3-20) this can be seen nicely; note that putting an focus on 'Mick' creates a situation in which focus clash is pending. This clash is deviated by left dislocation.

Another reason why the focus on one of the focalized elements (most often the object, as was the situation in 12 of the 17 cases) was not realized could be that both elements, *Most* (cider) and *Wein* (wine) had been very prominent in the discourse; note that here, in contrast to the first and second set, the clash example was preceded by the non-clash example such that the two alternatives were already established. Therefore, although defocusing is perhaps not a serious candidate for clash avoidance in general, it might be employed, however, if the contrast as such has been introduced already (and therefore the participants were aware that there was a focus on these constituents even though it was not overtly emphasized).

(3-19) S (= shift)

Schema: Pitt → gerne basteln

Easiest realization: PITT BASTELT gerne

Pitt tinkers gladly

'Pitt likes to do things'

Produced clause: PITT bastelt SEHR sehr gerne (#15)

Pitt tinkers very very gladly

'Pitt likes to do things very, very much'

(3-20) D (=defocusing)

Schema: Mick: Most → mögen

Wein → hassen

Easiest realization: Mick MAG MOST und HASST WEIN

Mick likes cider and hates wine

Produced clause: der MICK der MAG den Most,

the Mick this likes the cider

und ähm des Problem is er HASST den Wein (#14).

and hm the problem is he hates the wine

'Mick, he likes cider, but, uh, the problem is, he hates wine.'

Focus clash could also be avoided by a more drastic reformulation. A relatively easy strategy is to build the sentence in such a way that the critical proposition is realized by a subordinate clause. This is especially successful if the foci are on the subject and the verb respectively, since in German the verb stands at the end of the clause in subordinate clauses, creating a convenient distance between the subject and the verb; therefore, subordination can be used to rectify the situation (3-21; coded C).

(3-21) Schema: Bert → gerne brüllen

Easiest realization: BERT BRÜLLT gerne

Bert shouts gladly

‚Bert likes to shout'

Produced sentence: und ähm ja leider isch es eher so

and hum yeah unfortunately is it rather so

dass BERT gerner BRÜLLT (#12)

that Bert gladlier shouts

'and, well, unfortunately it is the case that Bert rather likes to shout.'

If the foci are on the subject and the object, the subordinate clause order would be fatal, as both constituents would be adjacent in the *mittelfeld*. I prompted the participants using a sentence fragment in these cases, to try to bias them into producing a subordinate clause. Nevertheless, about a quarter of participants realized the proposition as a main clause. In main clauses, one of the focalized arguments can be put into the *vorfeld*, so that the unfocused verb intervenes (3-22; coded M). This example is interesting, because the participant started with a subordinate clause (and avoided the pending CAR-violation by inserting a nominal element), then reconsidered and changed the construction into a main clause.

(3-22) Schema: ‚Ja doch, ich weiß genau, dass…

Pitt → Blau

Bert → Gelb

Easiest realization: dass PITT BLAU mag und BERT GELB (mag).

that Pitt blue likes and Bert yellow likes

'that Pitt likes blue and Bert likes yellow'

Produced sentence: dass ähm PITT die Farbe BLAU,

that hm Pitt the colour blue

und BERT zieht die Farbe GELB vor. (#8)

and Bert draws the colour yellow in-front.

'that Pitt [likes] the colour blue; and Bert prefers the colour yellow'

Another option which was observed several times was that the participant uttered the first focused element, said something about the referent, and only then uttered the proposition that s/he was supposed to produce. By that time the referent of the first focused element had achieved topic status and could be referred to by a mere pronoun (3-23; coded P).

(3-23) Schema: Bert → gerne brüllen

Easiest realization: BERT BRÜLLT gerne
Bert shouts gladly
‚Bert likes to shout'

Produced sentence: der BERT, ähm naja, eigentlich kann man vor allem sagen,
the Bert, hm well actually can one before all say
dass er gerne BRÜLLT (#14).
that he gladly shouts
'Bert, well, actually, one can mainly say that he likes to shout'

Sometimes the participants got into the scenario to such an extent that they formulated a new structure extremely freely (coded F; 2-24). It is hard to say anything about these cases, as they are too different from the easiest realization, but, as we are interested in focus clash, I only want to mention that they were all of such a form that the focused constituents were separated by unfocused ones.

(3-24) Schema: Pitt → gerne basteln

Easiest realization: PITT BASTELT gerne

Pitt tinkers gladly

'Pitt likes to do things'

Produced sentence: naja, also der PITT, das ist ein Kind,

well well the Pitt that is a child

das HANDwerklich begabt ist (#10)

that handicraftswise gifted is

'well, Pitt is a child that is gifted mostly practical.'

Code Z (= zero) means that no data was available.

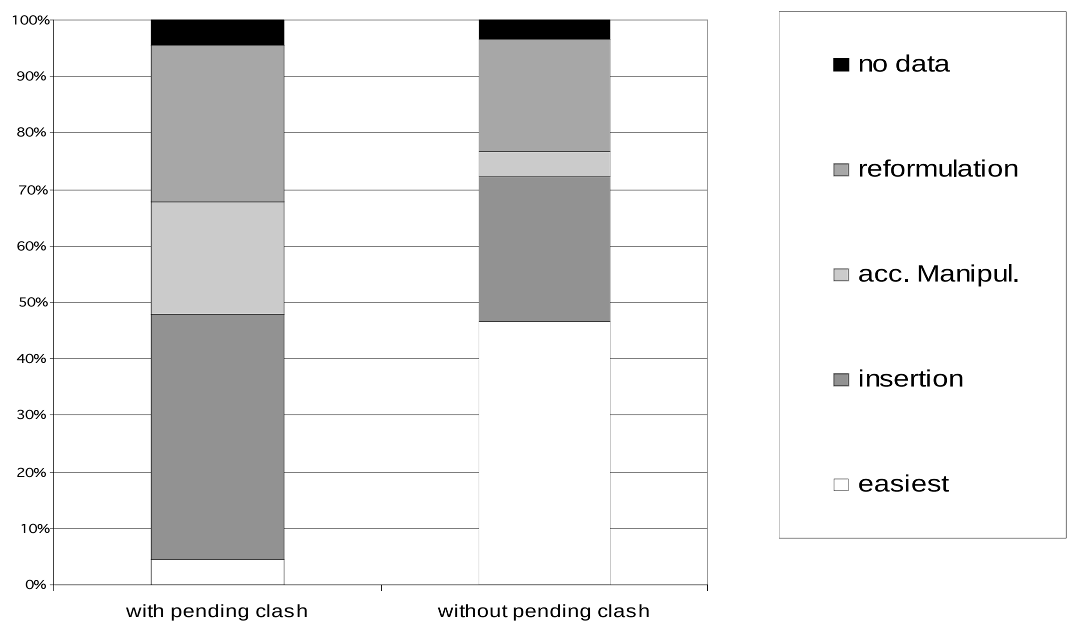
Table 3.5: Formulation types, cases with pending focus clash.

		insertion of material				manipulation of focus		reformulation				
	U	I	L	N	V	D	S	M	C	P	F	Z
SV	3	3	7	4	4	0	1	0	2	3	3	0
SO	1	0	2	12	0	0	0	7	0	5	1	2
OV	0	0	3	4	0	17	0	0	2	1	1	2
tot.	**4**	**3**	**12**	**20**	**4**	**17**	**1**	**7**	**4**	**9**	**5**	**4**
%	**4.4**	**3.3**	**13.3**	**22.2**	**4.4**	**18.8**	**1.1**	**7.8**	**4.4**	**10.0**	**5.6**	**4.4**
tot./ %	**4 4.4**	**39 43.3 %**				**18 20.0 %**		**25 27.8 %**				**4 4.4**

Table 3.6: Formulation types, cases without pending focus clash.

		insertion of material				manipulation of focus		reformulation				
	U	I	L	N	V	D	S	M	C	P	F	Z
SV	11	0	4	1	3	0	0	0	0	4	5	2
SO	12	1	2	2	2	1	0	2	0	2	5	1
OV	19	0	4	2	2	2	1	0	0	0	0	0
tot.	**42**	**1**	**10**	**5**	**7**	**3**	**1**	**2**	**0**	**6**	**10**	**3**
%	**46.7**	**1.1**	**11.1**	**5.6**	**7.8**	**3.3**	**1.1**	**2.2**	**0**	**6.7**	**11.1**	**3.3**
tot./ %	**42 46.7**	**23 25.6**				**4 4.4**		**18 20.0**				**3 3.3**

Figure 3.1: Formulation main types, comparison of cases with and without pending focus clash.



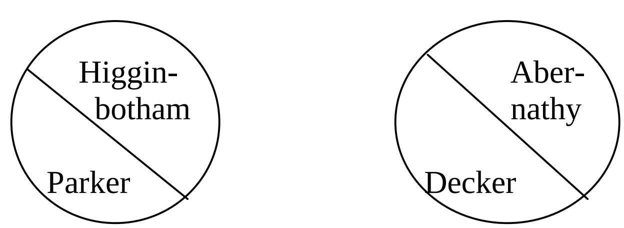
The main noticeable result is that the ‘easiest realization’ (coded U in the tables, ‘easiest’ and white in the figure) has been chosen significantly more often in cases that were not potential clash cases. This data suggests strongly that speakers avoid clashing foci naturally and prefer to alter their utterance slightly. On the whole the most ‘popular’ clash avoidance mechanism is the insertion of weak material; under this heading the most common strategies were the use of a noun-verb combination instead of a simple verb, or to some extent the use of left dislocation. Whereas left dislocation is not uncommon in non-critical cases, there is a clear preference for noun-verb-predicates in pending clash cases in contrast to non-critical cases. Defocusing and focus shift was employed rarely in general, with the exception of the OV-cases mentioned above. In the cases of easiest realization with pending clash, a pause between the two foci was present in all clashing cases.

A similar experiment has been done for English. In English it is more complicated to test production than in German as there are stricter constraints on word order and fewer avoidance mechanisms available. The experiment was conducted in November 2006 in the Phonetics Lab of the Department of Linguistics of the University of Pennsylvania. I had 12 participants, 5 male and 7 female, aged 20 to 45.

The English experiment was structured in the following way: The general approach was as in the German experiment, i.e. scenarios were presented in which the participants had to utter sentences. The schemata were not simply structures of the type ‘A → B’ as in the German experiment, but circles. The reason was to avoid forcing the participants into a particular biased word order. Thus the circles were separated such that it was not immediately clear which word is to the left, which to the right, which on top, which on bottom (see (3-13b), repeated below).

(3-13) b. A: “Is it now Higginbotham or Abernathy? Whom did the bosses choose?”

B:



The two cases that were tested were: focus on the subject and the object (OS; here the scenario was constructed in a way that suggested sorting-key-topicalization of the object) and focus on the subject and the verb (SV). In order to extend the database I gave the participants two scenarios for each case, so that they each uttered eight sentences in total. This gave a total database of 96 sentences, 48 for each case. In the first case the clash could be avoided by using ‘normal’ word order, in the second the ‘normal’ word order necessarily would lead to a clash. The two scenarios for SV were slightly different in that in the second scenario the predicate consisted of a verb plus adverb (SV/Adv; ‘talk well’ versus ‘act well’) As English does not offer as many escape hatches as German, I thought practically the only possibility for the speakers was to live with the focus clash and insert a pause. Table 3.7 and Figure 3.2 show the result. The codes are similar to the ones in the German experiment; they are repeated and illustrated below.

U = easiest realization, all with pause. In the OS cases ‘easiest realization’ would be O – S – V, in the SV cases it would be S – V – etc.

O = change of word order in comparison to ‘easiest realization’.

V = change of a simple verb form into a complex verb form (3-25).

N = change of a simple verb form into a noun-verb-complex (3-26).

S = shift of the focus to another part of the predicate (3-27).

D = defocusing of one of the focused elements (3-28).

P = reformulation so that the first focused element is discussed alone initially, allowing it to be replaced by a pronoun when the second focus is mentioned (3-29).

(3-25) Easiest realization: BOB PANTED

Produced sentence: BOB was PANTING (#4)

(3-26) Easiest realization: HIGGINBOTHAM TALKS well and ABERNATHY ACTS well.

Produced sentence: HIGGINBOTHAM is a good TALKER, ABERNATHY is a good SELLER. (#12)

(3-27) Easiest realization: HIGGINBOTHAM TALKS well

Produced sentence: HIGGINBOTHAM talks WELL (#1)

(3-28) Easiest realization: HIGGINBOTHAM TALKS well

Produced sentence: Higginbotham TALKS well (#10)

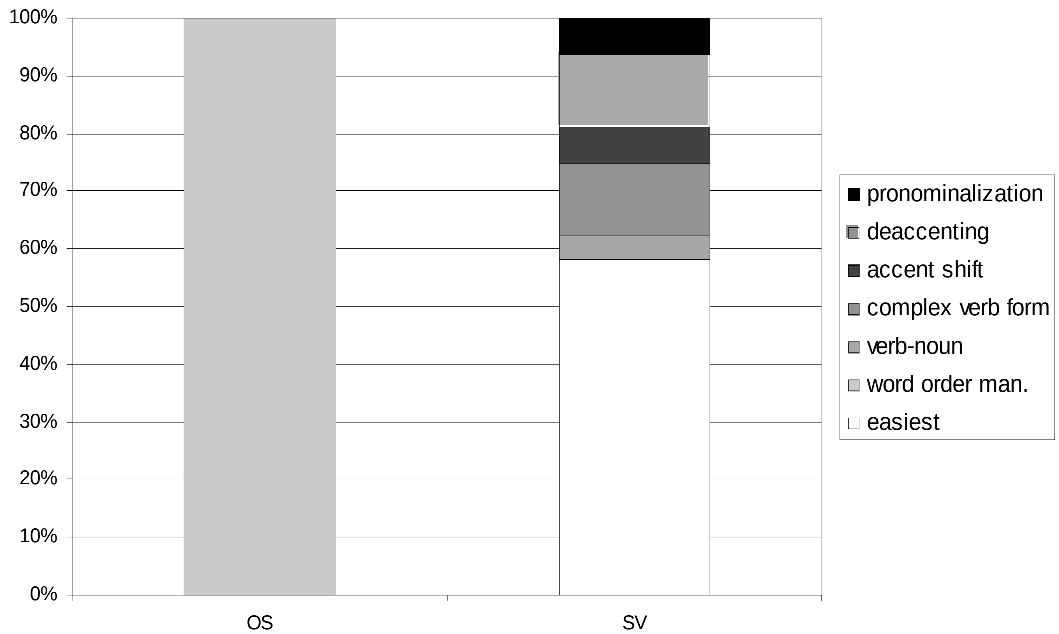
(3-29) Easiest realization: BOB PANTED

Produced sentence: BOB did WELL, he was only PANTING (#9)

Table 3.7: Formulation types in English sentences with pending focus clash.

n = 96	clash	no clash					
	U	O	N	V	S	D	P
OS	0	48	0	0	0	0	0
% OS	**0**	**100**	**0**	**0**	**0**	**0**	**0**
SV	13	0	0	5	1	2	3
SV/Adv	15	0	2	1	2	4	0
SV total	28	0	2	6	3	6	3
% SV total	**58.3**	**0**	**4.2**	**12.5**	**6.3**	**12.5**	**6.3**
% SV total	**58.3**	**41.8**					

Figure 3.2: Formulation types in English sentences with pending focus clash.



No one employed topicalization in the cases in which its use would have led to focus clash (for (3-13b) this would be *Higginbotham, Parker chose, but Abernathy, Decker chose*). This is exactly what the Middle English discussion of part 2 suggested. In the cases in which focus clash was unavoidable (clash between subject and verb), we see that sometimes the speakers still tried to avoid the clash by using the periphrastic continuous form, or by reformulating completely. In the cases of easiest realization with pending clash, a pause between the two foci was present in all clashing cases.

In this experiment we have seen what happens if participants are relatively free in their production: They avoid focus clash by formulating the sentence in such a way that no focused phrases are adjacent, or they manipulate slightly the focus structure of their clause. What happens now if they are forced to utter sentences in such a way that foci are adjacent to each other?

It has been observed in earlier experiment that speakers insert a small pause in such cases (cf. Schmerling 1976:96). This is what Selkirk (1984 etc.) calls demibeat insertion, that is: Insertion of a timing unit, or a silent grid position (Selkirk 1984:300). The production experiments which I am about to present later showed clearly that a measurable pause existed whose length was in the region of ca. 70 milliseconds.

We know that speakers can insert pauses anywhere and for a variety of reasons, but the pause at this juncture seems to be systematic. This pause is sometimes viewed simply as the marking of a prosodic boundary (e.g. Cooper & Paccia-Cooper 1980; Taglicht 1998). Whereas I do not want to claim that pauses never serve to mark prosodic boundaries, I want to claim that in this special case it cannot be purely the prosodic boundary. The reason for this is the following: In a sentence like (3-30a), a pause is perceived between ‘beans’ and ‘John’, whereas in a sentence like (3-30b) no pause is perceived. Note that both sentences have an identical syntactic structure. If we assume that prosodic constituency is in some meaningful way dependent on syntactic constituency, then there should be no difference between these two sentences. Otherwise we would have to stipulate ontologically non-syntactic causes such as the different lexical filling of the subject NP. But then prosodic phrasing would no longer be derivable from syntax, but rather be an ad-hoc mechanism. From a syntactic viewpoint, it makes more sense to assume that there is always a prosodic boundary (phonological phrase or the like) between a topicalized constituent and the rest of a sentence. But the prosodic boundary only sometimes corresponds to a pause, sometimes not. This means that there is at least no 1:1-relation between pause and prosodic boundary. The cases in which a pause is realized are regular, the regularities cannot be described with reference to syntax, but only with reference to purely intra-phonological regularities (the CAR, for instance). Consequently, we are forced to conclude that the realization of a pause has nothing to do with intonational phrasing in this case, but is a separate phenomenon.

(3-30) a. BEANS, JOHN likes

b. BEANS john LIKES

In the light of the Clash Avoidance Requirement, the purpose of the pause is obvious: An extra timing slot is inserted between the two focused noun phrases to keep them apart, by that producing an alternating rhythmic pattern. It has been observed before (e.g. Ladd 1980:43f.) that pauses can serve to increase the space between two beats; I want to go further and claim that the pause increases the space between two strong beats so that the equivalent of a weak beat intervenes between the two strong beats.

Of course, such a pause can be merely a subjective impression that researchers like myself perceive because they want to prove the existence of the CAR. So the question is: Is there really a measurable pause between adjacent foci or do we only perceive that something is there?

In order to test this I conducted two experiments, one for German and one for English. In this experiment I constructed a text in which I contrasted topicalized and non-topicalized sentences with focus clash and examples without focus clash that were otherwise similar with respect to structure and wording.

Let me first talk about the English experiment which was conducted in March 2006 at the Phonetics Lab of the Department of Linguistics of the University of

Pennsylvania. 6 participants (3 male, 3 female, aged 20 to 50) were given the task to read these sentences out loud. In reading the sentences they were in some cases forced to articulate sentences with adjacent foci. The sentences were in contexts in which the desired focalization was natural.

Apart from the main parameter (focus clash or not) the secondary parameters that varied were:

- Sentences with topicalization (focus on topicalized object and subject; T) against sentences without (focus on subject and verb; N).
- Length of the first word in clash (1 syllable = 1s versus 2 feet = 1l).
- Length of the second word in clash (1 syllable = 2s versus 1 foot = 2l).

I did not realize all combinations of the secondary parameters, partly to keep the experiment to a manageable length for the participants, and partly because it was not possible to construct sentences that sounded halfway natural in every case (e.g. N1l2l).

I tried to construct the examples in such a way that before and after the spot where the expected pause was to occur two non-homorganic stops would be adjacent. I did not succeed in the case of *Abernathy* and *Higginbotham*, because I could not find reasonable two-footed initially stressed proper names ending in a stop. Despite this, some effect was still noticeable (which would probably be larger if these two-footed words ended in a stop).

It turned out that there was a measurable difference regardless of what the length of the words in clash was and regardless of whether it was an instance of topicalization or not. Table 3.8 below gives the exact measurements.

Table 3.8: Pauses in focus clash in English.

Type	Example	Pause mean	Difference
T1s2s	RICK THEY RICK they	0.2350 0.1600	0.0750
	BOB THEY BOB they	0.2037 0.0900	0.1137
N1s2l	RICK DASHES RICK dashes	0.2396 0.1598	0.0798
	BOB DAWDLES BOB dashes	0.1437 0.1001	0.0436
T1l2s	ABERN. THEY ABERN. they	0.0570 0.0344	0.0226
	HIGG. THEY HIGG. they	0.0689 0.0218	0.0471
N1l2l	ABERN. DASHES ABERN. dashes	0.0917 0.0599	0.0318
	HIGG. DAWDLES HIGG. dashes	0.0459 0.0104	0.0355
T1s2l	RICK DENNIS RICK Dennis	0.2251 0.1549	0.0702
T1l2l	ABERN. DECKER ABERN. Decker	0.0964 0.0597	0.0367

The length of the words (or the fact that they did not end in a stop) can compensate for the pause effect somewhat (therefore the examples with ‘Abernathy’ and ‘Higginbotham’ show less of a pause, about 30 milliseconds, than the ones with Rick and Bob, about 70 milliseconds), but it is still present.

This illustrates another important point, namely that we are dealing here not necessarily with a ‘normal’ stress clash. If it was simply a matter of adjacent stresses, it would have been unnecessary at all to insert a pause in the 1l-cases, as there is a whole unstressed foot between the clashing elements. For keeping equal stresses apart this would have been enough.[28]

---

[28] I know that this is idealized to some extent; stress clash resolution mechanisms such as retraction occur also if there is no immediate need of repair, for instance in case unstressed material intervenes. *Christine Tódd Whítman* would easily be repairable to *Christíne Todd Whítman*, which is well-formed, but what happens instead is that extra retraction occurs and we get *Chrístine Todd Whítman*. I think, however, that in a more comparable case, such as e.g. *Jane Tódd Whítman*, where no extra retraction can occur, probably *Jáne Todd Whitman* would be produced, with simple retraction, and not *Jáne _ Todd Whítman*, that is,

A similar experiment was done for German in June 2006 at the Phonetics Lab of the University of Tübingen, and it produced similar results (10 participants; 4 male, 6 female, aged 20 to 40). Here the topicalization versus non topicalization parameter could not be used, instead another parameter (main clause with clash between subject in the *vorfeld* and verb in second position versus subordinate clause with clash between verb and object in the *mittelfeld*) was used.

The problem of concern is similar to that of topicalization: In both cases the preposed constituent could in theory form a high-order prosodic constituent of its own and thus the pause could be accounted for by viewing it as the marker of the prosodic constituent boundary. Therefore it is necessary to contrast cases in which the two foci are in the *vorfeld* and the area immediately following, i.e. the left sentence bracket (the place of the finite verb form) with cases in which both focused constituents are unscrambled in the *mittelfeld*.

There were pauses in all tested cases. The secondary parameters were as given below; the results are in Table 3.9.

- Position of foci in *vorfeld* / left sentence bracket (P) or in *mittelfeld* (M)
- First focused word short (1 syllable; 1s) or long (2 feet; 1l)
- Second focused word short (1 syllable; 2s) or longer (2 – 3 syllables; 2l)

---

retraction plus pause. So the fact that exactly this happens – 'Ábernathy _ Décker' instead of 'Ábernathy Décker' – indicates that we are not dealing with 'normal' stress clash.

Table 3.9: Pauses in focus clash in German.

Type	Example	Pause mean	Difference
M1s2s	PIT BIER… PIT Bier…	0.1553 0.1206	0.0347
	BERT BIER BERT Bier	0.1558 0.1229	0.0329
M1s2l	PIT BRAUSE PIT Brause	0.1113 0.0811	0.0302
	BERT BRAUSE BERT Brause	0.1544 0.0688	0.0856
M1l2s	AUENMAHLERT BIER AUENMAHLERT Bier	0.1459 0.0869	0.0590
	VANDEGORET BIER VANDEGORET Bier	0.1824 0.1024	0.0800
M1l2l	AUENMAHLERT BRAUSE AUENMAHLERT Brause	0.1045 0.0818	0.0227
	VANDEGORET BRAUSE VANDEGORET Brause	0.1270 0.1004	0.0266
P1s2s	PIT BAT PIT bat	0.1291 0.0974	0.0318
	BERT BAND BERT band	0.1267 0.0858	0.0410
P1s2l	PIT BETTELT PIT bettelt	0.1425 0.1267	0.0158
	BERT BRUDDELT BERT bruddelt	0.1396 0.1183	0.0213
P1l2s	AUENMAHLERT BAND AUENMAHLERT band	0.1186 0.0955	0.0231
	VANDEGORET BAT VANDEGORET bat	0.1451 0.1055	0.0396
P1l2l	AUENMAHLERT BRUDDELT AUENMAHLERT bruddelt	0.1419 0.0932	0.0487
	VANDEGORET BETTELT VANDEGORET bettelt	0.1215 0.0950	0.0265

On the whole the effects tend to be bigger if both focalized constituents are in the *mittelfeld*. This is slightly surprising as one would rather expect the boundary between *vorfeld* and sentence bracket to be more pronounced. But be that as it may, the important finding is that we can measure objectively a pause between the two focused constituents, whose length varies between 20 and 80 milliseconds.

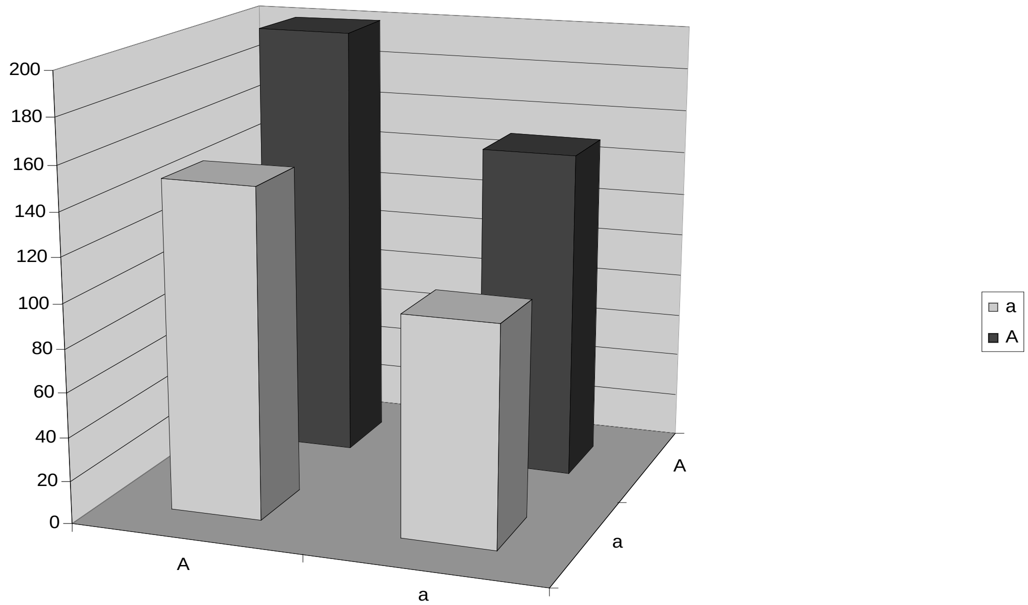
A word on the measurement of the pause: As I mentioned above, I tried to choose the wording such that the sounds at the end of the first word in clash and at the beginning of the second word were both stops; I furthermore monitored that the stops were not homorganic. The reason for this was the following: there is always a small pause for articulatory reasons between two stops. If an additional timing slot is inserted (as this is what the pause really is), it needs to be ascertained that it is really realized as a period of silence and not linked to e.g. a fricative. If two stops are on either side of the position in which we expect the insertion of the extra timing slot, we can be sure that the timing slot will be filled with silence. If one of the sounds had been a fricative, the timing slot could be used for an overlong realization of the fricative, which makes the measurement much more complicated.[29] So, if we use stops as the sounds that frame the critical point, we get a measurable pause in both cases (i.e. with or without clash), but the pause is longer if it also has to realize an empty timing slot.

One could object that there might be a different reason why there is a pause: Words tend to be pronounced more slowly and more diligently if they bear focal emphasis; therefore the pause could simply be an overemphasis of the word boundary.

I tested this assumption with two other experiments. My hypothesis was the following: If the pause was simply an overemphasis of the word boundary of the focused word, it should be observable also with single focused words alongside non-focused words. That means, if we have four pairs of otherwise identical two-word sequences that only differ in prominence (if A = focused and a = not focused, the four permutations are 'A A' (case 1), 'a A' (case 2), 'A a' (case 3) and 'a a' (case 4)), we should get relative durations of the pauses roughly like those depicted in Figure 3.3.

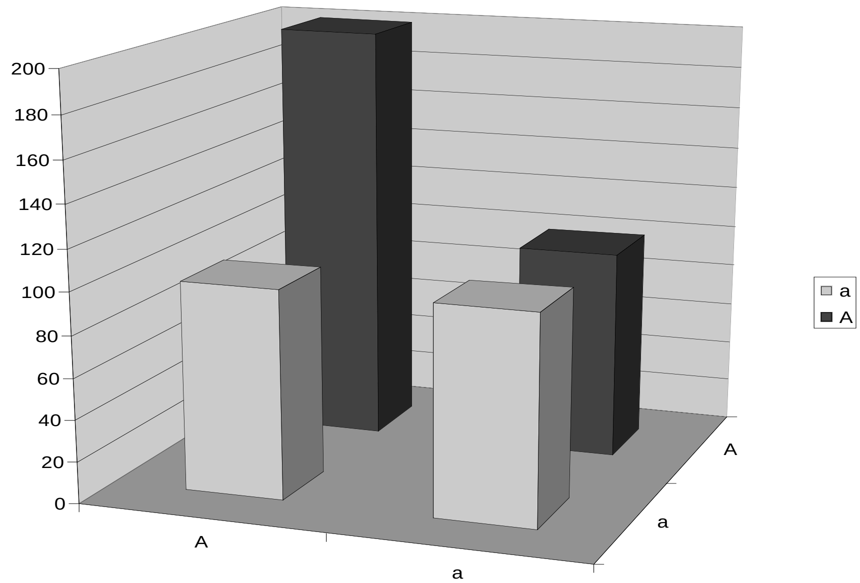
[29] I conducted a pilot study in March 2005 in which I chose words which ended in a fricative for instance, and found that it added considerably to the complexity of measurement, as there many other factors (length of the fricative in different environments etc.) had to be considered.

Figure 3.3: Hypothetical duration of the pause under the ‘pause = emphasized word boundary’ hypothesis (in milliseconds).



If this is not the case, that is, if the hypothesis underlying this study is correct and the pause is a tool to separate two adjacent foci, we would expect relative durations roughly as in Figure 3.4.

Figure 3.4: Hypothetical duration of the pause under the 'pause = empty timing slot' hypothesis (in milliseconds).



The experiments were designed in the following way: The participants were given a sheet with sentences in several groups of four. They were told that these groups of four were fragments of conversations, and that they should put prominence on the boldfaced words. These groups of four were built in such a way that all four cases (A A, a A, A a, a a) of a target two-word sequence were represented. Example (3-31) gives an example of such a quadruple.

(3-31) A: "**Rick discovers** things, but **Bob detects** things."

B: "That's not true. Rick **discovers** many things, and Rick **detects** many things."

C: "Well, I'd rather say: **Rick** discovers **nothing**, but **Bob** discovers **everything**."

D: "Rick discovers **this**, Rick discovers **that** – what's the point?"

This experiment was done for German and for English. The English experiment was conducted in November 2006 at the Phonetics Lab of the University of Pennsylvania (10 participants, 4 male, 6 female, aged 20-45), the German experiment was conducted in December 2006 with a portable digital voice recorder and an external microphone (same equipment as with the production experiment; 10 participants, 4 male, 6 female, aged 20-70). For each case the average duration of the pause was measured.

In the German experiment there were two groups of four: One where the critical words were the subject in the *vorfeld* and the finite verb (SV), and the other where the critical words were the object in the *vorfeld* and the finite verb (OV). Tables 3.10, 3.11, 3.12, visualized in Figure 3.5, give the results.

Table 3.10: Average pauses and differences to doubly unfocused case, German, SV.

SV	average pause	difference to ‚a a’
A A	0.2201	0.1122
a A	0.1258	0.0179
A a	0.1220	0.0141
a a	0.1079	0

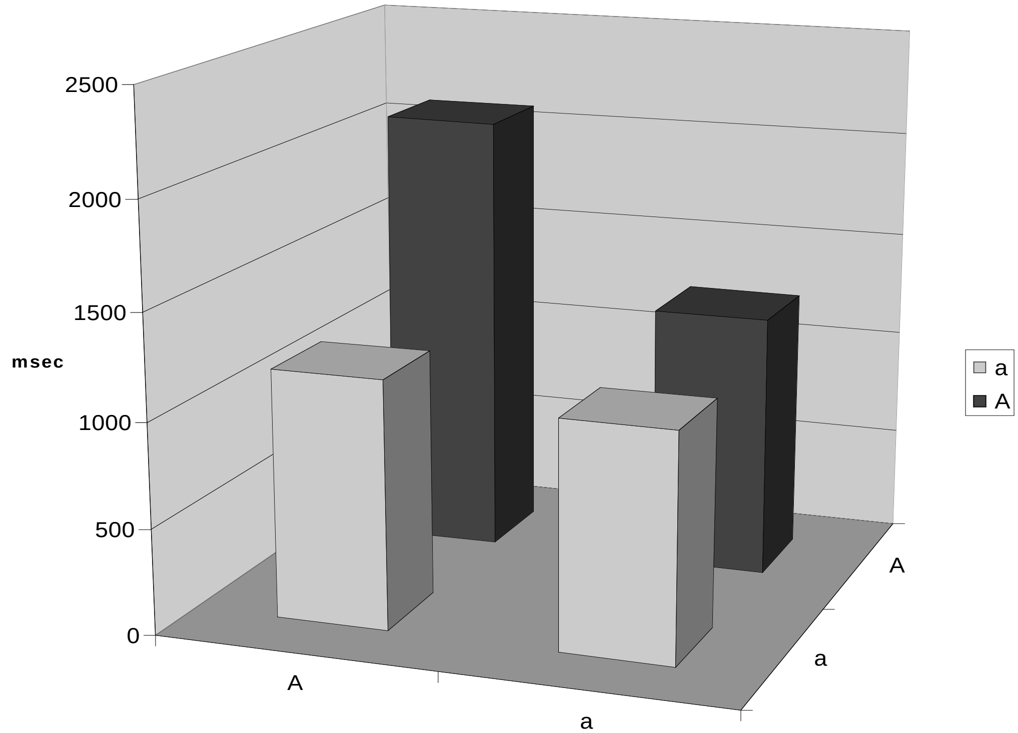
Table 3.11: Average pauses and differences to doubly unfocused case, German, OV.

OV	average pause	difference to ‚a a’
A A	0.1996	0.0914
a A	0.1095	0.0013
A a	0.1273	0.0191
a a	0.1082	0

Table 3.12: Average pauses and differences to doubly unfocused case, German, all cases.

	average pause	difference to ‚a a’
A A	0.2099	0.1018
a A	0.1177	0.0096
A a	0.1247	0.0166
a a	0.1081	0

Figure 3.5: Average pauses and differences to doubly unfocused case, German, all cases.



In the English experiment, there were three groups of four: One where the critical words were the subject and the verb, the verb being initially stressed (SV1), one where the critical words were the subject and the verb, the verb having the stress on the second syllable (SV2), and one with topicalization, where the critical words were the topicalized object and the subject (OS). Tables 3.13, 3.14, 3.15, 3.16, visualized in Figure 3.6 give the results.

Table 3.13: Average pauses and differences to doubly unfocused case, English, SV1.

N1s2s	average pause	difference to ‚a a'
A A	0.1872	0.0565
a A	0.1433	0.0126
A a	0.1346	0.0039
a a	0.1307	0

Table 3.14: Average pauses and differences to doubly unfocused case, English, SV2.

N1s2l	average pause	difference to ‚a a’
A A	0,1506	0.0525
a A	0,1069	0.0088
A a	0,1008	0.0027
a a	0,0981	0

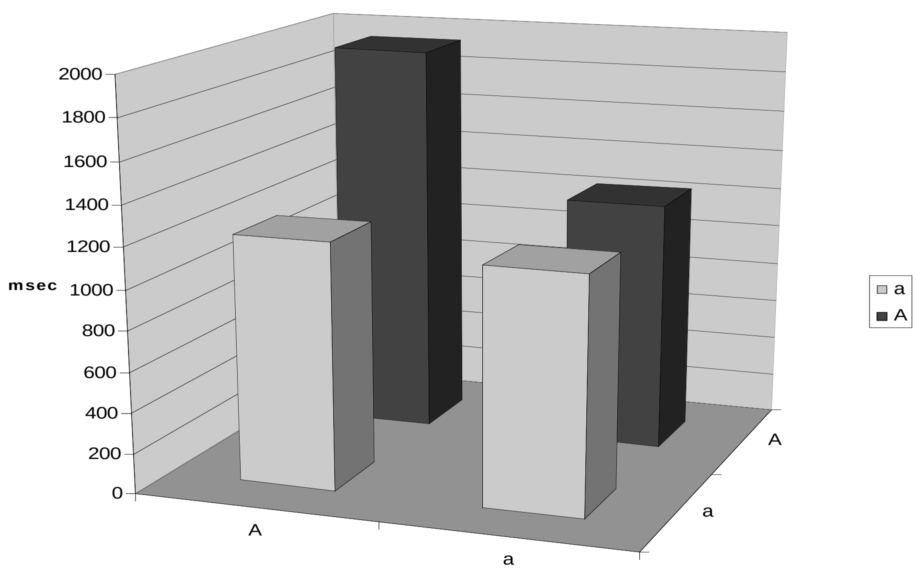
Table 3.15: Average pauses and differences to doubly unfocused case, English, OS.

T1s2s	average pause	difference to ‚a a’
A A	0,2434	0.1242
a A	0,1144	- 0.0048
A a	0,1356	0.0164
a a	0,1192	0

Table 3.16: Average pauses and differences to doubly unfocused case, English, all cases.

	average pause	difference to ‚a a’
A A	0.1937	0.0777
a A	0.1215	0.0055
A a	0.1237	0.0077
a a	0.1160	0

Figure 3.6: Average pauses and differences to doubly unfocused case, English, all cases



The main results of these two experiments are as follows:

First, it does not make much difference for the pause whether there is one focused word adjacent to an unfocused word, or two unfocused words adjacent to each other. The observed average pauses closely resemble model 3.4. The differences are never higher than 20 milliseconds, that is, roughly a fifth of the pause in the ‘a a’ case. The only real difference is when two foci are adjacent: Here, the difference to the ‘unmarked’ ‘a a’ case is between 50 and 130 milliseconds in English, between 90 and 120 milliseconds in German. That means, in focus clash situations the ‘normal’ ‘a a’ pause is extended by half or nearly doubled. This result strongly suggests that the pause corresponds to an empty timing slot and is not simply due to general lengthening because of the focal emphasis.

Second, the differences between the subcases in German – object versus subject in the *vorfeld* – are negligible. This is not true of English, where it seems to make a considerable difference whether we have topialization or not. It is possible to interpret the result in such a way that intonational phrasing after all does play a role, but not as it was claimed earlier: The prosodic boundary does not automatically get realized as a pause; if a pause is produced anyway, it may be additionally lengthened because of the adjacent prosodic boundary.

What do these experiments tell us? They confirm that there is a measurable pause between two clashing foci. The next question must be: What is the purpose of this pause?

Here the CAR comes into play again. We said earlier that the only way to secure the CAR in cases in which narrow syntax produces a sentence with focus clash is to insert meaningless weak material between the clashing foci. We have seen that the preferred way is to insert extra material or to otherwise manipulate the structure of the sentence so that, in the end, no foci are in clash. If people are barred from doing this – e.g. if they have to read a given text or if there is no surface syntactic escape hatch – the only way to rectify things is to insert a pause that corresponds to a weak mark on the relevant level of the grid. So the CAR is restored (3-32). This was already seen by Selkirk (1984: 300); the pause applied here corresponds to her 'silent grid position'.

(3-32)
$$\begin{array}{lll} \boxed{* \quad\; *} & . & \\ \text{RICK THEY like} & \rightarrow & \end{array} \begin{array}{llll} * & . & * & . \\ \text{RICK} & \varnothing & \text{THEY} & \text{like} \end{array}$$

To sum up this section: Topicalization is still acceptable in Modern English in most cases; the only case that is less acceptable is exactly the case where violations of the Clash Avoidance Requirement can occur, namely double focus topicalization with the topicalized element and the subject both being full noun phrases. A short survey of a corpus of naturally occurring topicalization cases showed that pronominal subjects are overproportionally frequent in topicalization cases. This was interpreted as evidence for the Clash Avoidance Requirement. Further evidence has been produced by three experiments for German and three for English. They have proven that

- speakers avoid focus clashes,
- speakers insert measurable pauses when they are forced to utter a focus clash,
- the pauses are due to clash avoidance and are not simply a matter of emphatic lengthening.

3.2 The Clash Avoidance Requirement, the Rhythm Rule and the OCP

The contemporary data from 3.1 strongly corroborates the Clash Avoidance Requirement that we tentatively introduced earlier. This chapter offers a more general formulation of the CAR and puts it in a broader context and relates it to well-known rules and principles, such as the Rhythm Rule and the Obligatory Contour Principle.

First, we have to define the CAR more precisely. We have seen in the preceding sections that double focus topicalization in Modern English is clearly marked and is employed notably less freely than in German. Let us begin by recapitulating the link between English double focus topicalization, German main clauses and the CAR.

One property which both the naturally occurring examples of English topicalization and a normal German main clause have in common is that there is a relatively weak element in second position, in English the subject pronoun, in German the verb. They are ‘relatively weak’ in that they usually do not bear focus: the subject pronoun usually does not bear focus, because it is topical and/or represents old information that is characteristically unfocused. In contrast to other noun phrases, it does not even bear word stress. The (finite part of the) verb in German usually does not bear focus because communicative circumstances in which verbs have to receive focal emphasis are rather rare.

Returning to English topicalization and looking at it from this angle, we have seen that a possible reason why topicalization with full noun phrase subjects is done only rarely is that a full noun phrase subject has a certain likelihood of receiving focal emphasis, as opposed to pronouns.

This has to do with the fact that speakers tend to realize noun phrases pronominally if they can. Reference by means of a pronoun is successful and felicitous if the referent is easily identifiable. This is the case with topics, but also other pieces of old information that are sufficiently salient. Reference with pronouns is not felicitous if the referent is new information and as such introduced into the discourse, or if the referent is put in contrast to another, comparable entity (we spoke about this situation already in more detail in section 2.4.1). As it is more economical to use pronouns for reference, we can conclude that speakers will use preferably pronouns, but full noun phrases only in situations in which reference by pronoun is infelicitous, as mentioned above. That however means that in double focus constructions, if the second focus (i.e. the one that is not associated with the topicalized element) is on a referential expression, this referential expression will normally be realized by a full noun phrase, since as we have seen that pronominal reference is as a rule infelicitous.[30] As the number of referring expressions in a sentence is limited, there is a high chance that the second focus lies on the subject, which then is normally realized with a full noun phrase. If we have a double focus construction where we know that one focus is on the object, and we see that the subject is realized as a full noun phrase, topicalization of the focalized object will lead to a violation of the CAR in many if not most cases, since the chance that the second focus lies on the subject is rather high. Sentences like in (3-33) would be the result.

[30] The well-known case discussed by e.g. Schmerling 1976, Ladd 1980, namely ‘John called Bill a Republican, and then HE insulted HIM’ is an exception; here we could argue that it works only under the premise that calling someone a republican is an insult (I guess the meaning of the sentence would not be understood if uttered on e.g. a local Republican convention in Texas), and under this premise, the ‘HE insulted HIM’ partis only a relevant contribution if the reference is he = Bill and him = John. Such sentences can only be understood after the classical implicature decoding procedure developped by e.g. Grice (1989), which indicates that under normal circumstances pronominal reference in contrast cases is infelicitous.

(3-33) a. BEANS JOHN likes, but PEAS MARY likes.

b. BEANS JOHN cooked for Mary yesterday, but PEAS BOB cooked for her.

Note that, if the subject is realized as a pronoun, we can be almost sure that there will be no focus on it, because pronominal reference cannot be done unambiguously in such a context, as we have seen. Consequently, the second focus in such sentences will lie on some other element, like some adverbial (3-34a) or the predicate itself (3-34b)

(3-34) a. BEANS he cooked YESTERDAY, but PEAS he cooked last WEEK.

b. BEANS he LIKES, but PEAS he HATES.

Thus, from a prosodic point of view, the most conspicuous difference between examples like (3-33) (which do occur only rarely) and the majority of naturally occurring topicalization cases (which are like the ones in (3-34)) is, that in sentences of the type (3-33) an focused element immediately follows the topicalized object, whereas in sentence of the type (3-34) this is not the case. We can thus infer that the property which renders sentences with full noun phrase and topicalization infrequent and often unacceptable is the likely adjacency, or 'clash', of the two foci.

This is of course not to say that full noun phrase subjects always bear focal emphasis. A problem is, however, that topicalization with full noun phrase subjects is in general disfavored and applies also to cases in which the subject does not bear focal emphasis. There are several ways to approach the problem. One way is to assume that language learners misinterpreted the constraint against focused subject as a general constraint against full noun phrases. Another, more promising, one is to assume that a strength hierarchy of the type ‘focal emphasis >> phrasal stress >> unstress’ is at work and that also relatively high non-focal metrical prominences are disallowed in the vicinity of foci, as Tony Kroch pointed out to me. Under the view of the CAR this makes sense, as we would have a clash if not on the topmost line, but still on a line below which is also presumably prosodically ill-formed, judging from the literature on stress clash (e.g. Hayes 1995). Another, empirical, argument is that often radical destressing in the strings immediately adjacent to a focalized element has been observed (e.g. Welby 2003).

Obviously a topicalization sentence is well-formed only if a prosodically weak element follows the focused preposed phrase. We have hinted above that the requirement is not confined to topicalization cases but really applies in all cases of two foci. It might be worth looking into whether there are any restrictions on the scope of the CAR, but so far the generalization seems to hold.

Under this view, we can abstract from the data in this section, the data in 3.1 and the data in part 2 the already known generalization that we have called the Clash Avoidance Requirement (=CAR):

> *The Clash Avoidance Requirement (descriptive form for focal emphasis; see 2.4.4):*
> If there is more than one focused element in a clause, at least one non-focused element must intervene.

We have assumed the CAR already in part 2, but there it could only function as a working hypothesis. Only the synchronic data presented in chapter 3.1 gives direct evidence that the CAR in this form is correct. Remember that in part 2 it looked as if an alternative definition that is less far-reaching (of the form: Two adjacent foci are ill-formed, without giving justification or introducing the alternation motif) would do the job equally well.

The data in chapter 3.1, however, suggests that the essence of the CAR is its urge for alternation.

Another hint that this is the 'right' form of the requirement is that similar constraints are well-known in phonology, especially in the field of Prosody. The obvious parallel is Liberman & Prince (1977)'s Rhythm Rule (=RR), which I do not quote here *verbatim* as they are not very explicit about it. They treat it more as a repair rule, in that it is a rule that repairs a metrical pattern into an alternating pattern (Liberman & Prince 1977:310). More technically, they formulate special transformation rules such as Iambic Reversal that ensures that a weak and a strong branch on a metrical tree are reversed if the strong branch is adjacent to another strong branch (Liberman & Prince 1977:319). This is the line also taken by Halle & Vergnaud (1987).[31] Later, in Prince (1983), the status of the Rhythm Rule is promoted to a higher level; here it is a well-formedness condition that constitute 'eurhythmicity' (Prince 1983:32f.), and underlies grid construction rules that yearn for the 'perfect' (= strictly alternating) grid (Prince 1983:47ff.). From Prince's treatment it is implicitly clear that the Rhythm Rule is rather the same as the CAR, or, more precisely: That both rules, RR and CAR, are based on the same underlying principle. In fact, the two real differences are firstly that Liberman & Prince only talk about the rule-generated word and phrase metrical prominence, whereas the CAR, at least in its provisional, descriptive form, is concerned with focal emphasis. The second difference is that the RR under the view of Liberman & Prince is a secondary process that

[31] Their treatment has another quality important for this discussion namely that the Rhythm Rule is extended in their applicability from word sequences to words; we might assume that it could then also be extended to all kinds of constituents (1987:235). If that were so, the CAR and the RR would be versions of the same rule, the CAR being the application to cases in which the constituent is the whole intonational phrase. But the problem is, again, that Halle&Vergnaud (1987) treat it only as a repair rule. The 'repair' of clashing metrical prominences and clashing foci however would be different, if we believed that it were a repair rule, retraction being applicable with metrical prominence but not with foci, so there is a problem with Halle & Vergnaud's (1987) rule as it stands.

repairs an ill-formed output locally, whereas I, following Prince (1983), regard the CAR as a condition to which outputs must conform, and in accordance to which they are generated. It is thus a rather a principle of how grid generation works, rather than a well-formedness condition, which can alter or 'repair' outputs. This has been seen already by Prince (1983) and Hayes (1984), who proposed a similar principle to the RR. Hayes' approach is perhaps more radical than Prince's approach. Hayes' impression was that the RR in Liberman & Prince (1977) was not far-reaching enough and he suggested that rhythmic organization always follows an alternating principle that he dubbed 'eurhythmy', a more general principle on metrical structures.[32]

Note that both, rule-generated metrical prominence and focal emphasis, are kinds of prominence and realized in a similar manner. That implies that they are ultimately generated by the same system, and the CAR or RR are properties of this system, rather than being properties of metrical prominence per se and focal emphasis per se, respectively. This is important, as we will see later when we talk about the relationship between metrical prominence and focus (chapter 3.3).

If we use a metrical system such as Hayes (1995)'s Metrical Stress Theory which is concerned with forming rhythmic units on the grid, and does not make use of the metrical tree[33], we can reformulate the CAR in a more technical manner that includes also implicitly the RR:

---

[32] One should perhaps point out that Liberman & Prince implicitly seem to think of alternation as an underlying principle (see 1977:310: 'the desire to maintain an alternating pattern'). In the later discussion, however, they 'rule out the possibility that it is some sort of phonetic universal (1977:311)'. As has been mentioned earlier, Prince (1983) explicitly promotes the Rhythm Rule to a universal condition on grid construction.

[33] A grid-only approach is to be preferred (see also Prince 1983), since in a tree we necessarily come to a point where only one highest prominence is to be given – namely at the highest node that necessarily has a weak and a strong branch – but with focus we can get more than one emphasis, and they are presumably equally strong. A tree-theory could not deal with that.

> *The Clash Avoidance Requirement (technical form):*
> On any level of rhythmic representation, strong and weak beats must alternate such that there is at least one weak beat between two strong beats.

Note that the requirement in this form looks like an application of the Obligatory Contour Principle to the problem of prominence. The OCP had been originally proposed for tones (Leben 1973), but soon it was realized that it really applies to all segmental and suprasegmental material, (e.g. McCarthy 1981; Yip 1988). The idea that the OCP could be applied to prominence was explicitly developed by Yip (1988:90ff.). And, in fact, the alternating requirement in grid structures could be described in terms of the OCP by a ban on two adjacent strong grid marks.

To summarize this section: We have found that foci are subject to prosodic constraints very similar to the Rhythm Rule, which again has conceptual similarities to the Obligatory Contour Principle. This led us to the assumption that they are all really instantiations of an underlying general principle that Yip (1988) decribes as a generalized form of the OCP, which is a well-formedness principle on all phonological structures.

3.3 The Clash Avoidance Requirement in German

German is a language in which prominence is encoded by the use of a pitch accent, as in English. If the Clash Avoidance Requirement is not a speciality of English but a phenomenon that all pitch accent languages show, we should see evidence for the Clash Avoidance Requirement also in German.

The potential evidence can be separated into two parts: diachronic evidence and synchronic evidence. Examples for diachronic evidence could be changes that altered the language in such a way that CAR-violations are not generated to begin with, or changes in the usage of certain structural configurations that are in conflict with the CAR. This was the type of evidence that we had for the Clash Avoidance Requirement in Middle and Early Modern English: Topicalization generated outputs that were not CAR-conforming after the subject-before-verb-constraint gained importance in English. Synchronic evidence would be observational data that indicates that speakers avoid violations of the CAR.

Diachronic evidence that is as compelling as the English evidence cannot be produced for German. The reason for this is simply that German, from its earliest attestations, has been a language whose syntax has been set up in such a way that it regularly generated sentences which happened to be CAR-conforming. There are especially two factors that are responsible for the fact that CAR-violations do not easily arise from the beginning:

- German follows the V2-constraint from its earliest attestations and never lost it (in contrast to English). On the contrary: German became even more restrictive in that respect: Whereas in Old High German subject pronouns could occupy a position between the *vorfeld* constituent and the finite verb (thus producing a V3-order) and still in Early New High German V3-sentences of several kinds were possible (3-35a), in Modern German V3 is highly restricted and extremely rare (3-35b, c).[34] This means that an focused phrase in the *vorfeld* is very unlikely to form a clash with the following constituent since the following constituent (if you want to call it that) is the finite part of the verb form, which is the element least likely to stand in focus. This is in direct contrast to English, where the 'CAR-friendly' V2-option disappeared from the language and thus situations with focus clash at the left periphery were on the rise.

(3-35) a. O Kindelein, [$_{\text{vorfeld}}$ [von Herzen] [dich]] will ich lieben sehr

O child from heart thee will I love very

'O child, I will love you deep from my heart'

(Friedrich Spee, Carol ‚Zu Bethlehem geboren', 1638)[35]

b. [[Großes Gewicht] [für die Geschworenen]] hatte ein aufgezeichnetes

great weight for the jury had a recorded

[34] There have been attempts to analyse Modern German V3 as a special case of V2, in that the two constituents before the verb are analysed as one constituent (Müller 2003; 2005). This is not the place to discuss this in detail; while many examples brought forward by Müller could be analysed as one complex constituent (such as 3-35b), but with many others this becomes difficult (3-35c). On the other hand, perhaps double *vorfeld*-filling is simply a historical remnant. If we analyse the German *vorfeld* as Frey (2004) does, namely as a cluster of several Split-C-projections, of which normally only one is overtly filled, we do not need to constrain the *vorfeld* categorically to one constituent only: If we have double *vorfeld*-filling, this is a lapse on the side of the speaker in that more than one C-specifiers are overtly filled; the normal case would be that only one is overtly filled and the others covertly with operators or silent expletives or whatever. I put forward a more detailed proposal in Speyer (2007b) within the framework of Stochastic Optimality Theory. This lapse is comparable to the double-C-filling of subordinate clauses in South German dialects (i). Here we have the same case: Two positions are available, normally only one is realized overtly, the other has a silent element in it; in clauses like (i) both positions are overtly realized.

(i) Sok ma, wann dass d' kiimsch. (Tyrolian Bavarian)

tell me when that you come

'Tell me, when you are coming'

[35] This carol is very popular in German; here cited after the German Catholic Hymnal 'Gotteslob' (Freiburg 1976).

Telefongespräch des Scheichs mit den Bombenlegern des World Trade

phone conversation of-the sheikh with the bombers of-the World Trade

Centers

Center

'A phone conversation of the sheikh with the WTC-bombers had a great impact on the jury.'

(taz, October 4, 1995, p.8; cited after Müller 2003:35)

c. [[In Züpfners Box][der Mercedes]] bewies, dass Züpfner zu Fuß gegangen

in Z's garage the Mercedes proved that Z. to foot gone

war.

was

'The Mercedes in Züpfner's garage proved that he had gone by foot.'

(Böll, p.165)

- German had, from its beginning, a 'free' word order; that is, constituents could be moved from their base-generated position, and this movement was motivated not by requirements of narrow syntax, but by information structure and the like ('scrambling'). Consequently, focus clashes could always be avoided by changing the word order. Since scrambling was just as possible in Old High German as it is in today's German, there is no starting point for a CAR-motivated language change.

So there was no obvious language change or change of usage in German that could be CAR-related. The only change is that the V2-syntax became successively more restrictive, though this is probably not related to the Clash Avoidance Requirement. The reason for the lack of CAR-related changes is simply that from its earliest attestations, German has been a language in which the syntax is built in such a way that CAR-violations do not easily arise. To put it bluntly: If a system is optimal, why further optimize it? Never change a winning combination. If the syntax does not generate CAR-violations anyway, there is no need to meddle with it, at least under this aspect.

How about synchronic evidence, then? Here the picture is clearer: We can observe that speakers of contemporary German in fact avoid violating the Clash Avoidance Requirement. The interesting difference between English and German in that respect is the following: English, having strict constraints on word order, came into problems with double focus topicalization and solved the problem by *forgoing* this movement operation. German, on the other hand, has less strict constraints on word order, and is free to solve CAR-problems by *applying* movement. Keep this in mind for the remainder of the section.

Most of the evidence has already been presented in section 3.1.2, but I nevertheless repeat it here briefly. The experiment which is relevant is the production experiment. Here we could see that the speakers took pains to form their sentence in such a way that the CAR is observed. They achieved this goal mostly by operations that on the surface look like the insertion of semantically empty material, e.g. Left Dislocation, in which an unemphasised pronoun intervenes between the targeted clashes. The relevant data has been presented in Tables 3.5, 3.6 and Figure 3.1.

Let us return to the first point which rendered German so immune against CAR-violations, i.e. the strict V2-constraint. A logical consequence of this would be that in double-focus constructions, there should be a strong preference for one of the focused elements to move to the *vorfeld*; otherwise speakers would not profit from the V2-constraint as a prophylacticum against CAR-violations.

This in fact seems to be the case. A well-known subtype of Hat Contour sentences are the so-called I-topicalization sentences (cf. Jacobs 1997; Steube 2003). The 'I' stands for 'intonation', and in fact I-topicalization sentences can be identified, roughly speaking, by the Hat Contour and the fact that one of the focused elements is in the area before the finite verb (Jacobs 1997:92). I-topicalization sentences have special properties which distinguish them from other types of Hat Contour sentences, but I will not go into details here (see Büring 1997; Jacobs 1997). I-topicalization is the exact counterpart of English topicalization in that it is the preposing of a contrastive and thereby focalized constituent (see Jacobs 1997:105). Moreover, the topicalized element has to conform to the conditions, under which topicalization in English is possible, namely a poset-relationship in the sense of Hirschberg (1986:122) and Prince (1986:208ff.; 1999:7ff.). In addition to this it seems as if the I-topicalized constituent must stand in a real contrastive relationship with another member in the same set, which should be spelled out explicitly or be otherwise so salient that it can be easily and unambiguously referred to by the addressee (cf. Jacobs 1997:95f.; Steube 2003:165). Examples for I-topicalization are in (3-36).

(3-36) a. A: Kann man alle Romane von Grass empfehlen?
can one all novels by Grass recommend
'Are all novels by Grass to be recommended?'
B: Naja, /ALle kann man sicher \NICHT empfehlen,
well all can one surely not recommend

aber den /ERsten muss man \SCHON gelesen haben.

but the first must one indeed read have

'well, certainly not all are to be recommended, but the first one you must read.'

(after Jacobs 1997:92)

b. A: Kann man alle Romane von Grass empfehlen?

B: Also, die /NEUeren sind eher \NICHT so der Hit

well the newer are rather not so the hit

(, aber die /ÄLteren sind ein absolutes \MUSS.)

but the older are an absolute must

'well, the newer ones aren't so good, but the older ones are definitely a must.'

The spelling-out of other members of the set is not a necessary condition, though; we find I-topicalization also with a trivial case of poset, namely that only one member is in the set (3-37).[36]

(3-37) A: Wo hast du denn die Schuhe her?

where have you ptc. the shoes from

'Where did you get your shoes from?'

B: /DIE hab ich in \MÜNchen gekauft.

them have I in Munich bought.

[36] It is not entirely clear whether this is really an example of I-topicalization or a 'normal' bridge contour that only looks identical to an I-topicalization. The emphasis in such cases is due to deixis.

'I bought them in Munich.'

Furthermore I-topicalization is also possible with *vorfeld*-free sentence moods such as imperatives (Jacobs 1997:93), by adjunction of a *vorvorfeld* (3-38a). This is not possible in the case of other types of Bridge Contours, where it is a topical element that has the first 'accent' (3-38b).

(3-38) a. A: Ich wollte mir heute den neuen Roman von Grass kaufen.

I wanted me today the new novel by Grass buy

'I wanted to buy the new novel by Grass.'

B: Den /NEUen Roman kauf dir lieber \NICHT.

the new novel buy you rather not

(Aber den /ERsten solltest du \UNbedingt lesen.)

but the first should you unconditionally read

'Better not to buy the new novel, but you definitely should read the first one.'

(after Jacobs 1997:93)

b. A: Was soll ich in München kaufen?

what shall I in Munich buy

'What should I buy in Munich?'

#B: In /MÜNchen kauf dir neue \SCHUhe.

in Munich buy you new shoes

'Buy new shoes in Munich.'

If the *vorfeld* is not available as a landing site for other reasons, and movement to the left of the *vorfeld* is not possible either, e.g. because it is a subordinate clause, the first focalized constituent is in a position as far to the left as possible, that is, right at the beginning of the *mittelfeld* (3-39; Jacobs 1997:95).

(3-39) A: Der erste Grass-Roman wurde ja begeistert aufgenommen.
the first Grass-novel was ptc. enthusiastically received
'The first novel by Grass was received enthusiastically.'
B: Also, ich hab aber gehört, dass /ALlen Kritikern das Buch keines\WEGS
well I have but heard that all reviewers the book not-at-all
gefallen hat.
pleased has
'well, but I have heard that not all reviewers liked the book.'
(after Jacobs 1997:95)

Steube (2003:173) conducted an experiment in which sentences with Hat Contours were read to participants who subsequently rated them. She found that sentences in which the I-topic was in the *vorfeld* received better grades than sentences in which the I-topic was elsewhere. So it seems as if the *vorfeld* is preferred for such elements.

In this context I wish to mention the phenomenon of split topicalization.[37] What is special about split topicalization from the point of view of this study is that it is a double focus construction, but a very special case: The two foci lie on a noun and a quantifier modifying the same noun (3-40). That means that under normal circumstances, that is, in

[37] For a comprehensive treatment of Split Topicalization see Nakanishi 2004.

base-generated word order, the two foci would be in the same phrase and hopelessly adjacent (3-40b). The mere fact that there is this slightly irregular kind of *vorfeld*-movement at all (it is irregular because it is not an immediate constituent but only part of an immediate constituent that is moved), indicates that there is a strong desire to avoid the CAR-violation in the base-generated version, even at the cost of a complicated syntactic operation.[38] In German, the split version is even slightly more acceptable than its non-split counterpart (3-40c), which again shows that the CAR-violation associated with the base-generated word-order is bad enough for the speakers that they prefer to use a complicated constructional device over living with a CAR-violation.

An additional factor is of course the urge to move the sorting-key up front (see the discussion in chapter 2.2); in this respect split topicalization is similar to other kinds of double focus topicalization constructions. There is however one difference: Split topicalization is confined to the *vorfeld*; it would not do to use a similar movement operation in sentences without *vorfeld* (3-40d). This is in contrast to the ‘normal’ sorting-key-topicalization: It is possible (however marked) to scramble in the *mittelfeld* such that the sorting key is at the beginning of the *mittelfeld* (3-40e); we saw this already when discussing I-topicalization (3-39). Since the *vorfeld* is the only position where we can be sure to have both foci separated, it is not far-fetched to assume that the main motivation

[38] See Nakanishi 2004:158ff. for a critical overview. The standard assumption, which is at the base of my discussion, is that the quantifier AP is moved out of the NP; the remnant of the NP is subsequently moved to the front. Nakanishi (2004:158ff.) argues that this view is incorrect and cites numerous examples that indicate that split topicalization cases are not simply derived from non-split versions. She adopts van Geenhoven’s (1998) analysis, under which the quantifier and the noun phrase are not base-generated in the same phrase but that each is a phrase of its own, both being immediate constituents of the VP. The quantifier AP attaches higher than the NP. The NP has to topicalize in order to be able to c-command the AP. This analysis has a lot of appeal; it is relatively complicated, though, whereas my argument that a CAR-violation is avoided even at the cost of a complex syntactic analysis remains valid, if van Geenhoven’s analysis is adopted. Note that this does not exclude from the grammar the ‘normal’ version of a quantified noun phrase where the AP is inside the NP; it is only the case that the split topicalization version is not directly derived from the non-split one.

for split topicalization is the certain avoidance of a CAR-violation rather than sorting-key-ordering.

(3-40) a. (question: Wieviel haben wir von was?)

how-much have we of what

/ÄPfel haben wir \DREI. Und /oRANgen \FÜNF.

apples have we three and oranges five

'We have three apples and five oranges'

b. dass wir /DREI \ÄPfel haben, und /FÜNF \oRANgen.

that we three apples have and five oranges

c. (#) Wir haben /DREI \ÄPFEL. Und /FÜNF \oRANgen.

we have three apples and five oranges

d. (question: Wieviel haben die Leute von was?

how-much have the people of what

\# Ich glaube, dass /ÄPfel die Leute \DREI haben und /oRANgen \FÜNF.

I think that apples the people three have and oranges five

e. (question: Weißt du, wo Hanna was gekauft hat?)

know you where Jane what bought has

'Do you know where Jane bought what?'

Ich glaube, dass in /MÜNchen Hanna \HOsen gekauft hat

I think that in Munich Jane trousers bought has

und in /GARmisch \SCHUhe.

and in Garmisch shoes.

'I think that Jane bought trousers in Munich and shoes in Garmisch.'

So we can say: The *vorfeld*, or more generally, the area left of the finite verb, is indeed a preferred position for one of the foci in double focus constructions. If the *vorfeld* is not available, we still sense the desire to keep the focused phrases as far apart as possible, and thus CAR-violations are less probable.

English has basically lost the possibility to use Split Topicalization. As Tony Kroch (p.c.) pointed out to me, we see in phrase with double focus within an NP a grading effect which can be probably interpreted in such a way that one of the foci is not realized as a focal emphasis but as a 'normal' phrase with phrasal prominence (3-41).

(3-41) He bought many textbooks, but only few novels.

One of the two elements, the quantifier or the lexical noun, sounds 'weaker' than the other; which of the two foci is selected for realization is probably dependent on the context, which contrast is more important than the other. The fact that defocusing occurs supports the claim that two adjacent foci are avoided although the resolution is different from what we would expect given the German data.

Speakers follow the CAR, as we have seen in our production experiment and the observations about I-topicalization. The CAR does not only play a role in sentence production, however, but also in sentence judgment. We have seen (3.1.1) that the main problem with the acceptability of topicalization cases in Modern English is related to the CAR: Exactly those cases that are in danger of violating the CAR, i.e. double focus topicalization with focused full noun phrase subject, are often judged unacceptable.

People do not have such strong feelings about cases in which the subject is not in focus but is a full noun phrase. They are slightly uneasy though, which may be tied to the fact mentioned earlier that there is still a clash situation on a lower line of the clausal level.

In order to test whether CAR-violations have an impact on the acceptability judgments of German topicalization, I conducted another experiment (14 participants, 4 male, 10 female, aged 30-75). The experiment was conducted in January 2007. Here the participants were given a sheet with three sentence groups; each sentence group consisted of three or four permutations of the same sentence, but with different word orders. The sentences all contained a subject, an object, and a scene-setting element and had both sentence brackets filled. The two foci were on the scene-setting element and the object in the first sentence group, on the subject and the object in the second group and on the subject and the scene-setting element in the third group. At least one of the permutations in each group was built in such a way that the two foci in the sentence were adjacent. I instructed the participants to give each sentence a ‘grade’, indicating whether they regarded it as a natural answer to the question which I read aloud before each sentence group.[39] The grades to be given were 1 (fully acceptable), 2 (kind of acceptable, but weird), and 3 (not acceptable). Then I read the sentences aloud with a hat contour on the two focused elements. In (3-42) I give the questionnaire with the questions added, the results are in Tables 3.17 and 3.18. Only the questions and the last versions of each sentence group are provided with an interlinear translation; the others can easily be

[39] I am aware that there are many factors that can influence the judgments. German word order can be subject to a wide variety of constraints, partly of a grammatical nature, partly of a pragmatic nature and partly of a cognitive nature (see e.g. Zubin & Köpcke 1985; Hoberg 1997). So the results in this experiment have to be taken with a grain of salt. I find it nevertheless remarkable that, although there are various factors that influence the judgment, that the data is still very suggestive for the prosodic problem we are interested in.

derived. Table 3.18 is a collapsed version of Table 3.17 with only bipartite distinctions, viz. ‘fully acceptable – yes/no’ and ‘focus clash – yes/no’.

(3-42) “Was hat Hanna gestern wo gekauft?”

what has Jane yesterday where bought

‘what did Jane buy where yesterday?’

Hanna hat gestern in **München Schuhe** gekauft, aber in **Garmisch Hosen**.

Gestern hat Hanna in **München Schuhe** gekauft, aber in **Garmisch Hosen**.

Schuhe hat Hanna gestern in **München** gekauft, **Hosen** in **Garmisch**.

In **München** hat Hanna gestern **Schuhe** gekauft, in **Garmisch Hosen**.

in Munich has Hanna yesterday shoes bought, in Garmish trousers

‘Jane bought shoes in Munich yesterday, and trousers in Garmisch.

„Wer hat letzte Woche was gelesen?“

who has last week what read

‘Who did read what last week?’

Hannes hat letzte Woche **Bücher** gelesen, **Uller** aber nur **Zeitschriften**.

Letzte Woche hat **Hannes Bücher** gelesen, **Uller** aber nur **Zeitschriften**.

Bücher hat **Hannes** letzte Woche gelesen, **Zeitschriften Uller**.

books has John last week read journals Uller

‘John read books last week, Uller read journals.’

„Wer hat dem   Uller wann eine geklebt?“

who has the.$_{DAT}$ Uller when one glued

‘who slapped Uller when?’

Petra hat **gestern** dem Uller eine geklebt, und **Hanna vorgestern**.

Dem Uller hat **Petra gestern** eine geklebt, und **Hanna vorgestern**.

Gestern hat **Petra** dem Uller eine geklebt, **vorgestern Hanna**.

yesterday has Petra the Uller one glued,   before-yest. Jane

‘Petra slapped Uller yesterday, Jane the day before.’

Table 3.17: Numbers of grade assignments for the sentences (bold italic: clash cases).

grades:	1	2	3
1a	***2***	***4***	***8***
1b		***3***	***11***
1c	6	3	5
1d	4	6	4
2a	14		
2b		***13***	***1***
2c		1	13
3a	6	8	
3b	***1***	***2***	***11***
3c	7	4	3

Table 3.18: Numbers and percentages of grade assignments for clash and no-clash-sentences.

	fully acceptable		not fully acceptable	
	number	**%**	number	**%**
1: clash (n=28)	2	**7.1**	26	**92.9**
2: clash (n=14)	0	**0**	14	**100**
3: clash (n=14)	1	**7.1**	13	**92.9**
1: non-clash (n=28)	10	**35.7**	18	**64.3**
2: non-clash (n=28)	14	**50**	14	**50**
3: non-clash (n=28)	13	**46.4**	15	**53.6**

Although we know that prosody is not the only factor for movement of constituents to the *vorfeld* – scene-setting elements are highly preferred as *vorfeld*-constituents, likewise topics (in the first sentence group realized as subject) are among the elements that have a certain preference for the *vorfeld* (see Speyer 2004; 2006a) – we see a certain impact: On the whole the versions where one of the focused constituents was in the *vorfeld* (and therefore not in clash) have been judged more positively than the ones where both focused constituents were in the *mittelfeld*, adjacent to each other. This finding is in accordance with Steube's (2003:173) findings. Note that the few 'lapses', i.e. the cases where sentences with focus clash were judged acceptable, are ones in which the 'topic-first' constraint interferes: In sentences 1a and 3b, the topic is in the *vorfeld*; here the participants regarded the topic-first constraint as more important than the desire to have lexical material between the foci.

Summarizing we can say: There is much evidence that the CAR is a serious factor in German as it is English. This evidence is primarily synchronic evidence, because diachronic data is not available since German – in contrast to English – never changed in a way that could have jeopardized the CAR. The evidence gathered from several different experiments is that speakers avoid focus clashes and judge sentences with focus clash as worse than sentences without.

3.4 The relationship of metrical prominence and focus

After having established the core of the thesis, namely, the existence of the Clash Avoidance Requirement, a few questions still need to be answered. One is the question of the relationship between (rule-governed) metrical prominence and (semantically motivated) focal emphasis. After defining the terms metrical prominence and focal emphasis as they are used here I show in section 3.4.1 on descriptive grounds that focal emphasis cannot be simply a continuation of the metrical prominence system.

In section 3.4.2 the consequences are drawn for the application of the relevant rules: Focal emphasis is encoded by a focus indicator, which is implemented here as a ‘strong credit mark’ that automatically adds a line to the grid containing a strong mark on the focused element. The metrical calculus, that is, the system of rules that generate metrical prominence and of whose rules the CAR is a part, has to build a well-formed grid ‘around’ the focus indicator, and sometimes has to resort to methods like pause insertion, if nothing else helps.

Section 3.4.3 is devoted to the nature of the nucleus (the highest peak in the clause). It is not comparable to focal emphasis, either conceptually or empirically, but is definitely part of the rule-governed metrical prominence system, as will be demonstrated in this section. As has been pointed out earlier, though, it is of course subject to the CAR as all prominence is.

Section 3.4.4 draws the conclusion that focal emphasis and rule-governed metrical prominence are two rather different entities which, however, both encode their information both by using the same phonological subsystem, namely the prosodic/intonational system. The apparent similarities between metrical prominence and focal emphasis – e.g. the urge to follow the Rhythm Rule / Clash Avoidance Requirement – are properties that are imposed by the phonological subsystem.

3.4.1 Comparison of metrical prominence and focal emphasis

In section 1.2.3 a distinction was introduced between metrical prominence and focal emphasis. The discussion implied that there is a difference between these two kinds of prominences, either on the phonological level, or on the acoustic level, or on both. This leads us to several important questions, namely: what is the relationship between these two kinds of prominence – metrical prominence and focal emphasis? Is this difference reflected in the phonetic correlates of both kinds of prominence? How does the nucleus fit in? I will turn for the rest of this section to the first and second question by offering a comparison between metrical prominence and focal emphasis on descriptive grounds which shows that there are certain differences, in their origin and in the way the prosodic rules deal with them. The question of the clausal prominence peak, that is: the nucleus, is discussed in the section following that.

From what we have seen so far, it is not immediately clear whether the relationship between metrical prominence and focal emphasis is trivial or not. Before we go on, it might be appropriate to discuss whether we would expect metrical prominence and focal emphasis to be directly related in the first place, or one to follow from the other. For this it might be helpful to recall the conflicting definitions of ‘accent’ or focal emphasis given at the end of section 1.2.3. It turns out that we can capture a greater generalization if we make the cut between metrical prominence and focal emphasis in such a way, that only the definition (2) of focal emphasis (short: focus) – as prominence associated with narrow focus – is the sufficient condition on focal emphasis. The other property (definition (1)) which we saw in section 1.2.3, viz. that focal emphasis represents the highest peak in a clause, is simply a consequence of that: Any prominence associated with narrow focus will end up being the highest prominence in the clause (see e.g. Selkirk 1984), therefore definition (1) just comes for free. But the reverse does not hold, as I will argue, namely that the highest prominence in a clause is always associated with focus. Therefore I will make the cut between focal and other prominence in such a way that only narrow focus generates a focus indicator that is realized as focal emphasis, following Ladd (1996:160f.) and Sluijter (1995:3). We will see in the next section that wide focus does not generate a focus indicator, therefore the metrical calculus can build a grid undisturbed.

The generalizations that we see when separating focus and metrical prominence in the way described above are listed in the following paragraphs. Under this light, we see that the differences between metrical prominence and focal emphasis are substantial compared to their shared properties. The only thing which they have in common is that they both are marked by prominence. This means that in the end they will have to interact in some way; later we will see that the focus indicator(s) set the frame within which the metrical calculus can construct the grid, always trying to conform to its rules, which are built in such a way that the CAR is respected. This means, of course, that focal emphasis is not simply the highest level of the ‘normal’ prominence assignment (as suggested by definition (1)), but a different process. It is not: First metrical prominence, then focus, but just the other way round.

Let us now turn to the generalizations. First, there is the question of how many 'peaks' are admitted within a given unit. Trivially, any 'highest' prominence can only be assigned once within a unit, be it a foot, a word or a phrase. Hayes (1995:24) calls this phenomenon *culminativity*. So we can clearly determine for each unit on which smaller unit (ultimately the syllable) the main prominence peak falls.

Foci, on the other hand, can be assigned several times within a given domain, say a clause. And it is not at all possible to judge which of two or three foci is of higher prominence in e.g. a sentence with Hat Contour, because multiple foci are usually realized with different contours (one rising, one falling, for instance), and trying to compare them is like comparing apples and oranges. The fact that the realization of a focus in a given clause is dependent on the realization on another focus in the same clause, however, suggests that they are in the same domain, as they influence each other, which they probably would not do across domain boundaries. So we can say that we can have multiple foci within a clause, not just one unit of highest prominence as on the lower levels.

Second, whereas with metrical prominence we can have an arbitrary number of assignments levels – which are realized phonetically as primary, secondary, tertiary etc. stress, each differing in what height the column of strong grid marks corresponding to it has – we see that focal emphasis is either present or not (cf. Ladd 1980:46). There is no such thing as a secondary focus. A rising accent (as the realization of focus) may be perceived as less prominent than a falling accent, but, as I said before, different gestures are really not comparable. This, of course, follows from the first point.

Thirdly, the relative prominence of metrical prominence is clearly assigned within a syntactically describable domain. This is true especially in a direct reference approach such as e.g. Cinque (1993), Seidl (2001), but even in indirect reference approaches such as Selkirk (1984) or Nespor & Vogel (1986). That means, we can identify different levels for metrical prominence assignment that correspond on the phonetic side to domains for e.g. word-stress assignment (within the word as it is inserted in spell-out or within the phonological word, depending on your approach) and phrasal stress assignment (within a syntactically describable constituent, e.g. a major constituent or a minor phase (as in Seidl 2001), or within a phonological phrase, as in Selkirk 1984; Nespor & Vogel 1986).

If we now try to determine the domain within which focal emphasis is assigned we encounter a problem as soon as we allow for contrastive foci (which we have to do, of course, especially under the assumption that contrastive foci are the archetypical foci). Contrastive foci – and by that also the emphasis that they trigger – can be assigned to arbitrarily small units, words, even morphemes (3-43c-e; Bolinger 1961; van Heuven 1994; Sluijter 1995:5), and since they can end up adjacent to each other, as in sentences (3-43a-d), we have to admit that in these cases it is indeed the word/morpheme that is the relevant domain for focus assignment (cf. also Drubig 2003). Note that in (3-43a) the contrast is between –*teen* and –*ty*, which is the reason why the Rhythm Rule does not apply. In (3-43b) the Rhythm Rule is free to apply because the names *Tenessee* and *Kalamazoo* are monomorphemic in English; here the contrast is between the words and not the final morphemes, as in (3-43a).

(3-43) a. I told you to call thirTEEN MEN and NOT thirTY WOmen!

b. You should have contacted the TEnessee LEgislators, NOT the KAlamazoo OPerators.

c. He keeps insisting that we COUNtersign it, but there's nothing TO countersign (from Bolinger 1961:88)

d. Du sollst das Bier hinAUF-TRAgen und nicht hinAUS-WERfen

You shall the beer up carry and not out throw

'You're supposed to CARry the beer UP, and not to THROW it OUT.'

e. Sie sollen die Leute ENTschädigen und nicht noch zusätzlich

You shall the people recompense and not yet additionally

BEschädigen!

damage

'You're supposed to recompense the people and not further damage them.'

Note that morphemes that normally are unstressed, such as in (3-43e), can be focalized, if the context is right (what Bolinger (1961) calls 'contrastive stress'). Thus, metrical prominence is not in itself a precondition for focal emphasis, in other words: Not everything that can receive focal emphasis must also be able to bear metrical prominence under normal circumstances.

Fourthly, as we will see later in section 3.5 and as follows from the previous point, mechanisms that resolve clash of prominent elements have different domains of application. Clash resolution mechanisms for metrical prominence often do not apply across syntactic boundaries – which is what we would expect, given that metrical prominence assignment is confined to syntactically describable units (see also Liberman & Prince 1977:320; Hayes 1984:72 and Kager & Visch 1988:48).[40] A simplified account

[40] What I mean by this is simply that prominence assignment usually happens exhaustively within a given unit, say a word, then the next cycle assigns prominence between words in the next higher unit, saythe phrase, etc., until the parse is on the sentence level, and there is no higher unit left. These units are in some way definable with some reference to syntax, be it direct or indirect. Important here is that these units are

would be that the metrical calculus, when assigning word or phrase metrical prominence, only ‘sees’ one word or phrase at a time and therefore does not care about potential clashes between words or phrases. This is certainly true for German, but there is a tendency for this also in English. The evidence for this is that stress retraction happens independently from the stress pattern of the words that are immediately before or after the word. Let us consider (3-44). In (3-44a), stress retraction is very handy, as it resolves the stress clash in the famous compound TennesSEE LEGislators. In (3-44b), retraction leads to another stress clash with the verb that precedes this compound, but this does not seem to be so much of a problem.

(3-44)a. We’d rather have TENnessee LEGislators be drug-free.

b. We deMAND TENnessee LEGislators to be drug-free.

This indicates that the domain of stress clash resolution rules is really very limited, namely the domain of the phrase, or, more precisely, the main constituent. The resolution mechanism cannot ‘see’ what happens outside the domain, and therefore it is possible that it produces clashes as in (3-44b).

Focus clash resolution mechanisms, on the other hand, are of a different nature than the mechanisms that resolve clash in the domain of metrical prominence; focus clash is resolved much more often by pause-insertion than by other mechanisms such as retraction (or shift) or destressing (see Hayes 1995:36). This is clear from the data from chapter 3.1 and will be further developped in section 3.5. Stress clash, on the other hand,

---

discreet, this means that the word prominence assignment of one word does not influence how the word prominence assignment on the next word is proceeding.

typically is not resolved by pause-insertion. Secondly, because we are on the highest level of prominence assignment when talking about focal emphasis, we do not see this kind of discreetness that we have seen in clashes of metrical prominence. Most often the focus clash resolution mechanisms apply exactly across syntactic boundaries. It is true that they apply within the highest domain, the clause, but the content of all smaller domains is irrelevant for focus clash resolution. This is of course a consequence of the way we believe focal emphasis to work; in later sections I will argue that focal emphasis is technically realized by a 'credit mark' which will become relevant for the parse only on the highest line of the highest level.

Fifthly, metrical prominence and focal emphasis differ fundamentally in their function. It is perhaps not too far-fetched to say that the function of metrical prominence is to facilitate the parsing of syntactic constituents.[41] This is not only important for the addressee in a conversation so that s/he is (better) able to parse the sentence spoken to him/her, but is especially important for the child that acquires the language; the scansion of speech by means of metrical prominence gives valuable clues to the language learner as to how to break up the string of sounds into phrases and words. This means that the 'contribution' of metrical prominence is first and foremost of a structural nature; it facilitates the syntactic parsing. Metrical prominence in itself, however, does nothing to encode semantic or pragmatic parts of the meaning of the clause.

Focal emphasis, on the other hand, most decidedly encodes something that is important to the meaning of the sentence, namely the focus indicator which we might view as a phonologically interpretable representation of a syntactic focus feature (for a

[41] Cf. Hyman 1977. This can be deduced from the fact that there is a more or less direct relationship between phonological phrases and syntactic phrases. Cf. e.g. Price et al. 1991; Sluijter 1995:1; Hayes 1995:31.

theory about the possible shape of the focus indicator see 3.4.2). The contribution by the focus feature is of either a semantic or a pragmatic nature, depending on the viewpoint, but it is a contribution of meaning at any rate. Rochemont (1986) distinguishes between Contrastive and Presentational Focus, both having at least a pragmatic meaning associated with them (marking an entity as discourse-new in the case of presentational focus, marking that an element is in contrast to some other element, which may even be a semantic part of the meaning). Rooth (1985) associates focus with a semantic operation that produces a set out of the entity in focus plus comparable entities; the meaning can be paraphrased as ‘x as a member of a set, and note that it is x and not some other set member that has been picked out’. Höhle (1992) observes another kind of focus, the so called *verum*-focus which emphasizes the fact that the truth value of the sentence containing it is 1.

Because focal emphasis is the realization of focus, it is assigned to units of variable size – morphemes, words, but not larger constituents. As such it is quite unusable for the purpose for which we think that metrical prominence is used, namely for the indication of the (larger) constituent structure. So we can say that the functions of metrical prominence and focal emphasis have nothing to do with each other.

From this, and from the point that stress is assigned within syntacticalöly describable domains, follows the sixth point: Metrical prominence assignment is rule-governed and must be rule-governed in order to provide a useful tool for the language learner to deduce the syntactic structure. The language learner can only extract syntactic constituents from metrical prominence if s/he can rely on the knowledge that the speaker will always, infallibly, mark the syntactic constituents the same way. Since focal emphasis, on the other hand, is governed by semantic and/or pragmatic requirements, it is not predictable and cannot be rule-governed. This should sound familiar since it is the hypothesis of Bolinger (1972); in contrast to him, though, the statement here describes only focal emphasis associated with narrow focus. Because it is predictable, sentence stress is different It is often thought of as beingon a par with focus. We will treat it in more detail below.

Finally, the seventh point, which I will elaborate slightly in section 3.4.4, is that there are acoustic differences between metrical prominence peaks and focal emphasis (e.g. Sluijter 1995). Although both types of prominence use the same parameters – pitch, loudness, vowel quality and duration (Schane 1979; Sluijter 1995) – foci show more extreme deviations from the normal values of these parameters in general than metrical prominence peaks, especially when it comes to pitch. As I mentioned, we will return to this point in section 3.4.4. But for the purposes of this comparison, it is important to keep in mind that we really have evidence that the phonetic correlates of a focus indicator and of metrical prominence peaks, including the highest clausal prominence peak, are rather different phonetically. This is more in line with the 'structure-based account' of broad focus, as Ladd (1996:163f.) calls it, which essentially acknowledges that within a broad focus the assignment of prominence is rule-governed. I go one step further and say, as soon as any prominence is assigned by rules, it will differ phonetically from a prominence that is assigned by narrow focus. So, broad focus does not make much of a contribution in terms of the acoustic rendering of an utterance, and we should perhaps better say that focus only has a phonetic (pitch gesture) and phonological (a focus indicator, for instance the 'credit mark' to be introduced below) correlate if it is a narrow focus, but no phonetic or phonological correlate if it is wide focus.

In this light it is obviuous that it makes sense to abandon the idea that focal emphasis equals the highest prominence in the clause, as we can capture more crucial generalizations if we abandon this idea. The definitions are repeated below (similarly Ladd 1996:160; Sluijter 1995:3):

- Metrical prominence: Prominence that can be computed by a finite set of phonological rules. Its domains are typically the (prosodic) word (audible as word stress), the (phonological) phrase (audible as phrasal stress) or the clause as a whole (audible as sentence stress, or nucleus)
- Focal Emphasis: Prominence that is associated with semantic-syntactic features (narrow focus) and whose placement thus has an impact on the semantic computation.

These points are very indicative that metrical prominence and focal emphasis are different phenomena that are not reducible to one or the other and that are such that the one cannot be derived from the other (cf. Sluijter 1995). They have to interact, however, in that they are both some kind of prominence. From this it follows that the interaction has to be non-trivial in nature and thus that concepts associated with metrical prominence computation (prosodic constituency and hierarchy) are of a limited importance for questions of focal emphasis. Because they are both types of prominence, they are subject to similar rules and well-formedness conditions, such as the CAR.

3.4.2 The metrical calculus and the primacy of the focus indicator.

We now have to ask whether these descriptive differences are reflected in the rules. Do the same rules generate metrical prominence and focal emphasis? How do the rules apply differently if focus is present? These are the question which I wish to address in this section.

Let us begin with the metrical calculus. As has been mentioned in section 1.2.4, I assume that the metrical calculus assigns prominence completely according to rules, and that there are at least three discreet levels of prominence assignment whose rules can vary. We are only interested in the highest, the clausal level. Here, as has been mentioned repeatedly (e.g. section 1.2.4) the basic rule for grid construction is the Iamb Construction Rule, a metrical version of the Nuclear Stress Rule, repeated below.

Iamb Construction Rule:
Assign iambs from right to left.

At this point, it might be interesting to try to fit in the Clash Avoidance Requirement. Up to now, the Clash Avoidance Requirement (and the related notion of Eurhythmy) has been used only implicitly as a well-fomedness condition to which grids should adhere. It is now time to show how the Clash Avoidance Requirement contributes to grid-construction itself, that is: how it is translated into grid-construction rules.

The first question we have to ask is what the status of the CAR is. Is it a kind of repair mechanism, as the Rhythm Rule in Liberman & Prince (1977) and Prince (1983), or is it something more fundamental? The discussion in section 3.2 suggested that the

CAR is rather comparable to other ‘principles’ such as the OCP. But what is the OCP? One could look at it as an abstraction of how suprasegmental structure in general looks like. This means that the OCP directly conditions the formation of its objects, it does not just repair things. Structures do not get built somehow and are then made compatible to the OCP in a second step, but they are built from the very beginning in such a way that the OCP is observed. Consequently we should think that the CAR is also an integral part of the relevant rules of grid production and needs to be included into the rule itself.

So let us concentrate on how the CAR can be included in the nuclear stress rule, the Iamb Construction Rule.[42] This rule, as it stands, is not yet an implementation of the CAR. Note that an iamb can be binary (branching) or unary (non-branching) in Metrical Stress Theory (3-45).

(3-45) Iamb: (. *) or (*) (Hayes 1995:65)

Since non-braching feet are allowed in principle, this rule could theoretically produce a clause grid as in (3-46) which however does not conform to the CAR.

(3-46)	(.	*)	(*)	(.	*)	
	*	*	*	*	*	C
	*	*	*	*	*	P

Such grids are uncommon, for it is not clear what the motivations for a non-branching foot in the middle of the clause should be, excluding the possibility of focus (which is, of

[42] We are not interested in rules for prominence asssignment on the word or the phrase level, although for prominence assignment on these levels the same condition holds, viz. that the CAR has to be part of the parsing rules and of the foot construction rules themselves.

course, just the environment in which weird parses can arise, but see further below). We may assume that the normal way of affairs would be that binary feet are assigned until the level is exhausted, with a non-branching foot as last (= leftmost) element, if needed. A normal parse is given in (3-47). Beneath the grid in (3-47) I put a random example, the square brackets indicating syntactic constituents and pseudo-constituents (such as verbal forms).

(3-47)

```
(.                             *)
(*)    (        .              *)
(*)    (.       *)     (.      *)
 *      *       *       *      *      C
---------------------------------------
 *      *       *       *      *      P
 J.     m.      r.      s.b.   f.h.g.
```

[John] [madly] [requires] [some beans] [for his girl-friend]

If only binary feet are assigned, the structure conforms nicely to the CAR. The fact that at the end (the leftmost phrase) a unary foot is assigned is unproblematic, as it is adjacent only to the weak part of the binary foot to the right of it. So we can keep the following as a metrical version of the CAR:

> *Clash Avoidance Requiorement (metrical form):*
> Only binary feet are assigned, except at the end of the parse

This is essentially the Priority Clause of Hayes (1995:95) which is concerned with degenerate feet rather than unary feet; if we ban unary feet as ill-formed, however, most of the conditions which Hayes describes for degenerate feet hold for unary feet as well. This statement means the following: In principle, the assignment of unary feet is banned. The only place where they are allowed is at the end of the assignment domain, that is, on the left edge of the clause. The reason why they are allowed there is that otherwise no foot could be constructed (exhaustivity), and because they do no harm in this position, as mentioned above. We may assume that the assignment process strives for completion, i.e. that it is more important to have the whole string parsed than to avoid unary feet at all costs. The assumption that structures have to be exhaustively parsed on all levels is widely agreed upon (e.g. for the syllable level Steriade 1982; Harris 1983; Ito 1986; Hayes 1995:109; for higher levels Ito 1989; Mester 1994; Hayes 1995:149).

It is clear that the CAR is a priniciple independent from the foot construction rules. Any foot construction rule is however subject to it. It is reasonable to assume that the foot construction rules, like our Iamb Construction Rule, can only operate within the boundaries set by the CAR. To make the exposition easier and to provide a reminder that the Iamb Construction Rule is constrained by the CAR, we could include the CAR in our Iamb Construction Rule, which hitherto we will call Iamb Construction Rule *cum* CAR. It is not meant to indicate that the CAR is part of this special rule but to remind us that the CAR is a presupposition for the operation of the Iamb Construction Rule like all foot construction rules.

Iamb Construction Rule cum CAR:
§1: Assign iambs from right to left.
§2: Only binary feet are assigned, except at the end of the parse.

The avoidance of clash is a product of this constraint on foot minimality: If unary feet are banned, no clashes can possibly arise, as a clash would always presuppose that one of two adjacent feet is unary, more precisely: the foot which is adjacent to the strong side of the binary foot.

Having described the rule relevant for the metrical calculus on the clausal level, let us turn to focus. We will have to see later how the Iamb Construction Rule deals with focus indicators, but first we must think of how they are represented. This is the problem to be tackled next.

Somehow it must be verified that focus is realized faithfully, that is: that the mapping of syntactic focus feature and focal emphasis is correct. This leads directly to a technical question: How does the assignment of focal emphasis actually work? In the end it will be represented in the same grid in which the metrical prominences are represented, where it must be made certain that the highest prominence is on the focalized element. This leads to an obvious question: what is realized first in the grid, the normal metrical prominence or the focal emphasis, and how do they interact?

I assume primacy of focus (as does Selkirk 1984, though for different reasons).[43] There are several good reasons for this. One reason is that the focus feature has been present throughout the narrow-syntactic derivation, whereas the metrical prominence prominences are a purely PF-internal matter. So the focus feature is already present when the string comes to PF and the scanning process begins; as it is relevant for the metrical scanning it would make sense conceptually if the scanning process took this as a starting point.

[43] The other possibility would be that the stress-assigning mechanism does its work, not taking focus into account, and that only in a secondary process the structure is repaired in order to conform to the focal structure. The advantages of this version to the other described in the text are not immediately clear. It is not important at all whether focus comes first and the whole grid is constructed around the focal emphasis, or whether the grid is first constructed, then the focus is superimposed, then the grid is repaired.

Another reason is that contours as a whole are realized differently when there is a focus in the clause. Let us take a clause like (3-48) as an example. In (3-48a), we have the normal contour that would arise if the sentence is uttered with neutral intonation. The highest clausal prominence peak lies on the object; by Eurhythmy (see Hayes 1984) the subject is more prominent than the verb. In structuralist and early generative notation (e.g. Chomsky & Halle 1968), we would describe the contour as 2 3 1.

In (3-48b), focal emphasis is on the verb *bought*. The subject loses its relative prominence and sounds more as if it had no additional prominence at all. Likewise the object is by no means as emphasised as it would be under neutral intonation; it is even doubtful whether it is more prominent than the subject at all. So we have a contour like 3 1 3, or even 3 0 3, as the prominence associated with the focal emphasis is higher than the sentence stress would be.[44]

```
(3-48) a.         .        *
                  *    .   *
               .  *    *   *    .
             The kid bought ice-cream

       b.              *
                  .    *   .
               .  *    *   *    .
             The kid bought ice-cream (and didn't pinch it)
```

These points indicate that a phonological focus indicator must be present from the beginning, even before the metrical prominence assignment starts. But how would the assignment of metrical prominence with a pre-existing focus indicator look like from a technical point of view?

[44] I leave out the foot structure in these grids as we will later have to see how foot construction interacts with focus.

Let me propose a way how it could work. First the clause is scanned for elements with a focus-feature. These elements get a 'credit'-mark: a strong mark that automatically projects another strong mark to the next level.[45] This 'credit-mark' is the focus indicator. Afterwards, the remaining grid construction goes as usual, that is: it tries to build feet and at the same time keeps the grid eurhythmic, until the highest line of the highest level is reached on which a foot can be constructed. Note that the credit mark keeps being projected to the next higher line. This means that the credit mark that has been assigned in one parse of a line l_1 to the next higher line l_2 will be a pre-specified strong mark in the parse of the next higher line l_2. This strong mark projects a credit mark to the next higher line l_3, which will then be treated as a prespecified strong mark when line l_3 is parsed and so on. Note that the strong mark on the highest parsed level will still have the credit mark above it, so that the grid in the end will have an extra-high peak. This needs not bother us; on the contrary; from this it would follow straightforwardly why focal emphasis is always stronger than a highest peak derived by metrical prominence: because it possesses an extra-high peak, which a normal metrical prominence peak does not.

By means of the credit mark it is guaranteed that the focalized element ends up as the strongest element. If focus is assigned to a unit bigger than a syllable, the exact position of the credit-mark remains open until the metrical calculus on the word level has done its work and assigned the peak of that unit to which the credit-mark is added.

Let me demonstrate such a derivation with an example. In (3-49), the grid for a clause is derived without a focus-feature. Let us enter the derivation at a point when phrasal prominence has already been assigned. The lowest grid line is consequently the

[45] Similar 'perspecified metrical accents' have been proposed earlier. See Halle & Vergnaud 1995:415ff. and references. Another similar proposal is the Stress Equalization Convention in Halle & Vergnaud 1987.

first line of the clause level, which is simply a copy of the highest line of the phrase level. The assignment on the clause level follows the Iamb Construction Rule *cum* CAR. In (3-50), we see the same process, but this time with a focus on *ice-cream*. In this and other similar sample derivations, the credit-mark is represented by a bold-faced *x*. Note that the construction does not change, apart from the fact that an extra grid line is added at all stages of the derivation. In the end the two clauses are similar, but distinct in the number of lines the main peak stands out above: without focus, the peak is only one line higher than the next-highest peak, with focus, it is two lines higher. This element is thus phonologically more prominent than it would be without the extra-mark, and this translates into a phonetic distinction: the prominence on *ice* relative to the next-highest peak is higher in the focalized version ('This little guy has bought a lot of [$_f$ **ice**]-cream (but no lettuce, as his mother has told him)') than in the non-focalized version.

```
(3-49)  a.                  *         *             *         C
                This little guy has bought a lot of ice–cream

        b.                ( * )    ( .              *  )
                            *         *             *         C
                This little guy has bought a lot of ice–cream

        c.                ( .                       * )
                          ( * )    ( .              * )
                            *         *             *         C
                This little guy has bought a lot of ice–cream

(3-50)  a.                                             x
                            *         *                *      C
                This little guy has bought a lot of [f ice]–cream

        b.                                             x
                          ( * )    ( .                 *  )
                            *         *                *      C
                This little guy has bought a lot of [f ice]–cream
```

```
c.                                       x
            ( .                          * )
            ( * )    ( .                 * )
              *        *                 *     C
This little guy has bought a lot of [f ice]–cream
```

If the focus is on an element different from the one that ends up with sentence stress anyway, the whole grid construction is different. In (3-51) the focus is on *bought* (as in the similar (3-48b)). Consequently, the metrical prominence at the right edge of the clause can arise relatively undisturbed (if we assume iamb assignment from right to left, the parser will try to put a strong mark on the rightmost column, unless it produces a clash as it would in (3-52b). This could go on infinitely, as there will always be the credit mark just at the spot where the parser would like to put a weak mark; let us assume that the parser notices early that further assignment would be redundant and bails out on the next line. (3-51c) would thus represent the highest assignment. The metrical prominence on *guy*, on the other hand, cannot develop freely, as it is too close to the focus, similarly for the metrical prominence on *ice-cream*. As the Clash Avoidance Requirement in its metrical form is a ban on unary feet, we might assume that in such a case the (theoretically possible) unary iamb on *recently* in the highest line of (3-51c) is not realized properly but either treated as extrametrical, or assigned a degenerate foot (which at least observes the alternating principle). The outcome either way would be that the prominence on *recently* would be significantly less high than on the focalized *bought*.

```
(3-51) a.                 x
              *           *          *        *       C
This little guy has [f bought] a lot of ice–cream recently
```

b. **x**

(. *) (. *)

* * * * C

This little guy has [$_f$ bought] a lot of ice–cream recently

c. **x**

(*) (.)

(. *) (. *)

* * * * C

This little guy has [$_f$ bought] a lot of ice–cream recently

If there is more than one focus in the clause (this was the case which we were originally interested in), the same happens, only that it is not one but two or more credit marks that have to be assigned (3-52). On the highest level it looks like a clash, but there is enough material in between (the weak mark on the level directly below). The assignment soon becomes vacuous, as it is not possible to assign less than two feet, because of the two credit marks.

Multiple foci can also lead to adjacency of the credit marks, as we have seen, and here the mechanism of pause insertion jumps in, as we have seen earlier (3.1.2) and will see more in detail (3.5). In the end, the relevant line is not the one bearing the extra credit marks, but the line below, as this line is the last line that has been parsed by the metrical calculus and on which feet are assigned; as the CAR refers to foot structure in the form in which it is relevant here (as the Iamb Construction Rule *cum* CAR), it can operate only on lines parsed in feet. A sample derivation of clashing foci is shown in (3-53). Note that the assignment process would select *bought* as the strong part of the first iamb, but since the next possible assignee is already strong by virtue of the credit mark, the same happens as in (3-51). In going on, the parser cannot assign a weak mark to the right of *Rich* because of the second credit mark. The only possibility in accordance with the Clash Avoidance Requirement is to assign a weak mark nevertheless which does not correspond to segmental material – a pause (3-53d).

(3-52) a.

```
              x                         x
              *        *                *
```

This little [$_f$ guy] has bought a lot of [$_f$ ice] –cream

b.

```
              x                         x
             (* )    ( .                * )
              *        *                *
```

This little [$_f$ guy] has bought a lot of [$_f$ ice] –cream

(3-53) a.

```
 x              x
 *              *      *
```

[$_f$ Ice]-cream [$_f$ Rich] bought.

b.

```
 x              x
 *      … ←     *
 *              *      *
```

[$_f$ Ice]-cream [$_f$ Rich] bought.

c.

```
  x               x
 (* )      ( .    * )  (.)
  *               *     *
```

[$_f$ Ice]-cream _ [$_f$ Rich] bought.

We have seen situations in which the focus leads to prominence loss on other syllables. We assumed the assignment of a degenerate foot on a column if the right-adjacent column is headed by a credit mark. Let us look at a different, more complicated example, such as (3-54).

We have a focus on *madly.* The grid construction would be bound by the presupposed presence of a strong grid mark on a spot where the Iamb Construction Rule would not put strong prominence under normal circumstances. But the metrical calculus would produce a grid that is not what we would expect (3-54); there would be a unary iamb on *requires*, thus rendering it stronger than *some beans*. If one observes how one speaks this sentence, however, one realizes that *requires* in fact is produced with a striking lack of prominence, thus rather as in (3-55).

(3-54)

```
            x
           (*)     (.             *)
 (.         *)     (*)     (.     *)
  *         *       *       *     *      C
 ----------------------------------------
  *         *       *       *     *      P
  J.        m.      r.      s.b.  f.h.g.
```

[John] [**madly**] [requires] [some beans] [for his girl-friend]

(3-55)

```
            x
 (.         *)     (.      .      *)
  *         *       *      *      *      C
 ----------------------------------------
  *         *       *      *      *      P
  J.        m.      r.     s.b.   f.h.g.
```

[John] [**madly**] [requires] [some beans] [for his girl-friend]

The grid in (3-54) strikes one as immediately ill-formed if one has the CAR in mind. The second-lowest line of the C level has two strong marks in a row, what looks like a blatant violation of the CAR. The violation is somewhat remedied on higher lines, but it would be doubtful that the CAR would allow such an obviously ill-formed line anywhere, especially in the form in which it is implemented here: The Iamb Construction Rule *cum* CAR could never produce such a line. We saw earlier (when discussing topicalization with non-focused full-noun-phrase subjects) that the CAR is sensitive also to clashes on lower levels. The grid in (3-55), on the other hand, looks fine: The clash has been avoided by destressing (here: assigning a weak mark instead of the regular strong mark on requires, which gets incorporated into the iamb to the right of it, thus creating an anapest (compare Hayes 1995:97 and references to the dactylic and anapestic effect). As degenerate feet are generally regarded as problematic (see e.g. Hayes 1995:87), the extension of an iamb to an anapest is probably preferred over the option to construct an iamb plus a degegnerate foot to the right. Note that in (3-55) no non-branching foot is in the parse.

Another issue is that we should get vacuous assignment of non-branching feet on the leftmost element: On the level which is the highest level in (3-54), the assignment of a binary iamb is impossible since the focus feature has its credit mark at the spot where with normal assignment would be the weak part of the iamb. The Iamb Construction Rule would have to generate a non-branching iamb on *for his girl-friend*, which would produce a new level by the credit-mark process, on which a further non-branching iamb would have to be generated for *for his girl-friend* etc. We can prevent this by a general ban on vacuous assignment, which is needed anyway and which has been introduced earlier in this section.

If destressing is not available as an option, e.g. because two credit marks are too close to each other, a pause is inserted, as we know from earlier sections. This means basically that a unary foot is turned into a branching foot. (3-56) shows a focus clash case whose grid is ill-formed; (3-57) shows a focus clash with pause insertion whose grid is well-formed.

(3-56)

x	**x**				
(*)	(*)	(.	.	*)	
*	*	*	*	*	C
*	*	*	*	*	P
J.	**m**.	r.	s.b.	f.h.g.	

[**John**] [**madly**] [requires] [some beans] [for his girl-friend]

(and Bill only half-heartedly)

(3-57)

x		**x**				
(*)	(.	*)	(.	.	*)	
*	.	*	*	*	*	C
*	.	*	*	*	*	P
J.	_	**m**.	r.	s.b.	f.h.g.	

[**John**] _ [**madly**] [requires] [some beans] [for his girl-friend]
(and Bill only half-heartedly)

So we can say that the statement which was tentatively introduced above holds also in cases where prominence is assigned not only metrically, but also by focus. Therefore, we can include the statement given above as a general statement into the relevant rule, the *Iamb Construction Rule cum CAR*, repeated below, and be sure that the rule applies to all grids, with or without focus indicator.

Iamb Construction Rule cum CAR:
§1: Assign iambs from right to left.
§2: Only binary feet are assigned, except at the end of the parse.

3.4.3 The nucleus as a continuation of the metrical prominence system

What about the nucleus then? The nucleus is the strongest prominence in a clause that does not have a narrow focus. In English it is usually on the rightmost constituent, in German usually on the rightmost constituent of the *mittelfeld*. Since the nucleus does not have a semantic value, it clearly is not on a par with focal emphasis in the sense of Rooth (1985), as we have defined it. In the following I will show that the phonological correlate of the nucleus or sentence stress is not a focus indicator, but rather the highest metrical prominence peak on the clausal level. This is pretty obvious, as the assignment of the nucleus is certainly rule-governed, as already Newman (1946:176) saw, who formulated the Nuclear Stress Rule for phrases/clauses as such:

> When no expressive accents (= focal emphases in my terminology, A.S.) disturb a sequence of heavy stresses (= metrical prominence peaks in my terminology, A.S.), the last heavy stress in an intonational unit takes the nuclear heavy stress.

I follow him in thinking that sentence stress is assigned when no element in the sentence is marked with a focus feature. This is in conflict with the usual view (e.g. Selkirk 1984) that all sentences have focus which can be either narrow or wide. But let us stop for a moment to think what wide focus under the Roothean definition of focus means. If I have a clause with wide focus, e.g. ‘Charlie plays baseball with his friends on the lawn’ (as an answer to a question triggering wide focus, e.g. ‘What’s going on?’), and the whole sentence is in focus, the set of alternatives would be the set of possible propositions, which is infinite. We would actually find infinite sets for all ‘wide foci’ (e.g. on the verb phrase) and would get sets of the kind demanded by focus only if we are on the word level, but not on any of the higher levels. We may assume the sets that are relevant for focus to be finite, especially because they contain only contextual relevant alternatives, namely the limited numbers of entities in the discourse universe which furthermore have this or that property that makes them eligible for the set. Thus wide focus and narrow focus are rather distinct: With narrow focus, set construction is possible, with wide focus it is not, as the set soon becomes infinite. This is due to the fact that predicates do not need to be anchored in the discourse universe to such an extent as entities do, as the latter usually have to be introduced by some kind of reference. The choice of predicates, on the other hand, is principally unlimited. We may multiply this with the possible combinations of entities in the discourse universe (as arguments of the predicates). It is clear that we soon reach a very large number, even if we would assume that the set of potential predicates is finite. If we have to construe an infinite or at least very large set, we may well ask whether constructing such a set is not vacuous and in contradiction to the original idea of focus. If this is the case, and if we believe that the semantic operation associated with the focus feature is set construction we are forced to conclude that the so-called 'wide focus' is not associated with a focus feature.

Sentence stress is generated the usual way metrical prominence is generated, by adding more lines to the clausal level of the grid and parsing them for the appropriate feet until there is nothing left to parse. It is rule-governed and semantically insensitive, which would be difficult to explain if we took the line that the nucleus is associated with a focus feature. Let me demonstrate the semantic insensitivity with (3-58).

(3-58) Yesterday, little John has bought a big dish of ÍCE-cream.

=|=> there are other things besides ice-cream which John or somebody else has bought or could have bought.

(3-58) is nothing more than an assertion, without implying anything. There is the nucleus on 'ice', however. Note that it does not imply that there are other things which John has or could have bought. Moreover, it is entirely rule generated: In domains higher than the word the parsing is from right to left and it is such that it starts with a strong mark (=

roughly 'iambic').[46] By this rule, the highest mark will necessarily be at the rightmost stressable part.

The rules for metrical prominence assignment on the clause level are quite similar to the rules of assignment of metrical prominence on the phrase level in English and German, which indicates that the same rule that operates on the phrase level simply goes on parsing on the clausal level and eventually produces the highest prominence peak on the clause level. The rule for English is given below, in a more explicite form than in preceding sections in that it gives clear instructions as to how the parse works, and explicitly includes the possibility of applying to more than one level. As I said earlier, it is basically the Nuclear Stress Rule of Newman (1946), known from virtually all studies on stress higher than the word level (Chomsky & Halle 1968; Liberman & Prince 1977 etc.; critical Ladd 1980), in the guise of a Foot Construction Rule such as Hayes (1995) formulates it (hitherto referred to as Iamb Construction Rule):

> *Iamb Construction Rule:*
> Assign iambs from right to left until the domain is exhausted, then go on to the next-higher domain and repeat assignment.

In German, a different rule has to be formulated, and it is not easy to formulate the rule for the nucleus. It has, however, been recognized early on, that it is entirely rule-governed (Kiparsky 1966:79). Kiparsky formulates rules that assign the nucleus according to phrase category and he is forced to distinguish between 'Satz' (= CP) which has final prominence, and 'S' (=VP in clauses with complex verb form). This accounts for the descriptively correct fact that the nucleus in German is normally somewhere to the right,

[46] This is old news; the rule has to be reformulated though for languages like German, in which the nucleus assignment mechanism 'skips' verbal material at the right edge of the sentence.

on the last constituent if all verbal material is in C° (3-59a), or on the constituent immediately preceding the verbal material, if there is some verbal material stranded at the right edge (3-59b). This means that we can formulate a rule which is roughly similar to the English one, but with a proviso regarding verb forms. Either the whole verbal complex is extrametrical, or verb forms are intrinsically weak, or they escape high prominence assignment because they are not deeply embedded (see Cinque 1993). The rule, at any rate, would run as follows: Parse right to left (this goes for all units higher than the word) and don't include the verbal complex in the assignment. In (3-59c,d) it is treated as if it were extrametrical; this is a purely notational choice.[47]

(3-59) a. Gestern fuhren die Stadlers mit dem Auto nach MANNHEIM
yesterday drove the S. with the car to Mannheim
'The Stadlers went yesterday to Mannheim by car.'

b. Gestern haben die Stadlers mit dem Auto nach MANNHEIM fahren
yesterday have the S. with the car to Mannheim drive
wollen.
want
'The Stadlers wanted to go to Mannheim yesterday by car.'

[47] We could get the same result by the rules that Cinque (1993) formulates. In Cinque's (1993) system, prominence is attracted by deeply embedded elements. The nucleus then goes on the most deeply embedded phrase. This will be the rightmost non-verbal phrase. Note that I am not necessarily arguing against Cinque if I choose a different representation.

```
c.      ( .                                                    *)          ← ia
        ( *)    (             .                                *)          ← ia
        ( * )   (   .           * )       (      .             *)          ← ia
          *       *             *                *             *
        Gestern fuhren die Stadlers mit dem Auto nach Mannheim
```

```
d.
        (  .                                                   *)          < >← ia
        (  * )  (              .                               *)          < >← ia
        (  * )  (  .            * )      (       .             *)          < >← ia
           *       *             *               *             *              *
        Gestern haben die Stadlers mit dem Auto nach Mannheim fahren wollen
```

The real question is: Do we need a focus feature in such wide-focus (that is: neutrally stressed) sentences? I do not think so, and I have already explained why I think wide focus cannot be associated with a focus feature. But there are more arguments in favour of this view. Selkirk (1984), assuming that all sentences do have a focus feature, takes great pains to restrict the projectability of focus features such that it covers the observed English fact of phrasal prominence. Her Phrasal Prominence Rule (Selkirk 1984:207) basically says that a constituent can be a focus if its head or an argument of it is a focus. Wide focus would then be a whole VP that is a focus.

I see a problem here. How should a focus – if we assume that focus is a semantically interpretable feature on words – project from an argument to the phrase as a whole? But under the idea of wide focus this is definitely what would be needed, because we need a correlation between the most embedded / rightmost standing element – on which the focus feature would be phonologically realized in the end – and the VP, of which this element is an argument, as it is the VP as a whole in the end that would be the focus in semantic terms. We have here a mismatch, as we would expect that the focus feature percolates only along the head-line. If the verb phrase as a whole is a focus, it should have a focus feature, and it should get its focus feature from the head. But the phonological correlate of the focus feature would not be on one of the elements (head or maximal phrase) that bear the focus feature, but on some argument. I find it difficult to account for this mismatch, to say the least.

Let me illustrate this with her own (wrong) example from German. Her short discussion of German unfortunately misses the point, as she reduces background-focus to an old/new-distinction and comes wrongly to the conclusion, that e.g. a sentence like (3-60a), her (5.47) in 1984:230, is the default prominence pattern for a sentence that is *in toto* informationally new. But the indicated prominence pattern is far from default; it automatically implies narrow focus on *betrachtet*. The normal prominence pattern would be as indicated in (3-60b).

(3-60) a. Peter BETRACHTET das Buch.

Peter watches *the book*

'Peter looks at/through the book'

b.
```
( .                   *)
( *)    (  .          *)
  *        *          *
Peter betrachtet das Buch
```

It would be difficult in principle to formulate a rule of focus in which highest prominence is automatically associated with a focus feature. This focus feature must be part of narrow syntax as it corresponds to an LF-operation in the sense of Rooth (1985). Therefore it must follow the rules that features follow, such as case features or the like. One of the basic principles of feature projection however is that there is a direct correspondence between the features of a phrase and the features of its head. This would mean that the focus feature in wide focus, corresponding to 'default accent,' must be associated somehow to the verb as head of the relevant phrase. But why would it almost never be realized on the verb, then, but always on one of its arguments?

Selkirk (1984), as we have mentioned, bypasses the question by her Phrasal Rule (1984:207) in which she allows arguments in the VP to bear focus and the VP to be a focus if one of its arguments is a focus. In the light of Cinque (1993) she modifies it later to the statement that it is the inner argument that can focus-mark the head (and thus the phrase; 1995:561).[48] This is an *a posteriori* observation, which she arrives at by assuming that a pitch accent creates a focus and not the other way round, as it is assumed in this study (where does the pitch accent come from to begin with?).

Going back to her extended study from 1984, she says later in that work that focus is something that is associated with new information (1984:213), and this proviso saves the day, as it is more often an argument rather than the verb itself that offers new information. But we see here that her concept of focus is somehow vague. A strict correlation between new information and focus is stipulated which can be easily shown to be wrong. Contrastive foci, for instance, are typically on old or inferable elements, never on new elements. With contrastive foci, members of the same set are compared, and this set at least is usually already evoked in the discourse. But even if this were not the case, it is necessarily the case that at least the second element in a contrastive setting must be

[48] This is not to say that a rule more or less resembling the rule battery Selkirk (1995:561) gives could not be the correct one for focus in the general sense. If we have focus on a whole phrase, like in *What did she buy? – She bought [a book on BATS]*, the assignment might follow such rules. In such cases, i.e. if the focus feature is on a unit bigger than the word, we could however also argue that the focus feature just does not get realized at all and thus default clausal stress is assigned. This is in line with Jackendoff (1972). Phonetically, the prominence on BATS in the above example is not different at all from the prominence BATS would have if the sentence was in wide focus, i.e. as an answer to something like 'what happened next?'. It is however very much different from a narrow focus on the word, i.e. as an answer to the question 'A book about what did she buy?'. In the following discussion it is this idea that is adopted.

inferable, as it must be of the same set from which the first element was taken, and by that the set is already present in the discourse universe. So it is quite clear that newness is not a condition on focus, neither a necessary nor a sufficient one.

But now one could say, it is unfair to scold Selkirk if she simply has a different conception of focus. Maybe her rules work under the assumption that they are not for focused elements in Rooth's sense of the word, but for new elements. But this is not true either. Contrastive foci attract the main sentence prominence, by virtue of their focus feature, as we have seen. We could now say: Well, we often have double foci, the first focus being on something contrastive, the second focus on the predicate, which is new information. But this does not work either. Consider (3-61).

(3-61) A: Tell me something about vegetables.

B: BEANS are very GOOD for you. PEAS are NOT so good for you. CABBAGE is again GOOD for you.

Here the predicate in the third sentence of B is obviously old information, as it is nothing but (x is good for you), with the value of x being the focused constituent. It nevertheless bears one of the two main prominence peaks of the sentence. We know that it does so because it is in a contrastive relationship to the other potential predicates, but it cannot attract focus because it is 'new'.

There are more things to say about Selkirk's Phrasal Rule, as it stands. For instance, it does not cover all English examples. The restriction on heads and arguments, for instance, would not cover a perfectly natural sentence such as (3-62), which is derived from her example (5.3) on (1984:208). Here the highest prominence falls on an adverb, that is, an adjunct. Note that it can serve as a felicitous answer to a question soliciting wide focus, as shown in the example.

(3-62) (question: And what happened next?)

She sneezed HORRIBLY.

Note that this must be metrical prominence, as there is no contrast intended between the adverb *horribly* and other ways regarding how one could sneeze. If we take a simple rule like the Nuclear Stress Rule mentioned above, which blindly assigns the highest prominence to the rightmost constituent, we should get exactly what we observe: insensitivity to the type of constituent that bears the highest prominence peak, as long as it is the rightmost one. Note that in the case of (3-62) the wide focus interpretation goes away if the highest prominence is on the verb. By putting prominence on *sneezed*, a listener understands the sentence as if the speaker emphasizes the act of sneezing in contrast to other things she could have done. This is not exactly wide focus; note that it is less felicitous as an answer to a question such as the one in (3-62), but it is a felicitous answer to a more precise question that focuses on the verbal action rather than the whole proposition (3-63).

(3-63) a. (question: And what happened next?)

She SNEEZED horribly.

b. (question: What did she do next?)

She SNEEZED horribly.

And even if we would count adjuncts, the problems are not entirely solved. It is also not entirely clear in her discussion what to do with embedded phrases. Take the examples (3-64a), taken from Halle & Vergnaud 1987:264. Why should the focus feature be realised on the most embedded element in the argument? Or consider example (3-64b). Why should it be realized on the second conjunct rather than the first conjunct?

(3-64) a. Jesus preached to the people of juDEa.

b. Boston is the home of the bean and the COD.

All these problems disappear if we assume the nucleus in 'wide focus' cases to be entirely rule generated, by something like the Iamb Construction Rule or Cinque's (1993) rules. These rules can be formulated phonologically with reference only to syntax. Thus this process of metrical prominence assigment seems thus to be a process that is insensitive to semantics. The conclusion one is justified to draw from this is that sentence stress is not associated with a semantic focus feature.

But there are other objections one could make against the association of sentence stress with a focus feature. Let me repeat the main objection from the beginning of this section in slightly different words. If a unit the size of a verb phrase or a clause is associated with the focus feature, focus is trivialized and loses its actual contrastive force. Of course, one could always construct a set of possible clauses and say: only one of these set of possible clauses is realized and therefore it is in contrast to the rest of the set (and this is what underlies the idea of wide focus in the end), but this is trivial; under this view, it is easy to argue that everything is in contrast to something else.

Focus in the stricter sense seems to be a property of single words or even smaller units (cf. Selkirk 1984:208; 269ff.; Drubig 2003). That is, the membership set is constrained in that it can consist only of entities or predicates but not predicates applied to entities or even larger objects (3-65).

(3-65) $M = \{e_1, e_2, \ldots e_n\}$, e.g. oranges, bananas, …

$M = \{P_1, P_2 \ldots P_n\}$, e.g. to peel, to cut, …

$*M = \{P_1(e_1), P_1(e_2), P_2(e_2) \ldots P_n(e_m)\}$, e.g. to peel bananas, to cut bananas, to cut oranges…

The reason for this is the following: If I have a large object like P(e) in focus, it is not entirely clear where the exponent of focus, which would be some sort of prominence, should be realized. It should be realized on the head, the verb, but this is obviously not the case (3-66).

(3-66) (question: what did he do?)

a. *John CUT an orange and he PEELED a banana.

b. John cut an ORANGE and he peeled a BANANA.

One could rescue the idea of association of sentence stress with a focus feature and add a proviso that takes care that in a structure marked with a focus feature that contains a verb and an object, it is the object on which the prominence is realized. But this cannot be right. If we alter the word order slightly, for instance by embedding the predicates under ‘to have s.th. done’, we get the stress on the verb (3-67a). Note that, although the predicate is now structurally realized as a small clause, it is the same predicate all the same. And if we add an adverb, the sentence stress moves to the adverb (3-67b).

(3-67) a. John had an orange CUT and a banana PEELED.

b. John cut an orange for a MINUTE and he peeled a banana on TOP of that.

If we say that such wide-focus structures do not bear a focus feature at all, because it is contentless as soon as it is applied to units larger than a primitive (i.e. predicate or entity), the problem goes away; the sentence stress is not the realization of any focus feature and for want of a focus feature the different layers of higher prominence have to be rule-generated. The rule, scanning from right to left and assigning iambs, repeating the process until the whole string is scanned (in the schemata indicated by '←ia'), will select the rightmost constituent automatically as the one with the highest prominence, regardless of the constituent's type (3-68).

(3-68)

```
a.   ( .     *)
     she sneezed

b.   (.      *)           ←ia
      *      *
     Sue sneezed.

c.    (.           *)     ←ia
      (*)  ( .     *)     ←ia
       *     *     *
     Sue sneezed horribly

d.    (.           *)     ←ia
      (*)  ( .     *)     ←ia
       *     *     *
     John cut an orange.
```

```
e.   ( .                     *)      ←ia
     ( *     )(       .      *)      ←ia
       *     .        *      *
     John had an orange cut.
```

This rule is of course a bit oversimplified; we have to assume adjustment rules that make the grid more eurhythmic (in the sense of Hayes 1984) in order to generate for instance the metrical prominence pattern of (3-67b). It would lead us too far afield if we delved into the rules for phrasal metrical prominence assignment in detail (for an overview cf. Kager 1995); for the moment it suffices to say that the first (=rightmost) assignment of a strong mark resists all secondary readjustment processes and therefore the sentence stress is on the rightmost constituent. So the rule as it stands generates the highest prominence peak, which consequently is associated with the highest stress in a clause.

Thus we can formulate rules for the placement of sentence stress, and these rules make reference only to the relative position of constituents and not to the type of constituent. If sentence stress were generated by a focus feature, we would expect some sensibility to phrase types, which the observable rule obviously lacks.

If we get rid of sentence stress as a potential realization of a focus feature, the projection facts of the focus feature are much more straightforward. If focus is a feature on words, it is at the same time a feature on heads, as all words ultimately are a head of some phrase. The accent as realization of focus is on the head of the relevant phrase, and it might project up to the nearest phrase node or not; as the diagnostic for focus is the accent on the head word, the projection of the feature to its phrase node would have no visible effect.[49] Let me illustrate this with an example. Take a sentence like (3-69), for instance.

(3-69) Sue meets Lord Emsworth's nephew in a shady restaurant.

If the focus feature was on an adjunct like *shady* (promoting a reading of the sentence that it was not a nice or trendy or whatever restaurant where they met, but a shady one), it would project to the next phrase boundary, i.e. the AP whose head (and sole occupant) *shady* is. If it was on the NP *restaurant*, it would project to the next phrase node, i.e. the NP phrase node. Then the AP *shady* would be in the scope of the focus feature, but since the focus feature is realized on the head, with prominence as 'exponent', it has no consequences regardless of what else is in the scope of the focus feature.[50]

There is one fact that needs to be examined when speaking about focus and the nucleus. Often it looks as if sentence stress and focus coincide: Often the element bearing

---

[49] The reason for this is probably that the scanning mechanism gets the string as input, not the hierarchical structure. In the string the phrasal nodes as such are not present any more, and therefore any relevant features are associated with the respective relevant terminals. Likewise the focus feature: It is foremost associated to a word, but the information that it projects up to the nearest phrase is simply lost, as the projection is trivial, i.e. it is not the case that by the projection other processes (agreement, concord or the like) are triggered, as it would be at the projection of e.g. case features.

[50] A potential diagnostic at least for German that comes to mind are focus particles like *nur*, *gerade* etc. But they cannot be used as a diagnostic as they follow a stronger constraint, viz. that they can attach only to immediate constituents (Altmann 1976:1). Therefore it looks as if the PP *in einem zwielichten Restaurant* in (i) would be always in the scope of the focus particle, but it can easily be seen that the contrastive elements within the PP are variable (i,b-d). So it follows that not everything that is in scope of a focus particle really is in focus; the focus particle can only attach to the immediate constituent that contains the focalized word/phrase.

(i) a. So eine Unterredung kann man nur in einem zwielichtigen Restaurant führen
such a discussion can one only in a shady restaurant lead
'such a discussion can only be held in a shady restaurant'
b. So eine Unterredung kann man nur in einem zwielichtigen RESTAURANT führen
(focus on NP; in contrast to other potential meeting places)
c. So eine Unterredung kann man nur in einem ZWIELICHTIGEN Restaurant führen
(focus on AP; in contrast to other properties restaurants can have)
d. So eine Unterredung kann man nur IN einem zwielichtigen Restaurant führen
(focus on PP; in contrast to other positions relative to the restaurant)

sentence stress is at the same time associated with focus, and one might wonder whether the word order is accommodated such as to bring the focalized element into the position in which it would also receive sentence stress. It is probably this observation that lies at the bottom of the idea that the nucleus is associated with focus in general. A German example would be (3-70). The sentence (3-70a) is with contrastive focus on *gestern* ('yesterday'); the word order has been scrambled to bring this word to the end of the mittelfeld. The unscrambled version would be (3-70b).

(3-70) a. …weil ihnen Peter den Kühlschrank GESTERN gebracht hat (und nicht
because them Peter the fridge yesterday brought has and not
HEUTE)
today
'because Peter brought them the fridge yesterday (and not today)'

vs. unmarked
b. …weil ihnen Peter gestern den Kühlschrank gebracht hat
because them Peter yesterday the fridge brought has

In my opinion, this evidence is not conclusive at all. This is not the place to go into this in detail, but obviously the relationship between the nucleus and focus positions is a topic worth pursuing, whether a relationship really exists (which I am not sure about) and second, if so, how it could be explained.[51] First it should be noted that focal emphasis and

[51] Rolf Noyer (p.c.) indicated some interesting problems connected with this question; I will be happy to pursue them in another context.

movement of an element into a position favoured for focus do not entail each other. Note that the sentence with focal emphasis but no scrambling is grammatical as well (3-71).

(3-71) …weil ihnen Peter GESTERN den Kühlschrank gebracht hat (und nicht HEUTE)

and the counterpart with scrambling but without focal emphasis is awkward but acceptable (3-72).

(3-72) …weil ihnen Peter den Kühlschrank gestern gebracht hat

Second, the phonetic realization of the nucleus due to highest metrical prominence and the realization of the nucleus due to association with a focus indicator are quite different. This is obvious: If one reads for instance the sentences (3-70a) and (3-72) aloud, the prominence on *gestern* in (3-72) is phonetically quite distinct from the prominence in (3-70a). Especially the pitch is much higher in (3-70a). See the next section for a more detailed account.

Thirdly, the sentence-initial position is a preferred focus-position, too, but here it does not coincide with the nucleus position. Steube (2003) sees the *vorfeld* as the default position for I-topics, that is, the first focused element in a double focus construction; Speyer (2004; 2006a) shows that contrastive elements are preferred elements for *vorfeld*-filling in German. And the English constructions double focus topicalization and focus movement are also constructions in which a focalized element is moved away from the sentence stress position. Thus the correlation between focus and the sentence stress position is not obligatory.

But still often it seems that there is an apparent correlation in the position of sentence stress and the preferred position for focalized elements. Why is that?[52] There are two possibilities that need not exclude each other. First: The focus positions are at the edges of the clause for processing reasons, as it is easier, if one is to divide the sentence into focus and background parts, to do only one cut and have one focus part and one background part (3-73). This principle of 'domain constancy' seems to play a role in German in general; Musan (2002) was able to show for instance that given elements tend to be moved out of a newness-domain in order to preserve domain constancy.

(3-73) [$_{\text{background}}$ Weil: Peter brachte ihnen den Kühlschrank] ✂ [$_{\text{focus}}$ GESTERN]

For German this is of course problematic as in most circumstances one will end up with two background parts anyway, due to the fact that the focus (and sentence stress) position is before the right sentence bracket (3-74).

(3-74) [$_{\text{background}}$ weil Peter ihnen den Kühlschrank] ✂ [$_{\text{focus}}$ GESTERN] ✂
[$_{\text{background}}$ gebracht hat]

The second possibility would be that the grammar as a whole is more 'optional' than we are wont to think. Recent research on German word order (e.g. Müller 1999, Frey 2006) points clearly in that direction. We probably are forced to conclude that the syntactic module generates not one output, but several alternative candidates which are evaluated

[52] I assume for the moment that this is the case in all pitch accent languages; it would be interesting to see if this is actually true. It certainly is true for German and English.

in some way; and when we look at what is actually performed, what we get is not simply the set of all grammatical sentences but a set of grammatical sentences with a strong bias in favor of one or the other option, which happens to be chosen most often. An Optimality Theoretic approach is probably the most suitable for this kind of question, but, as I mentioned before, I would like to leave that for further research. I only want to point out that it is the same optionality that a Middle English speaker had in choosing between V2 and V3; here we saw that the choice was done clearly in accordance with a prosodic requirement (and by that had nothing to do with degrees of grammaticality per se); likewise, candidates in which the nucleus and the position of a focalized element coincide might be preferred by some constraint, which probably is also phonological in nature (perhaps a constraint to have only one high prominence peak in the utterance?) Obviously this is a wide field that deserves systematic investigation, which I leave for future work.

3.4.4 Metrical prominence, focus and the intonational system

We have seen that focal emphasis and rule-governed metrical prominence are both different aspects of prominence. One similarity holds for both, namely the desire to alternate prominent and non-prominent elements in the output. For metrical prominence on the word- and intra-phrasal level, this has been demonstrated e.g. by Liberman and Prince (1977) and Hayes (1984). For the clause domain it has been demonstrated earlier in this thesis (chapter 3.1). For the metrical prominence system the same considerations hold that underlie the Clash Avoidance Requirement: Clash is avoided best and eurhthmy is observed best if the system is set up in such a way that prominence increases toward the edges (see Hayes 1984). Therefore the highest prominence assigned by this system will be near one of the edges.

If metrical prominence and focal emphasis are quite different in descriptive terms, as we have seen earlier, it is perhaps surprising to see this core similarity. How can it be that metrical prominence assignment and focus placement are more or less independent of each other, but yet apply in a similar, if not identical fashion?

The reason for this is simply that both feed into the same phonological sub-system and are therefore realized by the same expressional means, namely prominence (see Steube 2003:174f.). Prominence is, acoustically speaking, a complex mixture of pitch, volume, vowel quality and duration (Schane 1979; Sluijter 1995); we perceive elements as emphasized that are slightly higher pitched than we would expect, slightly louder than we would expect, and slightly longer than we would expect. There are individual mixtures of these ingredients, and sometimes one or another ingredient can be missing (cf. e.g. Sluijter 1995). The important thing is that this basically holds for all prominence and not only for focal emphasis (the so-called ‘pitch accent’). It might be that with focus the pitch part is a little more emphasized than with ordinary metrical prominence, but this can be due to the fact that focus in general generates a higher prominence than any metrical prominence could do. In section 3.4.2 a possible way to account for the focus indicator was introduced; a nice side-effect of this account is that it leaves the focalized element with ‘extra-high’ prominence, which might be interpreted as the observable higher pitch in focal emphasis which is higher than a metrical prominence peak.

Let me illustrate this with a little data. I had a few German speakers say three sentences, containing the word *Kühlschrank* ‘refrigerator’. The sentences are given in (3-75); the focal emphases were indicated by capitals as in (3-75). Note that the word

Kühlschrank is not emphasized in the first sentence (that is, it has its normal word stress, but nothing else), bears the nucleus in the second sentence and is in contrastive focus in the third. In Table 3.19 several parameters – highest pitch, pitch range, volume and vowel duration – are given for each of the three realizations of the stress-bearing vowel [y:] in the word *Kühlschrank*, all taken from the same speaker in the same session. All contours were falling.

(3-75) a. ICH wollte den Kühlschrank SOWIESO nicht haben.

I wanted the fridge anyway not have

'I didn't want the fridge anyway.'

b. Sie sind froh, weil ihnen Peter gestern den Kühlschrank gebracht

they are glad because them Peter yesterday the fridge brought

hat.

has

'They are glad, because Peter brought them the fridge yesterday.'

c. Sie ist sauer, weil ihr Peter gestern einen KÜHLSCHRANK

she is sore because her Peter yesterday a fridge

gebracht hat und keine SPÜLMASCHINE

brought has and no dishwasher

'She is angry, because Peter brought her a fridge and no dishwasher yesterday.'

Table 3.19: Three realizations of [y:], several parameters.

	maximal pitch (Hz)	pitch range (Hz)	volume (dB)	vowel duration (sec)
[y:], no prominence	201.93	20.53	54.16	0.1131
[y:], sentence stress	206.55	17.86	59.83	0.1087
[y:], focal emphasis	256.68	45.09	61.56	0.1180

It is obvious that for this speaker both pitch and loudness are used to distinguish both types of prominence from the unstressed variant. Moreover, we see a gradation in both parameters from unemphasized to sentence-stressed to focalized. Although with focal emphasis the pitch parameters are much higher than with the (metrically computed) nucleus, it is not warranted to say that a focus indicator is realized by pitch and metrical prominences are realized by something else, e.g. volume. In each case, the observed prominence is a mixture of at least both parameters. (Sluijter 1995) The fact, however, that with focalized elements it seems as if an extra prominence (encoded by pitch) is added, as opposed to the nucleus, suggests that these two types of prominence ‘don’t play in the same league;’ that is, it won’t do to treat sentence stress and focal emphasis equally and say, both are generated the same way, i.e. solicited by a focus feature. Focal emphasis is definitely one step ‘higher’ than the nucleus. If we assume that the nucleus is not associated with a focus feature, and if we assume that a grid in which the highest peak is a focus looks different from a grid in which the highest peak is entirely rule-generated by the metrical calculus (as I have done in this chapter), this phonetic effect would be expected. This empirical fact fits nicely with the discussion on the relationship between sentence stress and focus which we denied on theoretical grounds.

Let us summarize this section: Although (rule-governed) metrical prominence and (focal) emphasis are descriptively different, they are both kinds of prominence and as such subject to the requirements posed on them by the phonological submodule that creates prominence, such as the CAR. The metrical calculus generates grids in accordance to the CAR because the CAR is part of the grid construction rules. Focus is represented by a credit mark, the focus indicator, in the grid that always projects a strong mark on the focused element on the next higher level. This constrains the metrical calculus as it has to take these pre-specified strong marks into account. A focus indicator is only present with narrow focus; the nucleus (as prominence associated with 'broad focus') is really part of the metrical prominence system that happens to be the highest prominence in the absence of a focused element.

3.5 Clash and pause

A second question that needs to be addressed has to do with the pause insertion strategy to resolve focus clash that was presented in chapter 3.1. The question is: Why is the clash resolved by the insertion of a pause and not some other mechanism? In Section 3.5.1 I give an overview of potential other candidates for clash resolution, viz. stress shift and destressing.

In section 3.5.2 I discuss why neither of these mechanisms can be used to resolve focus clash. The main reason is that the focus structure of the clause would be wrongly represented by either of these clash resolution mechanisms, as we will see later, and pause insertion is the only mechanism that preserves the intended focus structure.

Finally, section 3.5.3 addresses the question of the domain of focal emphasis and comes to the conclusion that it is variable.

3.5.1 Mechanisms for clash resolution

The repair mechanisms of stress clash are well studied (e.g. Liberman & Prince 1977; Hayes 1995). As the main strategies to repair stress clash two mechanisms are especially important (see also Hayes 1995:ch.9):

- Shift of one of the stresses to a syllable farther away from the other stress (‘Move X’; Hayes 1995:35; 2-76a),

- Destressing of one of the clashing stresses (Hayes 1995:37; 2-76b).[53]

(3-76)	a.	tennesSÉE LÉGislators	→	TÉNnessee LÉgislators
	b.	tennesSÉE LÉGislators	→	tenessee LÉgislators

Shift is only possible if there are landing sites available for the shifted stress. This is the case in (3-76a), but it would be impossible with monosyllabic words or words in which the only other vowels are schwas that cannot bear stress (3-77).

(3-77)	a.	TÉN WÒmen	→	?
	b.	deNOÚNCED CRÌminals	x→	DÉnounced CRÍminals

Destressing seems to be a less problematic strategy, but it is subject to rules that constrain it, such as the Textual Prominence Preservation Condition (Hayes 1995:392) which ensures that the gradation of the original prominence pattern remains intact.

If, as we have concluded in 3.2, the alternating principle, that is,the OCP in its general form and thus also the CAR, is a principle on phonological representation in general and if it is thus part of the phonological subsystem that encodes prominence in general, it would be possible in theory that the same mechanisms that repair clashing stresses could also repair clashing foci. That is, if focus clash arises, one of the focal emphases could in theory be shifted somewhere else, or cancelled.

[53] Beat Addition (Hayes 1995:373) has not been included as it is not a mechanism that typically resolves stress clashes.

A general problem with comparing stress shift mechanisms and the resolution mechanism for stress clash, which has to be borne in mind throughout this discussion, is of course that destressing or stress shift in stress clash really is a repair mechanism, whereas the pause insertion is not a repair mechanism in the same sense. The need for destressing or stress shift arises if two elements for which prominence has been determined on the word level independently, are put together into a higher constituent (as is the case in e.g. compounds, where two phonological words are combined). The situation is consequently as follows: the metrical calculus derives two grids G_1 and G_2 for two words independently, according to the lexical and idiosyncratic rules these words adhere to. If G_1 and G_2 are combined on the next level to G_3, a clash situation can arise if G_1 has its peak at the right edge and G_2 at the left edge. So G_3 has to be modified in order to conform to the CAR or the Rhythm Rule, however you want to call it, and a modified grid G_3' has to be derived.

If foci clash the situation is different in that we are already on the highest level of metrical prominence assignment, and the clashing strong marks are in the same assignment domain to begin with. The rules, as we have seen, are formulated in such a way that they can only derive CAR-conforming structures within their level of application. This means there is no repair or anything else; the grid is generated in a CAR-conforming way from the very beginning. If this is so, we should not be too surprised if other ‘clash resolution mechanisms’ are operating in focus clash rather than in stress clash, as perhaps some strategies lend themselves easily to ‘repair’, fixing of something that already exists, whereas other strategies lend themselves easily to building a well-formed grid.

As we have seen in the experiments in part 3.1.2, the classical stress clash strategies are in fact rarely applied in the case of focus (3-78a,b), if speakers are forced to utter adjacent foci (see also Selkirk 1984:277; 280). The application of e.g. defocusing is possible only in a limited number of special cases, such as if two foci are in the same phrase (which is the case which would lead to Split Topicalization in German, for instance). Apart from such special cases, the repair mechanism that is used most often in normal cases is the insertion of a pause between the clashing foci (3-78c).

(3-78)	a.	Good ol' BÍLL MÁry likes.	–‖→	Good ÓL' Bill MÁry likes
	b.	Good ol' BÍLL MÁry likes.	–‖→	Good ol' Bill MÁry likes
	c.	Good ol' BÍLL MÁry likes.	→	Good ol' BÍLL _ MÁry likes

If we look at the grid, we see that inserting a pause (3-79c) is an equally suitable means to restore eurhythmy (or resolve the danger of clash) as stress retraction (3-79a) and cancelling (3-79b) do. The representation of the pause with a silent grid position has been proposed by Selkirk (1984:300), whom I simply follow in that regard.

(3-79) a.

```
.       *   *       .              *       .   *       .
*   .   *   *   .   *   .          *   .   *   *   .   *   .
*   *   *   *   *   *   *          *   *   *   *   *   *   *
TennesSÉE LÉgislators         →    TÉNnessee LÉgislators
```

b.

```
                                   .       .   *       .
                                   *   .   *   *   .   *   .
                                   *   *   *   *   *   *   *
                              →    Tennessee LÉgislators
```

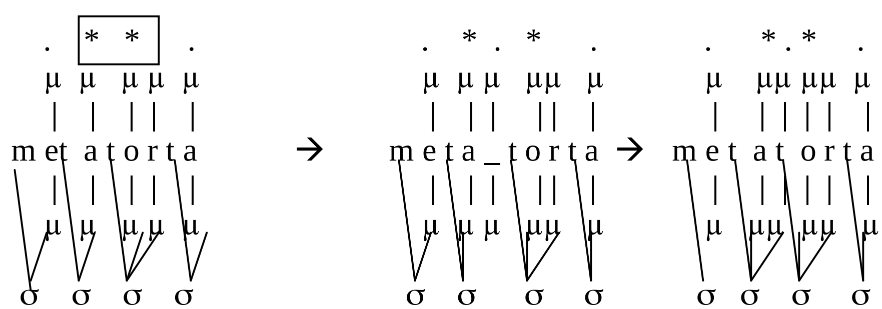
c.

```
.       *   *       .              .       *       *       .
*   .   *   *   .   *              *   .   *   .   *   .   *
*   *   *   *   *   *              *   *   *   *   *   *   *
Good ol' BÍLL MÁry likes.     →    Good ol' BÍLL _ MÁry likes
```

In fact, pause-insertion or lengthening phenomena have been described as resolution mechanisms also for clash of metrically derived prominences, although they seem to be applied less frequently than shift and destressing. A famous example for a lengthening phenomenon is *raddoppiamento sintattico*, as described by e.g. Nespor & Vogel 1979 and Yip 1988:92, whom this demonstration follows. If in (certain Northern local varieties of standard) Italian two words are adjacent to each other, where the first word ends in an open stressed syllable and the second starts with a stressed syllable, the consonant at the beginning of the second word is geminated (3-80a). Nespor & Vogel (1979:478f.) claim that by this gemination process a 'sufficient phonological distance' (1979:479) between the stresses is maintained.

(3-80) a. metà tórta 'half a cake' [me'ta't:orta]

b.



We can interpret the pause on an abstract level as an empty timing slot that is inserted between the clashing elements (cf. Yip 1988:92). It keeps them apart, just as an unstressed element would do. So the intonational parser can construct well-formed feet at the relevant places. With phenomena like *raddoppiamento sintattico*, the timing slot would not remain empty, but be linked to the nextbest consonant, creating a geminate that closes the open stressed final syllable of the first word (3-80b). Remember that coda consonants have a moraic value. The moraic tier is both the basis of metrical prominence assignment and of syllable structure; to make the picture less confusing I doubled the moraic tier in the schema in (3-80b).

3.5.2 Why a pause?

We have now encountered three clash resolution mechanisms: shift, destressing and pause-insertion and have seen that, whereas pause-insertion virtually plays no role with clashing stress, it is the repair mechanism of choice for the resolution of clash of focal emphases. Why is that?

The reason is that in the case of focus the exact position of the prominence matters because it encodes semantic information and serves as the exponent of the semantically interpretable focus feature. Under this view, it becomes immediately clear that a mechanism such as shift of focal emphasis cannot be applied felicitously, as it leads the listener on a wrong track with respect to the position of the focused word. Let us take ex. (3-79c) for illustration (3-81):

(3-81) a.

```
.        *    *    .              .        *      *    .
*    .   *    * .  *              *    .   *    . * .  *
*    *   *    * *  *              *    *   *    * * *  *
Good ol’ BÍLL MÁry likes.   →     Good ol’ BÍLL _ MÁry likes
```

Set of alternatives: good ol’ Bill and other guys

b.

```
.        *    *    .                   *   .     *    .
*    .   *    * .  *              .    *   *     * .  *
*    *   *    * *  *              *    *   *     * *  *
Good ol’ BÍLL MÁry likes.   →     Good ÓL’ Bill  MÁry likes
```

Set of alternatives: good ol' Bill vs. other Bills, e.g. Flashy Bill, Homey Bill, &c.

Phonologically, both repair mechanisms would produce a well-formed output. But semantically, there is a big difference. The focused word is *Bill* as head of its phrase. The intended contrast is between Bill and comparable entities, that is, some other guys that are salient in the discourse. In (3-81a), the position of the focal emphasis has not changed. It is consequently true to the intended meaning. In (3-81b) this is not the case; with shifting the emphasis from *Bill* to *ol'*, the listener gets the impression that the contrast is not between Bill and other guys that are salient, for instance Jonathan, Hans-Franz, Joel, Neville etc., but rather between different persons that all are called Bill but bear different epithets.[54]

And cancelling the emphasis would not do either because then the focus structure would not be properly represented. One could argue that in 'good ol' Bill' the focus is sufficiently represented by the fact that the phrase is topicalized. But this is clearly not true; first, we saw already that topicalization is optional, but the emphasis on the focalized element is compulsory. To topicalize this phrase without putting focal emphasis on it would not give the right interpretation as the listener would not perceive that the topicalized phrase is focalized. Note that topicalization does not automatically entail focus. In the case of anaphoric preposing, for instance, a phrase is fronted without being in focus. So the listener cannot deduce that a constituent is in focus from the mere fact that it is topicalized.

[54] This could entail, of course, that stress shift would not be available in languages in which word level prominence is used to encode semantical differences, e.g. between near-homonyms. But remember that I am talking only about languages with immobile stress; the case in languages with mobile stress may be completely different.

Another argument, and in fact the crucial argument, is that foci can also clash without topicalization (3-82). In this case cancelling the emphasis would deprive the focus feature of its phonetic realization, but in this case one could never recover it by taking positional oddities into account. Thus in the end the correct understanding of the focus structure by the listener would be seriously jeopardized.

(3-82) a. BÍLL LÌKES beans

(hearer expects that another subject has some other feelings about beans)

b. Bill LÍKES beans

(hearer expects that Bill has some other feelings too, probably about beans, possibly about other stuff. Or s/he interprets it as verum-focus)

We have seen in our experimental data that cancallation of emphasis is employed sometimes, but only in a subcase, namely if the foci are on the verb and its object, respectively. The fact that it is confined to this case shows that it is not a preferred strategy in general, but only, that there is something special about this case that allows for defocusing. I have hinted at potential reasons already in the discussion of the relevant data in 3.1.2.

Apart from these functional arguments we can give a technical reason why focus features cannot simply be moved around or cancelled. The way the system is set up, the focus feature is associated with a focus indicator before the metrical parse starts. That means, the metrical calculus has to accommodate to the preexistent focus indicator and has no power over it, like cancelling it or shifting it. The only way to build well-formed feet is to insert empty timing slots – vulgo: pauses – if necessary.

With the stress clash repair mechanisms, the whole picture is different. The peaks of words that come into clash by the combination of two independently derived grids are not associated with a focus feature and therefore can be moved around and cancelled freely, in accordance to the prosodic rules and well-formedness conditions of the relevant level.

So far we have seen that of the three possible clash resolution mechanisms, focus clash only makes use of one, namely pause insertion. This is one of the obvious points which distinguish focal emphasis from metrical prominence, and it could be derived from the nature of focal emphasis itself. We could now ask the question the other way: why does metrical prominence not use pause-insertion?

I have no conclusive answer to this. I assume that an important reason for this is that pause-insertion is 'costly' in that it requires building extra structure from the very bottom up, whereas the other mechanisms only involve local repairs, leaving the bottom line unaffected. It is reasonable to assume in general that the relevant mechanism only does as much work as is really needed, i.e. decides rather in favor of less costly alternatives to fix a given problem, clash resolution in this case.

There are other reasons why pauses could be dispreferred in general. They depend on what we think the exact nature of prominence is. Either we take the line that grid marks are representations of a [± stress] feature; in that case they would be autosegments, i.e. suprasegmental objects for which the rules of suprasegmental material would hold (cf. Halle & Vergnaud's (1987) view). Or we assume that they have merely a syntagmatic existence; they represent foot boundaries, and the foot construction rules give us a condition for each level which side of the foot is the more prominent one. I am not sure that we can decide between these options. For our present purposes it is however relevant to point out that under both conceptions the insertion of pauses is problematic.

Let us begin with the view that grid marks are autosegments, because things can be explained more easily that way. Under that view, a pause would violate a well-formedness principle of grid-marks that is actually rather trivial:

> *Anchoring Condition:*
> All suprasegmental material must have a segmental anchor

Anchoring is used in the sense in which it was used in the discussion of Autosegmental Phonology in the 1970s. This condition underlies much of the work in Autosegmental Phonology where suprasegments sometimes lose their segmental anchors but rather than being left floating they attach somewhere else. An example of this is compensatory lengthening. I illustrate it with a well-known example from Proto-Indo-European involving the loss of the First Laryngeal (here represented as [ç]). If a segment is lost by a sound change process or the like, it is completely natural that the mora that used to be associated with it looks for another segmental anchor (3-83a). Theoretically, another strategy could be possible, namely to associate with nothing and thereby create a pause, as the duration of the mora somehow has to be realized (3-83b). This possibility is employed in no language, as far as I know.

(3-83) a.

μ μμ μ		μ μμμ		μ μμ μ
| || |		| | |		| / |
di.d^{h}eç.ti	→	di.d^{h}e. ti	→	di.d^{h}e .ti

(cf. e.g. Greek *títhēsi*, Sanskrit *dádhāti*)

b.

* μ μμ μ		μ μμμ		μ μμ μ
| || |		| | |		| || |
di.d^{h}eç.ti	→	di.d^{h}e. ti	→	di.d^{h}e_.ti

This principle makes immediate sense, as suprasegmental material can only be heard (and thus has a chance of being picked up by language learners, listeners and the like) if it has a segmental anchor.

In the case under discussion, the inserted syllabic grid mark has no overt correspondent. It cannot have one, as there is no material left in the lexical array (to use the terminology of the phase-model of Chomsky 2001). In the case of stress clash we are speaking of a repair mechanism; repair of something that can be recognized as a problem only at PF. Therefore all material from the lexicon must have been used up already. For the syntactic module there are two possibilities: Send the whole derivation back to narrow syntax and have it fixed there – here new lexical material can be retrieved from the lexicon, but it is rather doubtful that we want to have a 'loop' in our grammar – or leave it in PF and try to solve the problem there. It is unlikely that the PF module has such unlimited access to the lexicon that it can simply pick lexemes not present in the narrow syntactic derivation; we believe that its access to the lexicon is limited to the association of concepts and feature clusters to phonological correlates ('Lexical Insertion'). This means that the extra grid mark enforced by the Principle of Alternation has to attach to a nothing. As grid marks correspond to timing slots, we perceive this 'nothing' as a pause, that is, as time in which no segmental material is produced.

If we assume that 'the grid' is essentially only a means to make the relative prominence more obvious to the eye of the reader, and that the only relevant mechanism is the grouping of syllables etc. into feet (that is: if strong marks have no independent existence as objects), basically similar considerations hold: In cases in which rule-conforming foot construction is impossible, it is presumably better to try to alter the lower constituency first in a way that eventually leads to the effect that higher constituents can be constructed according to the rules. What effects would the insertion of a pause have for foot construction? We can be sure that there is a mismatch; the pause would have to be counted as a given number of morae, but without segmental material associated with it. This means that at the place where the pause is inserted, an extra condition must hold that takes care that a foot is constructed out of segmental material plus a given number of extra morae. We may assume that such a foot is in general ill-formed, as it consists partly of ‘stipulated material’, that is, morae which are inserted ad hoc, without motivation from the segmental side. So the problem boils down to a similar situation as I described at length under the hypothesis that grid marks have an existence as autosegments: If a pause is inserted, a mismatch arises between the segmental material and the constituents into which it is grouped.

Another problem is the following: Most researchers, including me, assume that metrical constituency is assigned in a bottom-up fashion (e.g. Liberman & Prince 1977; Halle & Isdardi 1995; Hayes 1995). This means that first syllables or morae are grouped into feet; these feet are then grouped into larger constituents and so on. Note now that prominence clash is something that becomes apparent as a problem only on the higher levels of constituency. The remedy must therefore happen in a top-down fashion, meaning that already existent constituents must have to be altered. The changes that have to be done with destressing are relatively local: Only the constituent immediately below the level of complexity where the clash happens has to be altered, whereas lower levels remain unchanged. With stress shift all levels below the clash level might have to be altered, although well-formedness conditions such as the Continuous Column Constraint (Hayes 1995:35f.) strive to keep the alternation effort minimal. If, however, a pause is inserted, it is guaranteed that all lower constituents have to be altered, as the silent beat has to be represented down to the lowest foot, otherwise the Continuous Column Constraint would be violated. So this is by far the most costly operation and therefore the least preferred option for clash remedy.

3.5.3Morphemes as domain of focus

It was said earlier that focus is associated with words. This statement has to be slightly qualified. Consider (3-84).

(3-84) a. It doesn't matter whether you WANT to eat ice cream; the thing is, you simply WON'T do anything of the kind.[55]

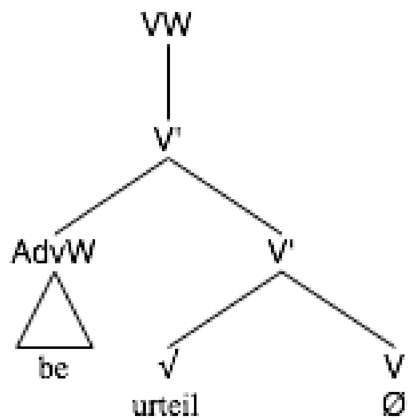
b. Du hättest sie BEurteilen und nicht gleich VERurteilen sollen.
you had her judge and not at-once condemn shall
'You should have judged her and not condemned her.'

In (3-84) obviously the focal emphasis goes on function words or even morphemes, and moreover ones that naturally would be unstressed. These cases are what Bolinger (1961) would call 'contrastive stress'. The property that everything, even normally unstressed elements, can be selected for focus holds for contrastive focus in general, as (3-84) shows. So we are compelled to say that focus can be associated with any lexical primitive, be it a word or a morpheme.

[55] It would be more natural to finish the sentence directly after 'won't'; I however put in that extra material so it is clear that it is not some clause-final nucleus we are talking about

How is this compatible with what has been said about the focus feature and its production further above (3.4.3)? We have seen that focus is associated with words and that it projects trivially up to the next phrase boundary, but as it is realized on the head, there is no way to detect this projection up to the phrasal node. Words, however, have a quasi-syntactic structure as well (see, among others, Selkirk 1982, Halle & Marantz 1994). Morphemes that are not category-changing have recently been analysed as word-internal adjuncts (Newell 2006, Speyer 2006b). This means that their features cannot percolate to the highest node of the word derivation (which is the reason why they do not influence the category of the derived word) but are stuck right at the adjoined morpheme itself, as there is a quasi-phrasal node (3-85). In (3-85) I treat the word as if it were a larger syntactic object, which is deliberate (as I believe, following Halle & Marantz (1994) and subsequent studies in the framewrk of Distributed Morphology, that the processes of syntax are the same within and between words); the node labels should be different (because the whole derivation is going to function as a 'head' for larger syntactic derivations; we would expect 'V' on the top, following e.g. Selkirk 1982), but I set it up as this in order to illustrate the quasi-phrasal character of the word derivation. X means head, X' means intermediate projection and XW means maximal projection (W for word, as the maximal objects here are words, not phrases).

(3-85)



We see that any focus feature that happens to be on the adjoined morpheme *be* rather than on the root *urteil* is captured in the ‘phrase’ of the adjunct and cannot percolate higher. Consequently, it is realized on the adjoined morpheme itself.

This has important consequences. It means that it is not always the syllable in the word with the highest prominence under normal circumstances on which the focal emphasis is realized. This is only the case if the word root is the element that bears the focus feature. We have to admit that this is by far the most common case. But focus on a word-internal adjunct is likewise possible, and here the normal stress-contour of the word is disturbed.

So it looks as if focus is associated with small elements like morphemes or words, but not with bigger units. This is surprising from the point of view of topicalization: Here, the whole constituent is moved; consequently we would expect the focus feature to be present on the constituent as a whole.

We have to be cautious however. The fact that whole constituents are topicalized may have nothing to do with their bearing the focus feature or not. In other words: If a phrase is topicalized, this does not mean that the phrase as a whole bears the focus feature (cf. Drubig 2003). It is simply the case that only maximal phrases, immediate constituents at that, can undergo topicalization. So, if some word or even morpheme is in focus and the speaker decides to topicalize it, the whole constituent containing the focused element has to undergo movement because otherwise the derivation would be ungrammatical.

We can even show that it is not the phrase as a whole that mysteriously inherits the focus feature and thus is in focus as a whole. Consider (3-86). Here the b-version is a more elaborate and verbose version of the a-version. By the verbosity the contrast is not between *beans* and *peas* any more, but on the properties *stringy* and *roundish*. If we assume that the noun phrase *these stringy little green things* as a whole is in focus, we would expect the focal emphasis to coincide with the highest natural prominence on the phrase, namely the phrasal prominence peak on *things*. This is because in objects larger than a word the focus would have to be associated with the syllable that would be most prominent under normal circumstances. Putting the emphasis on *things* sounds, however, awkward and hardly acceptable (3-86b). If the emphasis is on the elements that are really in contrast in these verbose versions, namely *stringy* and *roundish*, the sentence sounds much more acceptable (3-86c).

(3-86) a. [BEANS] [[he] LIKES _], but [PEAS][[he] HATES _]

b. # [These stringy little green THINGS][[he] LIKES _],
but [these roundish little green OBJECTS] [[he] HATES _].

c. [These STRINGY little green things][[he] LIKES _],
but [these ROUNDISH little green objects] [[he] HATES _].

From a functional point of view, topicalization is actually rather a bad focusing strategy, because of the mismatch between the target of movement (the phrase) and the target of focus (the word). By operating on the phrase as a whole it falsely leads the listener to believe that it is a property (namely: being focused) of the phrase as a whole that is the motivation for the topicalization operation. If the phrase remained in situ, nobody would ever try to associate the focus with some unit higher than the word.

To summarize this section: Contrastive focus can be assigned to any morphosyntactic primitive. The default assignees are words. The focal emphasis in topicalization is special only insofar as it looks as if it is assigned to the phrase as a whole, given the syntactic constraints on topicalization.

3.6 Summary

In section 3.1 we saw that topicalization is still acceptable in Modern English in most cases; the only case that is less acceptable is exactly the case where violations of the Clash Avoidance Requirement can occur, namely double focus topicalization with the topicalized element and the subject both being full noun phrases. A short survey of a corpus of naturally occurring topicalization cases showed that pronominal subjects are overproportionally frequent in topicalization cases. This was interpreted as evidence for the Clash Avoidance Requirement. Further evidence was produced by three experiments for German and three for English. They proved that speakers avoid focus clashes, that they insert measurable pauses when they are forced to utter a sentence with focus clash, as this is the only way to remedy the pending focus clash, and that these pauses are really due to clash avoidance and are not simply a matter of emphatic lengthening.

Section 3.2 put the Clash Avoidance Requirement in a broader conceptual perspective. We found that focal emphasis is subject to prosodic constraints very similar to the Rhythm Rule, which again has conceptual similarities to the Obligatory Contour Principle.

In section 3.3 we concentrated on potential evidence for the Clash Avoidance Requirement in German. There is much evidence that it is a serious factor both in German and in English. This evidence is, however, primarily synchronic evidence, as diachronic data is not available since German never changed in a way that jeopardized the CAR. The evidence gathered in several different experiments showed that speakers avoid focus clashes and judge sentences with focus clash as worse than sentences without focus clash.

Section 3.4 tried to illuminate the non-trivial relationship between metrical prominence and focal emphasis. Although (rule-governed) metrical prominence and focal emphasis are descriptively different, they are both subject to the requirements imposed on them by the phonological submodule that creates prominence, the metrical calculus. The difference is that in the case of focus a focus indicator is associated with elements that bear the semantic focus feature and that this association happens before metrical assignment takes place. This only holds for narrow focus: the nucleus as representative of 'wide focus' is really part of the metrical calculus that happens to be the highest prominence in the absence of a focused element.

In section 3.5 the domain of focus was identified and the question was investigated why focalized elements use pauses as a remedy for clash, whereas metrical prominences preferably use other means such as destressing and stress shift. Contrastive focus can be assigned to any morphosyntactic primitive. The default assignees are words. The focus in topicalization is special only insofar as it looks as if it is assigned to the phrase as a whole, given the syntactic constraints on topicalization.

4. Topicalization and the Clash Avoidance Requirement in Old English

In this chapter I will discuss the interaction of the Clash Avoidance Requirement with syntactic usage in Old English. I will argue that the well-known V2/V3 alternation (that is: the alternation of sentences with the word order X – V – S… (= V2) and X – S – V … (=V3)) is due to the CAR, and, following Haeberli 2002, that Old English syntax offers two structural landing sites for subjects. The choice between landing sites is made in accordance with the CAR and depends on whether the subject bears focal emphasis.

It is well known that V3 occurs with pronominal subjects and V2 with full noun phrase subjects, as described in section 4.1. According to Haeberli, the pronominal subjects occur in a the higher and leftmost position while the lower subjects occur in the lower position. Prosody and information structure require that pronominal subjects occur where we see them, these factors, combined with Haeberli's phrase structure, predict that , with full noun phrase subjects should occur in both positions, depending on whether they are focused or deaccented (see Haeberli & Pintzuk 2007). Because of the word order variation in Old English and the tendency for full noun phrase subjects to bear accent, it is difficult to determine whether this prediction is borne out. Section 4.2 tackles this problem and gives quantitative arguments that V3 does indeed occur with full noun phrase subjects.

In section 4.3 the Old High German data is examined. There are several indications that the Old High German sentence template and the Old English template differ from each other substantially; the Old English one being already rather similar to that of Modern English whereas the Old High German template also clearly prefigured modern German syntax. Crucially the existence of V3 in Old High German, which has often been used to demonstrate a basic similarity between the two languages, turns out to offer evidence that their syntax is almost as different as the syntax of their modern successor languages.

In section 4.4 I finally demonstrate that the choice between the two subject positions is predetermined pragmatically in such a way that CAR violations are minimized: Focalized constituents target the lower subject position, creating V2 sentences, while topical or discourse-old constituents target the higher subject position, creating V3 sentences.

4.1 V2 and V3 in Old English

The Clash Avoidance Requirement certainly was active in Old English. In fact, it was for an Old English text that one of the earliest observations concerned with it has been made, as we saw in section 2.4. John Ries (1907: 89ff.) found that in Beowulf after prominent elements a weak element has to follow. Whereas in a treatise on Beowulf it would be possible to downplay this observation as a mere poetical-metrical phenomenon (although Ries explicitly rules this out; 1907:90), it turns out that Ries actually discovered a fundamental principle not only of Old English poetry but of the rhythmic organization of English and German, perhaps of all pitch accent languages.

In recognizing the impact of the Clash Avoidance Requirement on Old English we are basically in the same position as we were for German in chapter 3.3: It is not possible to find evidence for it as compelling as the loss of topicalization in Early Modern English. The reason is that, as far as we can see, there was no change in Old English that was caused by or a reflex of the CAR, as was the case with the decline in topicalization in Middle English. But the whole of Old English syntax as it stands was CAR-compatible; hence we would not expect CAR-related changes.

Whereas in Modern English declarative sentences one phrase structure type is used (the Infl-medial, V-medial type), Old English permits considerable syntactic variation (cf. Pintzuk 1999; Trips 2002: 188; Haeberli & Pintzuk 2007). First, the head-complement ordering within the extended V-domain was variable. The two parameters that could vary are:

- In IP: the position of the VP relative to I
- In VP: the position of the complement within VP relative to V

The VP-internal variation need not concern us here. What is relevant to our concerns is the IP-internal variation. As a result of this variation we find main clauses that have the verb somewhere in the left area of the sentence (Infl-medial) and main clauses with the verb at the end (Infl-final). Often Infl-final clauses surface as verb-last (= VL); we will later see that unfortunately this is not always the case. An example of a VL clause is given in (4-1a) below.[56]

The position of the verb with respect to the subject in Infl-medial sentences varied systematically when some constituent was topicalized. As noted, in sentences with pronominal subject, the verb follows the subject, producing sentences in which the finite verb was the third 'constituent' of the clause (= V3; 4-1b), whereas in sentences with full noun phrase subjects, the verb preceded the subject, producing sentences in which the finite verb was the second 'constituent' of the clause (= V2; 4-1c; see Ries 1907:89). Sentences with the verb in initial position (=V1; 4-1d) were also possible, but they will not be discussed as they do not bear on our central hypotheses.

(4-1) a. VL: & þy geare Healfdene Norðanhymbraland gedælde.

and this year Halfdane Northumberland divided

'and in this year Halfdane divided Northumberland'

(cochronE,ChronE_[Plummer]:876.7.1190)

[56] To avoid confusion, I am choosing the following terminology (following Santorini 1992:612): Infl-medial and Infl-final refer to the underlying tree structures, that is: whether I° follows its complement VP (= Infl-final) or precedes it (Infl-medial). The terms V1, V2, V3 and VL (for verb-first, verb-second, verb-third and verb-last) refer to the surface serialization, that is whether the verb is the first, second, third, last overt constituent in the clause. The terminology does not entail a specific analysis; V2 sentences are mostly Infl-medial but need not be, etc. Especially it does not entail that V1/V2/V3 has been generated by moving the verb to C°, although this is commonly assumed in Germanic generative syntax. If reference is made to V2 as result of verb movement to C°, the term CP-V2 will be used.

b. V3: Ond eallum þam dagum buton Sunnandagum he afæste to æfenes,

and all the days except Sundays he fasted till evening

'and he fasted all days, except for Sundays, till the evening.'

(cobede,Bede_3:17.230.30.2368)

c. V2: þone wæterscipe beworhte se wisa cyning Salomon mid fif

the conduit constructed the wise king Salomon with five

porticon fæstum weorcstanum,

porticoes massive hewn stones

'The wise king Salomon constructed the conduit with five porticoes of watertight stones'

(coaelhom,+AHom_2:10.251)

d. V1: Hæfde þæt deor horse gelic heafod,

had that animal horse like head

'That animal had a head like a horse.'

(coalex,Alex:20.4.232)

The generalization that pronominal subjects lead to V3 sentences when topicalization takes place, and full noun phrase subjects to V2, needs to be qualified. The reason for this is that there are quite a few examples in which it looks as if we have V3 with a full noun phrase subject (4-2). Tables 4.1, 4.2, 4.3 and Figure 4.1 show that these examples are not especially infrequent; in period oe2, for instance, almost a quarter of all V3 sentences with a topicalized accusative noun phrase (most of which are objects) have a full noun phrase subject. In the same period, the proportion of full noun phrase subjects among all

V3 sentences is more than half with preposed dative NPs.[57] The tables are separated after type of preposed element; Table 4.1 shows the proportion in sentences with topicalized accusative noun phrase, Table 4.2 with topicalized dative noun phrase, Table 4.3 with topicalized prepositional phrase. Figure 4.1 combines the three tables.

(4-2) a. & hit Englisce men swyðe amyrdon.

and it English men fiercely prevented

'and the Englishmen prevented it fiercely'

(cochronE,ChronE_[Plummer]:1073.2.2681)

b. Forðon þa ærestan synne se weriga gast scyde þurh þa næddran,

For the first sin the wicked ghost incited by the adder

'For the wicked ghost worked the first sin with the help of the serpent'

(cobede,Bede_1:16.86.28.791)

c. Forðon hie nan monn ne dearr ðreagean ðeah hie agyltan,

Therefore them no man not dares punish even-if they sin

'Therefore no man dares to punish them, even if they commit a sin.'

(cocura,CP:2.31.12.138)

d. and ðas feower godspelleras God geswutelode gefyrn […] Ezechiele

and these four evangelists God revealed long-ago Ezechiel

[57] The evidence from preposed dative noun phrases is potentially misleading as there are a number of verbs in Old English that have a dative experiencer which usually precedes the verb, like the following example:

(i) ac him deriað bremelas þe him on weaxað, and wilde þornas,
and him hurt blackberries that him at grow and wild thorns
'and he was hurt by blackberries that grow towards him and wild thorny bushes'.
(coaelhom,+AHom_3:64.450)

Such cases are structurally different from 'normal' object topicalization, but as they are part of the dative data as searched by the computer, they obsure the real numbers.

‘and God announced these four evangelists to Ezechiel long ago

(coaelive,+ALS_[Mark]:174.3311)

Table 4.1: Proportion of sentences with full-NP-subj out of all V3 sentences; topicalized accusative NP.

	oe1/2	oe3/4	total
#V3 total	727	817	1544
#V3 with fNP-sbj.	170	72	242
%	**23.38**	**8.81**	**15.67**

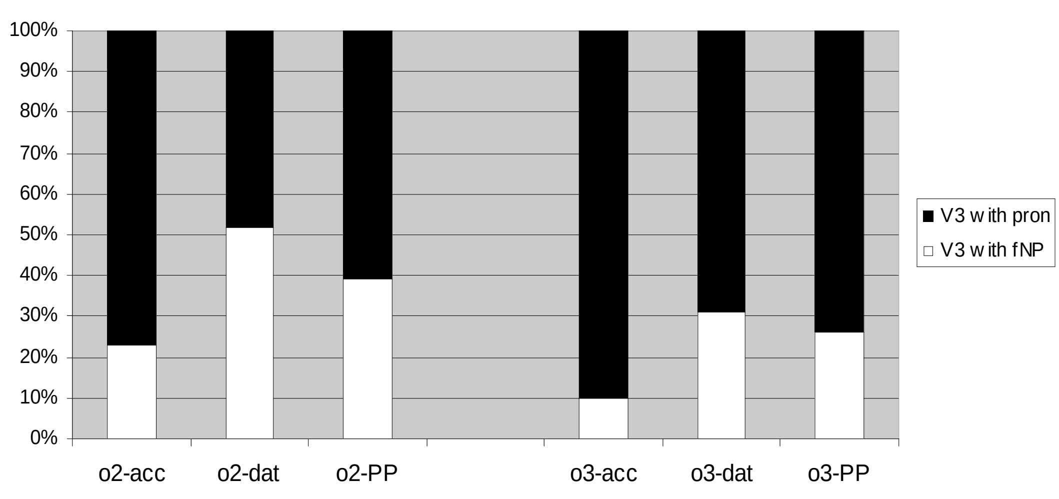
Table 4.2: Proportion of sentences with full-NP-subj out of all V3 sentences; topicalized dative NP.

	oe1/2	oe3/4	total
#V3 total	364	201	565
#V3 with fNP-sbj.	189	62	251
%	**51.92**	**30.85**	**44.42**

Table 4.3: Proportion of sentences with full-NP-subj out of all V3 sentences; topicalized PP.

	oe1/2	oe3/4	total
#V3 total	1448	1271	2719
#V3 with fNP-sbj.	569	335	904
%	**39.30**	**26.36**	**33.25**

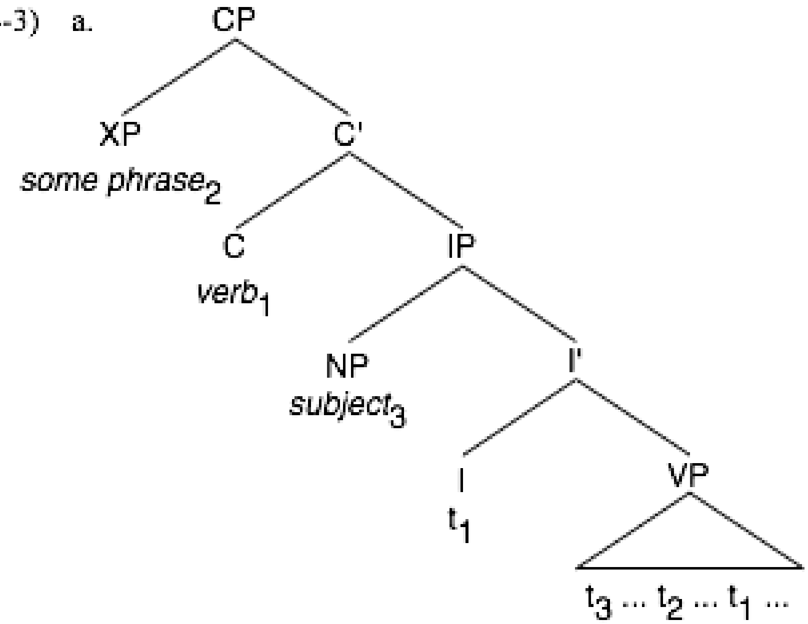
Figure 4.1: Proportion of sentences with full NP subject out of all V3 sentences.



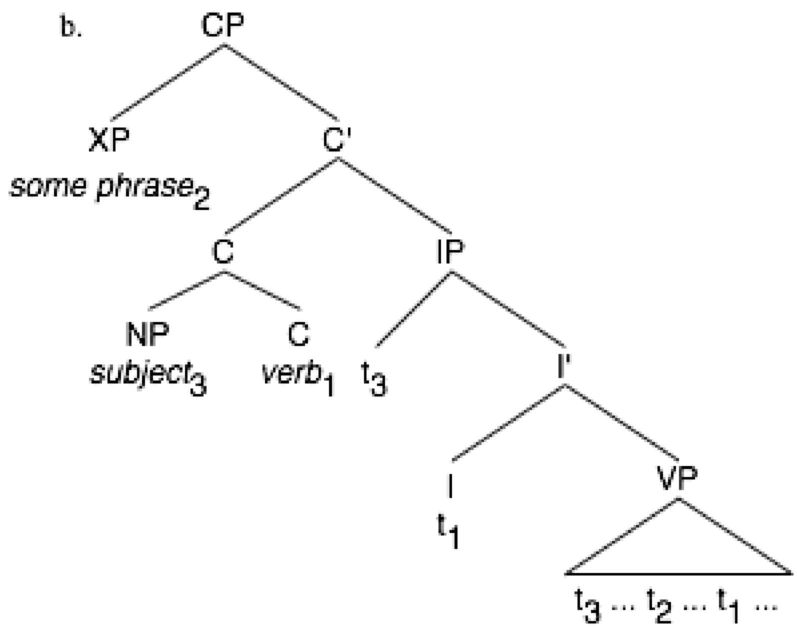
We will come back to these cases in the next section, as this data will turn out to provide the crucial evidence for the analysis pursued here.

How should we analyse this variation between V2 and V3? At least since van Kemenade (1987) – a view which she renounced later (e.g. Fischer, van Kemenade, Koopman & van der Wurff 2000: ch.4), by the way – the syntactic structure of Old English has been viewed as following the V2-constraint in a strict manner, that is: as CP-V2. In (4-3a) the sentence template of CP-V2 is given. CP-V2 can be regarded as a grammaticalization of the V2-constraint: With a CP-V2-syntax, the syntactic output is sure to follow the so-called V2-constraint (which probably has been nothing more than a preference in Proto-Germanic times) as the syntax cannot generate anything else but V2 sentences. For Old English, such an analysis is obviously not adequate. Van Kemenade (1987), however, proposed an analysis for Old English V3 sentences consistent with a uninform CP-V2 syntax. The analysis presumes that V3 is possible only with pronominal subjects, and pronominal subjects that appear to the left of the verb are analysed as syntactic clitics (van Kemenade 1987; Hulk & van Kemenade 1995). Thus there is no structural difference between V2 and V3 cases, and the grammar can be consistently V2 (4-3b).

(4-3) a.



b.



One could formulate this theory also without resorting to syntactic clitics, treating cliticization as a purely phonological operation. In this case one would treat subject pronouns as a kind of second position clitics. In a framework such as Distributed Morphology (cf. e.g. Halle & Marantz 1993), movement operations after narrow syntax are possible under a restricted set of circumstances (Embick & Noyer 2001). We are mostly interested in what Embick & Noyer (2001) call Local Dislocation. This is an operation on the linearized string after Lexical Insertion. It is basically that two elements can flip their places, if they are linearly adjacent. This is schematically demonstrated in (4-4): If Z in this structure is of a kind that has to be proclitic to an element outside its original domain, it can flip places with X and form a complex structure with X. The two important conditions by which the linear sequence of Z and Y are described, namely that Z and Y are adjacent and that Z immediately precedes Y, are still true on the dislocated string as X and Y form a unit and therefore the thing immediately preceding Y is not X but the whole Z+X complex (after Embick & Noyer 2001:562ff.; cf. also Speyer 2007a). Note that the original syntactic structure is obscured by this operation.

(4-4)	syntactic object:	$[_{XP}$ X $[_{YP}$ $[_{ZP}$ Z] Y]]
	after linearization:	[X * [Z * Y]]
	after local dislocation:	$[[_{X^\circ}$ Z+X] * Y]

The asterisk indicates 'linear adjacency'. The motivating forces for Local Dislocation are phonological requirements of lexical items, e.g. the requirement of a clitic to attach to the left or to the right of its target supporting element.

In the end it does not matter what the analysis of this cliticization should be, since a cliticization analysis of any kind faces serious empirical problems, as has been recognized repeatedly in the past (e.g. Allen 1990; van Gelderen 1991). I give a brief sumary of the major point. A cliticization account would make the wrong prediction as soon as it is extended to subordinate clauses and questions. This is easy to see: Under the cliticization view, pronouns are proclitic to the verb. Under the CP-V2 hypothesis, the verb is in C°. In subordinate clauses, however, C° is occupied by the complementizer. If the clitic hypothesis is correct, the pronoun should be proclitic to whatever is in C°, and shouls therefore appear before the complementizer in subordinate clauses. This is obviously never the case in Old English. A similar problem arises with questions. In Old English questions, we never have V3. A V3 wh-question of the relevant type would read as: 'what he will show to his friends in the afternoon?' and it is easy to see how unacceptable that sounds. But if interrogative and declarative sentences have an identical CP-V2 structure, we should expect no asymmetry between declarative and interrogative sentences, but we would expect that pronouns cliticize to the left of the verb regardless of the topicalized element is a wh-phrase, as in questions, or a normal noun phrase, as in declarative sentences. So the cliticization hypothesis is untenable on empirical grounds.

This is a pity, for there are several factors that seem to support van Kemeade's analysis. For instance, subject pronouns in English and German have at least one characteristic of clitics, namely the property of being obligatorily unstressed. Furthermore, they have strong positional preferences. In Modern German, they are regularly positioned directly to the immediate right of C. In Old English, on the other hand, they are regularly to the left of the finite verb. So, in principle, a cliticization account of subject pronouns is not far-fetched.

Furthermore, the explanation of V2 as CP-V2 works extremely well for languages closely related to English such as Modern German and Dutch. It is also assumed that this analysis, with slight modifications, accounts best for the syntactic structure of the Scandinavian languages, too (Vikner 1995).

The idea behind this is that some version of a CP-V2 structure was already the structure of Proto-Northwest-Germanic, that is, of Proto-Germanic after East Germanic

(Gothic etc.) split off.[58] Consequently English at some point should have had this structure too. As the split between English and the other West-Germanic languages occurred not such a long time before our first Old English documents set in (let us pinpoint the split at the time when the Anglo-Saxon started to immigrate to Britain, that is, in the fourth century AD, and the first Old English documents are from the time around 800 AD), it is conceivable that Old English still followed the CP-V2 structure in (4-3). And we do find numerous examples like (4-1c) of the type X – V – S, where X denotes any constituent, that are doubtlessly overt V2 sentences.

There are, however, serious problems with this line of reasoning. First, the premise that all Germanic languages were syntactically similar in the Early Middle Ages is an inference from the assumption that they started to differentiate only roughly 400 years before their respective earliest attestations, and therefore there was not much time for syntactic change. This view is over-simplified, as it does not take into account the possibility that languages (among which might have been English) might change rapidly due to language contact. Thus, the premise for the argument that all Germanic languages must have had a similar structure around 800 AD is based on shaky grounds, which makes the validity of the argument itself questionable. Second, the assumption that Proto-Germanic had a CP-V2 structure has no empirical basis but is a mere out-of-the-blue assumption. Third, not all surface V2 sentences must be the outcome of the strict CP-V2 structure of (4-3). Most Modern English declarative sentences of the form S – V – X, for instance, are V2 sentences on the surface, although it is assumed unanimously that

[58] Perhaps even of Proto-Germanic, as we cannot say for sure what the Gothic syntax looked like, as most of our Gothic texts are translations from Greek. Many Gothic sentences look like they follow the so-called V2-constraint, but this might be erroneous, as Greek had a similar constraint, too, so that in a typical Greek clause the verb is somewhere in the left part of the clause.

Modern English does not have CP in declarative clauses. This just goes to show that surface V2 can reflect several underlying syntactic structures.

A further problem is that Old High German of all languages shows overt V3 sentences with pronominal subjects. Either the surface V3 order is somehow derivable from the rigid CP-V2-structure in (4-3) – this point will be discussed in chapter 4.3 – or the rigid CP-V2-structure of German and Dutch is a later innovation. In that case the historical argument for CP-V2 in Old English would be undermined.

Finally, there is evidence against a CP-V2 analysis of Old English within Old English itself, viz. the rather frequent V3-sentences. In the light of van Kemenade (1987) and Pintzuk (1997) these cases have been interpreted as the result of a secondary cliticization operation, in which the subject pronoun has been cliticized to the position immediately to the left of the verb. But there are several V3 sentences with full noun phrase subjects, as we have seen (Tables 4.1, 4.2, 4.3), which cannot be the result of cliticization processes. In order to maintain the cliticization hypothesis we would have to assume that lexical nouns and even more complex noun phrases could cliticize, which is not a tenable assumption at all, or one would have to analyse all V3 sentences with full noun phrase subjects as underlying verb-last sentences. But we will see that this is not a tenable hypothesis either. There are severe problems with analysing these cases as verb-final, and that on empirical grounds this hypothesis has to be refuted.

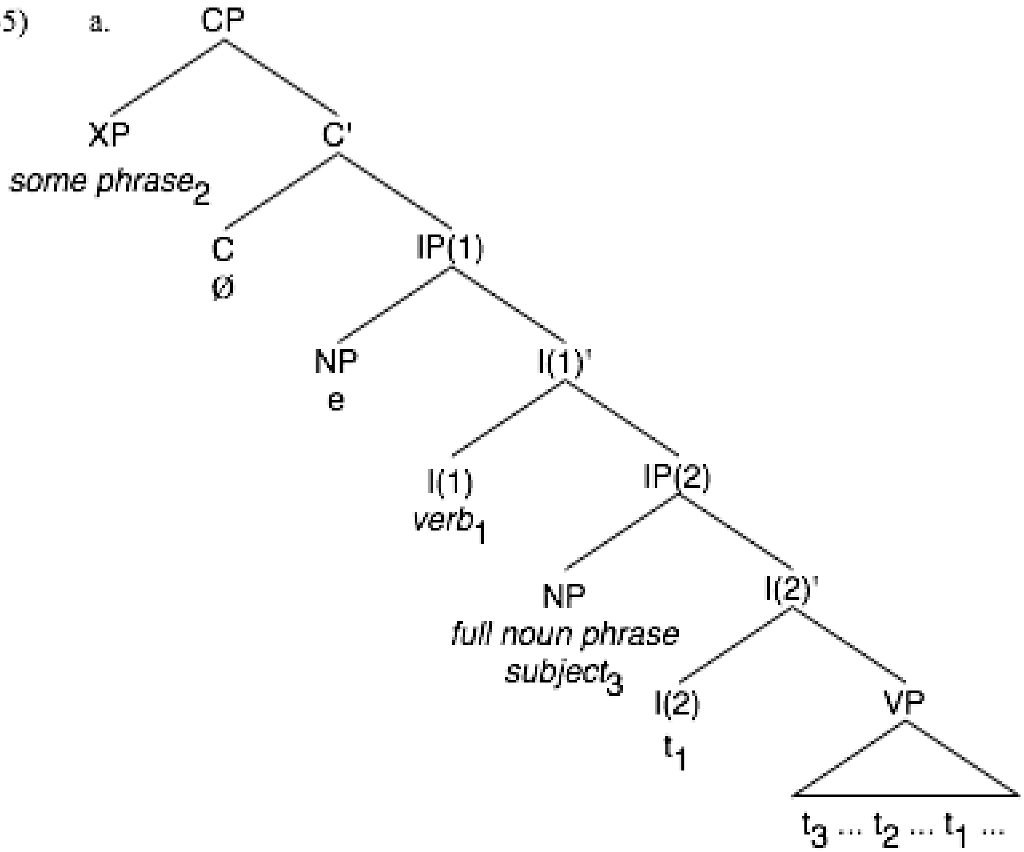
There is an alternative analysis of the V2/V3 alternation. This is the analysis in favour of which I am going to argue, so let me briefly introduce its main tenets. The idea is that the two surface structures, V2 and V3, are the direct reflex of different syntactic structures. The basic syntactic template of Old English thus has to be of a type that allows for both V2 and V3 sentences. This view has been endorsed e.g. by Kroch & Taylor

(1997) and Haeberli (2002). The Old English sentence structure thus would have to provide two subject positions, one lower for full noun phrase subjects and one higher for pronominal subjects. The landing site of the verb is between these two subject positions, and so we get the V2/V3 alternation without difficulties.[59] The structure would have to look like that in (4-5), (4-5a) being the structure with full noun phrase subjects, (4-5b) being the structure with pronoun subjects. Note that the V2 word order in (4-5a) is epiphenomenal under this view. It is not derived by CP-V2.

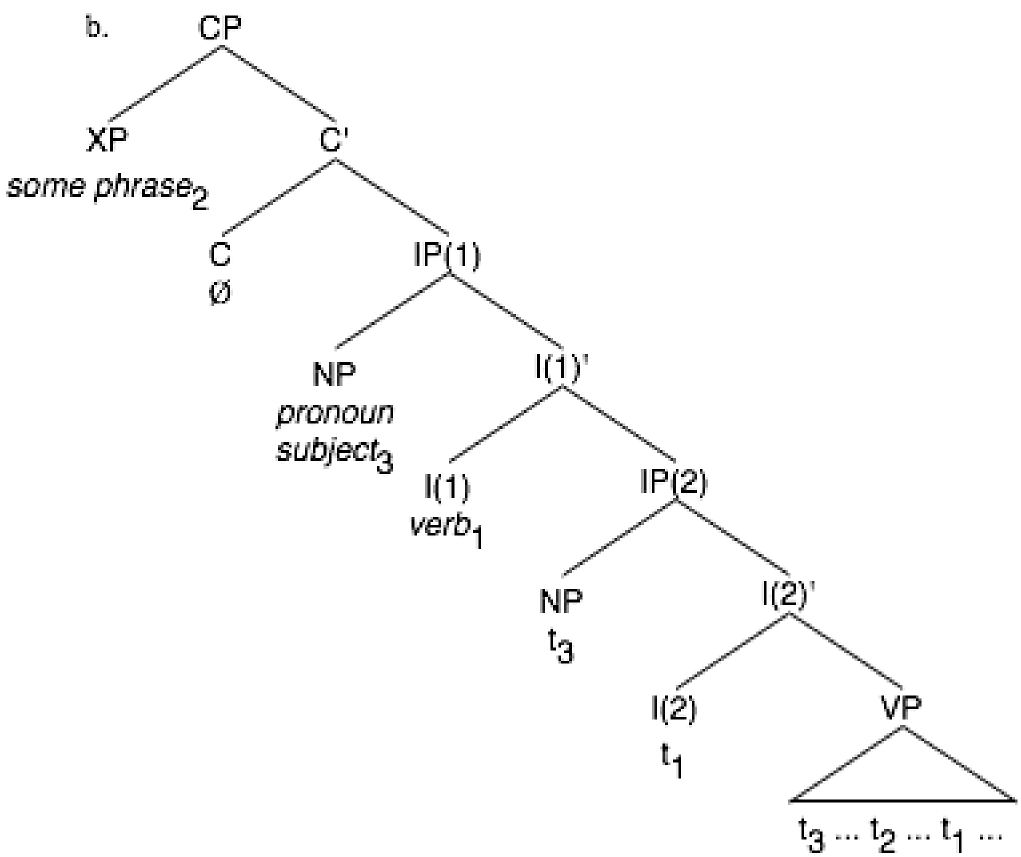
This analysis shows none of the problems that the cliticization analysis encounters. As a matter of fact, the mere presence of V3 sentences with lexical nouns as subject forces us to adopt the two-subject-positions-hypothesis. A die-hard advocate of a cliticization account could however argue that the examples of V3 with full noun phrase subjects are in reality verb last sentences, an alternative introduced earlier. Unfortunately, this objection is not as easy to refute as it might seem; yet it is possible and the issues connected with that and the ultimate evidence against this argument will be the topic of the next section.

[59] In fact there are some minor difficulties; there are issues such as what prevents the subject from rising always to the highest subject position. Haeberli (2002) assumes that a phonetically empty expletive occupies the higher position in such a case. I will return to this question in later sections.

(4-5) a.


CP
XP
some phrase$_2$
C'
C
Ø
IP(1)
NP
e
I(1)'
I(1)
verb$_1$
IP(2)
NP
full noun phrase subject$_3$
I(2)'
I(2)
t_1
VP
t_3 ... t_2 ... t_1 ...

b.


CP
XP
some phrase$_2$
C'
C
Ø
IP(1)
NP
pronoun subject$_3$
I(1)'
I(1)
verb$_1$
IP(2)
NP
t_3
I(2)'
I(2)
t_1
VP
t_3 ... t_2 ... t_1 ...

Let us summarize: The well-known V2/V3-alternation of Old English was exemplified, and two models to explain this variation were presented, the cliticization account following van Kemenade (1987) and the two-subject-positions account following Kroch & Taylor (1997) and Haeberli (2002). In contrast to the standard view that the two subject positions in Old English show a complementary distribution, in that pronouns stand in the higher position, and full noun phrases in the lower one, it could be shown that also full noun phrases can be found in the higher subject position, which weakens the cliticization hypothesis considerably.

4.2 Is V3 really V3?

In this section we investigate whether we can find evidence that there exist V3 sentences with Infl-medial structure in Old English. In the end we will find evidence, but it is not easy to come by. After having presented the intrinsic difficulties for V3 sentences, viz. that an Infl-final structure can be disguised by rightward movement of constituents and / or West Germanic verb raising (section 4.2.1) a method is presented that offers proof that at least some V3 sentences must be Infl-medial: We can calculate the expected number of V3 sentences under the assumption that they can be generated only by an Infl-final structure. These numbers are consistently lower than the numbers of observed V3 sentences, consequently at least some of the observed V3 sentences should be Infl-medial. This is the point of section 4.2.2. In 4.2.3, finally, a group of examples is presented that are unambiguously Infl-medial (sentences with particle verbs, where the particle follows the finite verb form) and it is demonstrated that, although they are rare, they are not so rare that they could be ignored as ‘slips of the pen’. Section 4.2.4 summarizes the preceding sections and demonstrates that a consequence of these findings is that Old English syntax was much farther away from the West-Germanic standard than previously assumed.

4.2.1 Verb-last sentences

Like all clauses, Infl-final clauses can exhibit a relative order X – S – … or, more specifically, O – S – … . In contrast to Infl-medial sentences (in which the leftmost constituent must have reached its place by topicalization), however, in Infl-final sentences this order can also be generated by scrambling.

Although the exact nature of scrambling is a matter of debate (for an overview of the different approaches see e.g. Trips 2002:174ff., or Corver & van Riemsdijk 1994 and the whole volume which the paper introduces), we can characterize it as a process in which the base-generated word order of the non-verbal constituents in a sentence is altered, so that the surface word order is not identical to the putative base-generated word order. In a generative framework it is most commonly assumed that scrambling is movement of phrases to positions below CP, either to specifiers of independent functional projections or Chomsky-adjoined positions (Haider & Rosengren 1998:7; Trips 2002:169).

German is a language in which scrambling is common and which in addition is closely related to English. Scrambling in German first means that the word order in the '*mittelfeld*', that is between C° and the clause-final V°/I°-complex, is not rigid, but allows for variation, often even without creating any special pragmatic effect (4-6; a preposed # indicates that a special context is needed if the utterance is to be felicitous). Note that in the examples (4-6) none of the permutations is ungrammatical.

(4-6) a. …dass Uller gestern Maria gesehen hat
that Uller yesterday Mary seen has
S – Adv – O

b. …dass Uller Maria gestern gesehen hat
S – O – Adv

c. …dass gestern Uller Maria gesehen hat

Adv – S – O

d. #…dass gestern Maria Uller gesehen hat

Adv – O – S

e. #...dass Maria gestern Uller gesehen hat

O – Adv – S

f. #...dass Maria Uller gestern gesehen hat

O – S – Adv

‘that Uller saw Mary yesterday’

There are constraints on scrambling in German (for an overview see Lenerz 1977; Eisenberg 1994:417ff.). This can be seen for instance in the fact that in general the examples with object before subject are more marked (which means: are felicitous in fewer contexts) than the examples with subject before object.

To find out what the constraints on scrambling were in Old English, the best way is to investigate Infl-final subordinate clauses, as here any non-canonical word order must be the result of scrambling. Topicalization is ruled out, since the target of topicalization, the C-projection, is already filled by subordinating conjunctions and complementizers.

We are mostly interested in sentences with full noun phrase subject and full noun phrase objects. As scrambling itself is not the focus of our interest, I confine myself to scrambling of full noun phrase objects over full noun phrase subjects, as this is the case that is most relevant for the line of research we are pursuing.

If we confine our search to subordinate clauses that are overtly Infl-final (because they have a complex verb form, whose members are in the relative order infinite – finite), we find no single example of full noun phrase objects before full noun phrase subjects (Table 4.4).[60]

Table 4.4: Rate of scrambling of full noun phrase objects over full noun phrase subjects in OE Infl-final subordinate clauses.

all Infl-final subordinate clauses containing a full noun phrase subject and object	108
thereof with relative order O – S	0
rate (%)	**0**

The number of clearly Infl-final subordinate clauses with fNP objects and subjects is not high, but there are enough for us to say that if none of them shows the order O – S, this order was heavily dispreferred. In other words: There was a strong constraint on scrambling of full noun phrases over full noun phrase subjects. Modern German, if we go back to examples (4-6), also shows a constraint on scrambling of a full noun phrase object over a full noun phrase subject: it is only possible if warranted by the context and only if there is a contrastive focus on the object. The German constraint does not seem to be as strong as the Old English constraint which, judging from the data alone may have barred scrambling of full noun phrase objects over full noun phrase subjects entirely.[61]

That however means, that we can rule out scrambling as the process underlying the order O – S in Infl-final main clauses with full noun phrase subjects and objects. Such sentences do occur, a search among unambiguous Infl-final sentences yielded 4 examples. This relative order must consequently be a consequence of topicalization. There is

[60] The search actually found two examples, one of which was a wrongly coded *man*, the other of which was a pronoun object with a putative trace of extraposed material.

[61] It might, however, be that this situation is simply so rare that it did not show up in the corpus by pure accident. It is not possible to state on this data that it was ungrammatical in Old English, of course.

nothing to hinder Infl-final sentences from exhibiting topicalization; in fact, strict Infl-final languages such as Japanese and Latin exhibit topicalization quite freely.

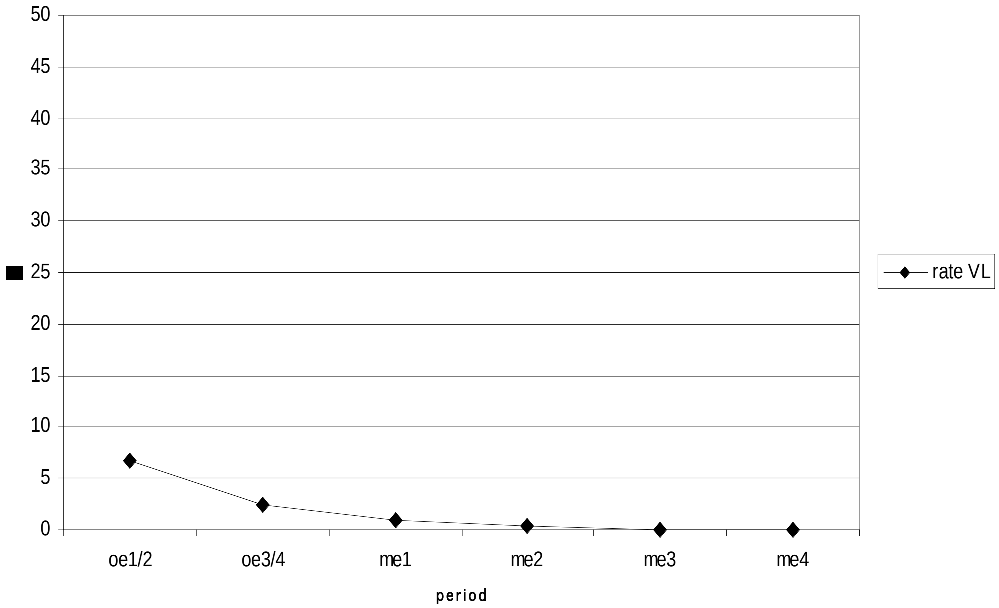
Let us now turn to Infl-final main clauses. We have seen that in Old English Infl-final main clauses are possible (4-1a). Such clauses are a heritage of Proto-Indo-European (which was Infl-final and where it was normal for the verb to remain in sentence-final position of any type of clause; cf. Lehmann 1974) but have died out in all of the Germanic languages at some point in their history. Old English is in a transitory state in that verb last main clauses have not entirely died out, but are in the course of being driven out in main clauses by the Infl-medial sentence type (cf. Pintzuk 1999). Table 4.5 and Figure 4.2 show an easy calculation, taking only the relative order of finite part of the verb form (I) and non-finite part of the verb form (V) into account. By the middle of the Middle English period, VL main clauses are gone.[62]

Table 4.5: Rate of unambiguous verb-last main clauses.

	oe1-2	oe3-4	**total OE**	me1	me2	me3	me4
all main cl. with V & I	4381	5537	**9918**	1576	1515	3441	1712
thereof order V-I	293	140	**433**	15	6	2	0
% V-I	**6.69**	**2.53**	**4.37**	**0.95**	**0.40**	**0.01**	**0.00**

[62] The two examples in me3 are not VL sentences but instances of gapping and verb topicalization that happen to conform to the query with which I searched for VL clauses in the Old English corpus.

Figure 4.2: Rate of unambiguous verb-last main clauses.



The examples where we can say for sure that we have a VL main clause have I° overtly filled and verbal material stranded in VP. The stranded material can be for instance the head of a lower verb phrase (4-7a) or a verbal particle (4-7b).[63]

(4-7) a. Ac […] þa studu […] þæt fyr gretan ne meahte.

But the pillar that fire challenge not could

'But that fire could not approach the pillar.'

(cobede,Bede_3:14.204.17.2076)

b. & hine se Godes monn up hof,

and him the God's man up lifted

'and the man of God lifted him up'

(cobede,Bede_2:9.132.22.1277)

[63] Probably also unstressed object pronouns, stranded prepositions, negative objects and possibly sentence adverbs (Santorini 1992; Haeberli & Pintzuk 2007).

It has turned out recently that this frequency is much too low (Haeberli & Pintzuk 2007). If other diagnostics are chosen, such as the position of verbal particles or negative objects relative to the verb, one arrives at percentages of verb-last between 16% and 56%. Later in the course of section 4.2 we will see that the phenomenon of verb raising considerably increase the apparently low rate to numbers similar to the ones reached by Haeberli & Pintzuk (2007). At any rate, we will be well advised not to use the low frequency of verb-last (and thus Infl-final) main clauses for any of our arguments.

If there is no verbal material stranded in VP, that is, if we have only one single verb form that has moved from V° to I°, we cannot be sure whether I° is to the left of VP or to the right of VP; in other words: whether the clause is Infl-final or Infl-medial. This however means that an overtly V3 sentence like (4-8) is ambiguous with respect to its analysis: It could either be an instance of an Infl-medial clause with topicalization or an instance of a verb/Infl-last main clause in which the object somehow has reached the left periphery, presumably by topicalization. With VL clauses it is in general hard to tell whether preposing is topicalization or scrambling, especially if pronouns are involved (for cases with only full nuon phrases scrambling has been shown essentially to be ruled out).

(4-8) & þa þing þe herunge wyrðe wæron, ic herede.
and the things that praise's worth were I praised
'and I praised the things that were worthy to praise'
(cobede,Bede_3:14.206.5.2090)

(4-9) a. & hit Englisce men swyðe amyrdon.
and it English men fiercely despoiled

‘and the Englishmen despoiled it fiercely’

(cochronE,ChronE_[Plummer]:1073.2.2681)

b. Forðon þa ærestan synne se weriga gast scyde þurh þa næddran,

For the first sin the wicked ghost incited by the adder

‘For the wicked ghost worked the first sin with the help of the serpent’

(cobede,Bede_1:16.86.28.791)

c. & usic þa ladteowas læddon þurh þa wædlan stowe wætres

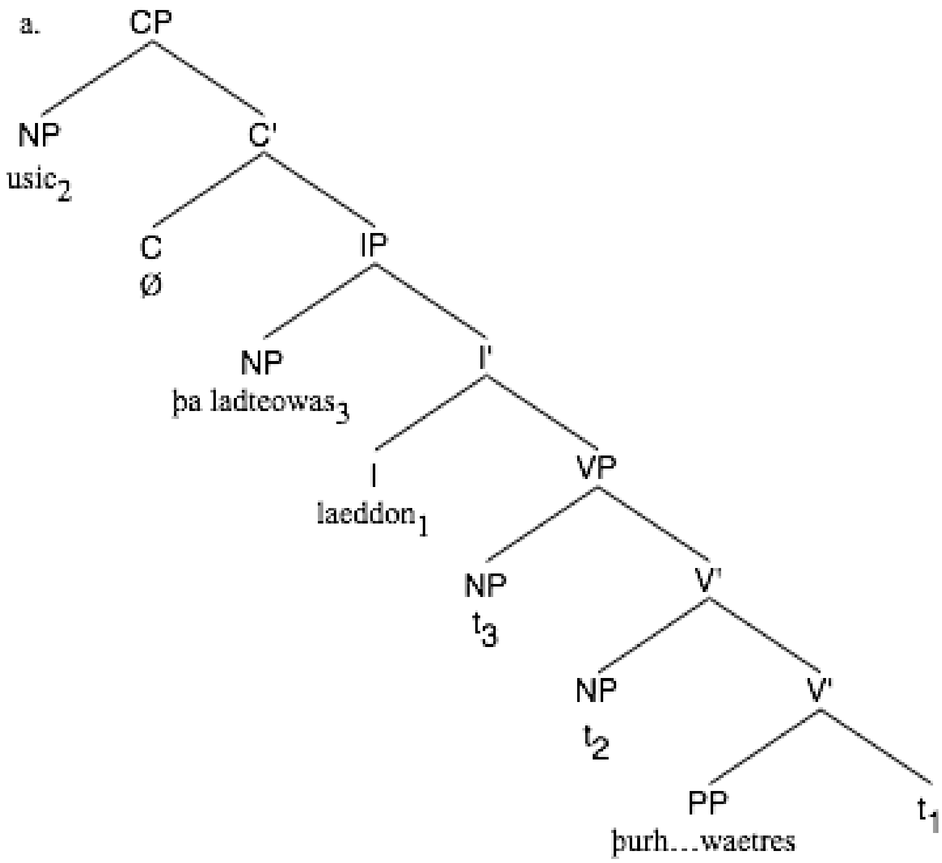
and us the teachers led through the poor place water’s

‘and the teachers led us through the place with little water.’

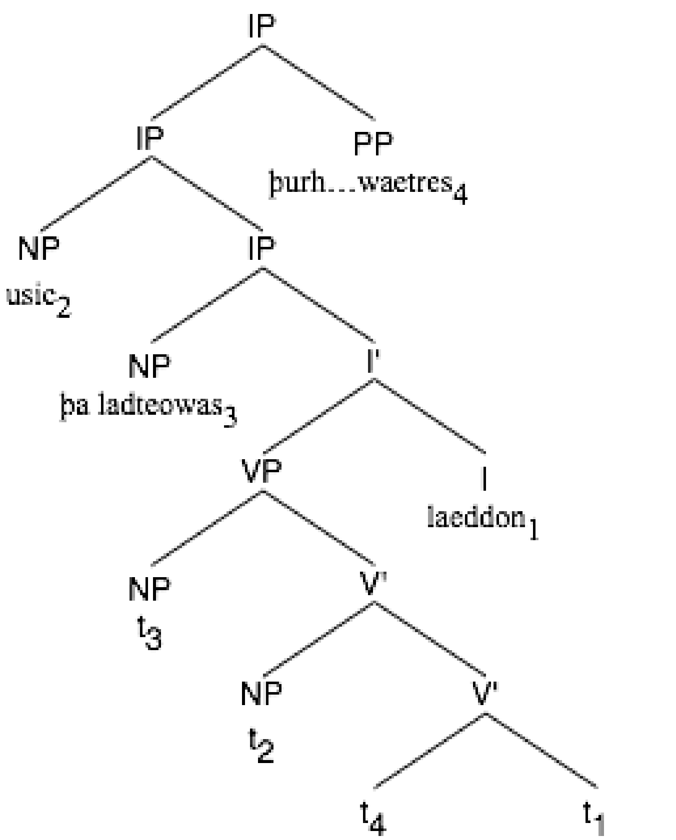
(coalex,Alex:33.9.424)

Sentences (4-2b) = (4-9b) and (4-9c), seem like ‘real’ V3-sentences, in that the verb stands in third place and is not the last element in the sentence. But such cases are in fact as ambiguous as the seemingly straightforward ‘overt’ VL cases like (4-8), which could be Infl-final or Infl-medial with fronting of the object together with its relative clause. What makes such cases ambiguous is the fact that there are rules of rightward extraposition in English, so that material appearing to the right of the verb can either be material in the VP that has remained in situ while the verb has moved up to I° (4-10a), or it can be rightward-moved material (4-10b; see also Haeberli & Pintzuk 2007). In this case, it is not possible to determine whether I° is to the right or the left of VP.

(4-10) a.


CP
NP
usic2
C'
C
Ø
IP
NP
þa ladteowas3
I'
I
laeddon1
VP
NP
t3
V'
NP
t2
V'
PP
þurh…waetres
t1

b.


IP
IP
PP
þurh…waetres4
NP
usic2
IP
NP
þa ladteowas3
I'
VP
I
laeddon1
NP
t3
V'
NP
t2
V'
t4
t1

In Modern as in Old English there are at least two quite distinct rightward movement operations, namely Heavy-NP-shift (4-11a) and rightward Extraposition (4-11b; cf. to the following Pintzuk & Kroch 1989). In Modern English they show distinct properties: Whereas Heavy NP-shift is restricted to prominent noun phrases and leaves a trace at the place where it was base-generated, Extraposition affects mostly prepositional phrases in Modern English. Heavy-NP-shift is also intonationally distinct from Extraposition: with Heavy-NP-shift the rightward moved constituent forms an intonational phrase of its own (as can be seen from the fact that the remainder of the sentence shows clause-final intonation), whereas with Extraposition the extraposed constituent stays in the same intonational domain as the remainder of the sentence.

(4-11) a. I gave t_i to Mary [several old records of mine]$_i$.

b. Rockefeller gave [a picture $\_\__i$] to the museum [of his late wife]$_i$.

Using the metrical properties of Beowulf, Pintzuk & Kroch (1989) were able to show that both constructions existed also in Old English and that Heavy-NP-shift showed the same intonational characteristics as today. Judging from the data of clear verb last sentences (i.e. with the order … V – I) only prepositional phrases and subordinate clauses were subject to rightward Extraposition, again just as today. An example of PP-extraposition in Old English is (4-12).

(4-12) ac he begyrded wæs [mid wæpnum þæs gastlican camphades].

and he girded was with weapons the.GEN holy warfare.GEN

‘and he was girded with weapons of holy warfare.’

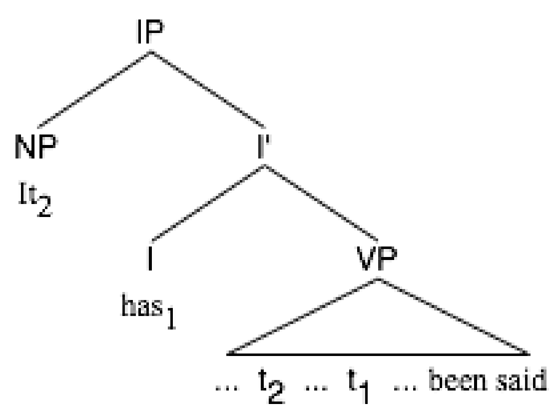
(cobede,Bede_1:7.36.10.291)

We can see that in all clauses in which a noun phrase, a prepositional phrase or a subordinate clause follows the verb the analysis is ambiguous between an Infl-medial structure with the material after the verb in situ and an Infl-final structure. Thus it looks as if most overtly V3 sentences will be structurally ambiguous. This means that all of the V3-sentences with full NP subject could easily be VL sentences and provie no evidence against a V2 analysis of Old English clauses.

Consider now the following line of thought. If a sentence contains a complex verb form, it looks as if we could directly see how VP and I° are serialized: VP (represented by the non-finite verb form) stands either before or after I° (represented by the finite verb form). That means, as the position of VP is fixed, we can see whether the finite part of the verb form, I°, is to the left or to the right of the non-finite part. If the order is 'finite – non-finite', as in the Modern English example (4-13a, b), the structure is Infl-medial, if the order is 'non-finite – finite', as in the Latin example (4-13c, d), the structure is Infl-final.

(4-13) a. It has been said, that the Gauls inhabit this part

b.



c. (Eorum una pars) quam Gallos obtinere dictum est,

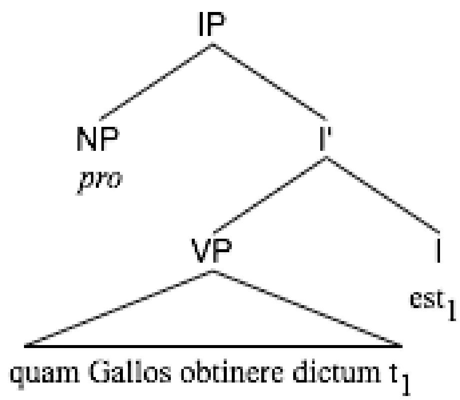
Of-those one part which Gauls hold said is

(initium capit a flumine Rhodano)

beginning takes from river Rhone

'One part thereof, the one which has been said that the Gauls inhabit, starts at the Rhone river.' (Caesar, Comm. de bello Gallico, 1.1.5)[64]

d.



Unfortunately, this scenario is too simple. Although it is true, that the order non-finite – finite always indicates an Infl-final structure, the reverse is not true: The order finite – non-finite is not always the outcome of an Infl-medial structure, but can be the result of West Germanic verb raising.

Verb raising is a process by which material from the inner VP-shells of a complex verb form is moved to an adjoined position to the right of the clause. Analysts universally agree that verb raising sentences occur in languages that are otherwise strictly Infl-final. Therefore verb raising is treated as a type of structurally Infl-final clauses rather than Infl-medial. We distinguish between pure verb raising, in which only the non-finite verb is moved, and verb projection raising, in which the non-finite verb pied pipes material from

[64] Cited from TITUS (http://titus.uni-frankfurt.de/indexd.htm)

its VP. Verb raising is common in the continental West-Germanic languages, such as Dutch (4-14a) and dialectal German (4-14b), and it also occurs in Old English (see on verb raising in general e.g. Haegeman & van Riemsdijk 1986; Kroch & Santorini 1991; Haegeman 1994; in OE e.g. van Kemenade 1987).

(4-14) a. …dat Jan het boekje wilde hebben (base-generated: hebben wilde)

that John the booklet want have

'that John wants to have the little book'

b. … dass Hannes das Buch wollte gelesen haben

that John the book wanted read have

(base-gen.: gelesen haben wollte)

'that John wanted to have read the book'

Let me illustrate with an example: In (4-15a) below it looks on first glance as if we have an unambiguous example of an Infl-medial V3 sentence, as the inflected part of the complex verb form, *scealt*, precedes the infinitival part, *gesettan*. This suggests an Infl-medial structure as in (4-16a) (on page 277).

It could however also be the case that the non-finite verb has been moved to the right. In that case, the sentence would then have an Infl-final structure as in (4-16b) and we could not say for sure whether the sentence was Infl-medial or Infl-final..[65]

[65] The syntactic details of verb raising are somewhat unclear; I do not wish to commit myself to the analysis as presented here. I adopt it merely for the sake of illustration, as an example for an analysis that would make an I-V sentence out of an Infl-final clause.

Such examples are not rare; (4-15b) = (4-2c) and (4-15c) are further examples. Example (4-15c) would be an example of verb projection raising, i.e. pied piping of VP-internal material along with the raised non-finite verb.

(4-15) a. Þas drohtunge & þis liif þu scealt gesettan […].

this conduct and this life thou shalt set

'You shall follow this conduct and this life.'

(cobede,Bede_1:16.64.21.604)

b . Forðon hie nan monn ne dearr ðreagean ðeah hie agyltan,

Therefore them no man not dares punish even-if they sin

'Therefore no man dares to punish them, even if they commit a sin.'

(cocura,CP:2.31.12.138)

c. Ac ða lufe mon mæg swiðe uneaðe oððe na forbeodan;

And the love one may very-much hardly or not-at-all refuse

'and the love one can refuse very hardly, if at all.'

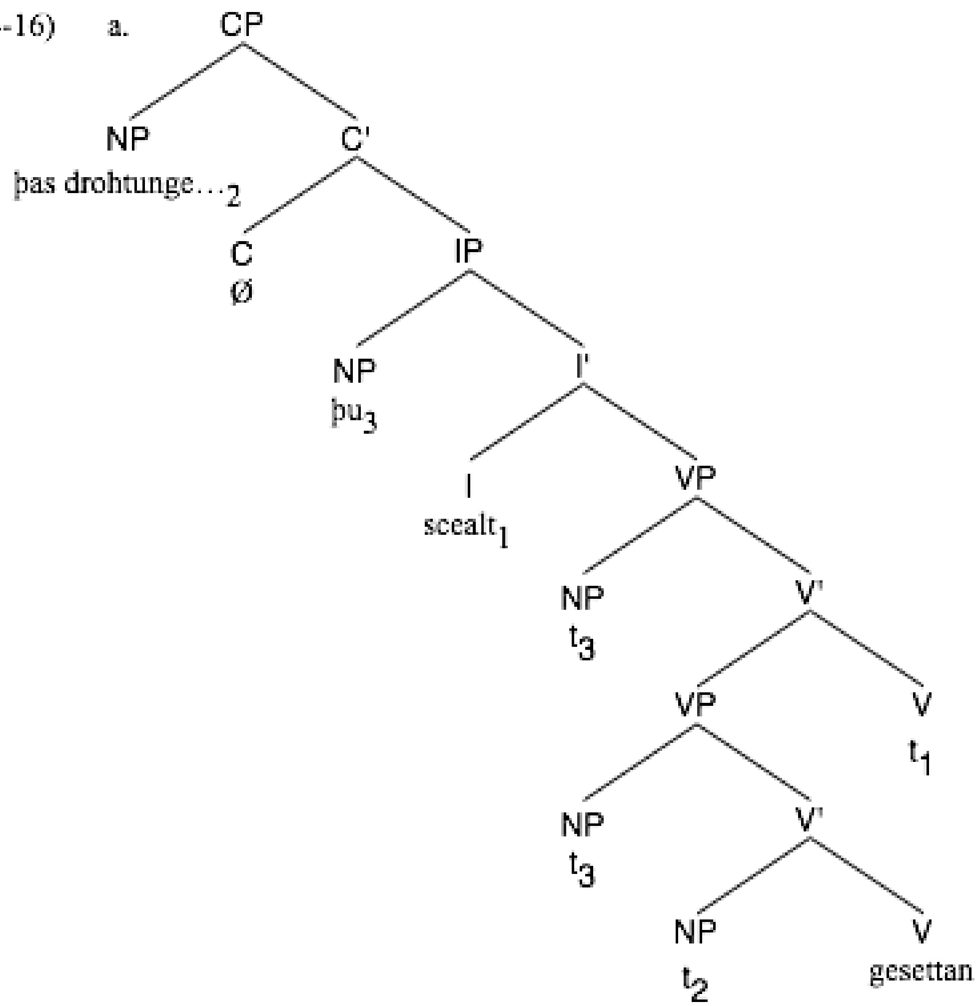
(coboeth,Bo:35.103.9.2001)

Our problem is the following: Verb Raising sentences are Infl-final sentences in disguise: They look like Infl-medial sentences, but they are not. This means that some portion of the roughly 95% of overtly non-Infl-final sentences is Infl-final in structure. It is important to know this proportion, as we would like to use this number as a corrective to our number of overtly Infl-final sentences. If the proportion is extraordinarily high, there might be more Infl-final main clauses in Old English than previously thought. This would however mean that the case for V3 with full noun phrase subjects would be considerably weakened, as with a high rate of Infl-final main clauses, one could say that all of the examples of V3 with full noun phrases could easily be Infl-final.

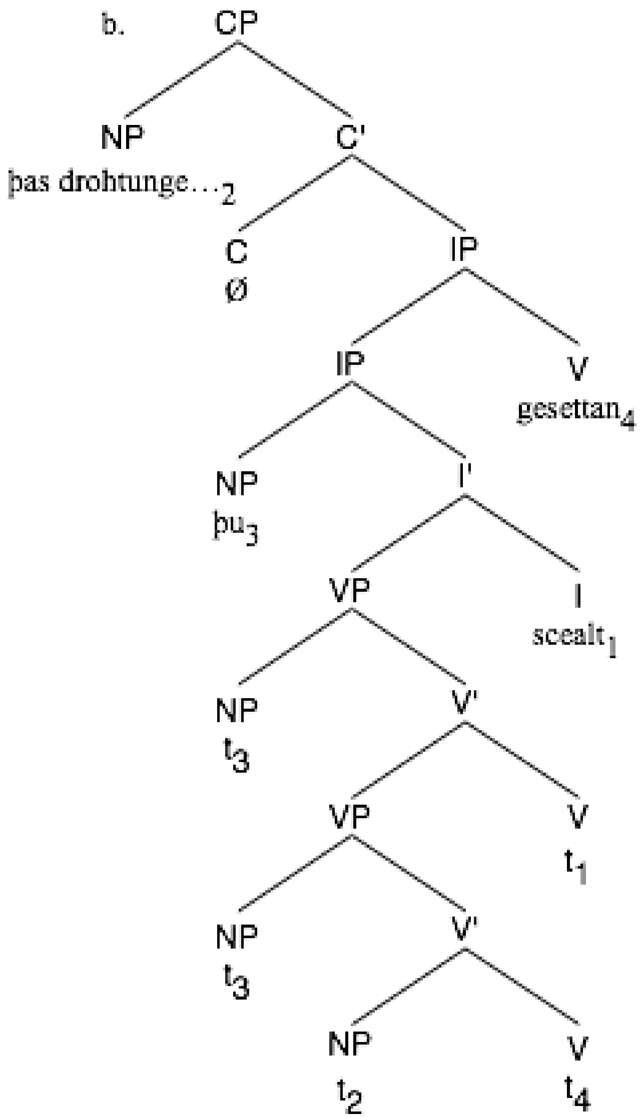
It seems clear that from the viewpoint of simple surface word order we cannot prove that the V3 sentences with full noun phrase subject are Infl-medial. For (almost) all of them an Infl-final analysis with verb raising and / or rightward dislocation is possible.

There are, however, other ways to make valid statements about the underlying structure of such ambiguous clauses. These are the subject of the next sections.

(4-16) a.



b.



4.2.2 Verbal particles

Fortunately, evidence exists that at least some V3 sentences must be Infl-medial. There is a small number of sentences that can receive only an Infl-medial analysis: sentences with a verbal particle. Some examples are given in (4-17).

(4-17) a. Þæne se geatweard læt in

The.ACC the doorkeeper allows in

'This one the doorkeeper admits'

(cowsgosp,Jn_[WSCp]:10.3.6596)

b. and him se innoþ eac geopenode ongean

and him the heart also opened again

'and for him the heart opened again'

(coaelive,+ALS_[Vincent]:170.7907)

c. Her Æþelheard cyng ferde forð,

In-this-year Ethelhard king went away

'In this year, king Ethelhard went away'

(cochronC,ChronC_[Rositzke]:740.1.331)

d. Ac þære ilcan niht […] wulfas atugan þa stacan up,

but the.DAT same night wolves drew the pins up

'But in the same night wolves (= devilish persons) drew up the pins'

(coorosiu,Or_5:5.119.25.2504)

Such examples can only have an Infl-medial structure. The reason for this is that the verbs of these sentences are particle verbs with the particle standing after the verb. When a verb moves leftward, its particle, for example *in* in (4-17a), does not move together with the verb but stays in its base-generated position. It also never moves rightward (cf. Haeberli & Pintzuk 2007). Thus, we can directly see the relative order of VP (indicated by the stranded verbal particle) and I° (indicated by the finite verb). This line of reasoning might sound familiar; in fact, we already used a similar argument when we discussed complex verb forms. In contrast to complex verb forms, however, we can be absolutely sure that the verbal particle is really in VP because there is no movement process similar to verb raising which might move the stranded diagnostic material out of VP. If we find a clause in which a verbal particle follows the finite verb form, we can be sure that VP, the position of which is marked by the particle, stands after I°, the landing site of the finite verb form.

Unfortunately, there are not many examples of V3 sentences with a verbal particle after the verb. In table 4.6 we see that there are only 20 examples with full noun phrase subject in the corpus, all but one of them with a preposed adverbial phrase or PP.

Table 4.6: Number of V3 sentences with particle verbs (X – S – V – Ptc.) in OE.

	full NP subject	pronoun subject
X = PP/AdvP	19	35
X = object (dat + acc)	1	10

One could argue that these few examples out of a million-word-corpus of Old English could be only slips of the pen again and could have no argumentative value. The smallness of this number is, however, less troubling if we compare it to the number of V2 clauses with verbal particles (X – V – S – Ptc.). If Infl-medial V3 were really only a marginal error, the number of V2 sentences, which would be the ‘correct’ way to form an Infl-medial sentence under this hypothesis, should be so large that the ‘erroneous’ V3 sentences should be occur in only a negligeable number of cases, say, less than one percent. This is however not the case; Table 4.7 presents the numbers for V2-sentences and gives the proportion of V3 sentences out of V2 plus V3.

Table 4.7: Number of V2 and V3 sentences with particle verbs (X – S – V – Ptc.) in OE.

	full NP subject	pronoun subject
numbers of V2		
X = PP/AdvP	70	6
X = object (dat + acc)	4	0
numbers of V3		
X = PP/AdvP	19	35
X = object (dat + acc)	1	10
numbers of V2 + V3		
X = PP/AdvP	89	41
X = object (dat + acc)	5	10
proportion of V3 (%)		
X = PP/AdvP	**21.3**	**85.4**
X = object (dat + acc)	**20**	**100**

The first three lines give the numbers of sentences with particle verbs in which we have V2 word order. Lines 4 to 6 give the numbers of sentences with particle verbs in which we have V3 word order. Lines 7 to 9 give the total number of sentences, V2 and V3 combined, and lines 10 to 12 give the proportion of V3 sentences among the cases. The first line of each group are cases in which an adjunct (prepositional phrase or adverb) is preposed, the second line of each group are cases in which an object is preposed. We see that about a fifth of all main clauses with particle verb, topicalization and full noun phrase subject are V3. This proportion is consistent, regardless of what the nature of the preposed element is, and it is decidedly too high to be due to error. With pronominal subjects V3 is almost the rule, but this is what we expected anyway.[66] So we can say that this calculation offers another indication for the correctness of the hypothesis that Infl-

[66] How to account for the 6 exceptions is hard to say, they probably have another analysis (for instance CP-V2).

medial V3 with full noun phrase subjects exists and that Old English V3 sentences therefore must receive an analysis with two subject positions.

4.2.3 Modelling

Even if we did not have the direct evidence of particle verbs, we could still prove that not all Old English V3 sentences can be Infl-final. There are several ways to determine quantitatively whether there are Infl-medial V3 sentences in Old English. One way would be to find a way to estimate the rate of Infl-final sentences and compare this to the number of ambiguous (Infl-final or V3) sentences hat we find. If the latter number is considerably higher than expected from the estimated rate of Infl-final clauses, then we might conclude that some of the ambiguous sentences must be V3. An important piece of information that Table 4.5 and Figure 4.2 offer is that the rate of unambiguous Infl-final sentences is quite low in Old English: 6.7% in earlier Old English, 2.5% in later Old English. As soon as both I° and a lower V° (that necessarily is adjacent to the V° where the verbal form now in I° has been base generated) are overtly filled, we can see directly where I° is relative to V°: If the relative order is V – I, the clause can only be Infl-final. So with such clauses it is possible to say for sure how many of them must be underlyingly Infl-final. We might then use this number to calculate an estimate of the rate of Infl-final word order and infer that the same proportion of Infl-final clauses also applies to the cases that are ambiguous on the surface.

If life were easy, this could mean that, without being able to detect which overt V3 clauses are Infl-medial and which Infl-final, we could say that of all ambiguous V3 sentences in the earlier periods of Old English, roughly 7% must be supposed to be Infl-final, and of all ambiguous sentences in the later periods, roughly 3% must be supposed to be Infl-final. If we now knew the rate of verb-last clauses, we could assume that the same rate applies to the ambiguous clauses, and we could compare this rate to the rate of V3 clauses with full NP subjects. If it turned out that the rate of verb last clauses was significantly lower than the rate of V3 clauses with full NP subject, which it is, we could conclude that at least some of the V3 sentences with full noun phrase subject could not be Infl-final, hence had to be Infl-medial (for the method cf. e.g. Santorini 1992; 1995). That would be enough to make our point.

Unfurtunately, this method is not valid because we cannot say that all clauses that are not certainly Infl-final (93% and 97% respectively of main clauses) are certainly Infl-medial. After all, the order I – V can be generated by verb raising and we do not know its rate. Haeberli & Pintzuk (2007) argue that the rate of Infl-final clauses is quite high. Hence, we cannot use the number of certain Infl-final clauses as the basis of our argumentation.

What we will have to do is to take the rate of verb raising explicitly into consideration in our estimates. This is possible, although the reasoning behind the estimation procedure is somewhat complex. Let us limit ourselves, for the moment, to sentences with complex verb forms, as there we can see better what is going on. If we know the number of overt Infl-final sentences and the rate of verb raising among the seemingly Infl-medial sentences, and moreover know the rate of topicalization in Infl-final clauses, we can calculate from these numbers, how many V3 sentences we would expect in general (that is: also with simple verb forms), under the assumption, that all V3 sentences were Infl-final. This number can be compared to the observed number of V3 sentences in the corpus. If the calculated number is equal or even higher than the number of observed sentences, we could conclude that all V3 sentences were underlyingly Infl-final. If the calculated number is significantly lower than the number of observed V3 sentences, we can conclude that not all V3 sentences can be underlyingly Infl-final, because there are just too many of them.

In the following I present these calculations. In order to increase the data material, all Old English texts from the YCOE, including those that cannot be assigned a period for sure, have been taken into account.

The crucial number that we need to know is, of course, the rate of verb raising, which can estimated in the following way:

Suppose a number of clauses which contain a subject (S), a verb in I° (I), another verb in a lower verb phrase (V) and some other constituent, preferably an argument (X). If we assume that the subject is in front, we get the following permutations of the remaining constituents:

1. S – X – I – V
2. S – X – V – I
3. S – V – X – I
4. S – V – I – X
5. S – I – V – X
6. S – I – X – V

Now let us look at how these different word orders can be generated.

Word order 1 can arise only by verb raising. If there is some argument or VP-internal adjunct between the subject and I, this clearly indicates that the sentence must be Infl-final. If that is so, however, V can end up to the right of the verb only by verb raising.

Word order 2, on the other hand, cannot be the result of verb raising, simply because V is to the left of I. The same goes for 3, but the OE grammar seems not to produce sentences of this type, that is, with medial V but final Infl. This parametrization is apparently excluded universally (see Pintzuk 1999). Word order 4 is clearly not the result of verb raising, either, for the same reason as 2 and 3 are not. Word order 4 can only be an Infl-final clause in which X has been moved rightward.

Finally, word orders 5 and 6 are ambiguous. 5 can be either the result of Infl-final syntax with verb raising and subsequent rightward extraposition/shift of X or of Infl-medial syntax with rightward movement of X. Finally, word order 6 can be either the product of verb projection raising out of an Infl-medial clause or an Infl-medial clause with V-final complement.

Let us now repeat the word order options and indicate which ones are sure cases of verb raising and which are not.

1. S – X – I – V   + VR
2. S – X – V – I   - VR
3. S – V – X – I   - VR
4. S – V – I – X   - VR
5. S – I – V – X   ± VR
6. S – I – X – V   ± VR

We cannot directly estimate the rate of verb raising because types 5 and 6 are a mix of verb rasing with Infl-medial in the first case and of verb projection raising with Infl-medial in the second. What we can do instead is to estimate the ratio of simple verb raising to unmodified Infl-final word order. The ratio is simply:

$$r = v / s$$

where v = number of clauses with word order 1 and s = number of clauses with word order 2. If the rate of topicalization in Infl-final main clauses is independent of the rate of verb raising, we can use this ratio to estimate how often XSIV word order should occur if it is always structurally Infl-final. We can then compare this expected frequency with the actual frequency of XSIV. If the actual frequency is substantially higher than the expected frequency, we can conclude that at least some cases of XSIV are Infl-medial.

Tables 4.8 and 4.9 give the rate *r* for X = accusative object (4.8), X = dative object (4.9); Table 4.10 combines accusative and dative objects. The numbers are given for main and subordinate clauses and for full NP and pronominal subject. To arrive at these numbers, only full noun phrase accusative and dative NPs that are not coded as adverbial in the corpus (and thus are most likely to be objects) have been taken into account.

Table 4.8: Rate of verb raising, full noun phrase accusative objects.

	main clause, full NP subject	main clause, pronoun subj.	subord. clause, full NP subject	subord. clause, pronoun subj.
number of SOIV	8	19	18	99
number of SOVI	15	39	102	434
rate *r*	0.53	0.49	0.18	0.23

Table 4.9: Rate of verb raising, full noun phrase dative objects.

	main clause, full NP subject	main clause, pronoun subj.	subord. clause, full NP subject	subord. clause, pronoun subj.
number of SOIV	6	4	8	32
number of SOVI	5	6	24	108
rate *r*	1.2	0.67	0.33	0.30

Table 4.10: Rate of verb raising, full noun phrase accusative and dative objects combined.

	main clause, full NP subject	main clause, pronoun subj.	subord. clause, full NP subject	subord. clause, pronoun subj.
number of SOIV	14	23	26	131
number of SOVI	20	45	126	542
rate *r*	0.7	0.51	0.21	0.24

To begin with, we observe several things that are interesting in themselves, but are not important for the argumentation in this thesis and therefore will be left for further research. Thus, we see that the rate of verb raising is dramatically higher in main clauses than in subordinate clauses, and it would be interesting to investigate what might cause this asymmetry. We also see that the rate of verb raising is consistently higher in clauses with dative object than in ones with accusative object. But this is not crucial for the line of thought of this thesis.

Let us now calculate the crucial numbers. The number *t* is the ratio of object preposing across the subject to non-preposing in Infl-final clauses. Note that in such clauses, the object can be moved to the left of the subject either by scrambling or by topicaliation. We have seen that with full noun phrase objects and full noun phrase subjects, the relative order O – S is, in practice, always generated by topicalization. The number *t* is calculated by dividing the number of attested SOVI clauses (o_1) by the number of attested OSVI clauses (o_2).

$t = o_2 / o_1$

The number is calculated in Table 4.11 below.

Table 4.11: Ratio of object preposing, full noun phrase accusative and dative objects combined.

	main clause, fNP subject	main clause, pronoun subj.	subord. clause, fNP subject	subord. clause, pronoun subj.
number of OSVI	4	5	12	5
number of SOVI	20	45	126	542
ratio *t*	0.2	0.11	0.10	0.01

If we know the number *i* of unambiguous Infl-final clauses with canonical word order, that is, the number of SOVI clauses, and the ratios *r* and *t*, we can now calculate, how many verb third clauses with topicalization (that is: OSIV) clauses we would expect under the assumption that V3 can only have an Infl-final structure. Let that number be *g*, the product of *i*, *r* and *t*.

$g = i \times r \times t$

We now compare the number *g* to the number of actual OSIV examples from the corpus, which we can call *h*. The results of our calculation are given in Table 4.12. The calculations are given separately for main clauses in which the subject is a full NP, and for main clauses in which the subject is a pronoun, ditto for subordinate clauses. Accusative and dative objects have been combined.

Table 4.12: Predicted and actual number of V3 clauses, complex verb form.

	main clause, fNP subject	main clause, pronoun subj.	subord. clause, fNP subject	subord. clause, pronoun subj.
f (number of SOVI)	20	45	126	542
rate *r* (verb raising)	0.7	0.51	0.21	0.24
ratio *t* (object prepos.)	0.2	0.11	0.10	0.01
g (predicted number of OSIV)	2.8	2.6	2.5	1.2
h (real number of OSIV)	22	219	2	7
h / g	7.9	84.2	0.8	5.8

It is obvious that the actual number of observed V3 main clauses is significantly higher than the number which we would expect if all V3 clauses had an underlying Infl-final structure. It is eight times higher with full noun phrase subjects and as much as 84 times higher with pronoun subjects. Thus it is clear that most of the V3 examples with full noun phrase subject must be Infl-medial. However, that means that they can only be generated by a syntactic system like the one suggested by Haeberli (2002), which offers two landing sites for the subject in the left periphery. Obviously, a cliticization analysis is impossible in V3 cases with full noun phrase subject, so this result confirms our rejection of a CPV2 analysis for Old English declarative sentences. The one aspect of a cliticization approach that is preserved in Haeberli's analysis lies in his assumption that pronominal subjects cannot occur in the lower position. This restriction is a reflection of the informational character of pronouns as denoting old information but it is also consistent with their prosodic weakness, as weak elements tend to move leftward in the Germanic clause.

Note that our model predicts the correct number of fronted objects in the case of full noun phrase subjects in subordinate clauses. This is not surprising, as we know that subordinate clauses in Old English are predominantly Infl-final. The fact that the number is in the same range in this case supports the model.

A similar calculation to the one just presented can be performed for sentences with a simple verb form. Recall that these sentences are, in general, multiply ambiguous with respect to their analysis as we cannot see the relative order of I° and VP. In consequence, any sentence can be Infl-final as almost any constituent occurring after the verb could have been moved there by one of the rightward movement operations described in section 4.1. The exceptions are particles, non-subject pronouns, and possibly adverbs (see Haeberli & Pintzuk 2007), as they are assumed not to be able to undergo heavy-NP-shift or Rightward Extraposition. That means: A sentence such as (4-18) must be structurally a Infl-medial sentence, as the verb is followed by the ethical dative *him*.

(4-18) þa scylde se Pascasius ne gelyfde na him to synne,
the.ACC offence the Pascasius not concede not-at-all him.DAT to sin
'Pascasius did not concede this offence to be a sin for himself.'
(cogregdC,GDPref_and_4_[C]:43.331.27.4999)

Such examples are, however, extremely rare. Example (4-18), for instance, is the only instance of a sentence with a topicalized accusative object, a full noun phrase subject, and a post-verbal pronoun or adverb in the period oe2.

Given the scarcity of the examples, the evidence for unambiguous Infl-medial V3-sentences from simple verbs is by itself inconclusive. A sentence like (4-18) could be a 'slip of the pen', not reflecting the real grammar of Old English faithfully. And, of course, it could also be the case that it is only by chance that there are no examples for extraposed adverbs in Old English. Thus we cannot say for sure that it was ungrammatical to have extraposition of adverbs (although it would be unlikely, given that it is impossible in Modern English and at least so rare in Old English that we find no examples). So we cannot use examples of this kind for a conclusive answer.

But we can use our quantitative model for such sentences; the argument is identical to the argument with complex verb forms: If we know f_s, the number of SOV-sentences (that is, sentences that are overtly Infl-final), and if we know t, the ratio of object preposing, we can calculate, how many OSV sentences – that is: V3-sentences with topicalized object – we would expect under the assumption that V3-sentences can only have an Infl-final structure. This number – let us refer to it as g_s – is the product of f_s and t, and can be compared to the observed number of OSV sentences, which we can refer to as h_s.

The value of the ratio t should be identical to the value that we calculated for sentences with complex verb forms. This is because it is reasonable to assume that the shape of the verb form, whether complex or simple, should be independent of the tendency to prepose an object. Santorini (1993) found exactly this to be the case with regard to the tendency to extrapose objects. So we feel justified in using the value of t which we have found with clauses with complex verb form.

Table 4.13 presents the necessary data and calculations for clauses with a simple verb form. The calculations have been done for main clauses in which the subject is a full NP, and for main clauses in which the subject is a pronoun, ditto for subordinate clauses. Accusative and dative objects have been combined.

Table 4.13: Predicted and actual number of V3 clauses, simple verb form.

	main clause, fNP subject	main clause, pronoun subj.	subord. clause, fNP subject	subord. clause, pronoun subj.
f_s (number of SOV)	649	1133	881	3836
ratio *t* (object prepos.)	0.2	0.11	0.10	0.01
g_s (predicted number of OSV) = $f_s \times t$	129.8	125.9	83.9	35.4
h_s (real number of OSV)	153	1090	76	53
h_s / g_s	1.2	8.7	0.9	1.5

Again, we see that for main clauses, the model consistently predicts lower numbers for topicalized objects than we can observe. Consequently the same conclusion holds that was reached at when we looked at clauses with complex verb forms: The difference can only be explained if we allow V3 clauses with Infl-medial structure. Although the ratio of h_s to g_s is not much above one in the crucial first column of the table, the number of excess cases is large enough to support our conclusions. Furthermore, the ratio in the second, pronoun subject column is also much lower than in the complex verb case, for reasons that are unclear. Since we know that the OSV sentences with pronoun subjects are largely topicalizations, whatever is causing the drop in the ratio is likely to be unconnected to our line of argument. Moreover, we can be confident that our procedure has underestimated the true ratio, because we have treated every case of SOV word order as structurally Infl-final, whereas we know that there was some scrambling of objects to a postion immediately before the non-finite verb in structurally SVO sentences in Old and Middle English. This scrambling, which occurred at least with negative and quantified objects, occurred at a rate that we cannot determine from our data but its effect on our estimates is clear. The number of structurally Infl-final sentences with simple verbs is somewhat lower than the number of surface instances. Hence, the true ratio of structurally topicalized to structurally Infl-final sentences (h_s/g_s) is somewhat higher than the one we give.

4.2.4 The consequences of Infl-medial V3

Summarizing this section we can say that we have seen strong evidence, if not proof, that overt V3 order (with full noun phrase subjects) can be generated by an Infl-medial structure. This means that we are forced to assume a sentence template with two subject positions, one on each side of the landing site of the finite verb in the style of Kroch & Taylor (1997) and Haeberli (2002). Now we should ask what consequences this finding has. The appeal of the cliticization account was that it allowed Old English syntax to follow a strict version of V2 (CP-V2, to be precise, or even IP-V2, if SpecIP is seen as a freely available topic-position as in Pintzuk 1999), because the examples of V3 word order could be explained away by the clitic behaviour of subject pronouns, which slip to the left of the verb, without affecting the underlying syntactic structure. It has become customary to treat the apparent ‘exceptions’, that is: V3 sentences with a full noun phrase subject, as Infl-final sentences.

But we have seen that not all of them can have an Infl-final structure. This means that Old English did not have a strict CP-V2 structure like Modern German does, but rather a structure with an optional CP that reminds us in some ways of the Modern English sentence template. This however means that the ‘split’ between English CP-optional syntax and the strict CP-syntax of the remaining West-Germanic languages must have occurred not at some time during the Old and Middle English period, but before our Old English sources were written, as the split is already completed when the transmission of Old English begins. The possibility is available, then, that the split might already have occurred at the time when English began to differentiate itself from the remaining Northern West-Germanic dialects, that is, at the time, when the first Saxon settlers came over to the British isles, and that we actually see here the result of imperfect language learning of Saxon by the (Celtic) inhabitants of the British isles. But, as this train of thought is too speculative for the moment, it is left for further research.

4.3 V3 in (Old High) German

We have now seen that Old English syntax never followed a strict CP-V2 constraint and that it is considerably different from the syntax of other West-Germanic languages. In this connection it is not out of place to discuss briefly another possible way to reconcile Old English syntax and the syntax of the remainder of the West-Germanic languages.

This other way can be characterized as follows: Instead of assuming that the West-Germanic languages started with a strict version of CP-V2 and that English drifted towards the subject-before-verb-syntax which it exhibits today, one could assume that the Germanic languages started much more similar to Old English and that the continental languages developed their strict CP-V2 syntax only later. This is the view that underlies e.g. works such as Tomaselli (1995). In other words: The West-Germanic languages had originally more than one 'vorfeld'-position, i.e. A'-positions to the left of IP, into which material could be moved that had been base generated somewhere within IP. Under this view, English would be more 'conservative' than the other West-Germanic branches, especially German, in that it preserved multiple vorfeld-positions, while German regularized the left periphery over the centuries and eventually reached a strict V2-state.

There is much to be said in favour of this hypothesis. For instance, it can be demonstrated that in Early New High German (ca. 1400-1700 AD), the strict CP-V2 syntax was not yet as strict as it is in Modern German. A phenomenon called *doppelte Vorfeldbesetzung* ('double prefield-filling'), which is extremely marginal in Modern German, is attested more frequently in Early New High German (cf. Speyer 2007b, on which the following discussion is based). Whereas in Modern German double vorfeld-filling can be estimated at somewhere below 0.07% (which is one divided by 1400, which is the number of declarative main clauses from the corpus that I used for my vorfeld-studies (Speyer 2004; 2006a), but most likely much lower, the rate of double vorfeld-filling in Early New High German, that is, in the time between 1350 and 1550, was much larger (Table 4.14)

Table 4.14: Rate of verb-third in Early New High German texts.

	number of lines	estimated word count	estimated clause count	thereof V3	**% V3**
Bavarian					
Kottanerin (Vienna, 1452)	1048	3700	820	5	**0,61**
Herberstein (Vienna, 1557)	682	7700	410	5	**1,21**
Alsatian					
Merswin (Straßburg 1370)	640	6500	510	10	**1,96**
Chirurgie (Straßburg 1497)	1857	13200	440	9	**2,05**
Cologne					
Nuwe Boych (Cologne 1396)	345	4100	130	5	**3,85**
Koelhoff (Cologne 1499)	393	5700	340	9	**2,64**
Upper Saxon					
Sermons (Leipzig ~1350)	507	6100	380	8	**2,11**
Tauler (Leipzig 1498)	569	3100	340	2	**0,59**

These texts have been taken from the Bonner Frühneuhochdeutschkorpus (Besch et al 1972-1985). The analysis of these cases in a generative framework is difficult, as multiple fronting is impossible under the standard account of the German V2-clause (den Besten 1977). Therefore multiple *vorfeldbesetzung* has either been analysed as underlyingly verb-last (that is, in German, without movement of the verb to C°), or as a movement process by which the multiple constituents in the vorfeld are actually only one constituent. The only generative account of Early New High German syntax that mentions multiple vorfeld-movement is Lenerz 1984. He, being heavily under the influence of den Besten (1977), views V3 as being always verb-last in disguise, sometimes blurred by rightward extraposition processes (Lenerz 1984:130), although he admits in other contexts that verb-last in matrix clauses was extremely rare in Early New High German (Lenerz 1984:132). Some examples of V3 cannot be analysed as verb-last with extraposed material, as we have elements to the right of the verb that cannot undergo rightward extraposition, such as the negation particle *nit* (modern: *nicht*; 3-19a) or the personal pronoun *mir* ('me.$_{DAT}$', 3-19b).

(4-19) a. Dar vm- du solt nit allein mercke- vnd verston

therefore you shall not alone notice and understand

'It is for this reason, that you should not be alone in making a diagnosis.'

(Chirurgie 25rB.20f.)

b. min herze in mime libe wolte mir zerspringen von rehter

my heart in my body wanted me.$_{DAT}$ burst from real

v·berswenkender fro^{e}iden.

effusive joy

'My heart wanted to pop in my body because of excessive joy.'

(Merswin 6.7f.)

Accounts of V3 in Modern German usually try to analyse multiple vorfeld-constituents as being part of one constituent. Müller (2003; 2005), for instance, analyses such a doubly filled vorfeld as a VP out of which the head and everything else has been moved, as demonstrated in (4-20).

(4-20) [[Großes Gewicht] [für die Geschworenen]] hatte ein aufgezeichnetes
large weight for the jury had a taped
Telefongespräch des Scheichs mit den Bombenlegern
phone coversation of-the sheikh with the bombers
des World Trade Centers
of-the WTC
'The jury assigned great importance to a taped phone call of the sheikh with the bombers of the World Trade Center.'
(taz, October 4, 1995, p.8; cited after Müller 2003:35)

$[_{CP}$ $[_{VP}$ t_2 $[_{NP}$ großes Gewicht] $[_{PP}$ für die G.] $t_1]_3$ hatte_1 $[_{IP}$ $[\text{ein…Centers}]_2$ t_1 t_3]]

This analysis cannot work for Early New High German, as there are numerous sentences in which the two *vorfeld*-constituents are in an order that is different from the order in which they would be base-generated in the *mittelfeld.* As under Müller's account the *vorfeld* is occupied by a remnant VP, the order of the constituents in the *vorfeld* must correspond to the one in the mittelfeld. Remnant movement of a scrambled VP should be ruled out, if we assume that scrambling is adjunction to IP. But in Early New High German we find numerous examples of the order temporal adverbial >> subject (4-21); this order does not correspond to the base-generated order, which is subject >> temporal adverbial.

(4-21) [Jm 6886. Jar] [der Großfuͤrst DEMETRI] hat den maͤchtigen Tatarischen

in 6886 year the archduke Demetri has the mighty Tatarian

Khuͤnig MAMAI geschlagen

king Mamai defeated

'In the 6886. year, the archduke Demetri defeated the mighty Tatarian king Mamai.' (Herberstein.B1r.11f.)

Müller's analysis thus is ruled out for Early New High German. Under a Split-CP-hypothesis the Early New High German data can be analysed without problems, provided that a mechanism is built in which prevents overgeneration. Such an analysis has been put forward in Speyer (2007b); as it is not relevant for our purposes, however, I do not present it here.

However, on the whole this discussion should not deceive us into believing that the constraint that only one constituent may occupy the vorfeld was only optional. It was close to categorical even in Early New High German, as we have seen, although today it is even closer to categorical. And even for Old High German, verb third is the exception rather than the rule.

In reading e.g. Tomaselli (1995), a reader who is not familiar with the Old High German data might get the impression that the order XP – S(pron) – V, as we see it in (4-22a,b), is all but categorical in Old High German.

(4-22) a. Erino portun ih firchnussu

bronze gates I destroy

‚I destroy gates of bronze'

(Isidor 3.2)

b. Dhes martyrunga endi dodh uuir findemes mit urchundin

this.$_{GEN}$ martyrdom and death we find with testimony

dhes heilegin chiscribes

of-the holy scriptures

'We find his martyrdom and death in the testimony of the Holy Scriptures.'

(Isidor 5.11)

c. umbi dhen selbun ir aer chiuuiso quhad

about the same he beforehand certainly said

'about this one he said before for sure: …'

(Isidor 3.10)

This impression is mistaken. First, Old High German is a pro-drop-language, as opposed to Old English. This means that we can assume that the few examples of overt personal pronouns somehow must bear some sort of prominence; otherwise they would not be realized overtly at all. By this, however, any cliticization account of Old High German subject pronouns is doomed to fail, as the main idea of cliticization is that the cliticized element does not have any intrinsic prominence. So we have to refuse an analysis such as the one by Tomaselli (1995) on conceptual grounds alone.

But how would an analysis with two subject position work? After all, the cliticization analysis has been refuted at length for Old English, and it looks as if it does not work for Old High German. But perhaps the two-subject-position-analysis could be applicable to Old High German.

This is not the case. There is a crucial difference between the Old High German and the Old English data: In Old English, pronouns always occur before the verb, in Old High German it is rather the exception. Let me illustrate this point.

Let us take the translation of Isidor. This is one of two Old High German texts that can be used for syntactic analysis.[67] If we count the positions of overt pronominal subjects in non-verb-last main clauses in this text passage, we see clear preferences for certain positions, as shown in Table 4.15.

Table 4.15: Positions of pronominal subjects in Isidor.

	alone in vorfeld	in vorfeld, preceded by other constituent	in first position of mittelfeld	later in mittelfeld
pos. of pro-s (n = 37)	21 (57%)	4 (11%)	12 (32%)	0

Being the second constituent in the vorfeld is clearly not the preferred option for a personal pronoun. The pronoun stands either alone in the vorfeld (which it can do also in Old English) or after the verb (which it never does in Old English), but rarely after a topicalized constituent, before the verb (as would be the rule for a pronoun in Old English). The preference for either the vorfeld alone or the first position of the mittelfeld (which is in German the archetypical position for topics; see Frey 2004a) is highly reminiscent of Modern German. The ratio even has to be qualified somewhat if we notice

[67] All Old High German prose texts are translations from Latin, which for the most part are slavishly dependent on the Latin pretext and therefore do not tell us anything about Old High German word order (cf. Lippert 1974). The only exceptions are the fragments of a translation of Isidor and some further translation fragments usually called collectively Monsee Fragments. The latter are two fragmentary to be usable for our purposes. In these texts the translator deviated considerably from his original and produced a word order that is clearly not dependent on the Latin pre-text, thus probably original Old High German word order.

that in two of the four examples the constituent preceding the personal pronoun in the vorfeld is a clause. It is not clear, how 'embedded' clauses in Old High German were, and if they count therefore as real subordinate clauses (in which case we had a V3 main clause) or not (in which case we had two clauses, the second clause starting with the personal pronoun and thus being a regular V2 clause). The only remaining examples are the ones given above in (4-22a,b), both of which could easily be verb-last main clauses in disguise (which are still possible in Old High German, see 3-22c)).[68]

The situation in Old English, on the other hand, is completely different. In Old English, as we have seen in the preceding sections, there is a systematic variation between verb second and verb third: With pronoun subjects, V3 is close to categorical, whereas with full noun phrase subjects, V2 is the predominant option.

The status of V3 in Old English is thus completely different from the status of V3 in Old High German: In Old English, V3 is regular, whereas in Old High German, V3 remains the exception, even with pronominal subjects. The occasional occurrence of Old High German sentences that correspond to the regular English type of sentences XP – S(pron) – V does not prove that this was the underlying or inherited structure of Old High German. Note that we find V3 in all stages of German (such as Early New High German, which is suitable for syntactic analysis since from this period a large number of prose

[68] In this context it is not surprising that verb third in general has been viewed as a marginal phenomenon in all traditional treatises on Old High German syntax. Lippert (1974:62), for instance, states that 'Späterstellung' (= V3) is very infrequent and even finds that the Old High German scribes are reluctant to translate a Latin sentence with V3 as a German V3 sentence, although they otherwise are very close to the Latin original text. If V3 occurs, it is a conscious assimilation to the original, in Lippert's view. In fact, if we find a random Old High German V3 sentence, the corresponding Latin sentence tends to be a V3 sentence, too. Robinson (1997:32), who re-evaluates Lippert's data, basically says the same, although he tries to incorporate Tomaselli's analysis. Note, however, that the case of V3 we are interested in, the form XP – S(pron.) – V, cannot be due to Latin influence as Latin is a pro-drop language to even a larger extent than Old High German.

texts have survived), but that V3 is not automatically a case of the order XP – S – V. There are several examples of completely different types of V3 (4-23).

(4-23) a. in der selben stvnden also knv̇wende v̇rschein mir die minnende
in the same hour so kneeling.$_{DAT}$ *appeared me.*$_{DAT}$ *the loving*
v̇rbermede gottes
mercy God.$_{GEN}$
'in the same hour the loving mercy of God appeared to me while I was kneeling.'
(Merswin 5.29ff.)

b. Dar vm- zů alle- zitte- sollen der hefften vngerad sin.
Therefore at all times shall the attachment unstraight be
'Therefore the suture has to be always curved.'
(Chirurgie 21vA.1f.)

c. Darna nyet lange vmb des besten wille so vnderwant sich
thereafter not long for the best will then constituted itself
eyn Rait zerzijt,
a council at-the-time
'Not long thereafter a council was quickly assembled for the best purpose'
(Nuwe boych 430.5f.)

It is clear that such examples cannot receive a cliticization analysis. At the same time it should be clear that they would not work under the analysis which I favoured for Old English in this chapter. V3 under Haeberli's analysis is necessarily always of the form XP

– S – V, because, besides the Ā-position to the far left, an A-position is involved which can be occupied only by a subject. For the Early New High German data however, we need an analysis which offers two (or more) Ā-positions in the left periphery and not an extra A-position, as the material which can be moved thither is essentially variable.

From this short overview it should be clear that V3 in German and English are different phenomena. (Old High) German V3 is the result of movement of the verb into the lowest of several split C-projections; the multiple vorfeld-constituents occupy some of the specifiers in the C-architecture. (Old) English V3, however, is the result of verb movement to a low position in the I-architecture, that is: below CP, movement of the subject to the specifier of the projection in which the verb has landed, and movement of a phrase into the C-architecture; whether there are several C-projections or only one is irrelevant.

4.4 The nature of the two subject positions

Working in a framework inspired by Rizzi (1997) in which functional projections are sensitive to the information status of potential specifiers, and given the standard assumption, i.e. that the lower subject position is for full noun phrase subjects, whereas the higher position is reserved for pronouns, it would be tempting to assign a pragmatic function typical for pronouns to the higher position. Pronouns often refer to the topic of a sentence, we have a Topic-Phrase in our arsenal of possible functional projections, and we know that in Modern German something resembling a Topic-Phrase in the strict sense is situated at the left edge of the *mittelfeld* (Frey 2004b). So we could assume that the higher subject position really is a topic position and happens to be filled predominantly by subjects so that everybody concludes that it is a subject position.

There are serious problems with this conclusion. First, it is problematic to confine the information-structural status of the higher subject position. The reason for this is that in the later history of English, this higher subject position by and by becomes the only possible subject position. This means that competition with the other subject position must have arisen, and the higher subject position eventually won in the competition. If it however had been reserved for elements with an unambiguous informational status, it would not be in direct competition with the other subject position: Language learners would learn that there is one position for subjects that are topics and one for subjects that are not. As long as distinct properties of both positions can be formulated, they will be perceived as different entities and therefore should not come into competition.

Whereas this first point is not a very strong argument (the information status which a position requires the element targeting this position to have might change over time), the second point is much stronger. The second problem is simply that the facts do not accord with that assumption: There are examples of non-topical elements in the higher subject position (4-24), although it must be conceded that most subjects in the higher position are at least discourse-old.

(4-24) a. Ðas ilcan geornfulnesse ðara hierda Sanctus Paulus aweahte, ða
the same eagerness of-the shepherd Saint Paul roused when
he cuæð:
he said
'The same pastoral zeal Saint Paul roused when he said: …'
(cocura,CP:18.137.25.939)

b. Ac þære ilcan niht […], wulfas atugan þa stacan up,
but the.DAT same night wolves drew the pins up
'But in the same night wolves (= devilish persons) drew up the pins'
(coorosiu,Or_5:5.119.25.2504)

Example (4-24a) is from a context where a sentence by Saint Peter was cited immediately before; Peter and Paul together form a set of potential authorities on the matter discussed in this passage. In the discourse from which (4-24b) is taken, the 'wolves' are newly introduced with this sentence.

To see what is going on I took a sample of both Old English V3 and V2 sentences with preposed accusative or dative noun phrase and identified the discourse structural properties of both the topicalized element and the subject. Since sentences with topicalized dative and topicalized accusative objects show a similar distribution, I felt justified in conflating them.[69] The results are in Tables 4.16 and 4.17.

Table 4.16: Pragmatic functions of topicalized dative and accusative full noun phrases and full noun phrase subjects in Old English V3 clauses.

n = 182	function of subj →	ϕ-topic	contrast	new	e-topic	old
function of top ↓						
ϕ-topic		-	13	5	8	25
contrast		-	10	1	13	28
new		-	-	-	3	5
e-topic		-	4	14	-	31
old		-	5	4	7	6

Table 4.17: Pragmatic functions of topicalized dative and accusative full noun phrases and full noun phrase subjects in Old English V2 clauses.

n = 197	function of subj →	ϕ-topic	contrast	new	e-topic	old
function of top ↓						
ϕ-topic		-	15	27	-	36
contrast		-	23	11	8	13
new		-	1	2	3	3
e-topic		-	9	27	-	11
old		-	2	5	-	1

These numbers show us several things, if grouped together in a meaningful way. For instance, we can see that the information status of the topicalized element does not matter for the choice between V2 and V3. In Table 4.18, the percentage of focus-bearing elements (that is, those labelled as ‘contrast’ and ‘new’) among the topicalized elements are compared in the sample of V2 and in the sample of V3 sentences. The rates are almost

[69] To give the reader a glimpse of the procedure of assigning and coding for informational structural properties, I have included the example sentences of V3-sentences with topicalized accusative NP with a small discussion of the assignments in an appendix.

identical. If we calculate the number of V2 among sentences with a focused topicalized constituent, we see that they amount to roughly half of the sentences that have V2 syntax (Table 4.19). So it looks as if the information status of the topicalized constituent does not influence the choice between V2 and V3 syntax.

Table 4.18: Proportion of focalized constituents among topicalized dative and accusative NPs of V2 and V3 main clauses.

	V2	V3
all sentences in sample	197	182
thereof topicalized element +foc	64	60
% (+foc)	**32.5**	**33.0**

Table 4.19: Proportion of V2 among sentences with focus on topicalized constituent.

all top. +foc	124
thereof V2	64
% (V2)	**51.6**

With subjects, it is different. If we look at Table 4.20, we see that a rather high percentage of V2 sentences have a subject with a high likelihood of being focused (that is, labelled as ‘contrast’ or ‘new’), but only a comparatively small number of V3 sentences have a focused subject. This is in line with the assumption that the higher subject position (that is, the one in V3 sentences) is a preferred landing site for topics and generally discourse-old elements. If we calculate the percentage of V2 among sentences with focused subject (Table 4.21), we see that almost three quarter of sentences with focused subjects are V2-sentences. We can contrast this with the percentage of V3 among sentences without focused subject (Table 4.22). Here, it is about two thirds of the sentences with a non-focused subject that show V3 syntax.

Table 4.20: Proportion of focalized subjects in sentences with topicalization; V2 and V3 main clauses.

	V2	V3
all sentences in sample	197	182
thereof subject +foc	122	56
% (+foc)	**61.9**	**30.8**

Table 4.21: Proportion of V2 among sentences with focus on subject.

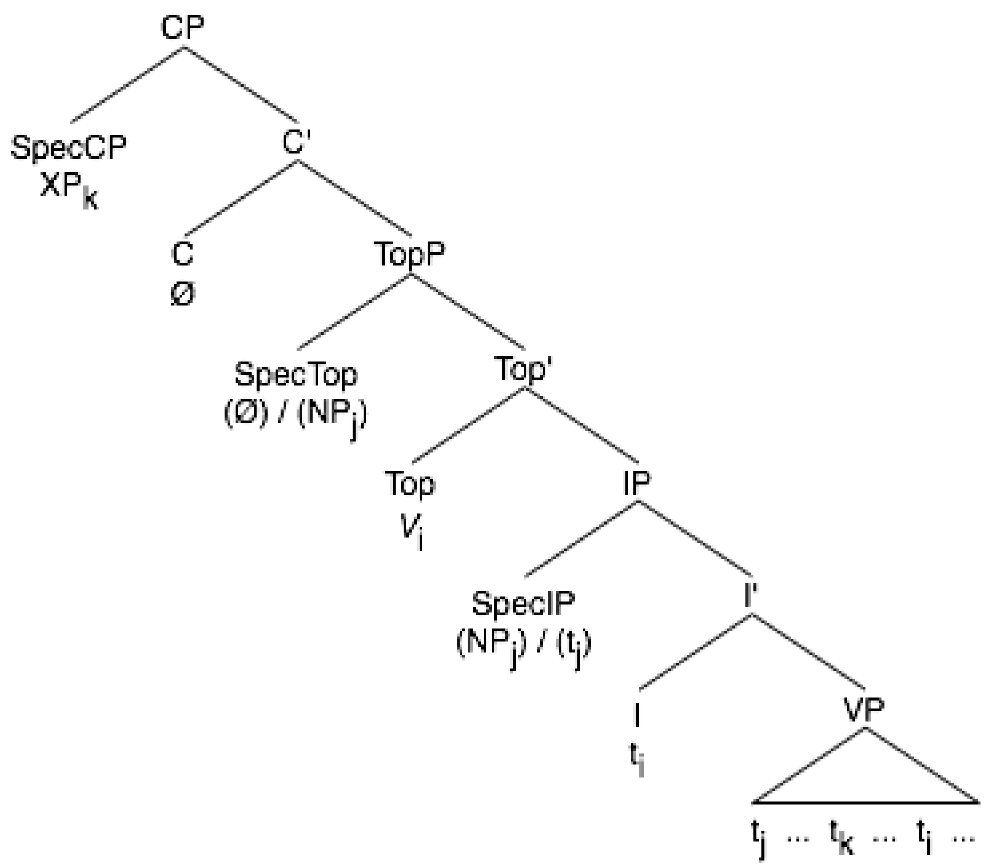
all subject +foc	178
thereof V2	122
% (V2)	**68.5**

Table 4.22: Proportion of V3 among sentences with no focus on subject.

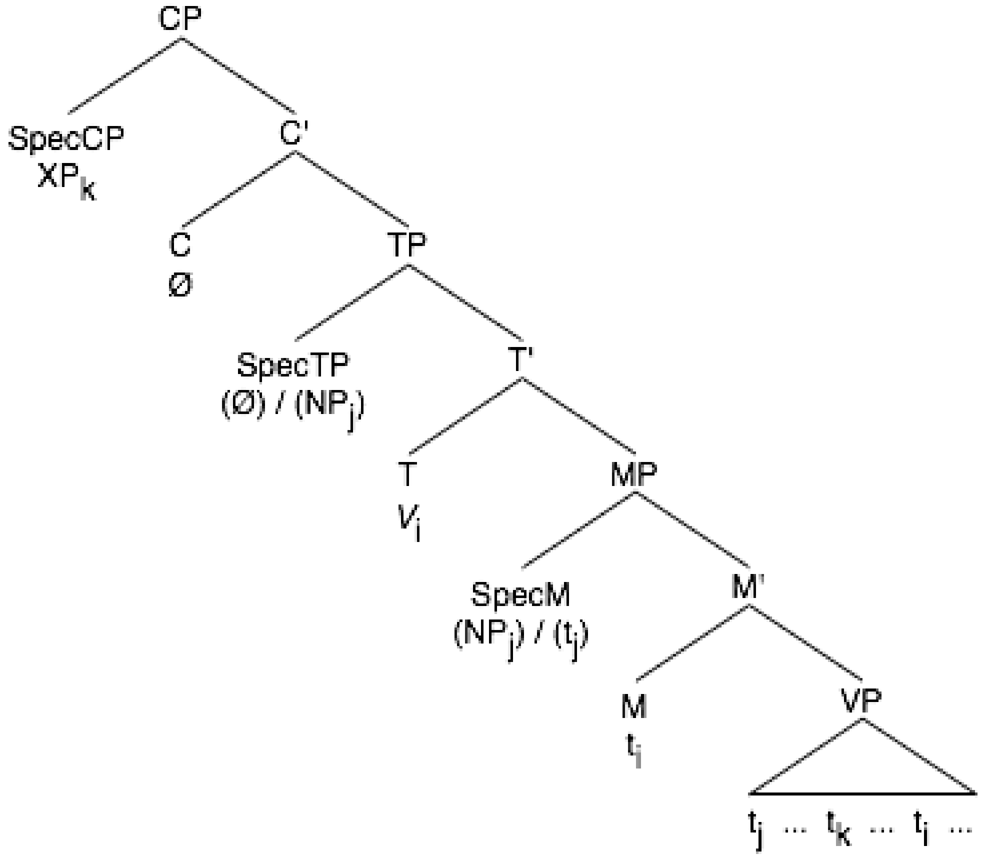
all subject -foc	201
thereof V3	126
% (V3)	**62.7**

These results can be interpreted in the following way: Both non-focused constituents (which are basically discourse-old elements, some of them topics) and constituents which are likely to be focused (that is, contrastive and discourse-new elements) show clear preferences for one of the two subject positions: focused constituents prefer the lower subject position, which is visible from the fact that over two thirds of sentences with a focussed subject are V2 sentences, whereas non-focused constituents prefer the higher subject position, thus they predominantly show up in V3-sentences. It might be the case that the higher position originally goes back to a specific Topic-Phrase in Proto-Germanic – and a remnant of this pedigree is probably the fact that pronouns (that is: guaranteed discourse-old, if not topical elements) regularly choose this position. By the time, homever, in which the Old English textual transmission sets in, this position has already changed its character into a more widely available subject position, with a high preference for discourse-old elements. The lower position, on the other hand, which in this model would go back to the ‘real’ subject position, SpecIP, has developed into a specialized subject position with a predilection for focused elements. This means that at some stage before 800 AD two projections at the left periphery were reinterpreted: The original TopicPhrase into a Phrase of the IP-architecture, the original highest (and probably only overt) phrase of the IP-architecture to the second-highest phrase in the IP-architecture. This reinterpretation is shown schematically in (4-25). TP and MP are Tense Phrase and Mood Phrase, respectively, as potential representatives of the two IP-projections we are looking for (cf. Han & Kroch 2000 for an I-split account of English).

(4-25) Pre-Old English



==> Old English



This reinterpretation was made possible by the fact that it is mostly subjects that are moved to the TopicPhrase; in today's English, the topic is almost by default realized as the grammatical subject (see e.g. Mathesius 1928). A discourse is regarded as maximally coherent if the equation subject = topic is preserved (see Walker, Joshi & Prince 1998).

The real point here is, however, not so much why the higher position has a predilection for old elements, but why the lower subject position acquires this specialization for focused elements. This is especially important if we assume that both subject positions are part of the IP-architecture. If this is so, we might assume that the speaker wants to move the subject as high up as possible, as subject movement is A-movement and certainly some features remain that can be checked only at the highest I-projection (in 3-25: TP). For the native speaker and learner, the evidence that in some cases it is not possible to move the subject to the highest possible landing site has to be indirect; the language learner can only posit that there is some zero element blocking the movement of the subject to the highest position when s/he hears his/her parents utter V2-sentences. A consequence of this might be the specialization of the lower position, for only if the language learner can sense a pattern governing the choice between V2 and V3 – or between the movement of the subject only to the lower or further up to the higher position – will s/he acquire the alternation between V2 and V3. So by Old English times, it looks as if the compromise several generations of language learners have found is: If the subject is in focus, it tends to move only to the lower position.

But the important question remains: How could this reinterpretation of the lower subject position to a position preferred for focal subjects come about? It presupposes that at some stage prior to the reinterpretation, focal subjects tended to appear in the lower position, which at that time of course was the only real subject position, and not moved further upwards to the higher position, which at that time was a TopicPhrase and therefore unsuitable for focalized subjects. This might be an accurate explanation from the perspective of English prehistory. After the higher position acquired some kind of general subject status, though, we have to ask why the lower subject position was kept in operation for such a long time. Strictly speaking, the logical consequence of the process, viz. the abolition of the lower subject position, only was completed some 700 years after the Old English period, in the Early Modern English period (see 2.4.5. and esp. Table 2.12 and Figure 2.8).

Here the Clash Avoidance Requirement comes into play once again. We have argued (for Middle English in chapter 2.4) that V2 was a handy way to avoid violations of the Clash Avoidance Requirement, as long as V2 was still a grammatical option in the English language. It is likely that the same considerations induced the speakers of Old English not only to keep the lower subject position in operation, once the situation with the two distinct subject positions had arisen, but also to regard it as a preferred landing site for focused subjects. The reason is this: if the focused subject is kept in the lower subject position, it will not cause focus clash with the topicalized constituent, because the finite verb, which is likely to be unfocused, intervenes between the topicalized constituent and the subject. Therefore, to move a focused subject only to the lower position can be regarded as a move by the speaker to avoid potential focus clashes with the topicalized constituent. We have seen that in Old English topicalization takes place, regardless of whether the topicalized constituent is focused or not. So the only way to avoid focus clashes is to manipulate the position of the subject. We have also seen that later in the language's history, when it became impossible to manipulate the position of the subject, the speakers had to manipulate the element which was to be topicalized and ceased to topicalize if it could lead to focus clash.

We cannot prove, of course, that it is the Clash Avoidance Requirement that determined the choice between V2 and V3 in Old English. We can, however, provide evidence which supports this assumption.

Let us look again at the numbers in Tables 4.16 and 4.17. If we extract the cases in which both the topicalized element and the subject are in focus, we see the following: whereas there are several examples of this configuration with V2 sentences, there are notably fewer with V3 sentences (Table 4.23).

Table 4.23: Proportion of focalized constituents among topicalized dative and accusative NPs and focalized subject of V2 and V3 main clauses.

	V2	V3
all sentences in sample	197	182
thereof with focus on top.el. & sub.	37	11
% (+foc)	**18.8**	**6.0**

If we look closer we note that several examples of the 11 V3 sentences with focus on both the topicalized element and the subject conform to the CAR because an adverb, a participial clause or an ethical dative intervenes between the topicalized element and the subject. As all of these elements count as phrases, the CAR is preserved in these cases. 5 examples conform to this category. We may add 2 more which are immediately following two of such CAR-conforming examples and are built exactly parallel to them for rhetorical reasons, but leaving the interveniung adverb out. In (4-26) we have an example of such a case: The first clause is CAR-conforming, because an adverb intervenes, the second is built parallel.

(4-26) Ðone ***UNGEÐYLDEGAN*** ðonne swiðe ***LYTEL*** scur ðære costunga mæg

the impatient then very little shower of-the temptation may

onhreran,swæ swæ lytel wind mæg ðone cið awecggean,

excite so so little wind may the altercation awake

ac ðone ***YFELAN*** *FÆSTRÆDAN* willan folneah ***NAN*** wind ne mæg awecggean .

and the evil constant will almost no wind not may awake

‘An impatient will, then, a very little shower of temptation can excite, just as a little breeze may arouse altercation, but the constantly evil will almost no storm may awake.’ (cocuraC,CP_[Cotton]:33.224.4.85f.)

So we can actually subtract 7 of those V3 cases. The corrected table is under 4.24. The remaining four sentences are such that a clash is inevitable either way. Example (4-27) shows two of those sentences; here each of the three constituents is in contrast to the corresponding constituent in the partner clause. It makes no difference, whether the sentence is V2 or V3.

Table 4.24: Proportion of focalized constituents among topicalized dative and accusative NPs and focalized subject of V2 and V3 main clauses, CAR-conforming V3 cases subtracted.

	V2	V3
all sentences in sample	197	182
thereof with focus on top.el. & sub.	37	4
% (+foc)	**18.8**	**2.2**

(4-27) Witodlice of þam twam wundrum, þe ic secgan wille,
Truly of the two miracles that I say will
oþer þæt folc ongeat,
other the people approached
oþer þa sacerdas oncneowon,
other the priests observed
'In fact, with respect to the two miracles that I will relate, the one happened to the laymen, the other was observed by the priests.'
(cogregdC,GDPref_and_3_[C]:30.235.14.3285)

If we look at the percentage of V2 in sentences with two foci on the topicalized element and the subject, we see that the overwhelming majority are V2 sentences (Table 4.25).

Table 4.25: Proportion of V2 among sentences with focus on topicalized constituent and subject.

all top. & sub. +foc	41
thereof V2	37
% (V2)	**90.2**

The observed distribution gives evidence that in Old English the lower subject position and thereby V2-syntax was used to avoid focus clash and that focused subjects tended to move not higher than the lower subject position as a precaution against focus clash. This means that the Clash Avoidance Requirement is powerful enough to influence the syntactic output directly: Old English syntax has developed into a system which, if left alone, produced CAR-conforming structures all by itself. The only critical parameter was the presence or absence of a focus feature on the subject. If a focus feature was present, movement of the subject to the higher position was dispreferred in most cases. Consequently it could not come into conflict with a focus feature on the topicalized constituent, if one such were present, as the weak verb intervened (4-28).

(4-28) $[_{CP}$ [$\mathbf{XP}_{[+foc]}$] – _ – $[_{TP}$ [_] Infl – $[_{MP}$ [$\mathbf{NP}_{[+foc]}$] – …]]]

If there was no focus feature on the subject, it was unnecessary to keep it in the lower position as it could not clash with the topicalized constituent. We may assume that the preferred option was to move the subject as high as possible. Subject pronouns which as a rule are not focalized undergo this movement to the highest subject position in all cases (4-29).

(4-29) $[_{CP}$ [$\mathbf{XP}_{[+foc]}$] – _ – $[_{TP}$ [$NP_{[-foc]i}$] Infl – $[_{MP}$ [t_i] – …]]]

Probably the actual position of the subject could be rearranged and further movement of the subject, even if focalized, could happen if it were clear that the topicalized element did not bear focus. This would happen in cases in which a focalized element appeared in the higher subject position (4-30).

(4-30) $[_{CP}$ [$XP_{[-foc]}$] – _ – $[_{TP}$ [$\mathbf{NP}_{[+foc]i}$] Infl – $[_{MP}$ [t_i] – …]]]

At first glance it looks as if the movement operations described here would only happen after it was clear what the prosody of the topicalized constituent was, that is: after Lexical Insertion. This would be a most undesirable result. But remember that the factor defining the final landing site of the subject is the focus feature which, as it is semantically interpretable, must be part of the whole derivation and must be present in narrow syntax. This means that the fact that the element on which the focus feature is situated eventually bears focus is already deducible before Lexical Insertion. All the syntactic component has to take into account is that it should not place two elements with focus feature adjacent to each other. It is an open question whether narrow syntax by itself has developed into being constrained in such a way that it produces structures in which no two elements with focus feature are adjacent, or whether it produces structures regardless of adjacency of any focus features but is forced to repair the structures because the phonological module has sent them back as ill-formed. The latter is the more likely assumption; otherwise it would be hard to conceive that focal emphasis clashes could happen at all. They do occur, but only very seldom, and if they occur, we can assume that for some reason the syntactic module did not succeed in fixing the structure, so that the phonological module had to output the structure with the limited means at its disposal, namely pause insertion.

Summarizing this section, we can say that Old English used the two subject positions it inherited (with slight reinterpretations) from Pre-Old English to minimize violations of the Clash Avoidance Requirement. Focalized subjects were mostly moved only to the lower subject position, where they were separated from any focalized or topicalized constituent by the verb. Other subjects (including all pronoun subjects) could move to the higher position, as they posed no threat of creating a CAR violation.

4.5 Summary

In section 4.1, the well-known V2/V3-alternation of Old English was exhibited. In contrast to the standard view that the two subject positions in Old English are in complementary distribution, in the sense that pronouns stand in the higher position and full noun phrases in the lower one, it was shown that full noun phrases can also be found in the higher subject position.

Section 4.2 offers a possible objection, namely that the examples of V3 with full noun phrase subject could be verb-last in disguise. It is shown, however, that several examples exist (mostly with verbal particles) that can only be analysed with an Infl-medial sentence pattern. Furthermore,the observable rate of main clauses with topicalization is significantly higher than we would expect if we assumed them all to be Infl-final clauses, therefore, at least some of them must be Infl-medial. So we conclude: Infl-medial V3 clauses are real in Old English.

Section 4.3 discusses V3 in the history of German. German always had a stricter version of the V2-constraint than English, although we can see this constraint being tigthened over the history of German: V3 in general was much more common as late as in Early New High German. Still, the structures of German and English were distinct from the beginning of their respective attestations, as ‘English-style V3’ with a subject pronoun between topicalized constituent and verb is not very common, at any rate not as common as it should be if Old High German and Old English really had an underlyingly similar syntax.

In section 4.4 the pragmatic properties of V3 versus V2 are investigated. We have seen that Old English used the two subject positions it inherited (with slight reinterpretations) from Pre-Old English in order to minimize violations of the Clash Avoidance Requirement. Focalized subjects were mostly moved only to the lower subject position where they were separated from any focalized topicalized constituent by the verb. Other subjects (including all pronoun subjects) could move to the higher position, as their appearance in this position could not produce a CAR violation.

5. Concluding remarks

In this final section I want to give a very brief view of the thesis as a whole; for more detailed summaries the reader is directed to the final sections of each chapter which sum up the chapters in more detail.

The purpose of this thesis was twofold: on the one hand I wanted to show compelling evidence for the Clash Avoidance Requirement. The Clash Avoidance Requirement is understood as a condition on the level of the highest points of prominence of the clause, that no equally strong prominences may stand adjacent to each other. On the other hand I wanted to demonstrate how the Clash Avoidance Requirement influenced syntactic usage in the history of English. The structure of the thesis was determined by this twofold aim. In the second chapter, the necessity for having something like the Clash Avoidance Requirement was deduced from a crucial set of data, the decline of topicalization in Middle and Early Modern English. After several potential explanations for this decline were rejected (especially rigidification of word order and loss of pragmatic contexts for topicalization), a model in which the decline was traced back to the loss of the V2 word order option and the rise of potential problem cases for the Clash Avoidance Requirement was shown to account successfully for the historical facts.

The third chapter was devoted to a demonstration of the reality of a Clash Avoidance Requirement. Its effects were demonstrated in a series of experiments. The relationship of the Clash Avoidance Requirement to other similar phonological well-formedness conditions was clarified. Other important issues in the context of the Clash Avoidance Requirement, like the relationship of metrical prominence and focus or the choice of different clash resolution mechanisms, were treated. In the end the Clash Avoidance Requirement was incorporated into a general theory of grid construction.

The fourth chapter was entirely devoted to demonstrating the effect of the Clash Avoidance Requirement on syntactic usage. The well-known alternation of V2- and V3-main clauses in Old English was directly linked to the presence or absence of focus on the subject; an Old English sentence template was developed following Haeberli's work that allows for this variation in a straightforward manner. This template is one with two potential subject positions, and a considerable effort expended in quantitative analysis of the Old English corpus to demonstrate that Old English V3 sentences have – or can have – an underlying Infl-medial syntax and are not Infl-final sentences in disguise or V2-sentences with cliticization of the subject.

The object of study in this thesis has for the most part been English. From time to time, I glanced at German as another, closely related language which has a similar system of prosodic prominence. In German no changes in syntactic usage can be attributed to the Clash Avoidance requirement, as German had, from its earliest attestations, a syntactic pattern that automatically produces sentences that conform to the Clash Avoidance Requirement. The Clash Avoidance Requirement is, however, at work in German just as much as in English, and in order to show this I treated German and English equally in the derivation and explanation of the Clash Avoidance Requirement in the third chapter.

I believe that I have achieved both of the aims of the thesis: I have been able to show that the Clash Avoidance Requirement exists and also that it influences syntactic usage. The experimental evidence, on the one hand, and the historical corpus data, on the other, clearly corroborate my claims.

Appendix: All Old English OSV-sentences with full noun phrase accusative object and subject

In order to give an example of how the prominence assignment was done, I decided to present a data set here with the assignment of information structural properties as I have done it, and the relevant context. I took the examples directly from the CorpusSearch query; I omitted, however, the tree structures to save space.

Legend:

`Examples from the query in Courier New.`

Context of example in Times New Roman.

Metatext in Arial.

`Emphasis marks:`

`Nucleus of topmost and second highest level marked by` **`boldface`**`.`

`Focal emphasis marked by` ***`BOLD ITALIC SMALL CAPS`***`.`

`Antecedents of topics occasionally underlined.`

`Contrastive elements occasionally in` *`italics`*`.`

`/~*`

Sum swiðe welig wif wæs swylce on wudewan hade coaelive,ÆLS_[Basil]:527.828

ac heo lyfde sceandlice , swa swa swin on meoxe , coaelive,ÆLS_[Basil]:527.829

and mid healicum synnum hi sylfe fordyde . coaelive,ÆLS_[Basil]:527.830

Heo wearð swa þeah æt nehstan wundorlice onbryrd þurh Godes mynegunge , coaelive,ÆLS_[Basil]:530.831

and ealle hyre manlican dæda awrat on anre cartan , coaelive,ÆLS_[Basil]:530.832

and beworhte mid leade . coaelive,ÆLS_[Basil]:530.833

Gesohte þa Basilium biddende and cweðende , Eale þu Godes halga beseoh me to are .

coaelive,ÆLS_[Basil]:533.834

EALLA mine synna ic, synfulle, awrat on þissere **cartan**,

(coaelive,ÆLS_[Basil]:535.835)

Number	object	subject
OS-1	contr	old

The quantifier induces a contrast (all vs. some, none etc.), therefore the object is in focus. The paper (*carta*) is old (everything she says in this sentence has been told in the preceding narrative), therefore it bears metrical prominence and no focal emphasis. There is no focus on the personal pronoun or its adjunct, as both are evoked/old.

/~*

Nu wylle we eow secgan hu se halga Hieronimus be ðam feower godspellerum , ðe Gode gecorene synd awrat

on ðære forespræce þaða he awende Cristesboc of ebreiscum gereorde , and sume of greciscum , to lædenspræce on þære ðe we leorniað . coaelive,ÆLS_[Mark]:106.3278

[…]

Ðas feower godspelleras syndon Gode gecorene , coaelive,ÆLS_

[Mark]:174.3309

and hi ealne middaneard mid heora lare onlihton , swa swa þa feower ean ðe yrna

ð of neorxnewange ealne þisne embhwyrft endemes wæteriað ; coaelive,ÆLS_[Mar

k]:174.3310

and ðas feower **godspelleras** God geswutelode gefyrn, on ðære ealdan æ

EZECHIHELE þam witegan .

(coaelive,ÆLS_[Mark]:174.3311)

number	obj	subj
OS-2	topic	old

The nucleus is on *ðas feower godspelleras*. The four evangelists are the topic of the whole paragraph. *God* is evoked, does not stand in contrast to anything. Focal emphasis on *Ezechiele*, as he is newly introduced into the discourse here.

/~*

De misericordia . coalcuin,Alc_[Warn]:132.103_ID

<u>Mildheortnysse</u> is swyðe hehlic god , coalcuin,Alc_[Warn]:132.104_ID

[…]

Ælc mildheortnysse wyrcð hire sylfre stowe æfter hire agenre geearnunge . coalcuin,Alc_[Warn]:152.115_ID

Se þe mildheortnysse deð , he brincð Gode swyðe gecweme lac . coalcuin,Alc_[Warn]:153.116_ID

On ælcen deme sceal beon æigðer gea mildheortnysse , gea wraca , for þan þe heora naðer ne mæig be on wel abuten oðren , þe $læs $te <TEXT:læste> on þære mildheortnysse ane forgyfe his heringmannen orsorhnysse to syne

gien , oððe eft for þære wraca anre þæs agyltendan mod seo gehwerfed on orwennysse , & se deme þonne fram Gode nane m

ildheortnysse ne geearneð , coalcuin,Alc_[Warn]:154.117_ID

ac þas **mildheortnysse** se mann sceal ærest on him **sylfen** aginnen . (coalcuin,Alc_[Warn_35]:154.118)

Number	object	subject
OS-3	topic	old

Nucleus on sylfen, 2ary sentence stress on mildheortnysse (by rule)

mildheortnys was the topic of the whole paragraph. *Se mann* is generic, not in contrast to anything.

/~*

De <u>penitentia</u> . coalcuin,Alc_[Warn]:360.263_ID

ðære soðen dædbote mæign ***SE HÆLEND SYLF*** on his godspelle æteowde, þuss cweðende,

(coalcuin,Alc_[Warn_35]:360.264)

Doð dædbote eower synnen , coalcuin,Alc_[Warn]:360.265_ID

þonne genelæceð eow heofona rice . coalcuin,Alc_[Warn]:360.266_ID

& *Johannes* se fulhtere cwæð , Doð medeme wæstmes dædbote . coalcuin,Alc_[Warn]:363.267_ID

Number	object	subject
OS-4	topic	contrast

Contrastive focus on subject; in contrast with *Johannes se fulhtere* in sent.267. Object not emphasized (topic)

/~*

Eac ðæm oþrum bisgum & geswencnissum þe us on becwom , þ

a cwom semninga swiðe micel deor sum mare þonne þara oðra ænig . coalex,Alex

:20.1.228

Hæfde þæt deor þrie hornas on foran heafde coalex,Alex:20

.2.229

& mid þæm hornum wæs egeslice gewæpnod . coalex,Alex:20.2.230

Þæt **deor** Indeos hatað dentes **tyrannum .**

(coalex,Alex:20.3.231)

Number	object	subject
OS-5	topic	old

Nucleus on *Þæt deor* and *dentes tyrannum*.

Indeos are salient throughout the text, therefore count as old. The paragraph is about this door. The name could arguably also be in focus.

/~*

Đa $gesawon we þær ruge wifmen , & wæpned men coalex,Alex

:29.2.344

wæron hie swa ruwe & swa gehære swa wildeor . coalex,Alex:29.2.345

Wæron hie nigon fota uplonge , coalex,Alex:29.3.346

& hie wæron þa men nacod coalex,Alex:29.3.347

& hie næniges hrægles ne gimdon . coalex,Alex:29.3.348

Đas **men** Indeos hatað **Ictifafonas**

(coalex,Alex:29.4.349)

Number	object	subject
OS-6	topic	old

Nucleus on *Þas men* and *Ictifafonas*.

Indeos are salient throughout the text, therefore count as old. The paragraph is about these *Ictifafonas*. The name could arguably also be in focus.

/~*

And he sylfa welwillendlice lifede mid his gemæccan seofon and hundseofonti geara

coapollo,ApT:51.29.586

and heold þæt cynerice on Antiochia and on Tyrum and on Cirenense , coapollo,ApT:51.29.587

and he leofode on stilnesse and on blisse ealle þa tid his lifes æfter his earfoðnesse .

coapollo,ApT:51.29.588

And ***TWA BEC*** he silf gesette be his **fare**

(coapollo,ApT:51.33.589)

and *ane* asette on ðam temple Diane , *oðre* on $bibliotheca . coapollo,ApT:51.33.590

Number	object	subject
OS-7	new	topic

Object in focus, as it is new information and names the set whose members are put into contrast in the following sentence. Nucleus on *fare*.

/~*

Ðær wæs þa heafde beslagen se strengesta martyr Sanctus

Albanus , cobede,Bede:7.40.4.328_ID

& þær he onfeng beah & sige eces lifes , cobede,Bede:7.40.4.329_ID

þone ylcan **sige** God behet ***EALLUM*** þam ðe hine lufian wyllað .

(cobede,Bede_1:7.40.4.330)

Number	object	subject
OS-8	old	old

sige is old (mentioned in preceding sentence), as is *God*. Nucleus (2ary) on *sige*. *eallum* induces focus.

/~*

& þa wæron Seaxan secende intingan & towyrde heora gedales wið Bryttas . cobede,Bede:12.52.20.481_ID

Cyðdon him openlice & sædon , butan hi him maran andlyfne sealdon , þæt hi woldan him sylfe niman & hergian , þær hi hit findan mihton . cobede,Bede:12.52.21.482_ID

& sona ða beotunge dædum gefyldon : cobede,Bede:12.52.23.483_ID

bærndon & hergedon & slogan fram eastsæ oð westsæ ; cobede,Bede:12.52.23.484_ID

& him nænig wiðstod . cobede,Bede:12.52.23.485_ID

[…]

Wæs in ða tid heora heretoga & latteow Ambrosius , haten oðre noman Aurelianus .

cobede,Bede:12.54.12.501_ID

Wæs god mon & gemetfæst , Romanisces cynnes mon . cobede,Bede:12.54.13.502_ID

In þisses monnes tid ***MOD & MÆGEN*** Brettas onfengon:

(cobede,Bede_1:12.54.14.503)

Number	object	subject
OS-9	new	old

The paragraph is about the settlement of the Saxons. The Brits have been mentioned before. Up to now they were rather on the receiving side of the fights, therefore *mod & mægon* introduces something new which is in focus.

/~*

Ono nu þæt wiif in blodes flownesse geseted hergendlice

meahte Drihtnes hrægle gehrinan , forhwon þonne , se þe blodryne þrowað mon

aðaðle , ne alefað hire in Drihtnes cirican gongan ? cobede,Bede:16.78.14.718_ID

Ac þu cwist nu : Heo nedde hire untrymnesse þæt heo Cristes hrægle gehrine ;

cobede,Bede:16.78.17.719_ID

þas wiif, bi þæm we sprecað, ***GELOMLIC GEWUNA*** getiið .

(cobede,Bede_1:16.78.17.720)

Number	object	subject
OS-10	topic	new

Topicality of object stated explicitly by relative clause. *gelomlic gewuna* is new in the discourse, therefore bears focus. The primary sentence stress that should be on *sprecað* is lowered adjacent to the focus.

/~*

INTERROGATIO VIIII . cobede,Bede:16.84.21.771_ID

Hwæðer æfter bysmrunge , seo þurh slæp wæpnedmonnum gelimpeð , oððo Drihtnes lichoman ænig onfoon mot , oðþo , gif hit sacerd bið , mot he þa halgan geryno mærsian mæssesonga ?

cobede,Bede:16.84.21.772_ID

RESPONSIO . cobede,Bede:16.84.24.773_ID

Ðeosne **mon** eac swylce seo cyðnis þære ealdan æ bismiten cwið, swa we ær in þæm uferan **kapitule** cwædon,

(cobede,Bede_1:16.84.24.774)

Number	object	subject
OS-11	old	new

The sentence is at the beginning of a section, therefore has no topic. The *mon* is taken over from the preceding question. The subject is new (for the section at least), but it is not directly presentational and therefore doubtful whether it is in focus. Metrical prominence (2ary) on *mon* and on *kapitule*.

/~*

Forþon þrim gemetum bið gefylled æghwilc syn , þæt is , ærest þurh scynnesse , & þurh lustfullnesse , & þurh geðafunge . cobede,Bede:16.86.25.788_ID

Seo scynis bið þurh deoful , cobede,Bede:16.86.27.789_ID

seo lustfulnes bið þurh lichoman , seo geðafung þurh gast . cobede,Bede:16.86.27.790_ID

Forðon þa ***ÆRESTAN*** synne se weriga gast scyde þurh þa ***NÆDDRAN***,

(cobede,Bede_1:16.86.28.791)

Number	object	subject
OS-11	contr	topic

Ordinals imply a contrast automatically, especially ‘the first’ (as opposed to ‘the second’ etc.), therefore *ærestan synne* is in focus. *Se weriga gast* has been introduced in the preceding sentence and functions as topic, is therefore not emphasized. The snake (*næddran*) is new and bears presentational focus.

/~*

On þam <u>gefeohte</u> eac swylce Ðeodbald Æþelfriþes broþ

or wæs ofslægen mid ealle þy weorode þe he lædde . cobede,Bede:18.92.19.853_ID

Þæt gefeoht **Æþelfrið** gefremede þy endlyftan geare his rices, þæt he hæfde feower & twentig **wintra** .

(cobede,Bede_1:18.92.21.854)

Þæt wæs þæt æreste gear Focatis þæs caseres , se þe hæfde Romana rice . cobede,Bede:18.92.23.855_ID

Siððan of þære tide nænig Sceotta cyninga ne dorste wið Angelþeode to gefeohte cuman oð ðys ne andweardan dæg . cobede,Bede:18.92.24.856_ID

Her endað seo æreste boc cobede,Bede:18.92.26.857_ID

& onginneð seo oðer . cobede,Bede:18.92.26.858_ID

Number	object	subject
OS-12	topic	old

The paragraph is about the *gefeoht*, which is, as a topic, unemphasized, but bears 2ndary clausal prominence. *AEþelfrið* was mentioned before and is unemphasized too. The ordinal *endlyftan* (11th) potentially could induce a contrast, but since no mention is made of other years of *AEþelfrið's* reign, I do not assume focal emphasis here. In the absence of focal emphasis there is primary metrical prominence on *wintra* (as marked in the text)

/~*

Wæs þis geworden þurh Osþryðe geornnesse Mercna cwene , seo wæs Oswiges dohtor his broðor , se æfter him feng to Norðanhymbra riice . cobede,Bede:9.182.11.1811_ID

Is æðele mynster in Lindesse ; cobede,Bede:9.182.15.1812_ID

is nemned Beardan ea . cobede,Bede:9.182.15.1813_ID

Ðæt **mynster** seo ilce cwen mid hire were Æþelrede **swiðe** lufade & arweorðade & beeode .

(cobede,Bede_3:9.182.15.1814)

Number	object	subject
OS-13	topic	old

The metrical prominence is regularly on *mynster* and *swiðe*.

Note that the phrase *seo ilce X* does not serve to introduce a contrast like ModE 'same', but rather serves to reintroduce an old entity and to establish it as topic of the following discourse (it is more like ModE 'the said queen', 'the queen mentioned above')

/~*

Wæs æfterfylgend his rices Anna geworden Eanes sunu of heora cyningcynne ;

cobede,Bede:14.208.30.2122_ID

wæs god monn & þæs betstan tudres $cennend , bi ðon her æfter in heora tiid is to secgenne .

cobede,Bede:14.208.30.2123_ID

[…]

Wæs fæger mynster getimbred in wuda neah sæ in sumre ceastre , seo is nemned on Englisc Cnoferesburg .

cobede,Bede:14.210.23.2139_ID

Þa **burg** eft æfter þon Anna þære mægðe cyning & monige æðele menn mid hearum getimbrum & **geofum** frætwade & weorðade .

(cobede,Bede_3:14.210.24.2140)

Number	object	topic
OS-14	topic	old

The topic *burg* was introduced in preceding sentence. *Anna* is present throughout the preceding discourse, often having been the topic. In the absence of focus metrical prominence is assigned on *burg* (2ary) and *geofum*.

/~*

Æþelwald þonne Oswaldes sunu þæs cyninges , se þe him on fultome beon sceolde , se wæs in þara wiðerweardra dæle cobede,Bede:18.236.7.2407_ID

& feaht & wonn wið his eðle & wið his fædran . cobede,Bede:18.236.7.2408_ID

Ða wæs sona , þæs þe heo þæt gefeoht ongunnon , þætte þa hæðnan wæron slegene & geflemde ;

cobede,Bede:18.236.10.2409_ID

ond þritig aldormonna & heretogena , þa ðe þam cyninge to fultome cwomon , lytesne ealle wæron ofslegene . cobede,Bede:18.236.10.2410_ID

[…]

Seo æfter twæm gearum gebohte tyn hida lond hire in æhte in þære stowe , seo is cweden Streoneshealh , cobede,Bede:18.236.31.2424_ID

ðær heo mynster getimbrode , in þæm seo gemyngade cyninges dohtor ærest wæs discipula & leornungmon regollices lifes , ond eft æfter þon wæs magister & lareow þæs mynstres , oð þæt heora daga rim gefylled wæs , þæt is anes wonþe syxtig wintra . cobede,Bede:18.236.31.2425_ID

Þa heo to clypnesse & to gemungum þæs heofonlican brydguman eadig fæmne ineode . cobede,Bede:18.238.2.2426_ID

In þæm mynstre heo & <u>Osweo</u> hire fæder & hire modor Eanflæd & hire modorfæder Eadwine & monige oðre æðele in Sancte Petres cirican þæs apostoles bebyrgde wæron . cobede,Bede:18.238.4.2427_ID

Þis **gefeoht** Osweo se cyning þy þreotteoþan geare his rices, in þæm londe þe Loidis hatte, þy seofonteoþan dæge Kalendarum Decembrium gefremede mid

micelre nytnisse æghwæðres **folces** . Forþon þe he his þeode alesde & generede from þære feondlican hergunge þara hæðenra, & eac swylce Mercna þeode & þara neahmægða, ofheawnum þy getreowleasan hæfde Pendan, to gife Cristes geleafan gecerde .

(cobede,Bede_3:18.238.7.2428)

Number	object	subject
OS-15	old	old

In the absence of focus, metrical prominence on *gefeoht* and *folces*. The CP introduced by *Forþon* counts probably an independent utterance (note the interpunction).

/~*

Se þridda biscop wæs Trumhær , se wæs Ongolcynnes , cobede,Bede:18.238.17.2434_ID

ac he wæs gelæred & gehadad from Scottum ; cobede,Bede:18.238.17.2435_ID

[…]

Wæron þær in þa tiid monige of Ongelþeode ge æðelinga ge oðerra , þa ðe in þara biscopa tide Finanes & Colmanes forleton heora eðelturf & þider gewiton , sume for godcundre leornunge , sume for intingan forhebbendran liifes . cobede,Bede:19.240.31.2465_ID

Ond sume sona in mynsterlicre drohtnunge in regollicum life getreowlice Drihtne þeowodon ; cobede,Bede:19.242.3.2466_ID

sume geond mynster eodon cobede,Bede:19.242.3.2467_ID

& him godcunde lareowas sohton . cobede,Bede:19.242.3.2468_ID

Ond ***EALLE HY*** Scottas lustlice onfengon

(cobede,Bede_3:19.242.5.2469)

Number	object	subject
OS-16	contr	old

The Scots are evoked throughout the chapter (see the ex. 2435). *Ealle hy* subsumes a set which was enumerated before and fulfils thus the requirements of contrastive focus.

/~*

Eft se papa nedde þone abbud Adrianus , þæt he biscophade onfenge . cobede,Bede:1.254.12.2582_ID

Þa bæd he hine eldenne & fyrstes , hwæðer he æfter fæce meahte oðerne findan , þe mon to biscope hadian meahte . cobede,Bede:1.254.13.2583_ID

Wæs in þa tid sum munuc in Rome , se wæs cuð $Adriane þæm abbude , þæs noma wæs Theodorus . cobede,Bede:1.254.15.2584_ID

Se wæs acenned in Tarso Cilicio : cobede,Bede:1.254.16.2585_ID

wæs se mon ge in weoruldgewreotum , ge in godcundum , ge in Grecisc ge in Læden wel gelæred ; cobede,Bede:1.254.16.2586_ID

& he gecoren wæs in his þeawum & arwyrðre ældo , þæt is þæt he hæfde syx & syxtig wintra . cobede,Bede:1.254.16.2587_ID

Þeosne **mon** Adrianus se abbud þæm **papan** cyðde, (cobede,Bede_4:1.254.20.2588)

Number	object	subject
OS-17	topic	old

In the absence of focus, metrical prominence is assigned to *mon* and *papan*.

/~*

Ond he hraðe þær mynster getimbrode , cobede,Bede:4.274.12.2786_ID

& him se gesið eac fultmade & ealle þa neahmen ; cobede,Bede:4.274.12.2787_ID

& he ðær þa Englescan men gesette & gestaðolede , cobede,Bede:4.274.12.2788_ID

ond Scottas forlet in þæm foresprecenan ealonde . cobede,Bede:4.274.12.2789_ID

Þæt **mynster** oð gen to dæge Englisce men þær in **elþeodignesse** habbað . (cobede,Bede_4:4.274.15.2790)

Number	object	subject
OS-18	topic	old

In the absence of focus, metrical prominence is assigned to *mynster* and *elþeodignesse*.

/~*

Þa he ða þas andsware onfeng , þa ongon he sona singan in herenesse Godes Scyppendes þa fers & þa word þe he næfre gehyrde , þære endebyrdnesse þis is : cobede,Bede:25.344.2.3451_ID

Nu sculon herigean heofonrices weard , meotodes meahte & his modgeþanc , weorc wuldorfæder , cobede,Bede:25.344.2.3452_ID

swa he wundra gehwæs , ece Drihten , or onstealde . cobede,Bede:25.344.2.3453_ID

<u>He</u> ærest sceop eorðan bearnum *heofon* to hrofe halig scyppend ; cobede,Bede:25.344.10.3454_ID

þa ***MIDDANGEARD*** monncynnes weard, ece Drihten, æfter teode firum foldan, frea

ælmihtig .

(cobede,Bede_4:25.344.10.3455)

Number	object	subject
OS-19	contr	topic

This is actually from a poetic insertion. It does not make much sense to assign peaks here; but note that *middangeard* is in contrast to *heofon*, therefore in focus, whereas the subject is the topic (*He* of preceding sent.)

/~*

Þa heo þa in þæt leoht bihygdelice locade & hit georne beheold , þa geseah heo þære foresprecenan Godes þeowe sawle Hilde þære abbudissan in þæm seolfan leohte , engla weorodum gelædendum , to heofonum up borenne beon . cobede,Bede:24.340.9.3409_ID […]

Ond eftsona æfter ðære abbudissan forðfore hio hwurfon to þam ærran unsefernessum ,

cobede,Bede:26.356.1.3583_ID

& ec micle manfulran fremedon . cobede,Bede:26.356.1.3584_ID

& mid ðy cwædon : Nu is sib & orsorhnes : hi ða instepe , ða hie laesest wendon , mid þy wiite ðæs foresprecenan wræces slægene wæron . cobede,Bede:26.356.3.3585_ID

ALL þas ðing me ðus gewurden se arwyrða min efenmæssepreost ***EEDGYLS*** sægde, se ða in ðam mynstre eardade & drohtode, & eft in ussum mynstre longe tide lifde & þær forðferde, æfter ðon monge ðara bigengena ðonon gewitan for þære burge tolesnesse .

(cobede,Bede_4:26.356.5.3586)

Number	object	subject
OS-20	contr	new

The whole paragraph is about the abbess Hilde (3409). *Eedgyls* is newly introduced (as can be seen from the rather elaborate description by the relative clause), but it is not clear whether he is in focus (he is not really presented, not playing a role in the subsequent discourse). The quantifier *all* on the anaphoric phrase induces focal emphasis. Note that a dative pronoun and an adjoined participial clause intervene between the two focused phrases.

/~*

Þær betweoh monige Ongelþeode , þa ðe oðþe mid sweorde ofslægene wæron oðöe þeodome bet æhte , oðþe of Peohta londe onweg flugon , ond eac swylce se arwyrða Godes mon Trumwine , se ðe heora biscop wæs , gewat mid his geferum , þa ðe wæron in þæm mynstre Æbbercurni , þæt is geseted in Engla londe ac hwæðre neah þæm sæ , þe Engla lond & Peohta tosceadaþ . cobede,Bede:27.358.16.3602_ID

[…]

Wæs þæs ilcan mynstres abbodesse in ða tid seo cynelice fæmne Ælflæd ætgædere mid Eanflæde hire meder , þære we beforan gemyngodan . cobede,Bede:27.358.29.3608_ID

Ah ða se biscop **þider** com, mycelne fultum gereces & somed hire lifes frofre

Gode seo wilsume **fæmne** in him gemette .

(cobede,Bede_4:27.358.31.3609)

Number	object	subject
OS-21	new	old

Fultum and *frofre* are new, but it is doubtful whether they are in focus. *Fæmne* is old. So we have nuclei on the subord. clause and on *fæmne*, all following material being of a sort that cannot be assigned prominence.

/~*

Ða feng Ealdfrið æfter Ecgfriðe to rice . cobede,Bede:27.358.33.3610_ID

Se man wæs se gelærdesta in gewreotum ; se wæs sægd , þæt he his broðor wære Osweos sunu þæs cyninges . cobede,Bede:27.360.1.3611_ID

& he ðone toworpnan steal þæs rices , þeah ðe hit wære binnan nearwum gemærum, eaðelice geedneowade . cobede,Bede:27.360.2.3612_ID

Þy geare þaa , þæt is ymb syx hund wintra & fif & hundehtatig from ðære Dryhtenlican menniscnesse , þæt Hloþhere Cantwara cyning ðis deadlice lif geendode & forðferde , se æftera Ecbyrhte his breðer , se ehta winter ricsade ; cobede,Bede:27.360.5.3613_ID

þa feng he to rice cobede,Bede:27.360.5.3614_ID

& þæt hæfde XII winter . cobede,Bede:27.360.5.3615_ID

Wæs he gewundad in Suþseaxna gefeohte , þa wið hine Eadric Ecgbyrhtes sunu gesomnade ; cobede,Bede:27.360.9.3616_ID

ond þa betweoh ðon þe hine man lacnade , he forðferde . cobede,Bede:27.360.9.3617_ID

& þa æfter him se ilca Eadric oðer healf gear þæt rice hæfde . cobede,Bede:27.360.11.3618_ID

Þa he ða **forðferde**, þa ðæt rice þa sum fæc ti de ***TWEONDE*** cyningas & fremde forluron & $towurpun, oþðæt heora riht cyning Wihtred, þæt wæs Ecgbyrhtes sunu, wæs in rice gestrongad .

(cobede,Bede_4:27.360.12.3619)

Number	object	subject
OS-22	topic	contr

The foreign and hesitant kings following Eadric are new and in contrast to the other kings beforehand and especially to *Wihtred* in the following clause. The focalized element is on one of the adjectives (probably the extraposed *tweonde*), as they describe the property which distinguishes these kings from the other mentioned kings. There is a 2ary metrical prominence on the *tha*-clause.

/~*

Þæs mynstres profost & regolweard wæs in ða tid Boisel , se wæs mycelra mægena mæssepreost & witedomes gastes . cobede,Bede:28.360.32.3628_ID

Ðisses discipulhada Cuðbyrht wæs eadmodlice underþeoded ; cobede,Bede:28.360.33.3629_ID

[…]

no æghweðerne gedwolan to gereccenne se Godes man wæs ut gongende of ðæm mynstre gelomlice : cobede,Bede:28.362.17.3635_ID

hwilum wæs on horse sittende , ac oftor on his fotum gangende ; cobede,Bede:28.362.17.3636_ID

for to þæm ymbsettum tunum cobede,Bede:28.362.17.3637_ID

& ðæm dwoligendum soðfæstnesse weg $bodode & lærde . cobede,Bede:28.362.17.3638_ID

Ðæt **sylfe** eac swylce Boisel his magister on his **tide** gewunelice dyde . (cobede,Bede_4:28.362.21.3639)

Number	object	subject
OS-23	anaphoric	old

In the absence of focus, metrical prominence is on *sylfe* and *tide*.

/~*

& þus cwæð : cobede,Bede:30.372.5.3718_ID

Ic ðe halsige þurh ðone lifigendan Drihten , ðæt ðu mec ne forlæte , ac þæt þu sie gemyndig getreowan geðoftan , ond bidde þa uplican arfæstnesse , ðæt þæm Gode , þe wit somod on eorðan ðeowedon ðæt wit eac somod moton to heofenum $feran his gife þær to geseonne & to sceawigenne .cobede,Bede:30.372.5.3719_ID

Forðon þu wast ðæt ic symle teolode to lifigenne to ðines muðes bebode , cobede,Bede:30.372.11.3720_ID

& swa hwæt swa ic for unwisnesse & for tydernesse agylte , ic þæt to dome ðines willan teolode hraðe to gebetenne . cobede,Bede:30.372.11.3721_ID

Ða aðenede se biscop hine in cruce cobede,Bede:30.372.14.3722_ID

& hine gebæd , cobede,Bede:30.372.14.3723_ID

& sona wæs in gaste gelæred , þæt he wæs from Dryhtne tigða þære bene , ðe he bæd ; cobede,Bede:30.372.14.3724_ID

ond cwæð . <T06900073100,30.372.16> Aris min broðor cobede,Bede:30.372.16.3725_ID

& ne wep þu , cobede,Bede:30.372.16.3726_ID

ah gefeoh & geblissa : forðon seo uplice arfæstnes unc forgeaf , ðæt wit bædon . cobede,Bede:30.372.16.3727_ID

Ðæs gehates & ðæs witedomes soð se ***AFTERFYLGENDA*** becyme ðara wisena geseðde & getrymede .

(cobede,Bede_4:30.372.19.3728)

se afterfylgenda becyme' introduces sentence 3729 etc.

Forðon ða hie betweoh him togangende wæron , ða hie ofer ðæt lichomlicum eagum ne gesawon ; cobede,Bede:30.372.20.3729_ID

Number	object	subject
OS-24	anaphoric	new

The cataphoric element *se afterfylgenda becyme* is presentational and therefore in focus. The word in focus is *afterfylgenda*, as it also is implicitly in contrast to the previous discourse that is subsumed by the topicalized anaphoric NPs.

/~*

Godcund gewrit us to eaðmodnesse myngað þus clypiende : cobenrul,BenR:7.22.10.333

ælc þe hine anhefð , he bið geneoþerad , cobenrul,BenR:7.22.10.334

and ælc þe hine mid eaðmodnesse geneoþerað , he bið mid weorðmynte onhafen .

cobenrul,BenR:7.22.10.335

Þurh þas clypunge is gesweotolad , þæt ælc upahafednes asprincð of modignesse cynrene ;

cobenrul,BenR:7.22.13.336

[…]

Nis butan tweon to understandenne se upstige and se niþerstige on nane oþere wisan , butan þæt heo

fona rices upstige mid eadmodnesse geearnod bið and mid ofermettum forwyrht .

cobenrul,BenR:7.23.6.344

Seo arærede hlædder tacnað ure lif on ðisse weorulde , ðæt bið mid eaðmodre heortan þurh Drihten arӕred

to heofonum ; cobenrul,BenR:7.23.9.345

þære hlædre sidan tacniað lichoman and saule ; cobenrul,BenR:7.23.9.346

on ðæm twam sidum, þæt is on saule and on lichoman, missenlice ***STÆPAS***

eaðmodnesse and þeawfæstnesse sio godcunde gelaðung to ðæm upstige

gefæstnode .

(cobenrul,BenR:7.23.9.347)

BE ÐAM TWELF STÆPUM EALRE EAÐMODNESSE . cobenrul,BenR:7.23.16.348

Number	object	subject
OS-25	new	old

The *gelaðunge* is the quote in 333f. It is therefore old. Whereas in the intervening discourse there is a lot of talk about ‘going up and down’, the introduction of the steps should count as new, and moreover as (presentational) focus, as this is going to be the topic of the next chapter. So there is focal emphasis on *stæpas*.

/~*

He cwæþ , Ga þu onbæcling , coblick,HomS_[BlHom_3]:31.78.412_ID

& gemyne þe sylfne hu mycel yfel þe gelamp for þinre gitsunga & oforhydo , & f

or þinum idlan gilpe ; coblick,HomS_[BlHom_3]:31.78.413_ID

& forþon ic þe ne fylge , coblick,HomS_[BlHom_3]:31.78.414_ID

forþon on þyssum þrim þu eart oforswiþed . coblick,HomS_[BlHom_3]:31.78.415_ID

Þas **cyþnesse** Drihten nam of þisse **wisan** .

(coblick,HomS_10_[BlHom_3]:31.81.416)

Number	object	subject
OS-26	anaphoric	topic

Drihten is actually the speaker of 412-415; he qualifies as topic as he was the topic (pronominalized) for the whole discourse before that; last mentioned explicitly (as full NP) in 406. In the absence of focus, metrical prominence is assigned.

/~*

Þysne dæg hie nemdon siges dæg ; coblick,HomS_[BlHom_6]:67.18.814_ID

se nama tacnaþ þone sige þe Drihten gesigefæsted wiþstod deofle , þa he mid

his deaþe þone ecan deaþ oferswiþde , swa he sylf þurh þone witgan sægde ;

coblick,HomS_[BlHom_6]:67.18.815_ID

he cwæþ , Eala deaþ , ic beo þin deaþ , coblick,HomS_[BlHom_6]:67.18.816_ID

& ic beo þin <u>bite</u> on helle . coblick,HomS_[BlHom_6]:67.18.817_ID

MYCELNE bite Drihten dyde on **helle** þa he þyder astag, & helle bereafode, &

þa halgan sauwla þonon alædde, & hie generede of deofles anwalde, þa to þeowdome þyder on fruman middangeardes gesamnode wæron .

(coblick,HomS_21_[BlHom_6]:67.22.818)

Number	object	subject
OS-27	contr	topic

Quantifiers tend to induce contrastive foci (*much* as opposed to *little*, etc.). In this case it is almost like focus-movement: 'bite' is salient, but it's not just some bite, it's a BIG bite. *helle* is sufficiently far away that it can have a normal metrical prominence.

/~*

Forþon se hindsið mancynnes & þæt heaflice gewrit þæt wearð þys dæge fordilegod , & se sarlica cwide eft oncerred , þe ure Drihten ær þurh eornesse to þæm ærestan men cwæþ :

coblick,HomS_[BlHom_11]:123.119.1539_ID

Terra es et in terram ibis ; coblick,HomS_[BlHom_11]:123.119.1540_ID

þu eart eorþe , he cwæþ , coblick,HomS_[BlHom_11]:123.119.1541_ID

& þu scealt on eorþan gangan & eft to eorðan weorðan . coblick,HomS_[BlHom_11]:123.119.1542_ID

On þa ***ILCAN*** menniscan gecynd þe he þæt ær þurh eornesse swa tocwæþ,

þa ***ILCAN*** he ure Drihten on þas halgan tid on him ***SYLFUM*** ahof, ofer heofonas

& ofer ealle engla þreatas .

(coblick,HomS_46_[BlHom_11]:123.124.1543)

Number	object	subject
OS-28	contr	old

The reference is a bit tricky. The passage is about Jesus as the new Adam, that is: the man who is mostly different from Adam by the absence of sin, and the discrepance between 'old' man as the outcast from Paradise, and the 'new' man Jesus who is higher than the angels. I think the thought goes back to St.Paul. The author refers with *þa ilcan* on man in a generic sense. As there is a contrast implied, it is justified to put contrastive focus on *þa ilcan*. There is then also one on the *ilcan* in the introductory PP. The intervening relative clause remedies the focus clash. Another focus is probably on *sylfum*. The subject *he ure Drihten* is (obviously) old.

/~*

Him þa se heora arwyrða bisceop eadiglice & halwendlice geðeaht forðbrohte ,

coblick,LS_[MichaelMor[BlHom_17]]:205.153.2624_ID

& hie lærede þæt hie raðost to Rome sendon to ðæm papan , & ðone papan & þæt papseld þæt hie befrinon

& beahsodan hwæt him þæs to ræde þuhte , hweþer hie þa ciricean halgian dorston on oþre wisan .

coblick,LS_[MichaelMor[BlHom_17]]:205.153.2625_ID

Þ***ISLIC*** ærende se eadiga papa ða ðær eft **onsende**

(coblick,LS_25_[MichaelMor[BlHom_17]]:205.158.2626)

Number	object	subject
OS-29	anaphoric	old

The word *Þislic* introduces a contrast (such and no other) and thus it bears focus. The assignment would be similar if there was no focus on *Þislic*.

/~*

Ða genam Sanctus Martinus hine be his handa , coblick,LS_[MartinMor[BlHom_17]]:219.147.2804_ID

& upheah arærde , coblick,LS_[MartinMor[BlHom_17]]:219.147.2805_ID

& hine lædde forð to þon cafortune þæs huses , coblick,LS_[MartinMor[BlHom_17]]:219.147.2806_ID

& hine eft þæm mannum halne & gesundne ageaf , þæm þe hine ær deadne leton .

coblick,LS_[MartinMor[BlHom_17]]:219.147.2807_ID

Ðas wundor & manig **oþer** ælmihtig God þurh þysne eadigan **wer** worhte, ær þon þe he æfre bisceop wære .

(coblick,LS_17.1_[MartinMor[BlHom_17]]:219.150.2808)

Number	object	subject
OS-30	anaphoric	old

Ðas wundor subsumes the preceding discourse. Not emphasized. *God* is salient throughout the text and counts therefore as old and thus unemphasized. In the absence of focus metrical prominence is assigned.

/~*

ða ongan se Wisdom singan coboeth,Bo:9.21.1.340

& giddode þus : coboeth,Bo:9.21.1.341

ÞONNE seo sunne on hadrum heofone beorhtost scineð , þonne aþeostriað ealle steorran , $forþam $þe <TEXT:forþamþe> heora beorhtnes ne beoð nan beorhtnes for hire . coboeth,Bo:9.21.1.342

þonne smylte blaweð suþanwestan wind , þonne weaxað swiðe hraðe feldes blosman ; coboeth,Bo:9.21.4.343

ac þonne se stearca wind cymð norðaneastan , þonne toweorpð he swiðe hraþe þære rosan wlite ; coboeth,Bo:9.21.4.344

swa oft þone to smylton sæ þæs norðanwindes **yst** onstyreð .

(coboeth,Bo:9.21.4.345)

Number	object	subject
OS-31	old	old

The contrast between south and northwind has been established in 344. The northwind is taken up. Calmness of sea is also a concept that has been introduced. In the absence of foci metrical prominence is assigned.

/~*

Hwæðer nu se anwald hæbbe þone þeaw ðæt he astificige *unðeawas* & $awyrtwalige *of ricra monna* $mode , $& plantige ðær cræftas on ? coboeth,Bo:27.61.7.1133

Ic wat ðeah þæt se eorðlica anweald næfre ne sæwð þa cræftas , ac lisð & gadrað *unðeawas* ; coboeth,Bo:27.61.9.1134

& þonne hi gegadrad hæfð , þonne eowað he hi , coboeth,Bo:27.61.9.1135

nallas ne hilð . coboeth,Bo:27.61.9.1136

forþam ðara ricra monna unðeawas ***MANIGE*** men geseoð, $forþam $ðe hi manige cunnon, & manege him mid beoð .

(coboeth,Bo:27.61.12.1137)

Number	object	subject
OS-32	old	contr

The *manige* introduces a contrast 'some…some…' in the sub. cl. Therefore it bears contrastive focus. As the paragraph is about sin and rich men, the object is old, if not topical.

/~*

& *God* þa geswefode þone Adam cocathom1,ÆCHom_1:181.88.82_ID

& þa þa he slep ða genam he an rib of his sidan . cocathom1,ÆCHom_1:181.88.83_ID

& geworhte of ðam ribbe ænne wifman . cocathom1,ÆCHom_1:181.88.84_ID

& axode Adam hu heo hatan sceolde . cocathom1,ÆCHom_1:181.88.85_ID

Þa cwæð Adam . heo is ban of minum banum . & flæsc of minum flæsce . cocathom1,ÆCHom_1:182.90.86_ID beo hire nama uirago . þæt is fæmne for ðan ðe heo is of hire were genumen .

cocathom1,ÆCHom_1:182.90.87_ID

Ða sette Adam eft hire oðerne naman . Aeua . þæt is lif for ðan ðe heo is ealra lybbendra modor .

cocathom1,ÆCHom_1:182.92.88_ID

EALLE gesceafta heofonas & englas . sunnan & monan . steorran & eorðan .

ealle nytenu & fugelas . sæ . & ealle fixas . & ***EALLE*** gesceafta God

gesceop &

geworhte on six **dagum** .

(cocathom1,ÆCHom_I,_1:182.95.89)

number	obj	subj
OS-33	contr	old

The all-quantifier induces focus, especially as several members of the relevant set are explicitly listed. God is old information. A metrical prominence is assigned to the right edge.

/~*

Hit wæs gewitegod þæt he on ðære byrig Bethl+e+em acenned wurde

cocathom1,ÆCHom_2:197.208.462_ID

& heo þearle wundrode þæt heo æfter þære witegunge þær acende . cocathom1,ÆCHom_2:197.208.463_ID

Heo gemunde þæt sum witega cwæð . se oxa oncneow his hlaford . & se assa his hlafordes binne :

cocathom1,ÆCHom_2:197.209.464_ID

þa geseah heo þæt cild licgan on binne . þær se oxa & se assa gewunelice fodan secað .

cocathom1,ÆCHom_2:197.209.465_ID

Godes heahengel Gabriel bodade <u>Marian</u> þæs hælendes tocyme on hire innoðe :

cocathom1,ÆCHom_2:197.212.466_ID

& heo geseah ða þæt <u>his bodung</u> unleaslice gefylled wæs . cocathom1,ÆCHom_2:197.212.467_ID

Ðillice **word** Maria heold aræfniende on hyre **heortan** .

(cocathom1,ÆCHom_I,_2:197.214.468)

number	obj	subj
OS-34	anaphoric	old

In the absence of focus, metrical prominence is assigned.

/~*

Soðlice he sylf ætbræd ure adlunga . cocathom1,ÆCHom_8:243.53.1435_ID

& ure sarnyssa he sylf bær .

(cocathom1,ÆCHom_I,_8:243.53.1436)

Number	obj	subj
OS-35	contr	old

adlunga (pain, sorrow) and *sarnyssa* (grief) are more or less the same concept, but it is most probably viewed as contrastive by the translator.

/~*

Crist cwæð to ðam apostolum . ***ÞÆRA*** manna synna þe ge ***FORGIFAÐ*** . þara beoð ***FORGIFENE*** .

(cocathom1,ÆCHom_I,_16:309.73.2982)

And þ*am* þe ge *ofteoð* þa forgifenysse : þam bið *oftogen* . cocathom1,ÆCHom_16:309.74.2983_ID

number	obj	subj
OS-36	contr	old

The subject, although coreferent with the object, counts as old, as the referent has been introduced by the topicalized object. The contrast is between the object (well, the possessive complement of the respective objects) and the full verb *paret* of the predicate.

/~*

We habbað þone geleafan þe Crist sylf tæhte his apostolum . & hi eallum mancynne :

cocathom1,ÆCHom_20:343.246.4075_ID

& þone **geleafan** God hæfð mid manegum **wundrum** getrymmed & gefæstnod .

(cocathom1,ÆCHom_I,_20:343.246.4076)

number	obj	subj
OS-37	topic	old

In the absence of focus, metrical prominence is assigned.

/~*

Se cyning eode inn . cocathom1,ÆCHom_35:480.142.7016_ID

& gesceawode þa gebeoras . cocathom1,ÆCHom_35:480.142.7017_ID

þa geseah he þær ænne mann þe næs gescryd mid gyftlicum reafe .

cocathom1,ÆCHom_35:480.142.7018_ID

Þæt giftlice reaf getacnað þa soþan lufe Godes & manna . cocathom1,ÆCHom_35:480.143.7019_ID

Þa **lufe** ure scyppend us geswutelode þurh hine **sylfne** þa ða he gemedemode þæt he us fram þam ecan deaþe mid his deorwurþan blode alysde: swa swa Iohannes se godspellere cwæð .

(cocathom1,ÆCHom_I,_35:480.144.7020)

number	object	subject
OS-38	topic	old

In the absence of focus, metrical prominence is assigned.

/~*

Be Cristes deaðe witegode se ylca Dauid . cocathom2,ÆCHom_1:8.195.172_ID

and cwæð be Cristes lice ; <T02520009100,8.196> Min lichama gerest on hihte . for ðan þe þu ne forlætst mine sawle on helle . cocathom2,ÆCHom_1:8.196.173_ID

ne ðu ne geðafast þæt min lichama gebrosnige ; cocathom2,ÆCHom_1:8.196.174_ID

Ðas **word** Crist geclypode to his **fæder**;

(cocathom2,ÆCHom_II,_1:8.198.175)

number	obj	subj
OS-39	anaphoric	topic

In the absence of focus, metrical prominence is assigned.

/~*

Ic cweðe Cristes fulluht ; cocathom2,ÆCHom_3:25.212.602_ID

Feawa manna Crist ***SYLF*** gefullode .

(cocathom2,ÆCHom_II,_3:25.214.603)

ac he forgeaf ðone anweald his apostolon . and eallum gehadedum mannum þæt hi sceoldon fullian mid Godes fulluhte . on naman ðære halgan ðrynnysse . cocathom2,ÆCHom_3:25.214.604_ID

and swa gefullod mann ne beo na eft oðre siðe gefullod . þæt ne sy foresewen þære halgan ðrynnysse toclypung ; cocathom2,ÆCHom_3:25.214.605_ID

number	obj	subj
OS-40	old	contr

Feawa functions here more like a indefinite pronoun and is therefore unemphasized; *Crist sylf* is probably focussed.

/~*

On ðære oðre ylde þissere worulde wearð eal middaneard mid <u>flodes</u> yðum adylegod for synna micelnysse . buton ðam rihtwisan Noe anum . and his seofan hiwon . þe on ðam arce belocene wæron to anes geares fyrste. cocathom2,ÆCHom_4:33.111.746_ID

[…]

Witodlice ða *eahta menn þe se arc on his bosme abær* wurdon ahredde wið þam yðigendum flode . cocathom2,ÆCHom_4:33.122.749_ID

and ealle ***OÐRE*** eorðlice gesceafta þæt brade wæter **adydde**;

(cocathom2,ÆCHom_II,_4:33.122.750)

number	obj	subj
OS-41	contr	topic

The object is clearly in contrast. The subject is the topic (diluvium).

/~*

Se trahtnere cwið þæt þæt gyftlice hus wæs ðryflere . for ðan ðe on Godes gelaðunge sind þry stæpas gecorenra manna ; cocathom2,ÆCHom_4:39.297.870_ID

Se nyðemysta stæpe is on geleaffullum læwedum mannum . þe on rihtum sinscipe wuniað . swiðor for bearnteame þonne for galnysse ; cocathom2,ÆCHom_4:39.299.871_ID

Se oðer stæpe is on wydewan hade þe æfter rihtre æwe on clænnysse wuniað . for begeate þæs upplican lifes ; cocathom2,ÆCHom_4:39.301.872_ID

Se hehsta stæpe is on mægðhades mannum . þa ðe fram cildhade clænlice gode þeowigende . ealle middaneardlice gælsan forhogiað ; cocathom2,ÆCHom_4:39.303.873_ID

Se drihtealdor cwæð to ðam brydguman . ælc man sylð on forandæge his gode win . cocathom2,ÆCHom_4:39.306.874_ID

`and þæt` **`waccre`** `þonne ða` **`gebeoras`** `druncniað .`

`(cocathom2,ÆCHom_II,_4:39.306.875)`

number	object	subject
OS-42	old	old

The meaning is a bit arcane. A 'guest' is among the things evoked by the wedding-setting in 874, so it can count as inferrable (actually, the whole passage is an interpretation of the wedding at Cana). 'Being vigilant' is kind of new, but probably also inferrable, as in other places in the Scriptures (the parabel of the foolish maiden, for instance) the wedding-setting and vigilance are combined and therefore vigilance can count as inferrable by association. I would be hesitant to put presentational focus on it, at any rate, as the metaphor is not followed up. Therefore I simply assign metrical prominence.

`/~*`

On sumere tide ða ða micel menigu samod com to ðam hælende . and fram gehwilcum burgum to him genealæhton . þa sæde he him þis bigspel ; cocathom2,ÆCHom_6:52.1.1061_ID

Sum sædere ferde to sawenne his sæd . cocathom2,ÆCHom_6:52.4.1062_ID

[…] (the parabel of the sower)

`Þas` **`word`** `Drihten clypigende` **`cwæð`** `.`

`(cocathom2,ÆCHom_II,_6:52.13.1071)`

number	obj	subj
OS-43	anaphoric	old

In the absence of focus, metrical prominence is assigned.

/~*

Ne geortruwige nan man hine sylfne for *his synna* micelnysse . cocathom2,ÆCHom_9:76.136.1537_ID

witodlice ða ealdan gyltas ***NINIUEISCRE*** ðeode . ðreora daga bereowsung adilegode .

(cocathom2,ÆCHom_II,_9:76.136.1538)

number	obj	subj
OS-44	contr	old

Penitence (*bereowsung*) is salient throughout the discourse. *gyltas Niniueiscre ðeode* is in contrast to his (= the random person's) guilt

/~*

Be þam þæt preostas sceolan forbugan & asceonian druncen . cochdrul,ChrodR:60.0.788_ID

Drihten on his godspelle cwyð , Begymað þæt eowre heortan ne wurðon gehefgode mid oferfylle . cochdrul,ChrodR:60.1.789_ID

& se apostol cwyð , Nelle ge eow oferdrincan on wine , on þam is gælsa . cochdrul,ChrodR:60.2.790_ID

[…]

Win & druncene wif gedoð hwilon þæt witon maffiað . cochdrul,ChrodR:60.7.796_ID

& eft hit cwyð , Ne gelaða þu to þire gesamnunge þa þe lufiað þæt hi hi sylfe an wine oferdrincon . cochdrul,ChrodR:60.8.797_ID

Druncene mæssepreostas & se apostol **genyðrað**,

(cochdrul,ChrodR_1:60.9.798)

number	object	subject
OS-45	topic	old

The apostle and priests are salient throughout the discourse. The passage is on drunkenness. In the absence of focus, metrical prominence is assigned.

/~*

Ne beo ge ofergyttole , cochdrul,ChrodR:79.89.995_ID

ne ne befeolan ge orsorhnysse , cochdrul,ChrodR:79.89.996_ID

ac gegearciað eower underþeoddum preostum ægðer ge lichaman bilyfne ge sawle , þæt hi bliðe mid eow wunian an þam Cristes þeowdome , þe hi on þeowian sceolon , þæt ge an þam towerdan mede underfon & gehyron Drihtnes stefne , þær he cwyð , þa þu wære an littlum þingum getrywe , ic gesette þe ofer manege þincg , cochdrul,ChrodR:79.89.997_ID

far an blysse þines hlafordes . cochdrul,ChrodR:79.89.998_ID

ME synfulne & ***EOW EALLE*** & ealle ure underþeoddan, þurh ealra haligra þingrædene, ure hælenda Crist gelæde ungewemmede an þa **blisse**, þæt is an ece lif, þær he lifað & rixað a butan ende .
(cochdrul,ChrodR_1:79.95.999)

number	object	subject
OS-46	contr	old

The *hælend* is salient throughout the discourse. The object is an exhaustive set {I, you guys} which by that probably attracts focal emphasis. On the right edge a (2ary) metrical prominence is realized.

/~*

þa com þæm Deniscum scipum þeh ær flod to , ær þa Cristnan mehten hira utascufan ,

cochronA2b,ChronA_[Plummer]:897.42.1148

& hie forðy ut oðreowon ; cochronA-2b,ChronA_[Plummer]:897.42.1149

Þa wæron hie to þæm gesargode . þæt hie ne mehton Suð Seaxna lond utan berowan , cochronA-2b,ChronA_[Plummer]:897.48.1150

`ac hira þær` ***`TU`*** `sæ on` **`lond`** `wearp .`

`(cochronA-2b,ChronA_[Plummer]:897.48.1151)`

& *þa men* mon lædde to Winteceastre to þæm cynge, cochronA-2b,ChronA_[Plummer]:897.50.1152

& he hie ðær ahon het . cochronA-2b,ChronA_[Plummer]:897.50.1153

& *þa men* comon on East Engle . þe on þæm anum scipe wæron . swiðe forwundode . cochronA-2b,ChronA_[Plummer]:897.51.1154

Number	object	subject
OS-47	contr	old

It is about two sets of Danes, the ones being led to Winchester, the others going to East Anglia. Sentence 1151 is a nice example of split topicalization. The numeral *tu* induces contrast (to the other ships that do not land). A 2ary metrical prominence is assigned on the right edge.

`/~*`

Þa com þam Deniscan scypum þeah ær flod to , ær þa cristenan mihton hira ut ascufan ,

cochronC,ChronC_[Rositzke]:897.38.997

and hi forði ut oþreowon . cochronC,ChronC_[Rositzke]:897.38.998

Þa wæron hi to ðam gesargode þæt hi ne mihton Suðsexanaland utan berowan ,

cochronC,ChronC_[Rositzke]:897.40.999

`ac hira þær` ***`TWA`*** `sæ on` **`land`** `wearp,` (cochronC,ChronC_[Rositzke]:897.40.1000)

and ða menn mon lædde to Winteceaster to þam cinge , cochronC,ChronC_[Rositzke]:897.40.1001

and he hi þær ahon het , cochronC,ChronC_[Rositzke]:897.40.1002

and þa menn comon on Eastengle þe on ðam anum scype wæron swyðe forwundode .

cochronC,ChronC_[Rositzke]:897.40.1003

Number	object	subject
OS-48	contr	old

It is about two sets of Danes, the ones being led to Winchester, the others going to East Anglia. Sentence 1151 is a nice example of split topicalization. The numeral *twa* induces contrast (to the other ships that do not land). A 2ary metrical prominence is assigned on the right edge.

/~*

context see last two exx.

ac hyra þær ***TWA*** sæ on **land** wearp,

(cochronD,ChronD_[Classen-Harm]:897.48.922)

Number	object	subject
OS-49	contr	old

It is about two sets of Danes, the ones being led to Winchester, the others going to East Anglia. Sentence 1151 is a nice example of split topicalization. The numeral *twa* induces contrast (to the other ships that do not land). A 2ary metrical prominence is assigned on the right edge.

/~*

Her com Eadward æþeling to Englalande , se wæs Eadwerdes broðor sunu kynges ,

cochronD,ChronD_[Classen-Harm]:1057.1.2087

Eadmund cing , Irensid wæs geclypod for his snellscipe . cochronD,ChronD_[Classen-Harm]:1057.1.2088

Þisne **æþeling** Cnut cyng hæfde forsend on **Ungerland** to beswicane,

(cochronD,ChronD_[Classen-Harm]:1057.5.2089)

number	object	subject
OS-50	topic	old

In the absence of focus (Cnut has not been mentioned for quite a while, but he is salient as being one of the main players in Old English history), metrical prominence is asigned.

/~*

Sio gedrefednes ðære *ungeðylde* on ðæm mode ðæt is se smala ciið , cocuraC,CP%[Cotton]:33.224.2.83

ac se *yfela* willa on ðære heortan ðæt is se greata beam . cocuraC,CP%[Cotton]:33.224.2.84

Ðone ***UNGEÐYLDEGAN*** ðonne swiðe ***LYTEL*** scur ðære costunga mæg onhreran,

swæ swæ lytel wind mæg ðone cið awecggean,

(cocuraC,CP_[Cotton]:33.224.4.85)

ac ðone *yfelan* fæstrædan willan folneah *nan* wind ne mæg awecggean . cocuraC,CP%[Cotton]:33.224.4.86

number	object	subject
OS-51	contr	contr

Note that an adverb is intervening which is furthermore on a weird position. Therefore the potential accentclash is remedied. Ungeðyld is forming a set (a scale, rather) with *yfel*; they are in a classic poset relationship. There is a second contrast: Whereas impatience can be shaken by a quite soft wind, no wind at all can shake an evil mindset. So this example is just like the Modern double focus topicalization examples.

/~*

ac ðone ***YFELAN*** fæstrædan willan folneah ***NAN*** wind ne mæg awecggean .

(cocuraC,CP_[Cotton]:33.224.4.86)

number	object	subject
OS-52	contr	contr

See last sentence. 86 is probably pronounced with a pause between the clashing foci (the word order is this way for stylistic reasons, viz. to create a parallelism).

/~*

Suaðeah mid ðy selflice se Dema bið genieded to ðæm ierre , cocura,CP:4.39.10.202

& se Dema se ðe ðæt inngeðonc eall wat , he eac ðæm inngeðonce demð . cocura,CP:4.39.10.203

We magon monnum bemiðan urne geðonc & urne willan , ac we ne magon Gode . cocura,CP:4.39.11.204

Hwæt se Babylonia cyning wæs suiðe upahafen on his mode for his anwalde & for his gelimpe , ða he fægnode ðæs miclan weorces & fægernesse ðærre ceastre , cocura,CP:4.39.13.205

& hine oðhof innan his geðohte eallum oðrum monnum , cocura,CP:4.39.13.206

& suigende he cwæð on his mode : Hu ne is ðis sio micle Babilon ðe ic self atimbrede to kynestole & to ðrymme , me selfum to wlite & wuldre , mid mine agne mægene & strengo ? cocura,CP:4.39.13.207

Ða suigendan **stefne** suiðe hraðe se diegla **Dema** gehirde,

(cocura,CP:4.39.18.208)

number	object	subject
OS-53	anaphoric	old

In the absence of focus, metrical prominence is assigned

/~*

Ðæt ilce ðæt he untælwyrðlice ondred to underfonne, ðæt ilce se ***OÐER***

swiðe hergeondlice gewilnode .

(cocura,CP:7.49.18.289)

Oðer ondred ðæt he forlure sprecende ða gestrion ðe he on ðære swigean geðencan meahte ;

cocura,CP:7.49.19.290

Oðer ondred ðæt he ongeate on his swygean ðæt he sumne hearm geswigode ðær ðær he freme gecleopian meahte , gif he ymb ðæt geornlice swunce . cocura,CP:7.49.20.291

Number	object	subject
OS-54	anaphoric	contr

se oðer is in contrast (as are the 'others' in the following sentences) to the *he* of the left dislocated subordinate clause, therefore bears contrastive focus. The anaphoric pronoun (object) is not emphasized.

/~*

Be ðæm geðence se sacerd , ðonne he oðre men healice lærð , ðæt he eac on him selfum healice ofðrysce ða lustas his unðeawa , $forðæm $ðe <TEXT:forðæmðe> he kynelic hrægl hæfð , ðæt he eac sie kyning ofer his agne unðeawas , & ða cynelican ofersuiðe ; cocura,CP:14.85.11.551

& geðence he simle sie sua æðele sua unæðele suæðer he sie ða æðelu ðære æfterran acennesse , ðæt is on ðæm fulluhte , cocura,CP:14.85.14.552

& simle atiewe on his ðeawum ða ðing ðe he ðær Gode gehet , & ða ðeawas ðe him mon ðær bebead .

cocura,CP:14.85.14.553

Be ðæm æðelum ðæs gæstes Petrus cuæð : Ge sint acoren kynn Gode & cynelices preosthades .

cocura,CP:14.85.17.554

Bi ðæm anwalde , ðe we sculon ure unðeawas mid ofercuman , we magon beon getrymede mid Iohannes cuide ðæs godspelleres , ðe he cuæð : cocura,CP:14.85.19.555

ða ðe hine onfengon he salde him anwald ðæt hie meahton beon Godes bearn . cocura,CP:14.85.19.556

Ða **medomnesse** ðære strengio se salmscop **ongeat**, ða he cuæð: Dryhten, suiðe suiðe sint geweorðode mid me ðine friend,

(cocura,CP:14.85.22.557)

number	object	subject
OS-55	topic	old

strengu is another word for *onweald*. The psalmist is cited every now and then as a source. In the absence of focus, metrical prominence is assigned.

/~*

For ðysum wæs geworden ðætte Paulus , ðeah ðe he wære gelæded on neorxna ong he arimde ða diogolnesse ðæs ðriddan hefones , ond suaðeah for ðære sceawungge ðara ungesewenlicra ðinga ðeah ðe he up aðened wære on his modes scearpnesse , ne forhogde he ðæt he hit eft gecierde to ðam flæsclican burcotum , & gestihtode hu men scoldon ðærinne hit macian , ða he cuæð : Hæbbe ælc monn his wif , & ælc wif hiere ciorl ; cocura,CP:16.99.6.643

& doo ðæt wif ðæm were ðæt hio him mid ryhte doon sceal , & he hire sua some , ðylæs hie on unryht hæmen . cocura,CP:16.99.12.644

& hwene æfter he cuið : Ne untreowsige ge no eow betweoxn , buton huru ðæt ge eow gehæbben sume hwile , $ærðæm $ðe <TEXT:ærðæmðe> ge eowru gebedu & eowra offrunga doon wiellen , & eftsona cirrað to eowrum ryhthæmede . cocura,CP:16.99.14.645

[…]

Bi ðon wæs gecueden on ðære æ : Ne forbinden ge na ðæm ðyrstendum oxum ðone muð . cocura,CP:16.105.7.690

Ðone **cwide** Paulus geryhte eft to **biscepum** ðara openlican weorc we gesioð,

(cocura,CP:16.105.8.691)

Number	object	subject
OS-56	anaphoric	old

In the absence of focus, metrical prominence is assigned.

/~*

Ic cuæð ðæt æghwelc monn wære gelice oðrum acenned , cocura,CP:17.107.18.709

ac <u>sio ungelicnes</u> hira geearnunga hie tiehð sume behindan sume , cocura,CP:17.107.18.710

& hira scylda hi ðær gehabbað . cocura,CP:17.107.18.711

Hwæt ðonne ða **ungelicnesse** ðe of hira unðeawum forðcymeð, se godcunda **dom** geðencð $ðætte ealle men gelice beon ne magon,

(cocura,CP:17.107.20.712)

Number	object	subject
OS-57	old	new

se godcunda dom is not mentioned or inferred before therefore counts as new. It is not used in the following discourse, therefore it is unlikely to have a focus. Metrical prominence is assigned.

/~*

Be ðæm se forma hierde *Sanctus Petrus* geornfullice monode , cocura,CP:18.137.15.934

& cuæð : Ic , eower emnðeowa & Cristes ðrowunge gewita , ic eow healsige ðæt ge feden Godes heorde ðe under eow is . cocura,CP:18.137.15.935

[…]

Ðas ilcan geornfulnesse ðara hierda Sanctus ***PAULUS*** aweahte, ða he cuæð: Se ðe ne gimð ðara ðe his beoð, & huru Godes ðeowa, he wiðsæcð Godes geleafan,
(cocura,CP:18.137.25.939)

Number	object	subject
OS-58	anaphoric	contr

Paulus is in contrast to Petrus and receives focal emphasis.

/~*

Forðæm is micel ðearf , ðonne se reða reccere ongiett ðæt he his hieremonna mod suiður gedrefed hæfð ðonne he scolde , ðæt he sona forðæm hreowsige , ðæt he ðurh ða hreowsunga gemete forgiefnesse beforan ðære Soðfæsðnesse ðæs ðe he ðurh ða geornfulnesse his andan gesyngade . cocura,CP:21.165.19.1130

Ðæt **ilce** Dryhten God us bisnade ðurh **Moysen**, ða he cuæð: Gif hwa gonge bilwitlice mid his friend to wuda treow to ceorfanne, & sio æcs ðonne awient
of ðæm hielfe, & sua ungewealðes ofslieð his geferan, he ðonne sceal fleon to anra ðara ðreora burga ðe to friðstowe gesette sint
(cocura,CP:21.165.23.1131)

Number	object	subject
OS-59	anaphoric	old

God is explicitly mentioned in 1123, but is salient anyway (speech of Paulus). In the absence of focus, metrical prominence is assigned.

/~*

Forðæm oft se welega & se wædla habbað sua gehweorfed hira ðeawum ðæt se welega bið eaðmod & sorgfull , & se wædla bið upahæfen & selflice . cocura,CP:26.183.9.1209

Forðæm sceal se lareow suiðe hrædlice wendan his tungan ongean ðæt ðe he ongiet ðæt ðæs monnes inngeðonc bið , forðæm ðæt se earma upahafena sie mid his wordum geðreatod & gescended , ðonne he ongiet ðæt hine ne magon his iermða geðreatigan & geeaðmedan . cocura,CP:26.183.11.1210

Ac sua micle liðelecor he sceal olecan ðæm welegan eaðmodan sua he ongiet ðæt he eaðmodra bið , ðonne hine ne magon ða welan forwlencean , ðe ælcne ofermodne oðhebbað . cocura,CP:26.183.15.1211

& oft eac mon sceal ðone welegan ofermodan to him loccian mid liðelicre olicunga , forðæm ðæt he hine to ryhte geweeme ; cocura,CP:26.183.18.1212

Forðæm oft hearda wunda beoð mid liðum beðengum gehnescode & gehælede , cocura,CP:26.183.20.1213

& eac ða wodðraga ðæs ***UNGEWITFULLAN*** monnes se læce gestilð & gehælð

mid ðæm ðæt he him olecð æfter his agnum willan .

(cocura,CP:26.183.20.1214)

Number	object	subject
OS-60	contr	old

leac evoked by *wunda* and *gehaelede* in 1213. The discourse before is about sensible men, and 1214 says: well, the doctor heals also unsensible men

/~*

he cuæð : Eft ic eow geseo , cocura,CP:27.187.20.1252

& ðonne blissiað eowre heortan , cocura,CP:27.187.20.1253

& eowerne gefean eow nan mon æt ne genimð .

(cocura,CP:27.187.20.1254)

number	object	subject
OS-61	old	contr

gefean is evoked by *blissiað*. The negative quantifier induces focus.

/~*

Ðone scamleasan mon mæg ðy bet gebetan ðe hine mon suiður ðreað & sciend , cocura,CP:31.207.4.1389

ac be ðæm scamfæstan hit is nyttre ðæt ðæt him mon on tælan wille , ðæt hit mon healfunga sprece , swelce hit mon hwon gehrine . cocura,CP:31.207.4.1390

Be ðæm *Dryhten* suiðe openlice tælde ða *scamleasan Iudeas* , cocura,CP:31.207.8.1391

& cuæð : Eower nebb sint sua scamleas sua ðara wifa ðe beoð forelegnissa . cocura,CP:31.207.8.1392

Ond eft he olehte ðam scamfæstan , ða he cuæð : ðære scame & ðære scande ðe ðu on iuguðe worhtes ic gedoo ðæt ðu forgietsð cocura,CP:31.207.10.1393

& ðæs bismeres ðines wuduwanhades ðu ne gemansð , forðæm ðæt is ðin Waldend ðe ðe geworhte . cocura,CP:31.207.10.1394

& eft ða scamleasan ***GALATHAS*** suiðe openlice Sanctus ***PAULUS*** tælde, ða he cuæð: Eala ge ungewitfullan Galatæ, hwa gehefegode eow? (cocura,CP:31.207.13.1395)

number	object	subject
OS-62	contr	contr

Double contrast between shameless Judaic men, whom the Lord tells something (1391) and the shameless Galatians that get scolded by Paulus. The focus clash is remedied by the intervening adverb.

/~*

Hwilum eac , ðonne ða fortruwudan & ða anwillan wenað ðæt hie nane scylde ðurhtogen næbben , ðonne magon we hi sua raðosð to ryhte gecierran ðæt we him sume opene scylde , ðe ær ðurhtogen wære , healfunga oðwieten , ðæt hie forðæm scamige , forðæm of ðære scylde ðe he hine ðonne bereccan ne mæge , he ongiete ða he ðonne deð , ðeah him ðonne ðynce ðæt he nan yfel ne do . cocura,CP:32.209.19.1412

Ða fortruwodnesse & ða anwilnesse an ***CORINCTHEUM*** Paulus ongeat suiðe wiðerweardne wið hine, & betweoh him selfum suiðe aðundene & upahæfene; Sua ðætte sume cuædon ðæt hie wæron Apollan, sume cuædon ðæt hi wæron Saules, sume Petres, sum cuæð ðæt he wære Cristes . (cocura,CP:32.211.1.1413)

Number	object	subject
OS-63	contr	old

The whole discourse is about St.Paul, so he is salient. This paragraph is about *fortruwudan* and *anwillan* in general, and 1413 picks as example the fort. and anw. of the Corinthians. Thus it is a poset relation.

/~*

Ðone ungeðyldegan ðonne suiðe lytel scur ðære costunga mæg onhreran, sua sua lytel wind mæg ðone cið awecgan,

(cocura,CP:33.225.4.1467)

number	object	subject
OS-64	contr	contr

Same context and situation as in cocuraC 85 (OS-51)

/~*

ac ðone yfelan fæsðrædan willan fulneah nan wind ne mæg awecgan .

(cocura,CP:33.225.4.1468)

number	object	subject
OS-65	contr	contr

Same context and situation as in cocuraC 86 (OS-52)

/~*

Be ðæm cwæð se witga : We lacnodon Babylon , cocura,CP:37.267.9.1738

& hio ðeah ne wearð gehæled . cocura,CP:37.267.9.1739

Ðonne bið Babylon gelacnad , nales ðeah fullice gehæled , ðonne ðæs monnes mod for his unryhtum willan & for his won weorcum gehierð sceamlice ðreaunga , & sceandlice suingellan underfehð , & ðeahhwæðre oferhygð ðæt he gecierre to bettran . cocura,CP:37.267.10.1740

Ðæt **ilce** eac Dryhten oðwat ***ISRAHELA*** folce, ða hie wæron gehergeode & of hiera earde alædde, & swaðeah noldon gesuican hiera yfelena weorca, ne hie
noldon awendan of hiera won wegum;
(cocura,CP:37.267.13.1741)

Number	object	subject
OS-66	anaphoric	old

There is a focus on *Israhela folce* (contrast with Babylon). There is also a 2ary prominence on *ilce*.

/~*

Be ðys ilcan wæs ðætte Gesaphað , se ðe ær on eallum dædum his lifes wæs to herigenne , fullneah mid eall forwearð for Achabes freondscipe . cocura,CP:46.353.24.2390

[…] (discourse follows about Josaphat and Achab)

ÆGÐER ðara Dauid sægde ðæt he swiðe wærlice beheolde, ða he cwæð:

Ic lufode ða ðe sibbe hatodon,

(cocura,CP:46.355.13.2399)

Number	object	subject
OS-67	contr	old

David would be salient in this story anyway, but he is explicitly mentioned in 2377. *AEgðer* (one of them both) automatically induces contrast to 'the other'.

/~*

& ða sprece ***NÆNIG*** mon uferran dogor on nænge oðre **halfe** oncærrende sie nymne suæ þis gewrit hafað .

(codocu1,Ch_1500_[Rob_3]:9.87)

number	object	subject
OS-68	anaphoric	contr

It is unclear what *ða sprece* refers to, but it is probable that it refers to the document as a whole (it is one of the last sentences of the document). In that case it is anaphoric. The negative quantifier induces contrast. A 2ary prominence is on *halfe*.

/~*

Gregorius him andswarode : sum wæs swyþe arwyrðes lifes wer , þam wæs nama Bonefacius .

cogregdC,GD_[C]:9.56.14.630_ID

Se hæfde biscopdom in þære cæstre , þe is haten Ferentis , cogregdC,GD_[C]:9.56.16.631_ID

& þone biscophad mid his þenunge & mid his þeawum rihtlice gefylde . cogregdC,GD_[C]:9.56.16.632_ID

Þyses manega wundru ***GAUDENTIUS*** se mæssepreost me sæde, se nu gyt leofað .

(cogregdC,GD_1_[C]:9.56.20.633)

Se wæs afeded on his þenunge , cogregdC,GD_[C]:9.56.23.634_ID

& he mæg swa mycele gewislicor secgan gehwylc þing be him , swa mycle ma him gelamp , þæt he sceolde betwyhhis þegnungum beon . cogregdC,GD_[C]:9.56.23.635_ID

Þisses ylcan Bonefacies cyrican gelamp , þæt þær wæs sume gære swiðe hefigu wædl & þearfednes , seo gewunað symble , þæt heo byþ þære eadmodnesse hyrde on godum & acorenum modum , & þær næs naht elcor to ealles geares andlyfne , buton þæt an , þæt he hæfde ænne wingeard .

cogregdC,GD_[C]:9.56.28.636_ID

number	object	subject
OS-69	topic	new, +f

The topic is embedded in the object NP. The discourse is about miracles by holy men. *Gaudentius* has not been mentioned at least for the preceding 150 sentences, so he should count as new. As he figures in the following discourse (at least an insertion of two more lines), it is justified to assign presentational focus to this phrase.

/~*

Be þan cwæð se æþela lareow Sanctus Paulus : ic wilnige , þæt ic sy tolysed & ic sy mid Criste .

cogregdC,GD_[C]:3.109.21.1253_ID

Þam þuhte , þæt him þæt Crist wære þæt he lifde , & gestreon þæt he swulte .

cogregdC,GD_[C]:3.109.24.1254_ID

Se þa gewin þære þ**rowunge** nalæs þæt an þæt he ***SYLFA*** gewilnode,

(cogregdC,GD_2_[C]:3.109.27.1255)

number	object	subject
OS-70	anaphoric	contr

þa gewin þære þrowunge (the profit gained by the passion, i.e. that men have the life of the world to come) rephrases the preceding sentence. The *sylfa* induces contrastive focus. A 2ary prominence is on the object.

/~*

Eac hit gelamp on oþre tid , þæt sum Gota , se wæs þearfa on gaste , com to þan halgan men to drohtnienne mid him . cogregdC,GD_[C]:6.113.15.1324_ID

Þone se Drihtnes wer sona onfeng swiðe lustlice . cogregdC,GD_[C]:6.113.16.1325_ID

Þa sume dæge sealde he him irengeloman , þæt is haten wudubill , cogregdC,GD_[C]:6.113.17.1326_ID

& het , þæt he sceolde on sumre stowe bremlas aweg aceorfan , þæt þær mihte beon wyrtun . cogregdC,GD_[C]:6.113.17.1327_ID

Seo ilce stow læg ofer þam stæþe sumes wæterseaðes . cogregdC,GD_[C]:6.113.19.1328_ID

Þas **stowe** se Gota underfeng to **clænsienne .**

(cogregdC,GD_2_[C]:6.113.21.1329)

number	object	subject
OS-71	topic	old

In the absence of focus, metrical prominence is assigned. *To clænsienne* counts as deverbal noun, therefore is not inherently weak.

/~*

Ac þa se ealda feond ne mihte adreogan þas wisan swigiende ne deogollice þurh swefn miðgian , swa swa is æfre his gewuna , cogregdC,GD_[C]:8.122.2.1459_ID

ac mid openlicre gesihþe he gebrohte hine sylfne & gelædde beforan eagum þæs arwyrðan fæder Benedictes . cogregdC,GD_[C]:8.122.2.1460_ID

& þa mid swa mycclum cleopungum he seofode , þæt he þrowode his ned , emne þæt þa oþre gebroþru eac gehyrdon his stefne , þe þær mid þam halgan were wunodon , þeh þe hi na ne gesawon his anlicnesse . cogregdC,GD_[C]:8.122.6.1461_ID

Ac swa swa se arwyrða fæder æfter þon his geongrum sæde , þæt se ealda & se swearta feond inæled hine æteowde his lichamlicum eagum , & þæt he gesawe þone feond reðian on hine mid his muþe & his byrnendum eagum . cogregdC,GD_[C]:8.122.9.1462_ID

Witodlice þa word, þe se feond gecwæð, ***EALLE*** þa gebroðru gehyrdon . (cogregdC,GD_2_[C]:8.122.12.1463)

Number	object	subject
OS-72	anaphoric	contr

Strictly speaking, *þa word* is not anaphoric; the content is only to follow. But it can be regarded as old, being evoked in 1462. *Þa gebroðru* is evoked by the setting of Benedict in his monastery; the universal quantifier induces focus.

/~*

Ac forþon þe his heorte hi selfe aheng in heanesse , ne feollon nanra þinga þas word on idelnysse of his muðe . cogregdC,GD_[C]:23.151.2.1804_ID

Ac swa hwæt swa he æfre gecwæð bodiende, þeah þe he hit na eorneste gecwæde, swa ***MYCELE*** mægnu & strengðe his word hæfde, efne swylce he þæt untweogendlice & buton yldinge & eac eall for rihtum dome forðbrohte, þæt
he þonne spræc .
(cogregdC,GD_2_[C]:23.151.6.1805)

Number	object	subject
OS-73	contr	topic

swa micel suggests focus. The concepts *mægnu* and *strengðe* are new anyway.

/~*

Sum Gota wæs , þam wæs nama Zalla , se wæs fylgende þa arrianiscan gedwolan treowleasnesse , se on Totillan dagum þæs cyninges & on þæra Gotena tidum abarn & aweoll mid þy bryne þære unmætestan wælhreownesse ongæn þa æfæstan weras þære æallæcan cyrican , swa þæt gif hwylc preost oþþe munuc com to him beforan his onsyne , ne eode he nanra þinga cwic fram his handum .

cogregdC,GD_[C]:31.162.18.1950_ID

[…]

& þa se Zalla feoll to eorðan afyrhted for þam weorce swa mycelre are

cogregdC,GD_[C]:31.164.19.1975_ID

& gebigde to þæs halgan mannes fotum þone sweoran his þære strecan wælhreownesse

cogregdC,GD_[C]:31.164.19.1976_ID

& hine sylfne þa befæste þam gebedum þæs arwyrþan Godes þeowes .

cogregdC,GD_[C]:31.164.19.1977_ID

Soðlice se halga wer Benedictus ne aras na fram his rædingce , cogregdC,GD_[C]:31.164.22.1978_ID

ac gecigde his broðra him to cogregdC,GD_[C]:31.164.22.1979_ID

& bebead , þæt hi hine sceoldon þær adune niman & to him gelædan , þæt he onfenge his bletsunge .

cogregdC,GD_[C]:31.164.22.1980_ID

& þa þone Gotan to him **gelædedne** Benedictus lærde & **manode**, þæt he scolde gestillan fram þære wedunge & ungewittignesse swa mycelre wælhreownysse .

(cogregdC,GD_2_[C]:31.164.25.1981)

Number	object	subject
OS-74	old	topic

In the absence of focus, metrical prominence is assigned.

/~*

þa geecte se Iudeisca man hit þus cweþende : cogregdC,GDPref_and_[C]:7.190.10.2408_ID

for hwon wiþsæcst þu þæs þe þu eart geacsod , þu þe gyrstan æfenne to þon gelæded wære , þæt þu mid þinre bradre hand þa nunnan ofer hire eaxle þaccodest ? cogregdC,GDPref_and_[C]:7.190.10.2409_ID

& þa butan tweon eadmodlice he andette , þæt he ær fæstlice wiðsoc , forþon þe se biscop wæs behealdende , þæt he in his agnum wordum arasod wæs . cogregdC,GDPref_and_[C]:7.190.14.2410_ID

Þæs hryre & his $scame þæs ylca Iudeisca man wæs afrefriende,

(cogregdC,GDPref_and_3_[C]:7.190.16.2411)

Number	object	subject
OS-75	new, +f	topic

The new object probably receives focus.

/~*

He hit þus sæde , þæt in Ualeria þam lande wære sum mæssepreost , se mid his preostum lædde swiðe clænlice þæt lif þæs halgan drohtoðes & wæs symble geornfull & behealden in Godes hyrnessum & godum weorcum . cogregdC,GDPref_and_[C]:22.224.11.3061_ID

Þa þa him to com se dæg his forðfore , he wæs bebyrged beforan ðære cyrcan . cogregdC,GDPref_and_[C]:22.224.14.3062_ID

on ðære ylcan circan wæron onfæste þa eowestran þara broðra sceapa . cogregdC,GDPref_and_[C]:22.224.15.3063_ID

& þonne seo ilca stow , in ðære þe wæs se mæssepreost bebyrged , wæs se weg þam mannum , þe to ðam sceapum eodon . cogregdC,GDPref_and_[C]:22.224.16.3064_ID

Þa sumre nihte singendum þam preostum in þære cyrican , com se þeof

cogregdC,GDPref_and_[C]:22.224.18.3065_ID

[Swa þa wundorlicum gemete $se $þeof, se þe ondred, þæt he sceolde beon gesewen & ongyten fram þam lifgendum mannum,] þysne **ylcan** se unlifgenda mæssepreost gehæfte & **gelette** .

(cogregdC,GDPref_and_3_[C]:22.225.1.3077)

number	object	topic
OS-76	old	topic

In the absence of focus, metrical prominence is assigned.

/~*

Witodlice of þam twam wundrum, þe ic secgan wille, ***OÞER*** þæt ***FOLC*** ongeat, oþer þa sacerdas oncneowon,

(cogregdC,GDPref_and_3_[C]:30.235.14.3285)

Number	object	subject
OS-77	contr	contr

This is a real clash case, but note that they are only disfavoured, not ruled out. In reading loud, there would be a pause between *oþer* and *þæt folc*

/~*

Witodlice of þam twam wundrum, þe ic secgan wille, oþer þæt folc ongeat, *OÞER* þa ***SACERDAS*** oncneowon,

(cogregdC,GDPref_and_3_[C]:30.235.14.3285)

Number	object	subject
OS-78	contr	contr

This is a real clash case, but note that they are only disfavoured, not ruled out. In reading aloud, there would be a pause between *oþer* and *þæsacerdas*

/~*

$Gregorius him andswarode : þæs þe we oncneowon of manigra manna sægene , þe comon of Ispanialandum , nu niwan hit gelamp , þæt <u>Erminigildus</u> se cyning <u>Leowigildes</u> sunu þæs cyninges wicode in Wauissi þære mægðe Gotena þeode . cogregdC,GDPref_and_[C]:31.237.16.3328_ID

<u>Se</u> wæs gecyrred fram þam arrianiscan gedwolan to þam allican geleafan bodiendum & lærendum þam arfullestan Leandro þam Ispania biscope , se gefyrn to me geþeoded wæs hiwcuðlice in manigfealdum freondscipum . cogregdC,GDPref_and_[C]:31.237.19.3329_ID

Þone **Erminigeldum** his fæder ongan læran & mid medum median & mid beotum **bregan**, to þon þæt he gecyrde to ðam arrianiscan gedwolan . (cogregdC,GDPref_and_3_[C]:31.237.22.3330)

Number	object	subject
OS-79	topic	old

In the absence of focus, metrical prominence is assigned.

/~*

Eac ic sylfa gewilnode , þæt ic gesawe þone wer swa mycles mægnes & geearnunge , cogregdC,GDPref_and_[C]:35.247.7.3493_ID

& ic wolde , þæt he to me gelæded wære & wunode feawa dagas in mettrumra manna huse , to þon þæt þær mihte beon swiþe hraðe gecunnod , hwæþer mid him wære ænig gifu þære lacnunge & gehælednesse . cogregdC,GDPref_and_[C]:35.247.7.3494_ID

Þa witodlice læg þær <u>sum man</u> on his mode gefangen mid ungewittignesse betwyh þam oðrum seocum mannum . cogregdC,GDPref_and_[C]:35.247.12.3495_ID

`Þone swylcne` **`seocne`** `læcas nemniað` **`gewitleasne`** `.`

`(cogregdC,GDPref_and_3_[C]:35.247.13.3496)`

Number	object	subject
OS-80	topic	old

læcas evoked by 3494. In the absence of focus, metrical prominence is assigned.

/~*

Witodlice þa þa þysne halgan wer nydde se deaþes dæg to ðam utgange of lichaman , manige men hi gesomnodon þa to swa haligre sawle leorendnesse of þysum middanearde .

cogregdC,GDPref_and_[C]:20.291.12.4312_ID

& þa þa hi ealle stodon æt his reste , þe þider gesomnode wæron , sume hi gesawon in gangende englas , cogregdC,GDPref_and_[C]:20.291.14.4313_ID

ac hi ne mihton nanum gemete aht cweþan , cogregdC,GDPref_and_[C]:20.291.14.4314_ID

sume hi naht eallunga ne gesawon , cogregdC,GDPref_and_[C]:20.291.14.4315_ID

`& swa þeh ealle, þe þær æt wæron, se swyþlica ege sloh, swa þæt nænig man ne mihte þær inne gestandan, þa þa seo halige sawl ferde of þam lichaman .`

`(cogregdC,GDPref_and_4_[C]:20.291.14.4316)`

number	object	subject
OS-81	anaphoric	contr

The universal quantifier does not automatically induce contrastive focus. Here it is purely anaphoric. *swyþlica*, on the other hand, induces contrast (as *such* in Modern English would do in comparable contexts).

/~*

Eac þa stefne þara gasta gehyrdon ealle , þe þær gehæftnede $wæron ,

cogregdC,GDPref_and_[C]:22.292.17.4328_ID

& wæron gewitan æfter þon $þæs ilcan $sanges þara muneca .

cogregdC,GDPref_and_[C]:22.292.17.4329_ID

Ac þa stefne þara **gasta** se ælmihtiga God **wolde**, þæt hi becomon to lichamlicum earum, to þon þæt gehwylce men, þe lifgende wæron in lichaman,

leornedon & ongæton, þæt gif hi Gode þeowiað in þysum andweardan life, hi lifiaþ soðlicur æfter þæs lichaman gedale in þam toweardan life .

(cogregdC,GDPref_and_4_[C]:22.292.19.4330)

Number	object	subject
OS-82	topic	old

In the absence of focus, metrical prominence is assigned.

/~*

$Soðlice þa þa hit nealæhte þære tide his deaþes , seo mæste hreohnes astah þære lyfte , þæt he ne mihte beon gelæded ut to bebyrgenne . cogregdC,GDPref_and_[C]:28.301.23.4479_ID

& hine þa on þære sawle bade acsode his agen wif mid swiðe hefigum sare & wope

cogregdC,GDPref_and_[C]:28.301.25.4480_ID

[…]

Þæt ilce wundor in þære spræce þæs æþelan weres eac ***OÐRE*** wundru wæron mid siðiende .

(cogregdC,GDPref_and_4_[C]:28.302.5.4489)

number	object	subject
OS-83	anaphoric	contr

The word *oðer* induces contrastive focus.

/~*

Ac ðonne hwæþre be sumum leohtum scyldum þæt clænsiende fyr is to gelyfanne , þæt hit sy ær þam dome , forþon seo soðfæstnes cwæð , þæt swa hwylc man swa yfelsacunge sæde on þone halgan gast , þæt hit him nære ne in þissere worulde forgifen ne in ðære toweardan . cogregdC,GDPref_and_[C]:41.328.5.4946_ID

[…]

& eac he cwæð: swa hwylc man swa ofer þisne staþol timbrað & $seteð gold oþþe seolfor oððe deorwyrðe stanas, treow oþþe hig oððe healm, anra gehwilces mannes **weorc** þæt fyr **acunnað** hwylc hit sy .

(cogregdC,GDPref_and_4_[C]:41.328.22.4952)

Number	object	subject
OS-84	old	old

In the absence of focus, metrical prominence is assigned.

/~*

Ac nu gyt þæt min mod onstyreð to acsunge be swa mycclum were & mærum *Pascasio* , forþon þe he wæs gelæded æfter deaðe to witiglicre stowe , cogregdC,GDPref_and_[C]:43.331.16.4994_ID

& his hrægl , þe ofer his bære læg , þa þa se deofulseoca man gehran þæt , sona fram him swa ofsetenum & geswenctum wæs geflemed & afyrred se awyrgda gast . cogregdC,GDPref_and_[C]:43.331.16.4995_ID

Gregorius him andswarode : in þissere wisan sceall mycel stihtung & swiðe mænigfeald ændebyrdnes þæs ælmihtigan Godes beon ongyten , þæs dome hit wæs gedon , þæt se ylca wer Pascasius se deacon , se gesyngode on him sylfum on sume tide , & swa þeah worhte æfter his deaðe sume wundorlice wisan þurh his lichaman beforan manna eagum , $se eac swylce ær his deaðe þam ylcan mannum ongytendum dyde manegu god & arfull weorc . cogregdC,GDPref_and_[C]:43.331.20.4996_ID

& þa nolde God ælmihtig , þæt þa $ðe his godan weorc $gesawon , wæron ungelyfende oððe ortreowe be þam wene þara ælmæssena þæs arwyrðan deacones , cogregdC,GDPref_and_[C]:43.331.27.4997_ID

ac wolde , þæt men wiston , þæt him nænig syn ungebeted butan wrace aleoðod wære ; cogregdC,GDPref_and_[C]:43.331.27.4998_ID

þa **scylde** se Pascasius ne gelyfde na him to **synne**,

(cogregdC,GDPref_and_4_[C]:43.331.27.4999)

Number	object	subject
OS-85	anaphoric	old

In the absence of focus, metrical prominence is assigned

/~*

For þam cwyde me is geþuht , þæt seo ufere hell sy on þyssere eorðan , & seo neoþere hell sy under ðissere eorþan . cogregdC,GDPref_and_[C]:44.332.19.5011_ID

& eac Iohannes stefn geþwæreþ þam ylcan andgyte $in þam ilcan $wene , se sæde , þæt he gesawe boc mid seofon inseglum geinseglode , & þæt nænig man wære gemeted wyrðe ne on heofonum ne on eorðan ne under eorðan , þe þa boc moste untynan & hire inseglu tobrecan

cogregdC,GDPref_and_[C]:44.332.21.5012_ID

þa **boc** swa þehhweþre **Iohannes** sæde, þæt heo wære æfter þon untyned

þurh þone leon of Iudan cynne .

(cogregdC,GDPref_and_4_[C]:44.332.21.5013)

number	object	subject
OS-86	topic	old

In the absence of focus, metrical prominence is assigned.

/~*

Þa þære ylcan nihte þe he on dæge bebyrged wæs , se eadiga martyr Sanctus Faustius , in þæs cyrican he bebyrged wæs , hine æteowde his ciricwearde þus cweþende :

cogregdC,GDPref_and_[C]:54.341.4.5153_ID

ga cogregdC,GDPref_and_[C]:54.341.4.5154_ID

& sæge þam biscope , þæt he aweorpe heonon ut þa fulan flæsc , þe he her lægde , forþon gif he swa ne deþ , he swelteð selfa þy $þrittegþan dæge . cogregdC,GDPref_and_[C]:54.341.4.5155_ID

Soþlice þa **gesihðe** se cyricweard ne dorste $geanddettan þam **biscope**, (cogregdC,GDPref_and_4_[C]:54.341.8.5156)

Number	object	subject
OS-87	old	old

In the absence of focus, metrical prominence is assigned.

/~*

Gregorius him andswarode , þa halgan weras swa swyðe swa hi beoð an samod mid Drihtne , swa swyðe hi witon Drihtnes andgit . cogregdH,GD_[H]:16.136.24.1329_ID

Soðlice eac se ylca apostol Paulus cwæð , hwylc man wat þæs mannes modgeþancas , butan þæs mannes gast , þe on him sylfum byð ? cogregdH,GD_[H]:16.136.28.1330_ID

Swa eac þa þing, þe beoð on Godes geþeahtunge, ***NAN*** mann ne cann butan se Godes gast .

(cogregdH,GD_2_[H]:16.137.3.1331)

Number	object	subject
OS-88	anaphoric	contr

The negative quantifier *nan monn* induces focus.

/~*

Nædderwyrt basilisca . coherbar,Lch_[Herb]:131.0.1926_ID

Đeos wyrt þe man basilisca & oðrum naman nædderwyrt nemneþ byþ cenned on ðam stowum þær seo nædre byþ þe man þam ylcan naman nemneð basiliscus . coherbar,Lch_[Herb]:131.1.1927_ID

Witodlice nys heora cyn an coherbar,Lch_[Herb]:131.1.1928_ID

ac hi sindon þreora cynna , coherbar,Lch_[Herb]:131.1.1929_ID

an ys olocryseis , þæt is on ure geðeode gecweden þæt heo eall golde scine ; coherbar,Lch_[Herb]:131.1.1930_ID

ðonne is oðer cyn stillatus , þæt is on ure geþeode dropfah , coherbar,Lch_[Herb]:131.1.1931_ID

seo ys swylce heo gyldenum heafde sy ; coherbar,Lch_[Herb]:131.1.1932_ID

þæt ðridde cyn ys sanguineus , þæt is blodread , coherbar,Lch_[Herb]:131.1.1933_ID

eac swilce heo gylden on heafde sy . coherbar,Lch_[Herb]:131.1.1934_ID

EALLE ðas cyn þeos wyrt basilisca hæfð .

(coherbar,Lch_I_[Herb]:131.1.1935)

number	object	subject
OS-89	anaphoric	topic

The universal quantifier subsumes a set that has formed a contrast. Therefore it is justified to assume that it has contrastive focus.

/~*

Gyf hwa mycelne hracan þolige & he þone him eaþelice fram bringan ne mæge for ðycnysse & to hnesce ,

genime of þysse wyrte wyrtruman ðæs dustes smæle gecnucudes tyn penega gewihte ,

coherbar,Lch_[Herb]:158.1.2339_ID

ylle drincan fæstende on liþon beore feower scenceas þry dagas oþðæt he sy gehæled .

coherbar,Lch_[Herb]:158.1.2340_ID

Ðam gelice þæt dust þysse sylfan wyrte on liþon beore geþiged ðone slep on gelædeþ

coherbar,Lch_[Herb]:158.1.2341_ID

& eac þæra innoþa astyrunge geliþigað . coherbar,Lch_[Herb]:158.1.2342_ID

Eac swylce þæt dust þysse ylcan wyrte næddrena slitas gelacnaþ . coherbar,Lch_[Herb]:158.2.2343_ID

[Þæt sylfe gemet þæt we her beforan cwædon þæs dustes ðysse ylcan wyrte

iris Illyrice foran mid ecede gemencged & gedruncen, hyt fremað þam þe

his

gecyndelice sæd him sylfwylles fram gewiteþ,] þone **leahtor** Grecas

gonorhoeam

nemneþ .

(coherbar,Lch_I_[Herb]:158.2.2344)

number	object	subject
OS-90	anaphoric	new, -f

The Greeks have not figured in this passage, are therefore new, but it is not to be counted as presentational. In the absence of focus, metrical prominence is assigned.

/~*

Þa Judees cwædon , Hwæt is se þe ne gelefð on God ? cojames,LS_[James]:20.19_ID

Jacobus andswerode , Se þe ne gelefð þæt on Abrahames cynne byð geerfeweard ealle þeode , & se þe ne gelyfð þæt Moyses cwæðe to his folca , God awecð eow mycelne wytega & mærne of eower gebroðren ,

cojames,LS_[James]:21.20_ID

& þone ge scylan geheren on eallen þingan swa swa me on þan þe he eow bebeott .

cojames,LS_[James]:21.21_ID

[…] (long list of prophecies)

EALLE þas þing, leofe gebroðre, Abrahames bearn foresædan þurh þone halgen **Gast** .

(cojames,LS_11_[James]:85.80)

number	object	subject
OS-91	contr	old

The universal quantifier induces contrastive focus. A 2ary metrical prominence is assigned to *Gast*.

/~*

(The context is the speech from which the preceding sentence is taken)

Þas þing & oðre **gelice**, Jacobus bodede þan folca ***SWA*** lange, þæt God ælmihtig him getyðede swa mycelne gefean, þæt eall þæt folc þe þær gegaderod wæs, anre stefne clypode, Eala þu halge Jacobus, mycel habbe we

gesynegod

(cojames,LS_11_[James]:96.89)

number	object	subject
OS-92	anaphoric	topic

A focus is on *swa lange*. A 2ary metrical prominence is on the object.

/~*

Gif hors bið gesceoten , Sanentur animalia in orbe terre et ualitudine uexantur in nomine dei patris et filii et spiritus sancti extinguatur diabolus per inpositionem manuum nostrarum , quis nos separabit a caritate christi , per inuocationem omnium sanctorum tuorum , per eum qui uiuit et regnat in secula seculorum , amen , Domine quid multiplicati sunt , III , iube , solue , deus , ter , catenis . colacnu,Med_[Grattan-Singer]:164.1.766_ID

Wið utsihte, þysne **pistol** se ængel brohte to **Rome**, þa hy wæran mid utsihte micclum geswæncte .

(colacnu,Med_3_[Grattan-Singer]:168.1.767)

number	object	subject
OS-93	anaphoric	old

The angel is obviously evoked (otherwise it wouldn't be definite, see Gundel et al 1993), but it has not been mentioned for several hundres lines. The epistle is longish, too. Metrical prominence is assigned.

/~*

Hwilum cnysseþ þæt sar on þa *rib* , colaece,Lch_[2]:46.1

.5.3021_ID

[…]

& eft ymb lytel ge þa ***GESCULDRU*** ge eft þone ***NEWESEOÞAN*** þæt sar gret

(colaece,Lch_II_[2]:46.1.5.3024)

Number	object	subject
OS-94	contr	topic

Both contrastive elements are enumerated in the conjunction. Both are moreover in contrast to *rib*, mentioned before.

/~*

Wiþ **lungenadle** læcedom **Dun** tæhte,

(colaece,Lch_II_[2]:65.2.1.3410)

number	object	subject
OS-95	old	new, -f

Dun has not been mentioned for at least 100 sentences. As he is only cited as a source and no role-player in this passage, he is not presentational. In absence of focus, metrical prominence is assigned.

/~*

And to Sancta Marian mæssan ælcere & to ælces apostoles mæssan fæste man , butan to Philippi & Iacobi mæssan we ne beodað nan fæsten , for þam easterlican freolse , colaw1cn,LawICn:16a.92

& ælces Frigedæges fæsten , butan hit freols sig . colaw1cn,LawICn:16a.93

And na þearf man na fæstan fram Eastran oð Pentecosten , butan hwa gescrifen sig oððe he elles fæstan wylle ; eallswa of middanwintra oð octabas Epiphanige þæt is seofen niht ofer Twelftan mæssedæge . colaw1cn,LawICn:16.1.94

And we forbeodað ordal & aðas freolsdagum & ymbrendagum & lengctendagum & rihtfæstendagum & fram Aduentum Domini oð se eahtaþa dæg agan sig ofer Twelftan mæssedæge & fram Septuagessima oð XV nihton ofer Eastron . colaw1cn,LawICn:17.95

And Sancte ***EADWEARDES*** mæssedæg witan habbað gecoren, þæt man freolsian sceal ofer eall Englaland þæt is on þam feowerteoðan dæge on Martige, XVIIII, kalendas Aprilis & Sancte Dunstanes mæssedæg on XIIII, kalendas Iunii þæt ys on þam þreotteoðan dæge þe byð on Mæge .

(colaw1cn,LawICn:17.1.96)

number	object	subject
OS-96	contr	old

The date is in contrast to other dates that are mentioned. *witan* is generic.

/~*

Ac gehwylc Cristen man do , swa him þearf is : colaw1cn,LawICn:19.109

gime his Cristendomes georne colaw1cn,LawICn:19.110

& gearwige hine eac to huselgange huru þriwa on geare gehwa hine sylfne , þe his agene þearfe wylle understandan , swa swa him þearf sig . colaw1cn,LawICn:19.111

& word & **weorc** freonda gehwylc fadige mid **rihte**,

(colaw1cn,LawICn:19.1.112)

number	object	subject
OS-97	anaphoric	old

The referent of ‘friends’ is not easy to see; it is probably generic. As such, it counts as evoked / old. In the absence of fous, metrical prominence is assigned.

/~*

And don nu eac lareowas & godcunde bydelas , swa swa hit riht is & ealra manna þearf is ;

colaw2cn,LawIICn:84.4.294

bodian gelome godcunde þearfe . colaw2cn,LawIICn:84.4.295

& ælc þe gescead wite , hlyste him georne , colaw2cn,LawIICn:84.4a.296

& godcunde lare ***GEHWA*** on geþance healde swyðe fæste, him sylfum to **þearfe**

(colaw2cn,LawIICn:84.4a.297)

number	object	subject
OS-98	topic	contr

univ, q. induces contrast. 2ary metrical prominence on *þearfe*.

/~*

& Sancte Eadwerdes mæssedæg witan habbað gecoren, þæt man freolsian sceal

ofer eal Englaland on XV kalendas Aprilis .

(colaw5atr,LawVAtr:16.47)

number	object	subject
OS-99	contr	old

see colaw1cn,96 (OS-96).

/~*

& word & weorc freonda gehwilc fadige mid rihte

(colaw6atr,LawVIAtr:28.72)

number	object	subject
OS-100	anaphoric	old

see colaw1cn,112 (OS-97).

/~*

And se Godes þeowe $Theothimus gefand þæt cild comargaC,LS_[MargaretCCCC_303]:3.4.14_ID

and he hit up anam comargaC,LS_[MargaretCCCC_303]:3.4.15_ID and hit wel befæste to fedenne . comargaC,LS_[MargaretCCCC_303]:3.4.16_ID

And þa hit andgeat hæfde , he him nama gesette , comargaC,LS_[MargaretCCCC_303]:3.5.17_ID

and þæt wæs Margareta , comargaC,LS_[MargaretCCCC_303]:3.5.18_ID

[…]

Ðis eadiga **mæden** se arwurða Godes þeowa $Theothimus fedde and lærde and **forðbrohte**, oðþæt hi XV wintre eald wæs .

(comargaC,LS_14_[MargaretCCCC_303]:4.1.21)

number	object	subject
OS-101	topic	old

In the absence of focus, metrical prominence is assigned.

/~*

Ða geherde seo eadiga Margareta and hi hit on bocum fand , þæt þa cinges and þa ealdormenn and þa yfela gerefan ofslogen æfre and bebyrodon ealle þa Godes þeowas þe þær on lande wæron .

comargaC,LS_[MargaretCCCC_303]:4.13.33_ID

Sumne hi mid wæpnum acwealdon and sumne mid hatum wætere .

comargaC,LS_[MargaretCCCC_303]:4.15.34_ID

Sumne hi onhengon be þan fotum and sumne be þan earmum .

comargaC,LS_[MargaretCCCC_303]:4.16.35_ID

Sumne hi pinedon mid wallende leade and mid hatum stanum .

comargaC,LS_[MargaretCCCC_303]:4.17.36_ID

Sumne heo mid sweorde ofslogen ; comargaC,LS_[MargaretCCCC_303]:4.18.37_ID

sumne mid spiten betweon felle and flæsce þurhwræcon . comargaC,LS_[MargaretCCCC_303]:4.18.38_ID

Eall **þæt** Godes þeowan geþafodon and geþrowodon for Godes deoran **lufan .**

(comargaC,LS_14_[MargaretCCCC_303]:4.19.39)

number	object	subject
OS-102	anaphoric	old

The univ. quant. does not automatically induce contr. focus. Here it looks like mere anaphoricity. Metrical prominence is assigned.

/~*

$On þone æfteran dæig Godes circean arworðiað Sanctus Stefanus gemind þæs ærestan diacones ond þæs ærestan martires æfter Cristes þrowunge . comart1,Mart_[Herzfeld-Kotzor]:De26,A.1.69_ID

Þone halgan **Stefanus** Cristes þegnas gehalgodon to **diacone**, (comart1,Mart_1_[Herzfeld-Kotzor]:De26,A.4.70)

number	object	subject
OS-103	topic	old

In the absence of focus, metrical prominence is assigned.

/~*

On ðone sexteoðan dæg ðæs monðes þonne bið Sancte <u>Marcelles</u> tid þæs papan .

comart3,Mart_[Kotzor]:Ja16,A.1.99_ID

Ðone papan ***MAXENTIUS***, Romeburge ealdormon, nedde þæt he his fullwihte wiðsoce ond deofolgeldum gelyfde .

(comart3,Mart_5_[Kotzor]:Ja16,A.2.100)

Number	object	subject
OS-104	topic	new

Maxentius is newly introduced and did not appear before. But even if he had, he should count as new, as each Saint's day entry has to be treated as separate text. Probably there is contrastive focus on each of these emperors/officials, as they continue to be important 'players' in the histories to come. This is the same in practically all similar examples in this text; I do not repeat this remark in these cases.

/~*

On ðone an ond twentigðan dæg bið Sancta Agnan þrowung ðære halgan fæmnan .

comart3,Mart_[Kotzor]:Ja21,A.1.193_ID

Seo geþrowade martyrdom for Criste þa heo wæs þreottene geara .

comart3,Mart_[Kotzor]:Ja21,A.2.194_ID

Þa fæmnan ***SIMFRONIUS***, Romeburge gerefa, ongan þreatian his suna to wife .

(comart3,Mart_5_[Kotzor]:Ja21,A.4.195)

Number	object	subject
OS-105	topic	new

s. OS-104

/~*

On þone æfteran dæg þæs monðes bið þæs biscopes geleornes Sancte Ceaddan .

comart3,Mart_[Kotzor]:Ma2,A.1.258_ID

Ond þæs wundor ond **lif** Beda se leornere wrat on Angelcynnes **bocum** .

(comart3,Mart_5_[Kotzor]:Ma2,A.2.259)

Number	object	subject
OS-106	topic	new, -f

Bede is new, but not presentational (does not play a role in the following), therefore no focus. Metrical prominence is assigned.

/~*

Đone **Ceaddan** ðyder se ercebiscop nam be norðan gemære on ðæm mynstre Læstenga **yge**

(comart3,Mart_5_[Kotzor]:Ma2,A.4.260)

Number	object	subject
OS-107	topic	new, -f

se ercebiscop is new, but not presentational (does not play a role in the following), therefore no focus. Metrical prominence is assigned.

/~*

On ðone ilcan dæg bið Sancte Cuthberhtes geleornes ðæs halgan biscopes , se wæs on þysse Brytene on þære mægðe þe is nemned Transhumbrentium , $ðæt is Norðanhymbra ðeod .

comart3,Mart_[Kotzor]:Ma20,B.1.376_ID

Đone wer oft ***ENGLAS*** sohtan

(comart3,Mart_5_[Kotzor]:Ma20,B.4.377)

ond him to brohtan heofonlico gereordo . comart3,Mart_[Kotzor]:Ma20,B.4.378_ID

number	object	subject
OS-108	topic	new, +f

The angles continue to play a role, therefore they can count as presentational and in focus.

/~*

On ðone þriddan dæg þæs monðes bið þara haligra fæmnena gemynd ond ðara eadigra gesweostra Sancta Agape ond Sancte Chonie ond Sancte Hirena . comart3,Mart_[Kotzor]:Ap3,A.1.503_ID

[…]

Þyssa fæmnena twa ***SISINNIUS*** se gesið het sendan on fyr, Agapan ond **Chonie**,

(comart3,Mart_5_[Kotzor]:Ap3,A.19.516)

Number	object	subject
OS-109	topic	new, +f

See comart3.100 (OS-104). 2ary metrical prominence on *Chonie*.

/~*

On ðone ylcan dæg bið Sancta Hirenan tid ðære halgan fæmnan . comart3,Mart_[Kotzor]:Ap5,B.1.536_ID

Ða fæmnan ***SISINNIUS*** se gesið sealde his **cæmpum** to bismrienne .

(comart3,Mart_5_[Kotzor]:Ap5,B.2.537)

Number	object	subject
OS-110	topic	new, +f

See comart3.100 (OS-104). 2ary metrical prominence on *cæmpum*. *Sisinnius*, although mentioned earlier, counts as new as we now are in a different saint's entry.

/~*

On ðone fif ond twentegðan dæg ðæs monðes bið seo tid on Rome ond on eallum Godes ciricum seo is nemned Letania maiora , þæt is þonne micelra bena dæg . comart3,Mart_[Kotzor]:Ap25,A.1.608_ID

On ðæm dæge eall Godes folc mid eaðmodlice relicgonge sceal God biddan þæt he him forgefe ðone gear siblice tid ond smyltelico gewidra ond genihtsume wæstmas ond heora lichoman trymnysse . comart3,Mart_[Kotzor]:Ap25,A.4.609_ID

Ðone **dæg** Grecas nemnað **zymologesin**, þæt is þonne hreowsunge dæg ond dædbote .

(comart3,Mart_5_[Kotzor]:Ap25,A.8.610)

Number	object	subject
OS-111	topic	new, -f

Grecas are new in the entry, but it is unusual in the text to cite the Greek names. This is an inserted explanation; as the Greeks are no players in the text to follow, they don't receive presentational focus. In the absence of focus, metrical prominence is assigned.

/~*

On ðone tu ond twentegðan dæg þæs monðes bið þæs apostoles ond þæs Godes ærendwracan gemynd þe on gewritum is nemned Iacobus Alphei . comart3,Mart_[Kotzor]:Ju22,A.1.1009_ID

[…]

Ðone Iacobum Iud+ea ***LEORNERAS*** ofslogan for Cristes læþþum mid webwyrhtan **rode**,

(comart3,Mart_5_[Kotzor]:Ju22,A.14.1019)

number	object	subject
OS-112	topic	new, +f

s. comart3.100 (OS-104). A 2ary metrical prominence is on *rode*.

/~*

On ðone teogeþan dæg þæs monðes bið seofon gebroðra ðrowung , þa þrowedon on Rome martyrdom for Criste on Antonius dagum ðæs caseres . comart3,Mart_[Kotzor]:Jy10,A.1.1121_ID

Hi wæron ðære mæran wudewan suna Sancta Felicitan . comart3,Mart_[Kotzor]:Jy10,A.3.1122_ID

Ða gebroðor ***Publius***, Romeburge gerefa, mid miclum witum wolde oncerran fram

Cristes **geleafan**,

(comart3,Mart_5_[Kotzor]:Jy10,A.5.1123)

number	object	subject
OS-113	topic	new, +f

s. comart3.100 (OS-104). A 2ary metrical prominence is on *geleafan*.

/~*

On ðone fif ond twentegðan dæg þæs monðes bið þæs apostoles gemynd ond þæs Godes ærendwreocan Sancte Iacobes , se ealra þara apostola ærest geþrowade for Criste .

comart3,Mart_[Kotzor]:Jy25,A.1.1258_ID

Ðone **Iacobum** se wælgrimma hyrde acwealde mid **sweorde**,

(comart3,Mart_5_[Kotzor]:Jy25,A.14.1265)

Number	object	subject
OS-114	topic	new, -f

It is not clear who the *wælgrima hyrde* (violent shepherd) is. In the day’s entry no potential referent could be found. By the logic of the text, it has to count as new, anyway. But since the person (I think it is a real person and not a ‘grim reaper’-like metaphor for Death) is not introduced properly, and does not continue to play a role, I would think that it is not in focus, although it is somewhat similar to the comart3.100-type of examples. Consequently, metrical prominence is assigned.

/~*

Þa nǣddran habbað twa heafda , ðæra eagan scinað nihtes swa leohte swa blacern . comarvel,Marv:5.3.27

[…]

On ðam londum byð piperes genihtsumnys . comarvel,Marv:6.5.33

Þone **pipor** þa nǣddran healdað on hyra **geornfulnysse** .

(comarvel,Marv:6.6.34)

Number	object	subject
OS-115	topic	old

In the absence of focus, metrical prominence is assigned.

/~*

Ne mæg ænig mann oþerne getimbrian buton he hine sylfne gelomlice behealde , and he mid syfrum andgyte þæt beo sylf wyrcende , God to gewitan hæbbende . comary,LS_[MaryofEgypt]:76.51_ID

Ac swa þeahhwæðere forþan þe þu cwæde þæt þe Cristes soðe lufu hyder us gelædde , eadmodne munuc us to gesecenne , ac wuna her mid us gif þu forðy come , comary,LS_[MaryofEgypt]:79.52_ID

and us ealle se goda hyrde ætgædere fede mid þære gife þæs halgan gastes

(comary,LS_23_[MaryofEgypt]:79.53)

number	object	subject
OS-116	contr	old

The Holy Spirit is in contrast to the other members of the Trinity. There is also a focus induced by *ealle* on the object. The subject is old (= Christ).

/~*

Ðysum þus $gecwedenum wordum fram þam abbode , Zosimus his cneowa gebigde , comary,LS_[MaryofEgypt]:82.54_ID

and onfangenum gebede on þam mynstre wunode . comary,LS_[MaryofEgypt]:82.55_ID

Þær he geseah witodlice ealle witon on þeawum and on dædum scinende , and on gaste weallende , and Drihtne þeowigende , comary,LS_[MaryofEgypt]:84.56_ID

þær wæs unablinnendlic staþolfæstnys Godes herunge æghwylcne dæg , and eac nihtes . comary,LS_[MaryofEgypt]:84.57_ID

And þær næfre unnytte spræce næron , ne geþanc goldes <MS:oðð> and seolfres , oþþe oþra gestreona ; comary,LS_[MaryofEgypt]:88.58_ID

ne furðon se nama mid him næs oncnawen . comary,LS_[MaryofEgypt]:88.59_ID

Ac þæt an wæs swiðost fram heom eallum geefst , þæt heora ælc wære on lichaman dead and on gaste libbende . comary,LS_[MaryofEgypt]:90.60_ID

Mid þam soðlice hi hæfdon ungeteorodne þæt wæron þa godcundan gespræcu . comary,LS_[MaryofEgypt]:91.61_ID

Heora lichaman witodlice mid þam nydþearfnyssum anum feddon , þæt wæs mid hlafe and mid wætere , to þam þæt hi þe scearpran on þære soðan Godes lufu hi æteowdon . comary,LS_[MaryofEgypt]:93.62_ID

Þas **weorc** Zosimus behealdende hine sylfne geornlice to fulfremednysse $aþenede gemang þam **emnwyrhtum**, þe þone godcundan neorxnewang butan ablinnendnysse geedniwodon .

(comary,LS_23_[MaryofEgypt]:95.63)

number	object	subject
OS-117	anaphoric	old

In the absence of focus, metrical prominence is assigned.

/~*

Da þa he soðlice sang and mid þære geornfullan behealdnysse up locode and þone heofon beheold , þa geseah he him on þa swiðran healfe þær he on gebedum stod , swa swa he on mennisce gelicnysse on lichaman hine æteowan , comary,LS_[MaryofEgypt]:165.104_ID

and þa wæs he ærest swiþe afyrht , forþan þe he wende þæt hit wære sumes gastes scinhwy þæt he þær geseah . comary,LS_[MaryofEgypt]:165.105_ID

Ða **wisan** Zosimus **georne** behealdende wæs,

(comary,LS_23_[MaryofEgypt]:177.114)

number	object	subject
OS-118	topic	old

In the absence of focus, metrical prominence is assigned.

/~*

Ða þa he swa neah wæs þæt heo mihte his stemne gehyran , þa ongan he forð sendan þyllice stemne mid hluddre clypunga wepende comary,LS_[MaryofEgypt]:189.123_ID

and þus cwæð , comary,LS_[MaryofEgypt]:189.124_ID

Hwi flihst þu me forealdodne syngigan , þu Godes þeowen ? comary,LS_[MaryofEgypt]:189.125_ID

Geanbida min for þam hihte þæs edleanes ðe þu swa micclum geswunce .

comary,LS_[MaryofEgypt]:192.126_ID

Stand comary,LS_[MaryofEgypt]:193.127_ID

and syle me þines gebedes bletsungan þurh þone God þe him nænne fram neawyrpð .

comary,LS_[MaryofEgypt]:193.128_ID

Ðas **word** soðlice Zosimus mid **tearum** geypte .

(comary,LS_23_[MaryofEgypt]:195.129)

number	object	subject
OS-119	anaphoric	old

In the absence of focus, metrical prominence is assigned.

/~*

Ða he þis þohte , þa geseah he hwær heo stod on oþre healfe þæs wæteres .

comary,LS_[MaryofEgypt]:676.446_ID

Zosimus soðlice hi geseonde mid micclum wynsumigendum gefean , and God wuldrigende up aras swaþeahhwæðere on his mode tweonigende , hu heo mihte Iordanes wæteru oferfaran .

comary,LS_[MaryofEgypt]:678.447_ID

Þa geseah he witodlice þæt heo mid Cristes rodetacne Iordanes wæteru bletsode ,

comary,LS_[MaryofEgypt]:681.448_ID

soðlice ealra þæra nihte þeostru þa ðæs monan byrhtnysse onlihte sona swa heo þære $rode $tacn on þa wætru drencte .

(comary,LS_23_[MaryofEgypt]:681.449)

number	object	subject
OS-120	contr	contr

It has not been mentioned that this happens by night. We could assume contrastive focus on both the object and the subject – they are in contrast to each other. The focus clash is remedied by the intervening *þa*.

/~*

Þa he þis ðohte , þa wæs þær an gewrit on þære eorðan getacnod þus gecweden ,

comary,LS_[MaryofEgypt]:748.503_ID

bebyrig abbud Zosimus , and miltsa Maria lichama ; comary,LS_[MaryofEgypt]:748.504_ID

ofgif þære eorðan þæt hire is , and þæt dust to þam duste . comary,LS_[MaryofEgypt]:748.505_ID

Geic eac $gebiddan þeahhwæðere for me $of þyssere worulde hleorende on þam monðe þe Aprilis , þære nigeþan nihte , þæt is idus APRELIS , on þam drihtenlican gereorddæge , and æfter þam huslgange . comary,LS_[MaryofEgypt]:751.506_ID

Þa se ealda þa stafas rædde þa sohte he ærest hwa hi write forþan þe heo sylf ær sæde þæt heo næfre naht swilces ne leornode , comary,LS_[MaryofEgypt]:754.507_ID

swaþeah $he on þam swiðe wynsumigende geseah þæt he hire naman wiste ,

comary,LS_[MaryofEgypt]:754.508_ID

and he swutole ongeat sona swa $heo þa godcundan gerynu æt Iordane onfeng þære ylcan tide þyder becom and sona of middanearde gewat . comary,LS_[MaryofEgypt]:754.509_ID

And se siðfæt þe Zosimus on xx dagum mid micclum geswince oferfor, þæt **eall** MARIA on anre tide **ryne** gefylde,

(comary,LS_23_[MaryofEgypt]:759.510)

number	object	subject
OS-121	anaphoric	old

In the absence of focus, metrical prominence is assigned.

/~*

Þa Scipia hæfde gefaren to ðære niwan byrig Cartaina , þe mon nu Cordofa hætt , he besætt Magonem , Hannibales broðor ; coorosiu,Or:10.104.29.2155_ID

& for þon þe he on þa burgleode on ungearwe becom , he hie on lytlan firste mid hungre on his geweald geniedde , þæt him se cyning self on hand eode ; coorosiu,Or:10.104.29.2156_ID

& he ealle þa oðre sume ofslog , coorosiu,Or:10.104.29.2157_ID

sume geband , coorosiu,Or:10.104.29.2158_ID

& þone cyning gebundenne to Rome sende , & monege mid him þara ieldestena witena .

coorosiu,Or:10.104.29.2159_ID

Binnan ðære byrig wæs micel licggende feoh funden . coorosiu,Or:10.105.3.2160_ID

SUM hit Scipia to ***Rome*** sende,

(coorosiu,Or_4:10.105.4.2161)

sum he hit het ðæm *folce* dælan . coorosiu,Or:10.105.4.2162_ID

number	object	subject
OS-122	contr	topic

The parsing looks weird to me. Be that as it may, there are two contrastive foci on *sum* and *to Rome* (in contrast to *sum* and *ðæm folce* in the following sentence)

/~*

& hit siþþan het Iuliuse onsendan , & his hring mid . coorosiu,Or:12.127.29.2700_ID

Ac þa hit mon to him brohte , he wæs mænende þa dæd mid micle wope , for þon he wæs eallra monna mildheortast on þæm dagum . coorosiu,Or:12.128.2.2701_ID

Æfter þæm Photolomeus gelædde fird wið Iuliuse , coorosiu,Or:12.128.5.2702_ID

& eall his folc wearð gefliemed , & he self gefangen . coorosiu,Or:12.128.5.2703_ID

& ealle þa ***MEN*** $Iulius het **ofslean** þe æt þære lare wæron þæt mon $Pompeius ofslog .

(coorosiu,Or_5:12.128.6.2704)

Number	object	subject
OS-123	contr	old

The universal quantifier induces contrastive focus. 2ary metrical prominence on *ofslean*.

/~*

Mid þæm bryne hio wæs swa swiþe forhiened þæt hio næfre siþþan swelc næs , ær hie eft Agustus swa micle bet getimbrede þonne hio æfre ær wære , þy geare þe Crist geboren wæs , swa þætte sume men cwædon þæt hio wære mid gimstanum gefrætwed . coorosiu,Or:1.133.15.2815_ID

Þone fultum & þæt **weorc** Agustus gebohte mid fela M **talentana .**

(coorosiu,Or_6:1.133.19.2816)

Number	object	subject
OS-124	anaphoric	old

In the absence of focus, metrical prominence is assigned.

/~*

& Moyses aðenode his gyrde upp , cootest,Exod:9.23.2720

& Drihten sende þunorrada & hagol & byrnende ligeta ofer eal Egypta land . cootest,Exod:9.23.2721

& Drihten let rinan hagol wið fyr gemenged cootest,Exod:9.24.2722

& hi ferdon ætgædere , cootest,Exod:9.24.2723

& swa micel he wæs swa næfre ær ne ætywde on eallum Egypta lande syþðan seo ðeod gesceapen wæs . cootest,Exod:9.24.2724

& ***EAL*** Egypta gærs se hagol **fordyde**,

(cootest,Exod:9.25.2726)

number	obj	subj
OS-125	contr	topic

The universal quantifier induces focus. 2ary metrical prominence on *fordyde*.

/~*

Ðæt halige ***REAF*** þæt Aaron werað, his suna habbað **æfter** him, þæt hig syn gesmyrede on þam & heora handa gehalgode .

(cootest,Exod:29.29.3401)

Seofon dagas he weraþ ðæt , se þe to bisceope bið geset for hine , þæt he þenie on ðam halierne .

cootest,Exod:29.30.3402

Þu nimst þære halgunge *ram* cootest,Exod:29.31.3403

& syþst his flæsc on haligre stowe . cootest,Exod:29.31.3404

& Aaron & his suna etaþ of ðam ; cootest,Exod:29.32.3405

number	obj	subj
OS-126	contr	old

Reaf in contrast to *ram* (3403). Aaron is discourse-old, as are his sons. 2ary metrical prominence on *æfter*.

/~*

Wið wifa earfodnyssum, þas **uncyste** Grecas hatað hystem **cepnizam** heortes hornes þæs smælestan dustes bruce þry dagas on wines drince .

(coquadru,Med_1.1_[de_Vriend]:3.7.95)

Number	object	subject
OS-127	anaphoric	new, -f

þas uncyste resumes the left dislocated constituent. The Greek name of any sickness has not hitherto be introduced; it is however not strictly presentational, therefore not in focus. In the absence of focus, metrical prominence is assigned.

/~*

Wið eagena sare , haran lifer gesoden ys god on wine to drincenne , & mid þam broþe ða eagan to beþianne . coquadru,Med_[de_Vriend]:5.18.217_ID

Ðam manum þe fram þære teoþan **tide** ne geseoð, þæs ylcan drinces smyc heora **eagan** onfon,

(coquadru,Med_1.1_[de_Vriend]:5.19.218)

number	object	subject
OS-128	old	topic

In the absence of focus, metrical prominence is assigned.

/~*

Ða ðæs binnon six dagum þa fullode Siluester þe papæ þone Iudam þe Sancta Helene þa rode tæhte corood,LS_[InventCrossNap]:526.553_ID

& hine bi nome Ciriacum nemde corood,LS_[InventCrossNap]:526.554_ID

& hine on ðone ilce dæge to arcebiscop halgode . corood,LS_[InventCrossNap]:526.555_ID

& ðes on morgen he wende to Ierusalem corood,LS_[InventCrossNap]:529.556_ID

& heo forð mid him þa twegen dæles sealde , corood,LS_[InventCrossNap]:529.557_ID

oðerne dæl he scolde don to Ierusalem , oðerne to Alexandriam , corood,LS_[InventCrossNap]:529.558_ID

& þone ***ÐRIDDE*** dæl þe papæ Siluester forþ mid him to ***ROMEBURIG*** hæfde,

(corood,LS_5_[InventCrossNap]:529.559)

number	object	subject
OS-129	contr	topic

A classical example of ‘modern’ double focus topicalization. The contrastive elements are the parts of the Cross on the one side and the locations to which they are brought on the other side, all mentioned in 558.

/~*

And he þa Malchus , þa he ful $gehende wið ðæs portes geate eode , þa beseah he þiderweard ,

cosevensl,LS_[SevenSleepers]:454.332_ID

and beseah to þære halgan Cristes rodetacne hwær heo uppan þam portgeate stod mid arwurðnysse

afæstnod ; cosevensl,LS_[SevenSleepers]:454.333_ID

[…] (it follows a long description of visions, to which the *eall* in 390 refers)

Eall he Malchus rehte his **geferum**, hu him gelamp on eallum þisum þingum, þa

he eft him to com on þam scræfe þe we ær foresædon, and þa heora seo

wundorlice ærest eallum mannum wæs geopened and heora þæt halige lif eall

geswutelod .

(cosevensl,LS_34_[SevenSleepers]:508.390)

number	object	subject
OS-130	anaphoric	topic

The author has referred to Malchus (the topic) for some 50 sentences exclusively by ‘he’. Here he specifies the pronoun with a parenthetical Malchus, in order to distinguish him from another potential referent (a young man who appeared to Malchus and spoke to him). In the absence of focus, metrical prominence is assigned.

/~*

Us is nu to witane , men þa leofestan , þæt for þan Cristes aldoras þy feowertigan dæge hine brohton to þam Godes temple for þan þe ðæt wæs þeaw & Godes bebod . coverhom,LS_[PurifMaryVerc_17]:28.2166_ID

[…]

Broðor mine , us is eac to witanne þæt þæt wæs þearfendra manna asægdnesse in þære ealdan æ þæt hie sceoldon þy dæge bringan twegen turturas oððe twegen culfran briddas Gode to æsægdnesse . coverhom,LS_[PurifMaryVerc_17]:64.2183_ID

Swylce **asægdnesse** Cristes aldoras hine mid brohton to þam Godes **temple** . (coverhom,LS_19_[PurifMaryVerc_17]:67.2184)

number	object	subject
OS-131	old	topic

In the absence of focus, metrical prominence is assigned.

/~*

Ic secge eow soðlice þæt furðon Salomon on eallum hys wuldre næs oferwrigen swa swa an of ðyson . cowsgosp,Mt_[WSCp]:6.29.341

Soþlice gyf æcyres weod þæt ðe to dæg is & bið to morgen on fen asend God scryt , eala ge gehwædes geleafan , þam mycle ma he scryt eow . cowsgosp,Mt_[WSCp]:6.30.342

Nellen ge eornustlice beon ymbhydige þus cweþende , hwæt ete we oððe hwæt drince we oþþe mid hwam beo we oferwrogene ? cowsgosp,Mt_[WSCp]:6.31.343

Soþlice ealle þas þing ***ÞEODA*** seceað . (cowsgosp,Mt_[WSCp]:6.32.344)

Witodlice eower Fæder wat þæt ge eallra þyssa þinga beþurfon . cowsgosp,Mt_[WSCp]:6.32.345

Eornustlice seceað ærest Godes rice & hys rihtwisnesse cowsgosp,Mt_[WSCp]:6.33.346

& ealle þas þing *eow* beoþ þærto geeacnode . cowsgosp,Mt_[WSCp]:6.33.347

number	object	subject
OS-132	anaphoric	contr

The contrast is between heathen and ‘you’ (= Christians). The univ. quantifier is here merely anaphoric.

/~*

Se forma ys Simon þe ys genemned Petrus , & Andreas hys broðor , Iacobus Zebedei , & Iohannes hys broður . <T06410028900,10.3> Philippus , & Bartholomeus , Thomas , & Matheus Puplicanus , & Iacobus Alphei , & Taddeus . <T06410029000,10.4> Simon Chananeus , & Iudas Scarioth þe hyne belæwde . cowsgosp,Mt_[WSCp]:10.3.581

Ðas **twelf** se hælynd sende him bebeodende & **cweþende**, ne fare ge on þeoda weg

(cowsgosp,Mt_[WSCp]:10.5.582)

number	obj	subj
OS-133	anaphoric	old

The phrase *Ðas twelf* subsumes the names that have been enumerated in the preceding sentence. In the absence of focus, metrical prominence is assigned.

/~*

He spræc to him oþer bigspel cowsgosp,Mt_[WSCp]:13.33.866

& þus cwæð , cowsgosp,Mt_[WSCp]:13.33.867

heofena rice is gelic þam beorman þone þæt wif onfeng & behydde on þrim gemetum melwes oð he wæs eall ahafen . cowsgosp,Mt_[WSCp]:13.33.868

Ealle þas þing se hælend spræc mid bigspellum to þam weredum;

(cowsgosp,Mt_[WSCp]:13.34.869)

number	obj	subj
OS-134	contr	old

The universal quantifier induces focus. 2ary metrical prominence on *weredum*.

/~*

And nelle ge secean hwæt ge eton oððe drincon , cowsgosp,Lk_[WSCp]:12.29.4690

& ne beo ge upahafene . cowsgosp,Lk_[WSCp]:12.29.4691

Ealle ***ÞAS*** þing ***ÞEODA*** seceað

(cowsgosp,Lk_[WSCp]:12.30.4692)

Eower fæder wat þæt ge þises beþurfon . cowsgosp,Lk_[WSCp]:12.30.4693

Þeahhwæþere seceað Godes rice cowsgosp,Lk_[WSCp]:12.31.4694

& ealle *þas* þing *eow* beoþ geihte . cowsgosp,Lk_[WSCp]:12.31.4695

number	object	subject
OS-135	contr	contr

Two contrasts, one on these things versus those things, one on ‘the heathen’ versus ‘you’.

/~*

Ða cwæð he to his leorningcnihtum , Sum welig man wæs hæfde sumne gerefan se wearð wið hine forwreged swylce he his god forspilde . cowsgosp,Lk_[WSCp]:16.1.4950

[…] (long speech)

Ðas ðing ealle þa ***FARISEI*** gehyrdon þa ðe gifre wæron,

(cowsgosp,Lk_[WSCp]:16.14.4981)

number	obj	subj
OS-136	anaphoric	new, +f

Farisei not introduced in that passage before. As they are role-players in the following section, this mentioning counts as presentational.

/~*

& cwæþ to him , soð ic eow secge , ne mæg se sunu nan þing don buton þæt he gesyhþ his Fæder don ;

cowsgosp,Jn_[WSCp]:5.19.6096

ða ***þing*** þe he wyrcþ se sunu wyrcð ***GELICE*** .

(cowsgosp,Jn_[WSCp]:5.19.6097)

number	obj	subj
OS-137	contr	topic

The contrast is between ‘the things that he makes’ versus ‘the things that the son makes’, not between ‘he’ and ‘son’. The realisation of the foci is a bit twisted, as the second *ða þing* is elided (sloppy reference), but the adverb *gelice* is there to repair the sloppy reference in some sense. Therefore the 2nd focus is there.

/~*

Ne fyligeaþ hig uncuþum cowsgosp,Jn_[WSCp]:10.5.6602

ac fleoð fram him forðam þe hig ne gecneowun uncuðra stefne . cowsgosp,Jn_[WSCp]:10.5.6603

Ðis bigspell se hælend him **sæde** .

(cowsgosp,Jn_[WSCp]:10.6.6604)

number	object	subject
OS-138	anaphoric	old

In the absence of focus, metrical prominence is assigned (with the help of the DAC).

/~*

Þa cwæð se hælend , nu gyt ys lytel leoht on eow . cowsgosp,Jn_[WSCp]:12.35.6846

Gaþ þa hwile þe ge leoht habbað þæt þystro eow ne befon ; cowsgosp,Jn_[WSCp]:12.35.6847

Se þe gæð on þystro he nat hwyder he gæð . cowsgosp,Jn_[WSCp]:12.35.6848

Þa hwile þe ge leoht habbon gelyfað on leoht þæt ge syn leohtes bearn ; cowsgosp,Jn_[WSCp]:12.36.6849

ðas **þing** se hælend him **sæde**

(cowsgosp,Jn_[WSCp]:12.36.6850)

number	object	subject
OS-139	anaphoric	old

In the absence of focus, metrical prominence is assigned.

/~*

Ðas þing ic eow sæde þæt ge habbon sibbe on me . cowsgosp,Jn_[WSCp]:16.33.7123

Ge habbað hefige byrðene on middanearde cowsgosp,Jn_[WSCp]:16.33.7124

ac getruwiað , cowsgosp,Jn_[WSCp]:16.33.7125

ic forswiðde middaneard . cowsgosp,Jn_[WSCp]:16.33.7126

Ðas **þing** se hælend **spræc**

(cowsgosp,Jn_[WSCp]:17.1.7127)

number	object	subject
OS-140	anaphoric	old

In the absence of focus, metrical prominence is assigned.

/~*

Witodlice manega ***OÐRE*** tacen se hælend worhte on his leorningcnihta

gesyhþe

þe ne synt an þysse bec awritene .

(cowsgosp,Jn_[WSCp]:20.30.7447)

number	object	subject
OS-141	contr	topic

oðre tacen induces contrast to the other miracles that are not described in the gospel. A 2ary metrical prominence is assigned on *gesyhþe*.

/~*

Ic eom ðin Drihten , he cwæð , þe gelædde þe ut of Egyptum . cowulf,WHomc:23.843_ID

Ne weorða þu fremde godas . cowulf,WHomc:24.844_ID

Ne þu þines Drihtnes naman ne namie on idel . cowulf,WHomc:25.845_ID

Wite þæt ðu þæne restedæg freolsige georne . cowulf,WHomc:26.846_ID

Weorða geornlice fæder & modor . cowulf,WHomc:26.847_ID

Ne beo ðu ænig manslaga . cowulf,WHomc:27.848_ID

Ne afyl þe mid forligere . cowulf,WHomc:27.849_ID

Ne sceaþa ðu . cowulf,WHomc:28.850_ID

Ne leoh þu . cowulf,WHomc:28.851_ID

Ne gyrn ðu oðres mannes wifes , cowulf,WHomc:28.852_ID

ne æniges þinges þe oðer man age ne gyrn þu on unriht . cowulf,WHomc:28.853_ID

Ðas tyn beboda God ***SYLF*** gedihte & awrat

(cowulf,WHom_10c:30.854)

number	object	subject
OS-142	anaphoric	contr

The *sylf* puts emphasis on the NP which it modifies (as today 'God himself' would do). *Sylf* induces a contrastive focus, emphasizing the fact that it is this entity and no other of the same set (here: potential authours of the 10 commandments) for which the proposition is true.

/~*

Leofan men , ic wille cyðan eow eallum , & þam huru þe hit ær nystan , hwanan seo bysn ærest aras þæt bisceopas ascadað ut of cyrican on foreweardan Lenctene þa men þe mid openan heafodgyltan hy sylfe forgyltað , & eft hy æfter geornfulre dædbote into cyrican lædað on þam dæge þe bið cena Domini , ealswa todæg is . cowulf,WHom:3.1317_ID

Ure Drihten gescop & geworhte Adam þone forman man haligne & clænne & synleasne him sylfum to gelicnesse , cowulf,WHom:8.1318_ID

& ða sylfan **gelicnesse** ure Drihten **eac** lærde

(cowulf,WHom_15:8.1319)

number	object	subject
OS-143	topic	old

In the absence of focus, metrical prominence is assigned.

/~*

And Godes þeowas sindon mæðe & munde gewelhwar bedælede ; cowulf,WHom:34.1659_ID

& gedwolgodan þenan ne dear man misbeodan on ænige wisan mid hæðenum leodum , swa swa man Godes þeowum nu deð to wide þar cristene sceoldon Godes lage healdan & Godes þeowas griðian . cowulf,WHom:34.1660_ID

Ac soð is þæt ic secge : cowulf,WHom:39.1661_ID

þearf is þare bote forðam Godes gerihta wanodan nu lange innan þisse þeode on æghwilcum ende , & folclaga wirsedon ealles to swiðe , & halinessa sindon to griðlease wide , & Godes hus sindon to clæne beripte ealra gerihta & innan bestripte ælcra gerisena , & godcunde hadas wæron nu lange swiðe forsawene , & wuduwan fornydde on unriht to ceorle , & to manega foryrmde , & earme men beswicene & reowlice besirwde & ut of þisum earde wide gesealde swiðe unforworhte fremdum to gewealde . & cradolcild geþeowode þurh wælreowe unlaga , & freoriht fornumene , cowulf,WHom:49.1662_ID

& Frige men ne motan wealdan heora sylfra , ne faran þar hi willað , ne ateon heora agen swa swa hi willað . cowulf,WHom:49.1663_ID

Ne þrælas ne moton habban þæt hi agon on agenan hwilan mid earfedan gewunnen , ne þæt þæt heom on Godes est gode men geuðon , & to ælmesgife for Godes lufan sealdon . cowulf,WHom:51.1664_ID

Ac æghwilc ælmesriht þe man on Godes est scolde mid rihte georne gelæstan

ÆLC man gelitlað oððe forhealdeð, forðam unriht is to wide mannum gemæne & unlaga leofe, &, raðost is to cweðenne, Godes laga laðe & lara forsawene .

(cowulf,WHom_20.2:54.1665)

number	object	subject
OS-144	old	contr

The universal quantifier *ælc* induces contrastive focus. *æghwilc* is a negative polarity item; as such it is rather like an existential quantifier and does not necessarily induce focus. If id did, the resulting clash would be remedied by the intervening relative clause.

References

Primary sources

Besch, Werner, Winfried Lenders, Hugo Moser & Hugo Stopp (1972-1985): *Das Bonner Frühneuhochdeutschkorpus*. Online: http://www.ikp.uni-bonn.de/dt/forsch/fnhd/

Böll, Heinrich (1963): *Ansichten eines Clowns*. Köln: Kiepenheuer & Witsch.

Böll, Heinrich (1965): *The Clown*. Translated by Leila Vennewitz. New York: McGraw-Hill.

Dürrenmatt, Friedrich (1955): *Der Richter und sein Henker*. Reinbek: Rowohlt.

Dürrenmatt, Friedrich (1983): *The Judge and his Hangman*. Translated by Eva H. Morreale. Boston: Godine.

Grass, Günther (1971 [1961]): *Katz und Maus*. London: Heinemann (Original: Neuwied: Luchterhand).

Grass, Günther (1963): *Cat and Mouse*. Translated by Ralph Manheim. New York: Harcourt.

Isidor of Sevilla (1964): *Der althochdeutsche Isidor*. Ed. Hans Egger. Tübingen: Niemeyer.

Kroch, Anthony, Beatrice Santorini & Lauren Delfs (2005): *Penn-Helsinki Parsed Corpus of Early Modern English*. University of Pennsylvania.

Kroch, Anthony & Ann Taylor (2000): *Penn-Helsinki Parsed Corpus of Middle English*. University of Pennsylvania.

Sayers, Dorothy L. (1923): *Whose body?* New York: Harper and brothers.

Sayers, Dorothy L. (1972): *In the teeth of the evidence and other stories*. London: Gollancz

Taylor, Ann, Anthony Warner, Susan Pintzuk & Frank Beths (2003): *The York-Toronto-Helsinki Parsed Corpus of Old English Prose (YCOE)*. University of York.

Ward, Gregory & Ellen Prince (no date) *Corpus of naturally occurring topicalization*, unpublished.

Wodehouse, Pelham G. (1924): *Bill the Conqueror*. London: Methuen.

Research literature

Abraham, Werner & László Molnárfi (2001): German clause structure under discourse functional weight: focus and antifocus. In: Abraham, Werner & C. Jan-Wouter Zwart (eds.): *On formal German(ic) typology*. Amsterdam/ Philadelphia: Benjamins, 1-43.

Allen, Cynthia L. (1990): Review of van Kemenade 1987. *Language* 66, 146-152.

Altmann, Hans (1976): *Die Gradpartikeln im Deutschen. Untersuchungen zu ihrer Syntax, Semantik und Pragmatik*. Tübingen: Niemeyer.

Asher, Nicholas & Alex Lascarides (2003): *Logics of Conversation*. Cambridge: CUP.

Barry, Roberta (1975): Topic in Chinese: An Overlap of Meaning, Grammar and Discourse Function. In: Grossmann, Robin E., L. James San & Timothy J. Vance (eds.): *Papers from the Parasession on Functionalism*. Chicago: Chicago Linguistic Society, 1-9.

Bing, Janet M. (1982): A Discourse Domain Identified by Intonation. *Minnesota Working Papers in Linguistics and Philosophy of Language* 8, 1-9.

Birner, Betty J. & Gregory Ward (1998): *Information Status and Noncanonical Word Order in English*. Amsterdam/ Philadelphia: Benjamins.

Bolinger, Dwight L. (1961): Contrastive Accent and Contrastive Stress. *Language* 37, 83-96.

Bolinger, Dwight L. (1972): Accent is predictable (if you're a mind reader). *Language* 48, 633-644.

Bolinger, Dwight L. (1983): Affirmation and Default. *Folia Linguistica* 17, 1-4; 99-116.

Brandt, Margareta, Marga Reis, Inger Rosengren & Ilse Zimmermann (1992): Satztyp, Satzmodus und Illokution. In: Rosengren, Inger (ed.): *Satz und Illokution*. Tübingen: Niemeyer, 1-90.

Brunner, Karl (1965): *Altenglische Grammatik*. Tübingen: Niemeyer.

Büring, Daniel (1997) *The Meaning of Topic and Focus. The 59th Street Bridge Accent*. London/ New York: Routledge.

Chafe, Wallace (1976): Givenness, Contrastiveness, Definiteness, Subjects, Topics, and Point of View. In: Li, Charles (ed.): *Subject and Topic*. New York: Academic Press, 25-55.

Chomsky, Noam (1981): *Lectures on Government and Binding*. Dordrecht: Foris.

Chomsky, Noam (1995): *The Minimalist Program*. Cambridge, Ma.: MIT Press.

Chomsky, Noam (2001): Derivation by Phase. In: Kenstowicz, Michael (ed.): *Ken Hale. A Life in Language*. Cambridge, Ma.: MIT-Press, 1-52.

Chomsky, Noam & Morris Halle (1968): *The Sound Pattern of English*. New York: Harper and Row.

Cinque, Guglielmo (1993): A Null Theory of Phrase and Compound Stress. *Linguistic Inquiry* 24, 239-297.

Cooper, William E. & Paccia-Cooper, Jeanne (1980): *Syntax and Speech*. Cambridge, Ma.: Harvard University Press.

Copperud, Roy H. (1960): *Words on Paper. A Manual of Prose Style for Professional Writers, Reporters, Authors, Editors, Publishers, and Teachers.* New York: Hawthorn.

Corver, Norbert & van Riemsdijk, Henk (1994): Introduction: approaches to and properties of scrambling. In: Corver, Norbert & van Riemsdijk, Henk (eds.): *Studies on Scrambling*. Berlin: Mouton de Gruyter, 1-15.

Daneš, František (1966): A Three-Level Approach to Syntax. In J. Vachek (ed.), *Travaux Linguistiques de Prague 1: L'Ècole de Prague d'aujourd'hu*i, 225–240.

den Besten, Hans (1977) On the Interaction of Root Transformations and Lexical Deletive Rules. Ms., MIT / University of Amsterdam (auch 1981 in *Groninger Arbeiten zur germanistischen Linguistik* 20, 1-78; online: http://www.dbnl.org/tekst/best005inte01/).

Drach, Erich (1937): *Grundgedanken der Deutschen Satzlehre*. Frankfurt/M.: Diesterweg.

Drubig, H. Bernhard (2003): Toward a typology of focus and focus constructions. *Linguistics* 41, 1-50.

Dryer, Matthew S. (2005): Three Types of Noun Phrase preposing in English. Ms., University of NY at Buffalo (http://wings.buffalo.edu/linguistics/people/faculty/dryer/dryer/np.preposing.three.pdf).

Eisenberg, Peter (1994): *Grundriß der deutschen Grammatik.* Stuttgart: Metzler.

Embick, David & Rolf Noyer (2001): Movement Operations after Syntax. *Linguistic Inquiry* 32, 555-595.

Fanselow, Gisbert (1991): Minimale Syntax. *Groninger Arbeiten zur germanistischen Linguistik* 32.

Féry, Caroline (1993): *German Intonational Patterns*. Tübingen: Niemeyer.

Féry, Caroline (to appear): The Prosody of Topicalization. To appear in: Schwabe, Kerstin & Susanne Winkler (eds.): *On Information Structure*, *Meaning and Form*. Amsterdam/ Philadelphia: Benjamins.

Firbas, Jan (1974): Some Aspects of the Czechoslovak Approach to Problems of Functional Sentence Perspective. In: Daneš, František (ed.): *Papers on Functional Sentence Perspective*. Prag: Academia, 11-37.

Fischer, Olga, Ans van Kemenade, Willem Koopman & Wim van der Wurff (2000): *The Syntax of Early English*. Cambridge: Cambridge University Press.

Frey, Werner (2004a): A medial position for topics in German. *Linguistische Berichte* 198, 153-190.

Frey, Werner (2004b): The grammar-pragmatics interface and the German prefield. *Sprache & Pragmatik* 52: 1-39.

Frey, Werner (2006): Contrast and movement to the German prefield. In: Molnár, Valéria & Susanne Winkler (eds.): *The Architecture of Focus*. Studies in Generative Grammar 82. Berlin / New York: Mouton de Gruyter, 235-264.

van Geenhoven, Verle (1998): *Semantic Incorporation and Indefinite Descriptions: Semantic and Syntactic Aspects of Noun Incorporation in West Greenlandic.* Stanford: CSLI.

van Gelderen, Elly (1991): Review of van Kemenade 1987. *Studies in Language* 15, 487-493.

Goldsmith, John A. (1976): *Autosegmental Phonology*. Ph.D. Dissertation, MIT (published 1979 Garland, New York).

Grewendorf, Günther, Fritz Hamm & Wolfgang Sternefeld (1987). *Sprachliches Wissen*. Frankfurt/M.: Suhrkamp.

Grice, Herbert P. (1989): Logic and Conversation. In: Grice, Herbert P. (ed.): *Studies in the Way of Words*. Cambridge/Mass: Harvard UP, 22-40.

Grosz, Barbara J., Aravind K. Joshi & Scott Weinstein (1995): Centering: A Framework for modelling the local coherence of discourse. *Computational Linguistics* 21, 203-225.

Gundel, Jeanette, Nancy Hedberg & Ron Zacharski (1993): Cognitive Status and the form of referring expressions in discourse. *Language* 69, 274-307.

Gussenhoven, Carlos (1992): Sentence accents and argument structure. In: Roca, Iggy M. (ed.) *Thematic structure. Its role in Grammar*. Berlin / New York: Foris, 79-106.

Haeberli, Eric (2002): Inflectional Morphology and the Loss of Verb-Second in English. In: Lightfoot, David W. (ed.): *Syntactic Effects of Morphological Change*. Oxford: OUP, 88-106.

Haeberli, Eric & Susan Pintzuk (2007): Head-final Structure in Old English Root Clauses. Ms., Université de Genève / University of York

Haegeman, Liliane (1994): Verb Raising as Verb Projection Raising. *Linguistic Inquiry* 25, 509-522.

Haegeman, Liliane & Henk van Riemsdijk (1986): Verb Projection Raising, Scope, and the Typology of Rules Affecting Verbs. *Linguistic Inquiry* 17, 417-466.

Haider, Hubert (1997): Projective Ergonomy. On the Minimal Functional Structure of the German Clause. In: Abraham, Werner & Elly van Gelderen (eds.): *Syntactic Problems – Problematic Syntax*. Tübingen: Niemeyer, 83-103.

Haider, Hubert & Inger Rosengren (1998): Scrambling. *Sprache und Pragmatik* 49, 1-104.

Haider, Hubert & Inger Rosengren (2003): Scrambling: Nontriggered Chain Formation in OV Languages. *Journal of Germanic Linguistics* 15(3), 203-266.

Halle, Morris & William Idsardi (1995): General Properties of Stress and Mettrical Structure. In: Goldsmith, John A. (ed.): *The Handbook of Phonological Theory*. Oxford: Blackwell, 403-443.

Halle, Morris & Alec Marantz (1993): Distributed Morphology and the Pieces of Inflection. In: Hale, Kenneth and Samuel J. Keyser (eds.): *The View from Building 20*. Cambridge, Ma: MIT Press, 111-176.

Halliday, Michael A.K. (1967): Notes on Transitivity and Theme in English. *Journal of Linguistics* 3: 37-81; 199-244; 4: 179-215.

Han, Chung-hye & Anthony Kroch (2000): The rise of do-support in English: implications for clause structure. In: *Proceedings of NELS 30.*

Harris, James (1983): *Syllable Structure and Stress in Spanish: A Nonlinear Analysis*. LI Monograph 8. Cambridge, Ma.: MIT Press.

Hayes, Bruce (1984): The Phonology of Rhythm in English. *Linguistic Inquiry* 15, 33-74.

Hayes, Bruce (1995): *Metrical Stress Theory*. Chicago: University of Chicago Press.

Heim, Irene (1982): The semantics of definite and indefinite noun phrases. Dissertation, MIT.

Herling, Simon H. A. (1821): Ueber die Topik der deutschen Sprache. *Abhandlungen des frankfurtischen Gelehrtenvereins für deutsche Sprache* 3, 296-363 (online available under http://www.linguistik.uni-tuebingen.de/hoehle/, select 'Materialien').

van Heuven, V.J. (1994): What is the smallest Prosodic Domain? In: Keating, P. (ed.): *Papers in Laboratory Phonology III, between the grammat and physics of speech.* Cambridge: CUP, 76-98.

Hinterhölzl, Roland (to appear): Information Structure and Unmarked Word Order in (Older) Germanic. In: Féry, Caroline & Malte Zimmermann (eds.): *Information Structure from Different Perspectives.* Oxford: OUP.

Hirschberg, Julia B. (1985): *A Theory of Scalar Implicature*. PhD dissertation, University of Pennsylvania.

Hoberg, Ursula (1997): Die Linearstruktur des Satzes. In: Zifonun, Gisela, Ludger Hoffmann & Bruno Strecker (eds.): *Grammatik der deutschen Sprache.* Band 2. Schriften des Instituts für deutsche Sprache 7.2. Berlin: de Gruyter, 1496-1680.

Hockett, Charles F. (1963): The Problem of Universals in Language. In: Greenberg, Joseph H. (ed.): *Universals of Language*. Cambridge, Ma.: MIT Press, 1-29.

Höhle, Tilman (1986). Der Begriff ‚Mittelfeld'. Anmerkungen über die Theorie der topologischen Felder. In: Weiss, Walter, Herbert Ernst Wiegand & Marga Reis (eds.): *Textlinguistik contra Stilistik? – Wortschatz und Wörterbuch – Grammatische oder pragmatische Organisation von Rede?.* Tübingen: Niemeyer, 329-340.

Höhle, Tilman (1992): Über VERUM-Fokus im Deutschen. *Linguistische Berichte, Sonderheft* 4: 112-141.

van Hoof, Hanneke (2003):The Rise in the Rise-Fall Contour: Does It Evoke a Contrastive Topic or a Contrastive Focus? *Linguistics* 41, 3 (385), 515-563.

Hulk, Aafke & Ans van Kemenade (1995): Verb Second, Pro-drop, Functional Projections and Language Change. In: Battye, Adrian & Ian Roberts (eds.): *Clause Structure and Language Change*. Oxford: Oxford University Press, 227-256.

Hyman, Larry M. (1977): On the nature of linguistic stress. In: Hyman, Larry M. (ed.): *Studies in Stress and Accent*. Los Angeles: Department of Linguistics, University of Southern California, 37-82.

Idsardi, William (1992): *The computation of prosody*. PhD dissertation, Massachusetts Institute of Technology.

Ito, Junko (1986): *Syllable Theory in Prosodic Phonology*. PhD dissertation, University of Massachusetts.

Ito, Junko (1989): A Prosodic Theory of Epenthesis. *Natural Language and Linguistic Theory* 7, 217-259.

Jackendoff, Ray (1972): *Semantics in Generative Grammar*. Cambridge, Ma.: MIT Press.

Jacobs, Joachim (1997): I-Topikalisierung. *Linguistische Berichte* 168, 91-133.

Jacobs, Joachim (2001): The dimensions of topic-comment. *Linguistics* 39, 641-681.

Kager, René (1995): The Metrical Theory of Word Stress. In: Goldsmith, John (ed.): *The Handbook of Phonological Theory*. Oxford / Malden: Blackwells, 367-443.

Kager, René & Ellis Visch (1988): Metrical constituency and rhythmic adjustment. *Phonology* 5, 21-71.

Kaisse, Ellen M. (1985): *Connected Speech*. Orlando: Academic Press.

van Kemenade, Ans (1987): *Syntactic Case and Morphological Case in the History of English*. Dordrecht: Foris.

van Kemenade, Ans (1997): V2 and embedded topicalization in Old and Middle English. In: van Kemenade, Ans & Nigel Vincent (eds.): *Parameters of Morphosyntactic Change*. Cambridge: CUP, 326-352.

Kiparsky, Paul (1966): Über den deutschen Akzent. *Studia Grammatica* 7, 69-98.

Kiparsky, Paul (1997): The rise of positional licensing. In: van Kemenade, Ans & Nigel Vincent (eds.): *Parameters of Morphosyntactic Change*. Cambridge: CUP, 460-494.

Kroch, Anthony S. (1989): Reflexes of Grammar in Patterns of Language Change. *Language Variation and Change* 1, 199-244.

Kroch, Anthony S. & Beatrice Santorini (1991): The Derived Constituent Structure of the West Germanic Verb Raising Construction. In: Freidin, Robert (ed.): *Principles and Parameters in Comparative Grammar*. Cambridge, Ma: MIT Press, 269-338.

Kroch, Anthony S. & Ann Taylor (1997): Verb movement in Old and Middle English: Dialect variation and language contact. In: van Kemenade, Ans & Nigel Vincent (eds.): *Parameters of Morphosyntactic Change*. Cambridge: CUP, 297-325.

Kuno, Susumu (1982): The Focus of the Question and the Focus of the Answer. In: Schneider, Robinson, Kevin Tuite & Robert Chametzky (eds.): *Papers from the Parasession on Nondeclaratives*. Chicago: Chicago Linguistics Society, 134-157.

Ladd, D. Robert (1980): *The Structure of Intonational Meaning*. Bloomington: Indiana University Press.

Ladd, D. Robert (1996): *Intonational phonology*. Cambridge: CUP.

Leben, William R. (1973): *Suprasegmental phonology*. Ph.D. Dissertation, Massachusetts Institute of Technology (published 1980 Garland, New York).

Lehmann, Winfred P. (1974): *Proto-Indo-European Syntax*. Austin: University of Texas Press.

Lehmann, Winfred P. (1976): Topic to Subject in Indo-European. In: Li, Charles N. (ed.): Subject and Topic. New York: Academic Press, 445-456.

Lenerz, Jürgen (1977): *Zur Abfolge nominaler Satzglieder im Deutschen*. Tübingen: Narr.

Lenerz, Jürgen (1984): *Syntaktischer Wandel und Grammatiktheorie. Eine Untersuchung an Beispielen aus der Sprachgeschichte des Deutschen.* Tübingen: Niemeyer.

Liberman, Mark Y. and Alan Prince (1977): On Stress and Linguistic Rhythm. *Linguistic Inquiry* 8, 249-336.

Liedtke, Frank (1997): Gesagt – getan: Über illokutionäre Indikatoren. In: Rolf, Eckhard (ed.): *Implikaturen und Sprechakte*. Linguistische Berichte, Sonderheft 8, 189-213.

Lippert, Jörg (1974): *Beiträge zur Technik und Syntax althochdeutscher Übersetzungen.* München: Fink.

Mahajan, Anoop (1994): Toward a unified theory of Scrambling. In: Corver, Norbert & van Riemsdijk, Henk (eds.): *Studies on Scrambling*. Berlin: Mouton de Gruyter, 301-330.

Mathesius, Vilém (1928). On Linguistic Characterology with Illustrations from Modern English. In *Actes du Premier Congrès International de Linguistes à La Have*, 56-63 (Here cited from the reprint: *A Prague School Reader in Linguistics*, Josef Vachek (ed.), 59-67. Bloomington: Indiana University Press, 1964).

McCarthy, John (1981): A prosodic theory of nonconcatenative morphology. *Linguistic Inquiry* 12, 373-418.

McCarthy, John (1986): OCP Effects: Gemination and Antigemination. *Linguistic Inquiry* 17, 207-264.

Mester, R. Armin (1994): The quantitative trochee in Latin. *Natural Language and Linguistic Theory* 12, 1-61.

Müller, Gereon (1999): Optimality, Markedness, and Word Order in German. *Linguistics* 37, 777-818.

Müller, Stefan (2003): *Mehrfache Vorfeldbesetzung*. *Deutsche Sprache* 31, 29–62.

Müller, Stefan (2005): *Zur Analyse der scheinbar mehrfachen Vorfeldbesetzung. Linguistische Berichte* 203, 29–62.

Musan, Renate (2002): Informationsstrukturelle Dimensionen im Deutschen. Zur Variation der Wortstellung im Mittelfeld. *Zeitschrift für germanistische Linguistik* 30, 198-221.

Nakanishi, Kimiko (2004): *Domains of Measurement: Formal Properties of Non-Split/Split Quantifier Constructions.* PhD dissertation, University of Pennsylvania.

Nespor, Marina and Irene Vogel (1979): Clash Avoidance in Italian. *Linguistic Inquiry* 10, 467-482.

Nespor, Marina and Irene Vogel (1986): *Prosodic Phonology*. Dordrecht: Foris.

Newell, Heather (2006): Bracketing Paradoxes and Particle Verbs: A Late Adjunction Analysis. In: *Proceedings of ConSOLE XIII*.

Newman, Stanley S. (1946): On the Stress System of English. *Word* 2, 171-187.

Odden, David (1980): Associative tone in Shona. *Journal of Linguistic Research* 1,2, 37-51.

Paul, Hermann (1875): *Principien der Sprachgechichte*. Halle/Saale: Niemeyer.

Petrova, Svetlana & Michael Solf (2007): On the Methods of Information-Structural Analysis of Texts from Historical Corpora. A Case Study on the OHG Tatian. In Hinterlhölzl, Roland & Svetlana Petrova (Hgg.): *New Approaches to Word Order Variation and Word Order Change*. Berlin: Mouton de Gruyter.

Pierrehumbert, Janet B. (1980): *The phonology and phonetics of English intonation*. PhD dissertation, Massachusetts Institute of Technology.

Pintzuk, Susan (1999): *Phrase Structures in Competition*. New York / London: Garland.

Pintzuk, Susan & Anthony S. Kroch (1989): The rightward movement of complements and adjuncts in the Old English of Beowulf. *Language Variation and Change* 1, 115-143.

Price, P., M. Ostendorf, S. Shattuck-Hufnagel & C. Fong (1991): The use of Prosody in syntactic disambiguation. *Journal of the Acoustic Society of America* 90, 2956-2970.

Prince, Alan (1983): Relating to the Grid. *Linguistic Inquiry* 14, 19-100.

Prince, Ellen F. (1981a): Toward a Taxonomy of Given-New Information. In: Peter Cole (ed.): *Radical Pragmatics*. New York: Academic Press, 223-255.

Prince, Ellen F. (1981b): Topicalization, Focus-movement and Yiddish Movement. *Berkeley Linguistic Society* 7, 249-264.

Prince, Ellen F. (1986): On the Syntactic Marking of Presupposed Open Propositions. In: A. Farley et al. (eds.), *Papers from the Parasession on Pragmatics and Grammatical Theory, 22nd Regional Meeting*. Chicago: Chicago Linguistic Society, 208-222.

Prince, Ellen F. (1999): How Not to Mark Topics: 'Topicalization' in English and Yiddish. In: *Texas Linguistics Forum,* chapter 8. Austin: University of Texas (ftp://babel.ling.upenn.edu/papers/faculty/ellen_prince//texas.ps).

Reinhart, Tanya (1981): Pragmatics and linguistics: an analysis of sentence topics. *Philosophica* 27, 53–94.

Reis, Marga (1987). Die Stellung der Verbargumente im Deutschen. Stilübungen zum Grammatik:Pragmatik-Verhältnis. In: Rosengren, Inger (ed.): *Sprache und Pragmatik. Lunder Symposion 1986*. Stockholm: Almqvist & Wiksell, 139-177.

Ries, John (1907): *Die Wortstellung im Beowulf*. Halle/Saale: Niemeyer.

Ringe, Donald A. (2006): *A Linguistic History of English. Volume I: From Proto-Indo-European to Proto-Germanic*. Oxford: OUP.

Rizzi, Luigi (1997): The fine structure of the left periphery. In: Haegeman, Liliane (ed.): *Elements of Grammar*. Dordrecht: Kluwer, 281-337.

Robinson, Orrin W. (1997): *Clause Subordination and Verb Placement in the Old High German Isidor Translation*. Heidelberg: Winter.

Rochemont, Michael S. (1986): *Focus in Generative Grammar*. Amsterdam/ Philadelphia: Benjamins.

Rosengren, Inger (1994): Scrambling – was ist das? In: Haftka, Brigitte (ed.): *Was determiniert Wortstellungsvariation?* Opladen: Westdeutscher Verlag, 175-196.

Rooth, Mats E. (1985): *Association with Focus*. PhD dissertation, University of Massachusetts.

Santorini, Beatrice (1992): Variation and Change in Yiddish Subordinate Clause Word Order. *Natural Language and Linguistic Theory* 10, 595-640.

Santorini, Beatrice (1993): The rate of phrase structure change in the history of Yiddish. *Language Variation and Change* 5, 257-283.

Santorini, Beatrice (1995): Two Types of Verb Second in the History of Yiddish. In: Battye, Adrian & Ian Roberts (eds.): *Clause Structure and Language Change*. Oxford: OUP, 53-79.

Schane, Sanford A. (1979): Rhythm, Accent and Stress in English Words. *Linguistic Inquiry* 10, 483-502.

Schein, Berry and Donca Steriade (1986): On Geminates. *Linguistic Inquiry* 17, 691-744.

Schmerling, Susan F. (1976): *Aspects of English Sentence Stress*. Austin: University of Texas Press.

Seidl, Amanda (2001): *Minimal indirect reference: a theory of the syntax-phonology interface*. New York: Routledge.

Selkirk, Elisabeth O. (1982): *The Syntax of Words*. Cambridge, Ma.: MIT Press.

Selkirk, Elisabeth O. (1984): *Phonology and Syntax: The Relation between Sound and Structure*. Cambridge, Ma.: MIT Press.

Selkirk, Elisabeth (1995): Sentence Prosody: Intonation, Stress and Parsing. In: Goldsmith, John A. (ed.): *The Handbook of Phonological Theory*. Oxford: Blackwell, 550-569.

Sgall, Petr, Eva Hajičová & Eva Benešová (1973): *Topic, Focus and Generative Semantics*. Scriptor, Kronberg.

Sluijter, Agaath (1995): *Phonetic Correlates of Stress and Accent.* Den Haag: Holland Academic Graphics.

Speyer, Augustin (2004): Competing Constraints on *Vorfeldbesetzung* in German. In: Shaer, Benjamin, Werner Frey & Claudia Maienborn (eds.): *Proceedings of the Dislocated Elements Workshop. ZASPiL* 35 vol.2, 519-541.

Speyer, Augustin (2005): A prosodic factor for the decline in Topicalisation in English. In: Stephan Kepser and Reis, Marga (eds.) *Linguistic Evidence,* Berlin: Mouton de Gruyter, 485-506.

Speyer, Augustin (2006a): German Vorfeld-filling as Constraint Interaction. To appear in: Benz, Anton & Peter Kühnlein (eds.) *Constraints in Discourse*, Amsterdam/ Philadelphia: Benjamins.

Speyer, Augustin (2006b): Bracketing Paradoxes und wordinterne Adjunktion. Paper read at the ggs-conference (Generative Grammatik des Südens), Stuttgart, May 26-28 2006.

Speyer, Augustin (2006c): On the change of word stress in the history of German. Ms., University of Pennsylvania, sent to *Beiträge zur Geschichte der deutschen Sprache und Literatur*.

Speyer, Augustin (2007a): The structure of the Prepositional Phrase in Umbrian. Ms., University of Pennsylvania.

Speyer, Augustin (2007b): Doppelte Vorfeldbesetzung heute und im Frühneuhochdeutschen. Talk given at the workshop ‚Satzstrukturen im Frühneuhochdeutschen', June 22, 2007, University of Saarbrücken, Germany.

Speyer, Augustin (2007c): Ist die lateinische Wortstellung wirklich beliebig? Invited talk at the Institutscolloquium of the Philologisches Seminar, University of Tübingen, June 12, 2007.

Steriade, Donca (1982): *Greek Prosodies and the Nature of Syllabification*. PhD dissertation, Massachusetts Institute of Technology.

Steube, Anita (2003): Bridge Contours in German assertive main clauses, *Folia Linguistica* 37, 163-190.

Taglicht, Josef (1998): Constraints on intonational phrasing in English. *Journal of Linguistics* 34, 181-211.

Thurneysen, Rudolf (1946): *A Grammar of Old Irish*. Translated by D.A. Binchy & Osborn Bergin. Dublin: Dublin Institute for Advanced Studies.

Tomaselli, Alessandra (1995): Cases of Verb Third in Old High German. In: Battye, Adrian & Ian Roberts (eds.): *Clause Structure and Language Change*. Oxford: Oxford University Press, 345-369.

Trips, Carola (2002): *From OV to VO in Early Middle English*. Amsterdam/ Philadelphia: Benjamins.

Truckenbrodt, Hubert (1999): On the relation between syntactic phrases and phonological phrases. *Linguistic Inquiry* 30, 219-255.

Truckenbrodt, Hubert (2006): Phrasal Stress. In: Brown, Keith (ed.): *Elsevier Encyclopedia of Languages and Linguistics*. 2nd ed., vol. 9. Amsterdam: Elsevier, 572-579.

Truckenbrodt, Hubert (2007): The syntax-phonology interface. In: de Lacy, Paul (ed.): *The Cambridge Handbook of Phonology*. Cambridge: Cambridge University Press, 435-456.

Vallduví, Enric & Maria Vilkuna (1998) On rheme and kontrast. In: P. Culicover and L. McNally (eds.): *Syntax and Semantics 29: The limits of syntax.* New York: Academic Press, 79–108.

Vikner, Sten (1995). *Verb Movement and Expletive Subjects in the Germanic Languages.* Oxford / New York: Oxford University Press.

Walker, Marilyn A., Aravind K. Joshi & Ellen F. Prince (1998). Centering in naturally occurring discourse: an overview. In *Centering Theory in Discourse*, Marilyn A. Walker, Aravind K. Joshi & Ellen F. Prince (eds.), 1-28. Oxford: Oxford University Press.

Welby, Pauline (2003): Effects of Pitch Accent Position, Type, and Status on ***Focus*** Projection. *Language and Speech* 46, 53-81.

Wells, J.C. (2006) *English Intonation. An introduction*. Cambridge: CUP.

Yip, Moira (1988): The Obligatory Contour Principle and phonological rules: A loss of identity. *Linguistic Inquiry* 19, 65-100.